

METHODOLOGICAL REPORT

Development of a clinical practice guideline for the pharmacological treatment of symptoms in the last days of life in nursing home care

Tim Biesbrouck (MD)^{1,2,3}, Maaïke De Roo (MD, PhD)^{1,2}, Dine AD Jennes (MD)^{4,5}, Nele Van Den Noortgate (MD, PhD)^{3,6}

1 Department of Geriatric Medicine, University Hospital Leuven, Leuven, Belgium

2 Department of Public Health and Primary Care, Gerontology and Geriatrics, KU Leuven, Leuven, Belgium

3 Department of Geriatric Medicine, Ghent University Hospital, Ghent, Belgium

4 Department of Geriatric Medicine, Antwerp University Hospital, Edegem, Belgium

5 Department of Geriatric Medicine, Antwerp University, Edegem, Belgium

6 End-of-life Care Research Group, Vrije Universiteit Brussel (VUB) and Ghent University, Laarbeeklaan, Brussels, Belgium

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Final report of the guideline development process and methodology

PREFACE

Throughout the guideline development process we have maintained a high scientific and methodological quality by following the “WOREL leidraad voor richtlijnontwikkeling”. Despite extensive searches for existing guidelines and evidence on the treatment of symptoms during the last days of nursing home residents only one Australian guideline of moderate scientific quality was found. Furthermore, six systematic reviews were conducted searching for evidence on the pharmacological treatment of pain, dyspnea, nausea and vomiting, death rattle, fever and psychological symptoms. However no articles were retrieved for fever or nausea and vomiting. Few articles were retrieved for pain, dyspnea and psychological symptoms, all of them descriptive with few implications for clinical practice. Lastly the systematic review for death rattle did provide some randomized controlled trials, however it remains unclear whether or not death rattle should be treated in general. Despite this lack of evidence, a guideline is needed on the pharmacological treatment of symptoms to help general practitioners in alleviating residents’ symptoms during the last days of life. Therefore the decision was made to adapt the previously mentioned Australian guideline to the Belgian context, supplemented with the available scientific evidence. All recommendations formulated were evaluated in an extensive Delphi procedure by Belgian experts. The result of this process is a practical guideline, mainly based on expert opinion, that can guide general practitioners in the treatment of symptoms in the last days of life of frail older people. This guideline is part of the developing ‘Nursing Home Care Program for the Last Days of Life’, which will focus on all aspects concerned with care in the last days of life of nursing home residents.

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Chapter 1: General declarations

1.1 Initiative and financing

This clinical practice guideline is part of the development of the PhD project called 'Development, implementation and evaluation of the Nursing Home Care Program for the Last Days of Life' and is accomplished by the End-Of-Life Care research group and funded by 'Kom op tegen Kanker' vzw.

This care program exists of:

- A clinical guideline for the pharmacological treatment of symptoms in the last days of life.
- A care guide for the last days of life, which includes key points about recognizing the dying phase, communication with family/relatives, monitoring symptoms, psychosocial and spiritual support, and interprofessional and transmutal communication and collaboration.
- An implementation strategy that should assist the staff in nursing homes to integrate the Care Program into daily practice.

This entire Care Program will be implemented and evaluated in eight nursing homes. This will be done through participatory action research, where the staff of the participating nursing homes will actively shape the implementation and put the care program into practice, adjusting it as necessary.

This methodological report focuses on the development of the clinical guideline for pharmacological treatment of symptoms in the last days of life in nursing home residents.

1.2 Objectives

The aim is to develop a CEBAM[^] validated evidence-based practice guideline for use in the 'Care Guide for the Last Days of Life'. The guideline will offer recommendations to health care providers about the pharmacological treatment of end-of-life symptoms in nursing home care. Where evidence is lacking, recommendations will be formulated based on expert consensus.

1.3 Accountability

The guideline will offer recommendations based on the synthesis of available literature supplemented with expert opinion. The use of the recommendations in clinical practice is the responsibility of the individual health care provider who weighs benefit and risk of harm for every individual patient.

[^] Belgian Centre for Evidence-Based Medicine –Cochrane Belgium

Chapter 2: Guideline development group and conflict of interests

2.1 Guideline development group (GDG)

Dr. Tim Biesbrouck (MD) is a geriatrics resident at University Hospital Leuven and joint-PhD student of the gerontology and geriatrics research group of KU Leuven and Geriatrics research group at Ghent University Hospital. His PhD focuses on the Care in the last days of life of geriatric patients.

Dr. Maaïke De Roo (MD, PhD) is the deputy head of the Geriatrics department at UZ Leuven and an associated senior clinical staff member within the Gerontology and Geriatrics research group at KU Leuven, focusing on the research line Palliative Care for Older Adults. She is a board member of the Belgian Young Geriatricians and serves on the Board of Directors of nursing home.

Dr. Dine Jennes (MD) is a geriatrician at University Hospital Antwerp. She graduated in 2015 as a general practitioner, during which time she frequently encountered nursing home residents in their last days of life. She later specialized in internal medicine and geriatrics.

Professor Dr. Nele Van Den Noortgate (MD, PhD) is the medical head of the care program for geriatric patients and palliative care at Ghent University Hospital, part-time professor at Ghent University, and an active member of both national (chair of the Belgian Society of Geriatrics and Gerontology; member of the palliative care and geriatrics working group of Palliative Care Flanders) and international scientific associations (executive board of the European Academy for Medicine of Ageing). She was a CRA for over ten years in a nursing home, and currently still is a member of the CRAgentseTehuizen working group, and serves on the board of directors of a non-profit organization managing thirteen nursing homes. She frequently provides advice and training on end-of-life care in nursing homes."

2.2 Expert group

A list of the experts invited for the consensus procedure can be found in Appendix A.

2.3 Stakeholder group

Stakeholders are all health care personnel involved in the care of nursing home residents at the end of life, health care personnel with expertise in both general and specialist palliative care, residents of nursing homes and family members. Nursing home residents and relatives will be involved in the implementation phase of the care guide, of which these pharmacological recommendation is only one section. In the development process of these pharmacological recommendations we opted not to involve nursing home residents and relatives, as these recommendations have very specific and specialistic character.

2.4 Declaration of interests

All members of the guideline development group declare to have no conflicts of interest. Conflict of interest documents have been signed and are available upon request.

Chapter 3: Timing and development

Table 1: Timeline	
August 2020	Scope and background Formulation clinical research questions
August 2020 ↓ October 2020	Searching for relevant guidelines in guideline databases
October 2020 ↓ January 2021	Screening relevant guidelines on content Selection of relevant guidelines
January 2021 ↓ March 2021	Determining methodological quality of relevant guidelines Identifying guideline for ADAPTE procedure
March 2021 ↓ April 2021	Development of the combination of an ADAPTE and DE NOVO procedure
April 2021 ↓ November 2021	Development review strategy and search strategy for DE NOVO procedure MEDLINE via PubMed
December 2021 ↓ January 2022	Translation of search strategy to EMBASE
March 2022	Literature searches
February 2022 ↓ November 2022	Screening of articles
November 2022 ↓ March 2023	Formulation of recommendations, development of Delphi procedure and obtaining ethics approval for Delphi procedure
March 2023	Update literature searches
April 2023 ↓ December 2023	Delphi procedure (existing out of 3 rounds)
December 2023 ↓ January 2024	Final development phase of guideline
19/2/2024 ↓ 14/05/2024	Submitted with CEBAM
11/05/2024 ↓ 31/10/2024	Revision process
04/11/2024	Resubmission with CEBAM

Chapter 4: Scope

4.1 Title

Clinical practice guideline for the pharmacological treatment of end-of-life symptoms in nursing home care.

4.2 Background, relevance and indication for the development of this guideline

This guideline is part of the development of a nursing home care program for the last days of life.

4.2.1 Background and relevance

The number of people dying in nursing homes is increasing, in 2019 this accounted for over 30% of all deaths in Flanders (1). A nursing home is considered a unique environment because of its differences and similarities with both ambulatory and hospital care (2). The average person dying in nursing homes is female, 85 years or older, with multiple comorbidities and suffering from moderate to severe cognitive impairment (3). Due to this population's medical complexities with physical frailty they are at an increased risk of medical complications near the end-of-life. The recent COVID-19 pandemic further emphasized the need for better medical guidance in nursing homes. Medications such as opioids and benzodiazepines are being extensively used to achieve symptom control in palliative care (4). This however without any foundational evidence or guidelines for the use in geriatric patients, causing a trend towards lower opioid doses in older patients (5). This combination puts our population at a high risk of unnecessary symptom burden, suffering and a decreased quality of dying, while this should be avoided at all costs.

4.2.2. Preliminary investigation

The development of this guideline is based on a preliminary investigation (part of a Master after Master thesis) that was conducted by 2 general practitioners in training, Thomas Vandembemt MD and Joost Van den Berg MD under the supervision of Nele Van den Noortgate PhD, MD and Jan De Lepeleire PhD, MD (6). The orientating research can be found in the thesis of Thomas Vandembemt MD which focused on the recognition of the end-of-life or terminal phase and symptom control (6). Written consent for the use of the above mentioned thesis was obtained and can be requested from the guideline development group.

Additionally, as part of the larger implementation project interviews were performed with relatives of nursing home residents, nursing home staff members and members of the palliative care geriatrics workgroup (PaGe).

4.2.2.1 Qualitative study – orientation

These two general practitioners in training conducted a qualitative survey based on semi-structured interviews of 20 different caregivers who are involved in the treatment and care of dying patients in nursing homes with different levels of experience. The persons interviewed were general practitioners (45%), general practitioners in training (35%) and nursing staff in nursing homes (20%). Especially younger general practitioners and practitioners in training expressed the need for guidance based on scientific evidence. They indicated that their medical interventions are predominantly experience-based and based on advice from senior practitioners (6).

The additional interviews confirmed the earlier findings. The current approach by general practitioners is perceived as highly variable, with some avoiding or delaying pharmacological symptom management in the final days of life,

while others may resort to sedation too quickly. Less experienced doctors often tend to leave the pharmacological management of symptoms largely to the nurses. The inclusion of a pharmacological guideline in the care program for the last days of life is viewed as a valuable addition to current practice, providing guidance and improving consistency in care.

4.2.2.2 Definition of the terminal phase: recognizing a dying patient

Timely recognition of dying patients is challenging. There are no specific signs or parameters to indicate the beginning of the terminal phase. Vandenbempt (2019) summarized the symptomatology seen in the terminal phase in his systematic review (Table 2). These symptoms in combination with scores such as the Palliative Prognostic Index can be useful to recognize dying patients (7).

In the thesis of Vandenbempt (2019) the terminal phase is not defined in clear time intervals (hours, days or weeks) but rather in the appreciation of the caregiver that the dying process has begun. A clear distinction is made between the terminal phase and the broader palliative phase. In the palliative phase focus shifts from curation or stabilization of morbidities to comfort without death being imminent. This larger palliative phase falls outside of the scope of this guideline that focuses on the pharmacological treatment of symptoms once the terminal phase has begun.

In consultation with the members of the guideline development group a time limited definition was chosen. The definition for the terminal phase that will be applied from here on will be: “the phase of hours and days prior to death”. This is in line with the definition that will be applied in the care guide and will add to uniformity in the selection and interpretation of recommendations.

Table 2: Symptoms indicating the beginning of the terminal stadium	
Agitation or delirium	Cheyne-Stokes breathing
Reduced consciousness or coma	Loud and rattling breath: death rattle
Decreased intake of fluids and food	Progressive loss of weight
Increased fatigue and somnolence	Impaired function to swallow
Marbled skin	Terminal fever with cold extremities
Low urinary output and dark urine	Impaired mobility and loss of strength
Impaired speech and decreased social interactions	

4.3 The clinical practice guideline

4.3.1 Population

Target population of this guideline

- Frail older patients (≥ 65 years) dying in nursing homes
- Frail patients will be defined as patients with loss of functionality and/ or poly-pathology

Population that is NOT a part of this guideline

- Frail older patients in a palliative phase preceding the terminal or end-of-life phase
- Robust older patients
- Patients younger than 65 years

4.3.2 Setting

Guidance that will be formulated in this guideline has the specific intention to be implemented in nursing homes.

Target setting for which this guideline is intended

- Nursing home: long-term care
- Nursing home: short-term care

Settings for which this guideline is NOT intended

- Assisted living accommodation
- Rehabilitation centers
- Service residence or long-term care facility without 24 hours nursing permanence
- Ambulatory care
- Hospitals

4.3.3 Key Aspects

Which aspects will be addressed in this clinical practice guideline?

- Pharmacological treatment of terminal or end-of-life symptoms
- If possible advice about dosage and routes of administration
- Aspects concerning safety and practical administration of the pharmacological treatment

Which aspects are NOT a part of this clinical practice guideline?

- Non-pharmacological treatments of end-of-life symptoms
- Communication aspects of end-of-life care
- After death care
- Palliative sedation or the treatment of medical therapy refractory symptoms
- Parenteral fluid administration or parenteral nutrition
- Dietary advice

4.3.4 Selection of the clinical research questions

The selection of the clinical questions is based on the preliminary work of Vandembemt (2019). In his thesis, literature review revealed which symptoms occur frequently at the end of life. The following symptoms were selected: pain, vomiting, dyspnea, death rattle, agitation and delirium. Based on the guideline development group's experience, fever and anxiety were added to the list of symptoms. Although fever and anxiety are less described in the literature, clinical experts from the guideline development group argued their relevance in clinical practice.

To facilitate practical implementation of the guidance, a final clinical question was added. In the terminal phase multiple symptoms can co-exist and practical issues about the co-administration of drugs can arise. The final clinical research question aims to address the best and safest way to administrate drugs alone or in combination in the setting of a nursing home. An overview of the different clinical questions and outcome measures can be found in Chapter 5.

Chapter 5: Objectives and clinical research questions

5.1 Synthesis of the objectives

5.1.1 Objective

- To offer support and guidance to the caregiver in regard to the pharmacological treatment of end-of-life symptoms in nursing home care
- To formulate pharmacological recommendations for the improvement of care and comfort of the frail and older patient, dying in a nursing home

5.1.2 Target group

- General practitioners
- Coordinating and advisory physicians in nursing homes (Belgian CRA)
- Nursing staff in nursing homes

5.1.3 Target population

- Frail and older residents of a nursing home in their last days of life
- 65 years or older
- In a terminal phase

5.2 Clinical research questions

5.2.1 Scientific basis (see Chapter 4)

- Preliminary research was done by 2 general practitioners in training with 20 semi-structured interviews with stakeholders
- Review of the literature for the Master after Master's thesis by Vandenbempt (2019)
- Clinical expertise of promoters of the above mentioned Master after Master's thesis, their clinical experience and the experience of the guideline development group

5.2.2 Background questions (see Chapter 4)

- How can we define the terminal phase or end-of-life phase?
- How can we recognize the beginning of the terminal phase?
- Which symptoms are common in the terminal phase?

5.2.3 Clinical research questions

5.2.3.1. Symptom control: clinical research questions 1-6

Table 3: PIPOH clinical research question 1-6	
Clinical research question 1: Which pharmacological treatment is advised for pain control in the last days of life in nursing home care?	
PIPOH	
Patient population / disease characteristics	Terminal nursing home resident ≥ 65 years
Intervention(s) of interest	Pharmacological treatment of pain
Professionals targeted by the guideline	Primary care physicians and coordinating and advisory physicians in nursing homes Nursing home staff
Outcomes and endpoints to be taken into consideration	Comfort and control of pain
Healthcare settings and context	Nursing home care
Clinical research question 2: Which pharmacological treatment is advised for control of dyspnea in the last days of life in nursing home care?	
PIPOH	
Patient population/ disease characteristics	Terminal nursing home resident ≥ 65 years
Intervention(s) of interest	Pharmacological treatment of dyspnea
Professionals targeted by the guideline	Primary care physicians and coordinating and advisory physicians in nursing homes Nursing home staff
Outcomes and endpoints to be taken into consideration	Comfort and control of dyspnea
Healthcare settings and context	Nursing home care
Clinical research question 3: Which pharmacological treatment is advised for control of nausea and vomiting in the last days of life in nursing home care?	
PIPOH	
Patient population/ disease characteristics	Terminal nursing home resident ≥ 65 years
Intervention(s) of interest	Pharmacological treatment of nausea and vomiting
Professionals targeted by the guideline	Primary care physicians and coordinating and advisory physicians in nursing homes Nursing home staff
Outcomes and endpoints to be taken into consideration	Comfort and control of nausea and vomiting
Healthcare settings and context	Nursing home care

Clinical research question 4: Which pharmacological treatment is advised for control of death rattle in the last days of life in nursing home care?

PIPOH

Patient population/ disease characteristics	Terminal nursing home resident ≥ 65 years
Intervention(s) of interest	Pharmacological treatment of death rattle
Professionals targeted by the guideline	Primary care physicians and coordinating and advisory physicians in nursing homes Nursing home staff
Outcomes and endpoints to be taken into consideration	Comfort and control of death rattle
Healthcare settings and context	Nursing home care

Clinical research question 5: Which pharmacological treatment is advised for control of fever in the last days of life in nursing home care?

PIPOH

Patient population/ disease characteristics	Terminal nursing home resident ≥ 65 years
Intervention(s) of interest	Pharmacological treatment of fever
Professionals targeted by the guideline	Primary care physicians and coordinating and advisory physicians in nursing homes Nursing home staff
Outcomes and endpoints to be taken into consideration	Comfort and control of fever
Healthcare settings and context	Nursing home care

Clinical research question 6*:

Which pharmacological treatment is advised for control of psychological symptoms in the last days of life in nursing home care?

6a: Which pharmacological treatment is advised for control of agitation and restlessness in the last days of life in nursing home care?

PIPOH

Patient population/ disease characteristics	Terminal nursing home resident ≥ 65 years
Intervention(s) of interest	Pharmacological treatment of agitation and restlessness
Professionals targeted by the guideline	Primary care physicians and coordinating and advisory physicians in nursing homes Nursing home staff
Outcomes and endpoints to be taken into consideration	Comfort and control of agitation and restlessness
Healthcare settings and context	Nursing home care

* The initial research question number 6 was: Which pharmacological treatment can be recommended for the management of agitation and delirium in the last days of life in nursing home care. During the development process this clinical question was divided in three different sub-questions because of better compatibility with the clinical question in the source guideline chosen for adaptation. Adding the 2 other questions (6b and 6c) assures the coverage of the full content of our initial clinical question without compromise.

6b: Which pharmacological treatment is advised for control of confusion and delirium in the last days of life in nursing home care?

PIPOH

Patient population/ disease characteristics	Terminal nursing home resident ≥ 65 years
Intervention(s) of interest	Pharmacological treatment of confusion and delirium
Professionals targeted by the guideline	Primary care physicians and coordinating and advisory physicians in nursing homes Nursing home staff
Outcomes and endpoints to be taken into consideration	Comfort and control of confusion and delirium
Healthcare settings and context	Nursing home care

6c: Which pharmacological treatment is advised for control of anxiety in the last days of life in nursing home care?

PIPOH

Patient population/ disease characteristics	Terminal nursing home resident ≥ 65 years
Intervention(s) of interest	Pharmacological treatment of anxiety
Professionals targeted by the guideline	Primary care physicians and coordinating and advisory physicians in nursing homes Nursing home staff
Outcomes and endpoints to be taken into consideration	Comfort and control of anxiety
Healthcare settings and context	Nursing home care

5.2.3.2. Practical Implementation: clinical research question 7

Clinical research question 7 was part of a master's thesis of pharmacy students, and can be found in Appendix O.

Table 4: PIPOH clinical research question 7: How can we administer different pharmacological agents for the treatment of symptoms in the last days of life in nursing home care in a safe and practical way?

Patient population/ disease characteristics	Terminal nursing home resident ≥ 65 years
Intervention(s) of interest	Combination of different pharmacological therapies.
Professionals targeted by the guideline	Primary care physicians and coordinating and advisory physicians in nursing homes Nursing home staff
Outcomes and endpoints to be taken into consideration	Practical aspects and safety
Healthcare settings and context	Nursing home care

5.3 Outcome measures

5.3.1 Outcome measures for clinical research questions regarding symptom control (research questions 1-6)

Critical outcome measures:

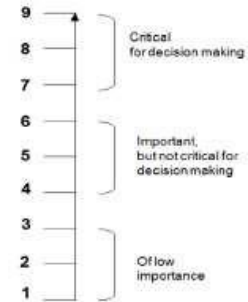
- Patient comfort (9)
- Symptom control (7)

Important outcome measures:

- Undesirable side effects (6)

Outcome measures of lesser importance:

- Life span prolongation (2)
- Life span shortening (2)



5.3.2 Outcome measures for practical implementation: clinical research question 7

Critical outcome measures:

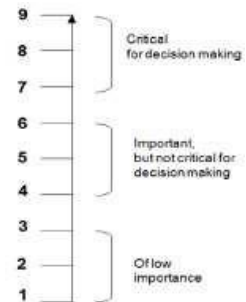
- Safety (9)
- Availability (8)
- User friendliness (7)

Important outcome measures:

- Pharmacological interaction with clinical significance (6)
- Cost per unit (5)

Outcome measures of lesser importance:

- Pharmacological interaction without clinical significance (1)



Chapter 6: ADAPTE procedure

The methodology for the development of this guideline is based on the manual for guideline development by the 'Working group development of primary care guidelines' ('Werkgroep Ontwikkeling Richtlijnen Eerste Lijn' (WOREL)).

6.1 Search strategy

6.1.1. Determination of the search strategy

Guidelines were searched in the summer of 2020 in three obligatory guideline databases according to WOREL guidelines:

- Ebpracticenet (www.ebpnet.be)
- EBM or Duodecim
- G.I.N. (Guidelines International Network)

Additionally searched international guideline databases:

- NICE (National Institute for Health and Care Excellence)
- SIGN (Scottish Intercollegiate Guidelines Network)
- American Guideline Central and SUM search
- NHG (Dutch Society of General practitioners)
- NVKG (Dutch Society of Clinical Geriatrics) and the Dutch Society of Medical specialists.
- Guidelines from IKNL (Integrated Cancer Centre) were searched via pallialine.nl and the Belgian version pallialine.be

Members of the guideline development group suggested a specific search for Australian guidelines because of the reputation of Australian physicians and their knowledge and attention for palliative care. The website Caresearch and the official website of the Australian government (www.clinicalguidelines.gov.au/) were hand searched.

6.1.2. Determination of in- and exclusion criteria

In the first phase of guideline identification, the inclusion criteria were applied broadly to maximize sensitivity. Guidelines concerning palliative care and guidelines containing palliative care recommendations were withheld for further evaluation. Context and target population were not taken into account in this stage of the screening. Searches varied slightly between databases (as presented in Appendix B) and were adapted according to the characteristics of the databases in order to optimize search results. In small databases that did not contain a 'search' function, hand searching of the available guidelines was conducted based on title. There were no limits applied to publication date. Only Dutch and English language databases were searched.

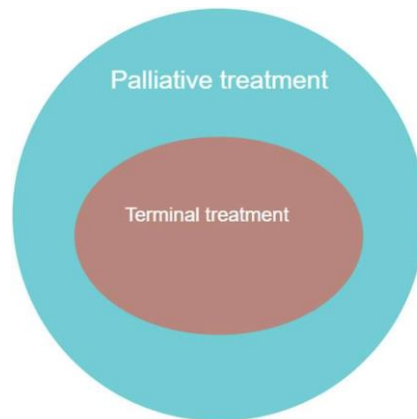
6.1.3. Guideline databases: summary of the search strategy and search results

Results can be found in Appendix B.

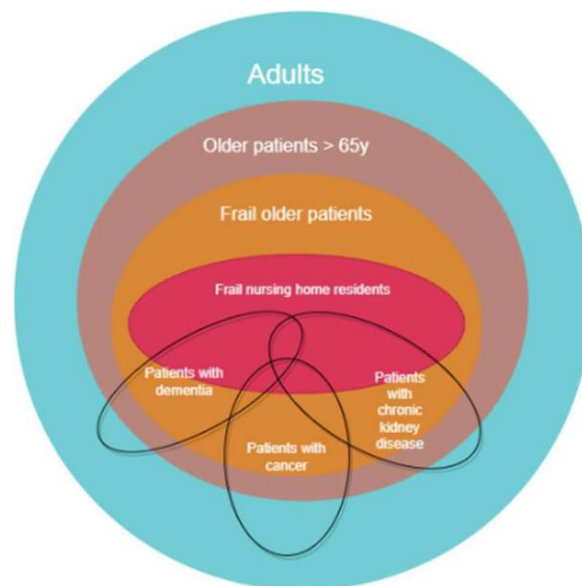
6.2 Guideline selection

6.2.3 Screening based on content

The guidelines that were identified through different guideline databases were screened for content and relevance. The search strategy and results can be found in Appendix B. As the terminal phase is part of the palliative phase and to optimize sensitivity, all guidelines in regard to the pharmacological treatment of symptoms in the palliative phase were selected for further evaluation. Protocols and articles that did not meet the criteria of 'guideline' were excluded in a second step. Synthesis of systematic reviews and clinical trials were excluded from further evaluation.



In a third step, pathology specific guidelines were excluded. There was consensus among researchers that pathology specific guidelines were not eligible for inclusion because of a significant difference of the population in these guidelines and the target population. See Appendix C: Guideline screening and selection for the results.



6.2.2 Evaluation of the development process

Because of the large number of guidelines that were selected based on broad content criteria, the decision was made to separate the qualitative evaluation in two separate rounds. In a first round a MINI-AGREE procedure was done to identify the guidelines for which the full 23 item AGREE was executed.

6.2.3.1 MINI-AGREE II

Following WOREL procedure, the MINI-AGREE used a three item score of the AGREE-II assessment (item 7, 9, 23) to easily and efficiently eliminate the less qualitative guidelines. A fourth screening criterion based on clinical content similarity between the source-guideline and the target-guideline was added. Because of the very specific target population in the target guideline, additional weight was attributed to content similarity (even when lacking methodological rigidity).

Scoring was done by two researches independently. In- and exclusion was based on the mean score by the two researches and the following predetermined criteria:

AGREE-II criteria	Minimal score for inclusion
Item 7: Systematic methods were used to search for evidence	Score of 4 or more
AND	
Item 9: The strengths and limitations of the body of evidence are clearly described.	Score of 4 or more
AND	
Item 23: Competing interests of guideline development group members have been recorded and addressed.	Score of 4 or more

OR

Additional criterium	Minimal score for inclusion
The content of the source-guideline is very close to the scope and clinical research questions in the target-guideline.	Score of 6 or more

Scores assigned were according to the 7-point scale as predetermined in the AGREE II instrument (Update 2017). The following guidelines achieved a sufficient high score to enter the complete AGREE-II evaluation:

- Palliative care (symptom control)/ Palliatieve zorg (symptoomcontrole) (KUL 2020)
- Dyspnea/ Dyspnee (GIN, IKNL 2010)
- Delirium in adults/ Delier bij volwassenen (NVKG, 2013)
- Palliative care for adults: strong opioids for pain relief (GIN/NICE 2012)
- Care of dying adults in the last days of life (GIN/NICE, 2015)
- Guide to the pharmacological management of end of life (terminal) symptoms in residential aged care residents (BSPCC, 2015)

A more detailed representation of the MINI-AGREE score results can be found in Appendix D.

Detailed domain scores for AGREE-II can be found in Appendix E.

Table 5: Agree II scores					
	Domain	R1	R2	Average	Difference
Palliative care (symptom control)/ Palliatieve zorg (symptoomcontrole) (KUL 2020)					
	Domain 1	17	21	19	-4
	Domain 2	14	16	15	-2
	Domain 3	34	34	34	0
	Domain 4	19	20	19,5	-1
	Domain 5	18	16	17	2
	Domain 6	14	14	14	0
					-5
Dyspnea/ Dyspnee (GIN, IKNL, 2010)					
	Domain 1	19	21	20	-2
	Domain 2	25	27	26	-2
	Domain 3	35	42	38,5	-7
	Domain 4	21	21	21	0
	Domain 5	18	13	15,5	5
	Domain 6	11	11	11	0
					-6
Delirium in adults/ Delier bij volwassenen (NVKG, 2013)[§]					
Palliative care for adults: strong opioids for pain relief (GIN/ NICE, 2012)					
	Domain 1	20	20	20	0
	Domain 2	28	27	26,5	1
	Domain 3	41	44	42,5	-3
	Domain 4	12	21	16,5	-7
	Domain 5	22	23	22,5	-3
	Domain 6	12	14	13	-2
					-14
Guide to the pharmacological management of end of life (terminal) symptoms in residential aged care residents (BSPCC, 2015)					
Care of dying adults in the last days of life (GIN/ NICE, 2015)[°]					
	Domain 1	20	19	19,5	1
	Domain 2	26	24	25	2
	Domain 3	46	46	46	0
	Domain 4	16	19	17,5	-3
	Domain 5	19	27	23	-8
	Domain 6	13	13	13	0
					-8

[§]During the AGREE II evaluation of the guideline for delirium we noticed that terminal delirium or delirium in a palliative setting was used as an exclusion criterium for the systematic review. With this knowledge, the evaluation was halted and the guideline was excluded.

[°] Authors of this guideline were contacted in order to obtain the available methodological data on the development of this guideline (See appendix H). A methodological report could not be retrieved which is the reason the AGREE-II evaluation could not be conducted.

6.2.3 Selection process of relevant and qualitative guidelines

6.2.3.1 Evidence based guideline: Care of dying adults in the last days of life (NICE)

Based on the AGREE II score system, the two NICE guidelines were the only two to meet quality criteria. However, their target population are adults. Although the adult population encompasses the frail and older population in a nursing home, the differences are significant. After discussion in the guideline development group, the discrepancy between target populations were considered too significant to consider the guideline for an ADAPTE procedure.

6.2.3.2 Consensus based guideline: “Guide to the pharmacological management of end-of-life (terminal) symptoms in residential aged care residents” from the Brisbane South Palliative Care Collaborative

Grey literature search identified an Australian guideline from 2015 called “Guide to the pharmacological Management of End Of Life (Terminal) Symptoms in Residential Aged Care Residents” (8). This guide is part of a residential aged care palliative approach toolkit that was developed for the specific use in nursing home facilities. The guideline was developed through a collaboration of Australian and New-Zealand associations for palliative care, the Queensland University and the society of pharmacists. The residential aged care association was considered a stakeholder. This source guideline was considered relevant because of content similarity with the clinical questions in the target guideline. Because of absence of an available methodological report, the guideline could not be evaluated by the AGREE-II. The researchers contacted one of the main authors, Prof. dr. L. Raymond, by telephone in order to collect additional information about the guideline development process. A summary of this telephone call can be found in Appendix F. The guideline was developed based on a literature review conducted by the Queensland University. A written report was no longer available. Because of lack of scientific evidence, the guideline was foremost the result of expert consensus. Despite its methodological limitations and after consulting CEBAM, the guideline development group decided that the content of this guideline was sufficiently qualitative to enter an ADAPTE procedure. To guarantee scientific quality, the decision was made to test the recommendations in this guideline against the recommendations in the methodological stronger NICE guideline and the results of a newly developed systematic review of available literature.

6.3 Selection of relevant recommendations from the source guideline

The recommendations responding to the formulated research questions that were identified in the “Guide to the pharmacological management of end-of-life (terminal) symptoms in residential aged care residents” of the Brisbane South Palliative Care Collaborative can be found in Table 6 (8). The level of Evidence-score attributed by the Australian researchers is also listed.

Levels of evidence

The levels of evidence are those designed by the National Health and Medical Research Council of Australia with the addition of a Level V.

I	Systematic review of all relevant randomized control trials (RCTs)
II	At least one properly designed RCT
III-1	Well designed pseudo-RCTs
III-2	Comparative studies with concurrent non-randomized controls, case control studies or interrupted time series with a control group
III-3	Comparative studies with historical control, two or more single arm studies, or interrupted time series without parallel control group
IV	Case series, either post-test or pre-test and post-test
V	Specialist expert opinion (the opinion of specialists with experience in the field of palliative medicine)

Table 6: Recommendations from the original Australian guideline		
References	Recommendations	Level of evidence
Which pharmacological treatment can be recommended for pain management in dying nursing home residents? (clinical question 1)		
(9) Won et al., 2004 (10) Mercadante & Acuri, 2007	Studies indicate that pain is a common problem experienced by elderly people living in RACFs. The prevalence of persistent pain in this population is estimated to be between 49% and 80%.	V
(10) Mercadante & Acuri, 2007	Opioids are effective and generally well tolerated in the elderly.	V
(11) Pergolizzi et al., 2008 (12) Elliot, 2006	Opioid naïve residents requiring opioids to manage pain should be commenced on the lowest opioid dose possible. Careful upward titration minimizes the risk of toxicity.	I, V
(13) King et al., 2011	Morphine should be avoided in residents with severe renal failure (eGFR<30) due to the build-up of toxic metabolites. Fentanyl has no active metabolites of relevance and has been identified as the opioid that is least likely to cause harm in residents with severe renal impairment when used appropriately.	I
(14) Palliative Care Expert Group, 2010	To optimize relief of persistent pain, opioids should be administered on an ‘around-the-clock’ basis according to the duration of action of the prescribed opioid.	V
(15) Portenoy et al., 2006	Breakthrough pain occurs commonly in people who are receiving opioids for persistent pain.	III-2
(14) Palliative Care Expert Group, 2010	In addition to the regular opioid dose, a PRN breakthrough opioid dose should be prescribed at 1/12th to 1/6th of the 24 hour dose.	V

(16) Rossi, 2013	Transdermal opioid patches (buprenorphine and fentanyl) are not suitable to commence in the last days of life. Transdermal opioid patches have a prolonged onset time and therefore rapid, safe dose titration to manage escalating symptoms is not possible.	I
(14) Palliative Care Expert Group, 2010 (X) Abrahm et al, 2012 (deleted page on UpToDate)	When initiating opioids in the last days of life or when oral route is no longer viable, a continuous subcutaneous infusion using a syringe driver is the preferred route of administration.	V, V
(14) Palliative Care Expert Group, 2010 (X) Abrahm et al, 2012 (deleted page on UpToDate)	Common side effects of opioid administration include constipation, nausea and vomiting, dizziness and sedation. Most side effects diminish with continued use except for Vconstipation	V, V
Developed using a Delphi method	Flowchart 2: Pharmacological Management of pain for Residents on the Residential Aged Care End of Life Care Pathway (RAC EoLCP)	V
Which pharmacological treatment can be recommended for the management of dyspnea in dying nursing home residents? (clinical question 2)		
(17) Dudgeon & Shadd, 2012 (18) Currow et al., 2010	Dyspnoea is a common symptom experienced in advanced disease irrespective of diagnosis. The prevalence and severity can increase over time particularly in the last days of life.	I, III-2
(14) Palliative Care Expert	Initiate simple measures to reduce dyspnea such as repositioning the resident, tepid sponge if febrile and air flow across the face using rotating fan or open window	V
(17) Dudgeon & Shadd, 2012	There is limited evidence to support the use of oxygen to manage dyspnea at end of life. Oxygen has not been shown to relieve dyspnea in non-hypoxic patients.	I
(19) Booth et al., 2004	If a resident is hypoxic, oxygen is recommended for provision of short term relief. Oxygen should be continued for residents who have required long term use for the management of breathlessness in chronic respiratory illnesses.	I
(20) Jennings et al., 2001	Systemic opioids administered in appropriate doses are safe and effective in managing dyspnea.	I
(13) King et al., 2011	Morphine should be avoided in residents with severe renal failure (eGFR<30) due to the build-up of toxic metabolites. Fentanyl has no active metabolites of relevance and has been identified as an opioid that is least likely to cause harm in residents with severe renal impairment when used appropriately.	I
Developed using a Delphi method	Flowchart 3: Pharmacological Management of dyspnea for Residents on the Residential Aged Care End of Life Care Pathway (RAC EoLCP)	V
Which pharmacological treatment can be recommended for management of nausea and vomiting in dying nursing home residents? (clinical question 3)		
(21) Glare et al., 2011	Factors contributing to nausea and vomiting in a resident with a life-limiting illness may include but are not limited to: drug toxicity, urinary tract infection, constipation, diseases of the gastrointestinal tract, metabolic and biochemical disturbance and organ failure. Cause(s) of nausea and vomiting in the last days of life may be unidentifiable and multi-factorial.	V
(22) Hardy et al., 2002	Nausea is often under recognised and under treated.	I
(21) Glare et al., 2011	There is limited evidence to guide the use of antiemetic therapy in the elderly.	V
(21) Glare et al., 2011 (22) Hardy et al., 2002 (23) Porreca & Ossipov, 2009	Opioids commonly cause nausea and vomiting. Metoclopramide has been shown to be effective in the management of nausea and vomiting in patients with cancer who are on opioid therapy.	V, I, V

(21) Glare et al., 2011 (22) Hardy et al., 2002	Haloperidol can be trialed to manage nausea and vomiting if metoclopramide is ineffective	V, I
(14) Palliative Care Expert Group, 2010 (24) CareSearch, 2013	Metoclopramide or haloperidol can cause extrapyramidal side effects. These medications need to be avoided or used with caution in residents with neurodegenerative disorders such as Parkinson's disease.	V, V
(24) CareSearch, 2013	Subcutaneous infusion of antiemetics delivered via a syringe driver has been shown to be effective in managing persistent symptoms of nausea and vomiting.	V
Developed using a Delphi method	Flowchart 1: Pharmacological Management of nausea and vomiting for Residents on the Residential Aged Care End of Life Care Pathway (RAC EoLCP)	V
Which pharmacological treatment can be recommended for management of agitation and restlessness in dying nursing home residents? (clinical question 6a)		
(14) Palliative care expert group, 2010	Restlessness and agitation occur commonly at end of life and can often be attributed to multiple causes. Investigating the underlying cause may not be appropriate in the last days of life.	V
(14) Palliative care expert group, 2010	It is important to assess and manage factors which contribute to restlessness and agitation such as pain, urinary retention, rectal impaction, hypoxia, environmental factors, psychological and spiritual distress.	V
(25) Kehl et al., 2004 (26) Breura et al., 2012	Restlessness and agitation at end of life can be caused by anxiety and distress. The addition of a low dose benzodiazepine can be effective in managing these symptoms.	I, V
(27) Lonergan et al., 2007	Low dose haloperidol is effective in managing restlessness and agitation associated with delirium.	I
(27) Lonergan et al., 2007	Extrapyramidal side effects (dystonia and akathisia) occur more commonly in doses of haloperidol above 4.5 mg per day.	I
Developed using a Delphi method	Flowchart 4: Pharmacological Management of Restlessness and Agitation for Residents on the Residential Aged Care End of Life Care Pathway (RAC EoLCP)	V
Which pharmacological treatment can be recommended for the management of confusion and delirium in dying nursing home residents? (clinical question 6b)		
(27) Lonergan et al., 2007	Low dose haloperidol is effective in managing restlessness and agitation associated with delirium.	I
Developed using a Delphi method	Flowchart 4: Pharmacological Management of Restlessness and Agitation for Residents on the Residential Aged Care End of Life Care Pathway (RAC EoLCP)	V
Which pharmacological treatment can be recommended for management of anxiety in dying nursing home residents? (clinical question 6c)		
(17) Dudgeon & Shadd, 2012 (28) Navigante et al., 2006	Anxiety is often associated with shortness of breath and benzodiazepines are effective in managing this symptom.	V, I
Developed using a Delphi method	Flowchart 4: Pharmacological Management of Restlessness and Agitation for Residents on the Residential Aged Care End of Life Care Pathway (RAC EoLCP)	V

6.4 Clinical evidence tables of literature in source guideline

The clinical evidence tables of the literature in the guideline “Guide to the pharmacological management of end-of-life (terminal) symptoms in residential aged care residents” of the Brisbane South Palliative Care Collaborative can be found in Appendix G.

6.5 Literature review

Because of the lack of a sound methodological report for the development of the “Guide to the pharmacological management of end of life (terminal) symptoms in residential aged care residents” of the Brisbane South Palliative Care Collaborative, a systematic review was constructed for all the clinical questions described in the guideline.

6.5.1 Review protocols

6.5.1.1 Pain: clinical research question 1

	Description
Review question	Which pharmacological treatment is recommended for pain management in the last days of life in nursing home care?
Context and objective	To be able to advise the most suitable pharmacological treatment for pain control in frail and dying older patients in the setting of a nursing home.
Search strategy	- Databases: Medline, Embase - Language restrictions: English, Dutch, French - Forward- and back search allowed - Period: from inception
Study type	Inclusion: - Quantitative and qualitative data are allowed - Prospective and retrospective studies - Observational and interventional studies Exclusion: - Case reports - Animal studies - Comments, opinions and letters
Population/Setting	Older people (>65j) in the terminal phase in a long-term care facility/ nursing home In the case of lack of quality evidence, inclusion of: - Frail and dying patient - Older and dying patient (Mean or median age ≥ 65 years, with lower limit of IQR and SD ≥ 55 years) Exclusion: - Older people in intensive care units - Older people in operating room or recovery rooms - Non palliative setting
Intervention	- Non-opioid analgesics - Opioid Analgesics - Anti-inflammatory Agents - Anesthetics Corticosteroids
Control	- Comparison to other/alternative therapy - Placebo - Non-pharmacological treatment - No treatment
Outcome	Critical outcome measures: - Symptom control defined as complete or partial pain control and/or discomfort related to pain. Important outcome measures: - Undesired side effects Minor outcome measures: - Life span lengthening - Life span shortening

Data-extraction and Quality control	• First screening of title and abstract • Exclusion of all articles without accessible full text • Second screening of full text and data extraction by 1 reviewer and repeated by second reviewer for all included full texts. Disagreements resolved by 3th reviewer • Quality control with MMAT instrument version 2018
Analysis of subgroups if available	For subpopulations with: • Chronic kidney disease • Cancer related pain • Opioid negative patients versus drug dependent patients • Crampoid pain • Neuropathic pain For following administration routes: • Oral • Rectal • Cutaneous, topical • Enteral (nasogastric tube, PEG) • Parenteral (subcutaneous) For different dosage

6.5.1.2 *Dyspnea: clinical research question 2*

	Description
Review question	Which pharmacological treatment is recommended for management of dyspnea in the last days of life in nursing home care?
Context and objective	To be able to advise the most suitable pharmacological treatment for control of dyspnea in frail and older dying patients in the setting of a nursing home facility.
Search strategy	• Databases: Medline, Embase • Language restrictions: English, Dutch, French • Forward- and back search allowed • Period: from inception
Study type	Inclusion: • Meta-analysis • RCT • Prospective and retrospective studies • Observational and interventional studies Exclusion: • Case reports • Comments, opinions, letters • Animal studies
Population/Setting	Frail older people (>65j) in the terminal phase in a long-term care facility/ nursing home In the case of lack of quality evidence, inclusion of: • Frail and dying patient • Older and dying patient (Mean or median age ≥ 65 years, with lower limit of IQR and SD ≥ 55 years) Exclusion: • Older people in intensive care units • Older people in operating room or recovery rooms • Non palliative setting
Intervention	• Opioid Analgesics • Corticosteroids • Diuretics • Inhalation therapy
Control	• Comparison to other/alternative therapy • Placebo • Non-pharmacological treatment • No treatment

Outcome	• Critical outcome measures: ◦ Symptom control defined as complete or partial relief of discomfort related to dyspnea. • Important outcome measures: ◦ Undesirable side effects • Minor outcome measures: ◦ Life span lengthening ◦ Life span shortening
Data-extraction and Quality control	• First screening of title and abstract • Exclusion of all articles without accessible full text • Second screening of full text and data extraction by 1 reviewer and repeated by second reviewer for 20% of included full texts. Disagreements resolved by 3th reviewer • Quality control with MMAT instrument version 2018
Analysis of subgroups if available	For subpopulations with: • Lung disease • Heart disease For following administration routes: • Oral • Rectal • Inhalation • Enteral (nasogastric tube, PEG) • Parenteral (subcutaneous) For different dosage

6.5.1.3 Nausea and vomiting: clinical research question 3

	Description
Review question	Which pharmacological treatment is recommended for management of nausea and vomiting in the last days of life in nursing home care? (clinical question 3)
Context and objective	To be able to advise the most suitable pharmacological treatment for control of nausea and vomiting in frail and older dying patients in the setting of a nursing home facility.
Search strategy	• Databases: Medline, Embase • Language restrictions: English, Dutch, French • Forward- and back-search allowed • From inception
Study type	Inclusion: • Quantitative as well as qualitative research • Prospective and retrospective studies • Observational and interventional studies Exclusion: • Case reports • Animal studies • Comments, opinions, letters
Population/Setting	Frail older people (>65j) in the terminal phase in a long-term care facility/ nursing home In the case of lack of quality evidence, inclusion of: • Frail and dying patient • Older and dying patient (Mean or median age ≥ 65 years, with lower limit of IQR and SD ≥ 55 years) Exclusion: • Older people in intensive care units • Older people in operating room or recovery rooms • Non palliative setting

Intervention	• Gastroprokinetic • 5-HT3 antagonists • Neuroleptics or antipsychotics: typical and atypical • NK1 antagonists • Corticosteroids • Somatostatin analogues
Control	• Comparison to other/alternative therapy • Placebo • Non-pharmacological treatment • No treatment
Outcome	• Critical outcome measures: ◦ Symptom control defined as complete or partial relief of discomfort related to nausea and vomiting. • Important outcome measures: ◦ Undesirable side effects • Minor outcome measures: ◦ Life span lengthening ◦ Life span shortening
Data-extraction and Quality control	• First screening of title and abstract • Exclusion of all articles without accessible full text • Second screening of full text and data extraction by 1 reviewer and repeated by second reviewer for 20% of included full texts. Disagreements resolved by 3th reviewer • Quality control with MMAT instrument version 2018
Analysis of subgroups if available	For subpopulations with: • Opioid induced nausea and vomiting • Metabolic causes • Abdominal obstruction • Elevated intracranial pressure For following administration routes: • Oral • Sublingual • Rectal • Enteral (nasogastric tube, PEG) • Parenteral (subcutaneous) For different dosage

6.5.1.4 Psychological symptoms: clinical research question 6 (a+b+c)

	Description
Review question	Which pharmacological treatment is recommended for management of psychological symptoms in the last days of life in nursing home care? (clinical question 6)[£] • a: agitation and restlessness • b: confusion and delirium • c: anxiety
Context and objective	To be able to advise the most suitable pharmacological treatment for control of agitation and restlessness in frail and older dying patients in the setting of a nursing home facility.
Search strategy	• Databases: Medline, Embase • Language restrictions: English, Dutch, French • Forward- and back search allowed • Period: From inception

[£] The mixed methods review of 'The pharmacological treatment of psychological symptoms in the last days of life in nursing home care' (clinical research question 6) is part of a Master after Master's thesis. To this objective, the inclusion criteria concerning age requirements were kept wider than the inclusion criteria for the reviews of physical symptoms.

Study type	Inclusion: - Quantitative as well as qualitative research - Prospective and retrospective studies - Observational and interventional studies - Evidence based and experience based guidelines and protocols Exclusion: - Case reports - Animal studies - Comments, opinions, letters
Population/Setting	Inclusion: Target population are frail older people (>65j) in the terminal phase in a long-term care facility/ nursing home In the case of lack of quality evidence, inclusion of: - Frail and dying patient - Older and dying patient (Mean or median age ≥ 65 years, with lower limit of IQR and SD ≥ 55 years) Exclusion: - Older people in intensive care units - Older people in operating room or recovery rooms - Non palliative setting
Intervention	- Benzodiazepines - Z-drugs - Neuroleptics or antipsychotics: typical and atypical - Cholinergic medication
Control	- Comparison to each other or drug therapy not included in our intervention - Placebo - Non-pharmacological treatment - No treatment
Outcome	- Critical outcome measures: - Symptom control defined as complete or partial relief of discomfort related to agitation and restlessness. - Important outcome measures: - Undesirable side effects - Minor outcome measures: - Life span lengthening - Life span shortening
Data-extraction and Quality control	- First screening of title and abstract by 2 independent reviewers - Exclusion of all articles without accessible full text - Second screening of full text and data extraction by 1 reviewer and repeated by second reviewer for 20% of included full texts. Disagreements resolved by 3th reviewer - Quality control with MMAT instrument version 2018
Analysis of subgroups if available	For subpopulations with: - Underlying cognitive disease For following administration routes: - Oral - Sublingual - Rectal - Enteral (nasogastric tube, PEG) - Parenteral (subcutaneous) For different dosage

6.5.2 Search strings

Search strings can be found in Appendix H.

6.5.3 Search results

Searches were performed in MEDLINE/Pubmed and EMBASE from inception until 08/03/2023.

6.5.3.1 Pain: clinical research question 1

- Records found through database searching: date 08/03/2023
 - Medline (08/03/2023): n = 2106
 - Embase (08/03/2023): n = 2024
- Records found through citation searching and grey literature: n = 11
- Records included: n = 4 (5,29–31)
- Prisma flowchart can be found in Appendix I

6.5.3.2 Dyspnea: clinical research question 2

- Records found through database searching: date 08/03/2023
 - Medline (08/03/2023): n = 988
 - Embase (08/03/2023): n = 1820
- Records found through citation searching and grey literature: n = 25
- Records included: n = 4 (31–34)
- Prisma flowchart can be found in Appendix I

6.5.3.3 Nausea and vomiting: clinical research question 3

- Records found through database searching: date 08/03/2023
 - Medline (08/03/2023): n = 248
 - Embase (08/03/2023): n = 795
- Records found through citation searching and grey literature: n = 9
- Records included: n = 0
- Prisma flowchart can be found in Appendix I

6.5.3.4 Psychological symptoms (agitation and restlessness, confusion and delirium, anxiety): clinical research question 6 (a+b+c)

- Records found through database searching: date 08/03/2023
 - Medline (08/03/2023): n = 602
 - Embase (08/03/2023): n = 743
- Records found through citation searching and grey literature: n = 64
- Records included: n = 4 (35–38)
- Prisma flowchart can be found in Appendix I

6.5.4. Clinical evidence tables

The clinical evidence tables were constructed to fit evaluations of different types of research and took into account information about frailty characteristics of the study population. The clinical evidence tables of the selected studies for the different review question can be found in Appendix J.

6.5.5 Quality of the selected evidence – MMAT

The quality of the selected evidence was evaluated with the Mixed Methods Assessment Tool (MMAT) version 2018. Individual evaluations of the selected evidence can be found in Appendix K.

6.6 Recommendations withheld from systematic review

The systematic reviews showed very little evidence on the treatment of pain, dyspnea, or psychological symptoms with no evidence for nausea and vomiting (39,40). Only four articles were included for pain, four for dyspnea and four for all psychological symptoms combined. Most studies are descriptive, retrospective and have no implications for clinical practice.

The evidence can be summarized as opioids being effective in the treatment of pain and dyspnea in the last days of life of older people and antipsychotics (haloperidol) and benzodiazepines being recommended in the treatment of psychological symptoms. No dose recommendations can be formulated from the available evidence. In depth description of these results together with discussion have been submitted and currently under review (39,40).

These management approaches align with the recommendations from the Australian guideline, therefore no new recommendations were formulated. The recommendations of the Australian guideline will be evaluated and assessed against this evidence, as well as against the previously mentioned NICE guidelines. These recommendations can be found in Chapter 11.

Chapter 7: DE NOVO procedure

The ADAPTE procedure could not identify a source guideline that covered the full extent of the scope of our target guideline. This led to the decision (in December 2020) to simultaneously start a 'de novo' procedure for the remaining clinical questions.

7.1 Search strategy

7.1.1 Review protocols

7.1.1.1 Death rattle: clinical research question 4:

	Description
Review question	Which pharmacological treatment is recommended for management of death rattle in the last days of life in nursing home care? (clinical question 4)?
Context and objective	To be able to advise the most suitable pharmacological treatment for death rattle in the setting of a nursing home or long term care facility for older dying patients.
Search strategy	• Databases: Medline, Embase • Language restrictions: English, French, Dutch • Forward- and back search allowed • Period: from inception to date
Inclusion and exclusion criteria	Inclusion: • Quantitative as well as qualitative research • Prospective and retrospective studies • Observational and interventional studies Exclusion: • Case reports • Animal studies • Letters, opinions and comments
Population/ Setting	Inclusion: Target population are frail older people (>65y) in the terminal phase in a long-term care facility/nursing home. In the case of lack of quality evidence: • Frail older people and dying patient • Older and dying patient. (Mean or median age ≥ 65 years, with lower limit of IQR and SD ≥ 55 years) Exclusion: • Older people in the intensive care units • Older people in operating room or recovery rooms • Non-palliative setting
Intervention	• Anti-secretory drugs • Anticholinergic drugs
Control	• Each other • Placebo • Non pharmacological treatment • No treatment
Outcome	• Critical outcome measures: ◦ Symptom control defined as complete or partial relief of discomfort related to death rattle. • Important outcome measures: ◦ Undesirable side effects • Minor outcome measures: ◦ Life extension ◦ Life span shortening

Data-extraction and quality control	• First screening of title and abstract by 2 independent reviewers • Exclusion of all articles without accessible full text • Second screening of full text and data extraction by 1 reviewer and repeated by second reviewer for 20% of included full texts. Disagreements resolved by 3th reviewer • Quality control with MMAT instrument version 2018
Analysis of subgroups	For following administration routes: • Oral • Sublingual • Rectal • Enteral (nasogastric tube, PEG) • Parenteral (subcutaneous) For different dosage

7.1.1.2 Fever: clinical research question 5

	Description
Review question	Which pharmacological treatment is recommended for management of fever in the last days of life in nursing home care? (clinical question 5)
Context and objective	To be able to advise the most suitable pharmacological treatment for death rattle in the setting of a nursing home or long term care facility for older dying patients.
Search strategy	• Databases: Medline, Embase • Language restrictions: English, French, Dutch • Forward- and back search allowed • Period: from inception to date
Inclusion and exclusion criteria	Inclusion: • Quantitative as well as qualitative research • Prospective and retrospective studies • Observational and interventional studies Exclusion: • Case reports • Animal studies • Letters, opinions and comments
Population/ Setting	Inclusion: Target population are frail older people (>65y) in the terminal phase in a long-term care facility/nursing home. In the case of lack of quality evidence: • Frail older people and dying patient • Older and dying patient. (Mean or median age ≥ 65 years, with lower limit of IQR and SD ≥ 55 years) Exclusion: • Older people in the intensive care units • Older people in operating room or recovery rooms • Non-palliative setting
Intervention	• Antipyretics and NSAIDs • Corticosteroids
Control	• Each other • Placebo • Non pharmacological treatment • No treatment
Outcome	• Critical outcome measures: ◦ Symptom control defined as complete or partial relief of discomfort related to fever. • Important outcome measures: ◦ Undesirable side effects • Minor outcome measures: ◦ Life extension ◦ Life span shortening

Data-extraction and quality control	• First screening of title and abstract by 2 independent reviewers • Exclusion of all articles without accessible full text • Second screening of full text and data extraction by 1 reviewer and repeated by second reviewer for 20% of included full texts. Disagreements resolved by 3th reviewer • Quality control with MMAT instrument version 2018
Analysis of subgroups if available	For patients with: • Terminal hyperthermia (set-point related) • Infectious fever • Paraneoplastic fever For administration routes: • Oral • Sublingual, lingual • Enteral (gastric tube, PEG) • Parenteral (subcutaneous) • Rectal For different dosage

7.1.2 Search strings

Search strings can be found in Appendix H.

7.1.3 Search results

7.1.3.1 *Death rattle: clinical research question 4*

- Records found through database searching: date 08/03/2023
 - Medline (08/03/2023): n = 44
 - Embase (08/03/2023): n = 61
- Records found through citation searching and grey literature: n = 4
- Records included: n = 7 ((41–47)
- Prisma flowchart can be found in Appendix L

7.1.3.2 *Fever: clinical research question 5*

- Records found through database searching: date 08/03/2023
 - Medline (08/03/2023): n = 77
 - Embase (08/03/2023): n = 197
- Records found through citation searching and grey literature: n = 7
- Records included: n = 0
- Prisma flowchart can be found in Appendix L

7.1.4 Clinical evidence tables

The clinical evidence tables were constructed to fit evaluations of different types of research and took into account information about frailty characteristics of the study population. The clinical evidence tables of the selected studies for the different review questions can be found in Appendix M.

7.1.5 Quality of selected evidence - MMAT

The quality of the selected evidence was evaluated with the Mixed Methods Assessment Tool (MMAT) version 2018. Individual evaluations of the selected evidence can be found in Appendix N.

7.2 Recommendations

As no studies were withheld for fever, no recommendations could be formulated based on scientific evidence. The existing studies on treatment of death rattle once it is present do not show superiority of one anticholinergic over another, nor are there studies that show a better effect of anticholinergics compared to placebo (41,42,44–46). Prophylactic treatment of death rattle does shown to be effective in reducing the occurrence of death rattle in two randomized controlled trails (43,47).

The following recommendations for death rattle were constructed. All recommendations were constructed in Dutch, as they will be scored in the consensus procedure. Scoring of these recommendations is necessary as there is much controversy and uncertainty on the treatment of death rattle. Experts are still debating whether or not treatment for death rattle is necessary (48).

Levels of evidence were appointed according to the level of evidence scale of the Australian guideline, as to make comparison between the different symptoms easier.

Table 7: Recommendations based on systematic literature search for research question 4 (death rattle)		
References	Recommendation	Level of evidence
Which pharmacological treatment can be recommended for management of death rattle in dying nursing home residents? (clinical question 4)		
(42) Wildiers et al. 2009	Doodsreutel is het geluid dat ontstaat doordat secreties in de luchtpijp of keel met de ademhaling meetrillen bij stervende bewoners die deze niet meer kunnen ophoesten.	II
(43) Van Esch et al. 2021	Doodsreutel treedt meestal op als de bewoner niet meer bij bewustzijn is. Het is dan ook onduidelijk of de bewoner hieronder lijdt. Voor de omgeving kan het echter zeer belastend zijn.	V
(43) Van Esch et al. 2023	Om doodsreutel minder belastend te maken voor de familie is een duidelijke uitleg over doodsreutel belangrijk.	V
(42) Wildiers et al. 2009	Indien medicatie wordt opgestart, wordt dit best zo snel mogelijk gedaan bij het ontstaan van doodsreutel.	II
(45) Heisler et al. 2013, (43)Van Esch et al. 2021	Bij het gebruik van anticholinergica moet men aandachtig blijven voor bijwerkingen zoals: droge mond, urineretentie, onrust, agitatie, delier, tachycardie...	II II
(42) Wildiers et al. 2009	Buscopan® (hyoscine butylbromide) vermindert luchtwegsecreties. Het passeert de bloedsheerbarrière niet en kan zo geen duizeligheid of delier veroorzaken.	II
(43)Van Esch et al. 2021, (47) Mercadante et al. 2018	Profylactische toediening van Buscopan® (hyoscine butylbromide) vermindert de kans op het ontwikkelen van doodsreutel.	II II
(42) Wildiers et al. 2009	Buscopan®(hyoscine butylbromide), Scopolamine®(hyoscine hydrobromide) en atropine hebben geen meetbaar verschil in bijwerkingen bij de behandeling van doodsreutel.	II
(46) Hugel et al. 2006	Robinul®(glycopyrronium) en Scopolamine® (hyoscine hydrobromide) in continue toediening zijn even effectief in de behandeling van doodsreutel.	III-3

(45) Heisler et al. 2013	Toediening van 2 druppels atropine 1% oplossing, sublinguaal, heeft geen voordeel in de behandeling van doodstreutel in vergelijking met placebo.	II
(43) Van Esch et al. 2021, (47) Mercadante et al. 2018	Voor de profylactische behandeling van doodstreutel kan Buscopan ® (hyoscine butylbromide) met een oplaaddosis van 20mg subcutaan, gevolgd door 60mg/24u via continue subcutane toediening of 4x20mg/d subcutaan gebruikt worden.	II II
(42) Wildiers et al. 2009	Bij de symptomatische behandeling van doodstreutel kan Buscopan ® (hyoscine butylbromide) veilig gebruikt worden: a. 20mg subcutaan “zo nodig”, dit mag iedere 2-4u toegediend worden b. Oplaaddosis van 20mg subcutaan, gevolgd door een continue toediening aan 60mg/24u of 10mg iedere 4u	II
(46) Hugel et al. 2006, Clark et al. 2009	Bij de symptomatische behandeling van doodstreutel kan Scopolamine ® (hyoscine hydrobromide) veilig gebruikt worden. a. Oplaaddosis van 0.25mg subcutaan, gevolgd door een continue toediening van 1.5mg/24u of 0.25mg iedere 4u.	III-3 III-1
(42) Wildiers et al. 2009, (45) Heisler et al. 2013	Bij de symptomatische behandeling van doodstreutel kan atropine veilig gebruikt worden: a. Oplaaddosis van 0.5mg subcutaan, gevolgd door een continue toediening van 3mg/24u. b. 0,5 mg tot 3 mg/dag subcutaan c. Oogdruppels atropine 1% sublinguaal toe gediend iedere 4 uur 2-3 druppels	II II
(46) Hugel et al. 2006	Bij de symptomatische behandeling van doodstreutel kan Robinul® (glycopyrronium) veilig gebruikt worden: a. Oplaaddosis van 0.2mg gevolgd door een continue toediening aan 0.6-1.2mg/24u.	III-3

Chapter 8: Methodology of clinical research question 7

8.1 Review protocol

Review protocol clinical research question 7: How can the use of different pharmacological agents be combined in a safe and practical way for administration in a nursing home?

	Description
Review question	How can different pharmaceuticals for the treatment symptoms in the last days of life be administered in a safe and practical way in the setting of a nursing home?
Context and objective	To advise the most practical way to administrate and combine pharmaceuticals in the treatment of end-of-life symptoms in nursing home care.
Search strategy	Separate search strategy developed by pharmacists in training, see Appendix O.
Inclusion and exclusion criteria	Separate search strategy developed by pharmacists in training, see Appendix O.
Population/ Setting	Inclusion: Target population are frail older people (>65y) in the terminal phase in a long-term care facility/nursing home. In the case of lack of quality evidence: • Frail older people and dying patient • Older and dying patient Exclusion: • Older people in the intensive care units Older people in operating room or recovery rooms Non-palliative setting
Intervention	• Drug interactions: ○ Pharmaceutical (physical/chemical) interactions: ▪ Between 2 or more drugs ▪ Between active drug and solute ○ Pharmacokinetic interactions: resorption speed, bio- disposability, metabolism, excretions, protein binding ○ Pharmacodynamical interactions: synergy, antagonism • Administration routes: subcutaneous, off label, crushed, magisterial syrups, orally • Technical attributes: ○ Syringe driver type Graseby, Micrel Infusion needle types
Outcome	<u>Critical outcome measures:</u> ○ Safety ○ Availability ○ User friendliness <u>Important outcome measures:</u> ○ Pharmacological interaction with clinical significance ○ Cost per unit <u>Minor outcome measures:</u> ○ Pharmacological interaction without clinical significance
Analysis of subgroups if available	For administration routes: • Oral • Sublingual, lingual • Enteral (gastric tube, PEG) • Parenteral (subcutaneous) • Rectal For different dosage

Chapter 9: Knowledge hiatus

After the ADAPTE and DE NOVO procedures there were no evidence based recommendations for clinical research question 5, the pharmacological management of fever. Therefore recommendations were developed based on the research group's expert opinion in combination with recommendations from "Ouderenformularium BCFI" and "Palliatieve Zorg Vlaanderen" which are based on expert opinion. These recommendations can be found in Table 8.

Table 8: Expert based recommendations on research question 5 (fever)		
References	Recommendation	Level of evidence
Which pharmacological treatment can be recommended for management of fever in dying nursing home residents? (clinical question 5)		
(49) Ouderenformularium BCFI	Terminale koorts (lichaamstemperatuursverhoging bij een stervende persoon) wordt meestal niet behandeld, tenzij de kans op stuipen reëel is. De ongewenste effecten eigen aan deze behandeling kunnen voor de bewoner meer belastend zijn dan de koorts zelf.	V
Developed by the GDG	Het is geen doel om de oorzaak van de koorts te achterhalen.	V
(49) Ouderenformularium BCFI	Terminale koorts wijkt meestal niet met koortswerende middelen.	V
(49) Ouderenformularium BCFI (50) Palliatieve Zorg Vlaanderen	Als de behandeling van koorts gewenst is, kan gebruik gemaakt worden van Paracetamol 1g 2-4 keer per dag per os of rectaal.	V V
(50) Palliatieve Zorg Vlaanderen	Als de behandeling van tumorkoorts gewenst is, kan gebruik gemaakt worden van Naproxen 250mg twee tot drie keer per dag per os of rectaal.	V
(50) Palliatieve Zorg Vlaanderen	Als de behandeling van tumorkoorts gewenst is, kan gebruik gemaakt worden van Diclofenac 25mg twee tot drie keer per dag per os of rectaal.	V
(50) Palliatieve Zorg Vlaanderen (49) Ouderenformularium BCFI	Als de behandeling van tumorkoorts gewenst is, kan gebruik gemaakt worden van Ibuprofen 200-400mg twee tot drie keer per dag per os.	V V
(49) Ouderenformularium BCFI	Als de behandeling van tumorkoorts gewenst is, kan gebruik gemaakt worden van Dexamethason 4-8 mg 1 keer per dag, per os of subcutaan.	V

Chapter 10: Adaptation to the Belgian health care context

To adapt the source guideline to the Belgian context all selected recommendations were translated to Dutch. However, the practical information on how to treat the symptoms in the last days of life was found in the flowcharts, and not expressed in recommendations. Therefore the flowcharts were converted into recommendations, and translated to Dutch. Small grammatical changes were made to ensure the meaning of every recommendation did not change due to translation. These flowcharts were developed in a Delphi procedure, based on expert opinion.

Only one recommendation cannot be implemented in Belgium due to the unavailability of fentanyl for subcutaneous injection in the nursing home setting. Therefore the following recommendation was adapted.

“Morphine should be avoided in residents with severe renal failure (eGFR<30) due to the build-up of toxic metabolites. Fentanyl has no active metabolites of relevance and has been identified as the opioid that is least likely to cause harm in residents with severe renal impairment when used appropriately”

Adapted and translated to:

“Achtergrond: Bij patiënten met ernstig renaal falen [eGFR<30], wordt beschreven in de literatuur dat morfine dient vermeden te worden o.w.v. stapeling van toxische metabolieten. In andere landen gaat daarom de voorkeur uit naar fentanyl en oxycodon subcutaan, die minder relevante toxische metabolieten hebben. [ref (13) – Level I] Fentanyl en oxycodon subcutaan zijn in België echter niet beschikbaar of heel duur buiten de ziekenhuissetting.

Stelling: De dosering van morfine bij ernstig renaal falen [eGFR<30] bij woonzorgcentrumbewoners in de laatste levensdagen dient niet aangepast te worden aan de nierfunctie. (Stelling opgesteld door de onderzoekers)”

In the Australian source guideline the treatment of agitation (research question 6a) is incorporated into the treatment of delirium and anxiety (research questions 6b and 6c). As the literature review did not provide additional information on how to treat agitation, delirium or anxiety and the guideline development group agreed with the treatment strategy of the source guideline, the decision was made to incorporate the treatment of agitation (research question 6a) into the treatment of delirium and anxiety (research questions 6b and 6c). Whereby in the case of restlessness and agitation the underlying pathology, delirium or anxiety, will determine the recommended therapy (antipsychotics or benzodiazepines).

Furthermore, to make sure the guideline is practical the researchers decided to translate the key principles and key points in the introduction of the source guideline and adapt them to the Belgian context. This way these introductory recommendations were incorporated in the consensus procedure, ensuring that stakeholders can give their view on these recommendations. For the two symptoms of the DE NOVO procedure, practical recommendations were added, such as explaining death rattle. All recommendations that were scored during the consensus procedure are shown in Chapter 11, Table 9.

From the flowcharts several recommendations were extracted, staying as close as possible to the initial wording. However, some adaptations were needed in order to make clear recommendations. Only a few content changes were made, these are highlighted in the comment section.

Table 9: Translation and adaptation of the original recommendations to the Belgian

	Original recommendation	Dutch translation	Recommendation translated and adapted to the Belgian context	Comments
Which pharmacological treatment can be recommended for pain management in dying nursing home residents? (clinical question 1)				
1	Studies indicate that pain is a common problem experienced by elderly people living in RACFs. The prevalence of persistent pain in this population is estimated to be between 49% and 80%.	Studies tonen aan dat pijn een belangrijk probleem is bij bewoners van WZC, de prevalentie van aanhoudende pijn in deze populatie ligt tussen de 49-80%	Pijn is een belangrijk probleem bij woonzorgcentrumbewoners, de prevalentie van aanhoudende pijn in deze populatie bedraagt 49-80%.	Content unchanged, textual changes to improve clarity.
2	Opioids are effective and generally well tolerated in the elderly.	Opioiden zijn effectief en worden goed getolereerd door ouderen	Opioiden zijn effectief en worden meestal goed getolereerd door ouderen.	Content unchanged, textual change to add nuance
3	Opioid naïve residents requiring opioids to manage pain should be commenced on the lowest opioid dose possible. Careful upward titration minimizes the risk of toxicity.	De laagste dosis opioiden moet worden gebruikt voor pijn management van opioid naïeve patiënten. Voorzichtig ophogen van de dosis, vermindert het risico op toxiciteit.	Bij bewoners die vooraf geen opioiden kregen, hierna opioid-naïeve bewoners genoemd, gebruikt men de laagste dosis voor pijn management. Het voorzichtig ophogen van de dosis vermindert het risico op toxiciteit of ongewenste nevenwerkingen.	Content unchanged, textual changes to improve clarity.
4	Morphine should be avoided in residents with severe renal failure (eGFR<30) due to the build-up of toxic metabolites. Fentanyl has no active metabolites of relevance and has been identified as the opioid that is least likely to cause harm in residents with severe renal impairment when used appropriately.	Morfine dient vermeden te worden bij patiënten met ernstig renaal falen [eGFR<30] o.w.v. stapeling van toxische metabolieten. Fentanyl heeft geen relevante toxische metabolieten en is dan ook het opioid dat wordt geacht het minste schade te veroorzaken bij bewoners van WZC met renaal falen, als het correct wordt gebruikt.	Achtergrond: Bij patiënten met ernstig renaal falen [eGFR<30], wordt beschreven in de literatuur dat morfine dient vermeden te worden o.w.v. stapeling van toxische metabolieten. In andere landen gaat daarom de voorkeur uit naar fentanyl en oxycodon subcutaan, die minder relevante toxische metabolieten hebben. Fentanyl en oxycodon subcutaan zijn in België echter niet beschikbaar of heel duur buiten de ziekenhuissetting. <u>Stelling:</u> De dosering van morfine bij ernstig renaal falen [eGFR<30] bij woonzorgcentrumbewoners in de laatste levensdagen dient niet aangepast te worden aan de nierfunctie. (Stelling opgesteld door de onderzoekers)	To our knowledge, morphine is the only opioid used as a subcutaneous injection during the last days of life in Belgium. It is frequently administered for pain or discomfort in patients with an eGFR of less than 30 ml/min. Therefore, we wanted to know if experts find it necessary to adjust the dosage in cases of severe renal failure.
5	To optimize relief of persistent pain, opioids should be administered on an 'around-the-clock' basis according to the duration of action of the prescribed opioid.	Om verlichting van aanhoudende pijn te bekomen, moeten opioiden toegediend worden 24u/24u volgens de duur van werkzaamheid van het opioid.	Voor de controle van aanhoudende pijn, is 24u/24u toediening van opioiden noodzakelijk, rekening houdend met de werkingsduur van het voorgeschreven opioid.	Content unchanged, textual changes to improve clarity.

6	Breakthrough pain occurs commonly in people who are receiving opioids for persistent pain.	Doorbraakpijn komt vaak voor bij bewoners met aanhoudende pijn [ref 22 – Level III-2]. Naast de regulier dosis opioïden, moet een zo nodig voorschrift worden voorzien voor doorbraakpijn, van 1/12 tot 1/6 van de 24uurs dosering.	Doorbraakpijn komt regelmatig voor bij bewoners met aanhoudende pijn. Naast de onderhoudsmedicatie, moet een “zo nodig” voorschrift worden voorzien voor doorbraakpijn. De dosering hiervan bedraagt 1/12 tot 1/6 van de 24-uurs dosering en kan elke 2u worden toegediend.	Two recommendations were merged to improve clarity, without altering their content.
7	In addition to the regular opioid dose, a PRN breakthrough opioid dose should be prescribed at 1/12th to 1/6th of the 24 hour dose.			
8	Transdermal opioid patches (buprenorphine and fentanyl) are not suitable to commence in the last days of life. Transdermal opioid patches have a prolonged onset time and therefore rapid, safe dose titration to manage escalating symptoms is not possible.	Transdermale opioïdpleisters (buprenorfine en fentanyl) zijn niet geschikt om te starten in de laatste levensdagen. Het pijnstillend effect laat namelijk even op zich wachten en daardoor kan er geen snelle en veilige dosis titrage gebeuren om escalerende symptomen te controleren.	Transdermale opioïdpleisters (buprenorfine en fentanyl) zijn niet geschikt om te starten in de laatste levensdagen. Het pijnstillend effect laat namelijk even op zich wachten, waardoor geen snelle en veilige dosistitraging kan gebeuren.	Content unchanged, textual changes to improve clarity.
9	When initiating opioids in the last days of life or when oral route is no longer viable, a continuous subcutaneous infusion using a syringe driver is the preferred route of administration.	Wanneer opioïden worden gestart in de laatste levensdagen of wanneer perorale intake niet langer mogelijk is, heeft continue subcutane toediening de voorkeur qua toedieningsweg.	Wanneer opioïden worden gestart in de laatste levensdagen of wanneer perorale intake niet langer mogelijk is, heeft continue subcutane toediening de voorkeur qua toedieningsweg.	Unchanged
10	Common side effects of opioid administration include constipation, nausea and vomiting, dizziness and sedation. Most side effects diminish with continued use except for constipation	Vaak voorkomende bijwerkingen zijn obstipatie, misselijkheid en braken, duizeligheid en sufheid. De meeste bijwerkingen verminderen bij doorgezet gebruik behalve obstipatie dat aanwezig blijft. Voorschrift voor een laxativum zo nodig, kan het probleem verkleinen	Vaak voorkomende bijwerkingen van opioïden zijn obstipatie, misselijkheid en braken, duizeligheid en sufheid. De meeste bijwerkingen verminderen bij doorgezet gebruik behalve obstipatie, die aanwezig blijft. Voorschrift voor een laxativum “zo nodig” verkleint de kans op obstipatie.	Content unchanged, textual changes to improve clarity.
Flowchart 2: Pharmacological Management of pain for Residents on the Residential Aged Care End of Life Care Pathway (RAC EoLCP)				
11	Review regularly for symptoms of pain → Symptom present: Administer appropriate medication as currently charted for pain. Request MO/NP to review immediately current drugs, both regular and PRN orders	Regelmatige evaluatie van signalen die uiting kunnen zijn van pijn: → Wanneer PIJN aanwezig is, dient de huidige medicatie die voorgeschreven is voor pijn te worden toegediend. De huisarts moet onmiddellijk worden gecontacteerd om de huidige medicatie, zowel de vaste pijnmedicatie als de “zo nodig” medicatie te herbekijken.	Wanneer pijn aanwezig is, dient de voorgeschreven medicatie voor (doorbraak)pijn te worden toegediend.	The necessity for regular symptom assessment is outlined in Recommendation 63. We removed the final sentence, as immediate contact with the general practitioner following every unplanned medication administration is unnecessary. Guidelines when to contact the general practitioner are provided in comment 15.

12	Are regular PRN opioids prescribed for pain? Yes → Ensure order written for PRN dose. PRN order = 1/12 q 2hrly of total daily subcut dose No → Write/request opioid order for pain management. Consider: Morphine 2.5 to 5 mg subcut PRN q 2hrly OR Fentanyl 25 to 50 mcg subcut PRN q 2hrly OR Hydromorphone 0.5 to 1 mg subcut PRN q 2hrly	Maken opioïden deel uit van de vaste of de “zo nodig” medicatie? JA: Zorg voor een voorschrift voor “zo nodig” medicatie, PRN order 1/12 per 2uur van de totale subcutane dosis. NEE: Voorzie voorschrift voor opioïden: bv. 2.5-5mg morfine zo nodig elke 2u. OFWEL fentanyl 25 tot 50 microgram subcutaan zo nodig iedere 2uur OFWEL hydromorfon 0.5 tot 1mg subcutaan zo nodig iedere 2 uur.	Voorzie een “zo nodig” voorschrift opioïden zelfs als er geen pijn aanwezig is: a. Morfine 2.5-5mg subcutaan dat elke 2u kan worden toegediend voor opioïd-naïeve bewoners. b. 1/12 tot 1/6 van de 24-uurs dosering dat iedere 2u kan worden toegediend voor niet opioïd-naïeve bewoners.	As fentanyl and hydromorfon are not available in the Belgian nursing home setting as subcutaneous injections these options were deleted. Furthermore a PRN order for opioid tolerant patients of 1/12 seemed very low according to the guideline development group, common practice is 1/6. Therefore this was adapted to ‘1/12 to 1/6’. During the Delphi study this was further adapted to 1/6 based on the input of the experts.
13	If resident using regular opioids and unable to swallow consider: converting regular oral opioids to appropriate subcut dose administered by CSCI using a syringe driver over 24 hours (see Opioid Conversion Chart)	Als de bewoner regelmatig opioïden gebruikt en niet meer kan slikken, zet de orale dosering opioïden om naar de overeenkomende subcutane dosis (zie omrekeningtabel) die via een spuitdrijver kan worden toegediend over 24u.	Indien de bewoner niet meer kan slikken, zet dan de orale dosering opioïden om naar de overeenkomende subcutane dosis, die via een spuitdrijver kan worden toegediend over 24u (Zie omrekeningtabel BCFI (die wordt toegevoegd in de richtlijn) of gebruik de Orthodose app).	Potential conversion tools were added.
14	If opioid patch in situ consider: - continuing patch at same dose and giving opioid PRN subcut dose for B/T pain (may require advice from specialist palliative care team to calculate appropriate PRN dose) OR - converting patch to appropriate subcut opioid dose administered by CSCI using a syringe driver over 24 hours (see Opioid Conversion Chart)	Als een opioïdpleister in situ is, kan gekozen worden voor a. Doorzetten van de pleister met dezelfde dosering en het geven van een subcutane “zo nodig” dosis (advies van een gespecialiseerd palliatief care team kan nodig zijn om de gepaste “zo nodig” dosis te berekenen) b. OF omzetten van de dosering van de pleister naar de overeenkomende subcutane dosis via spuitdrijver over 24u.	Als een transdermale opioïdpleister in situ is, kan gekozen worden voor: a. Doorzetten van deze pleister in combinatie met subcutane “zo nodig” dosissen, eventueel kan morfine geassocieerd worden via een subcutane spuitdrijver b. OF omzetten van de dosering van de pleister naar de overeenkomende subcutane dosis via spuitdrijver over 24u in combinatie met subcutane “zo nodig” dosissen.	Content unchanged, textual changes to improve clarity.
15	If pain causing distress and/or if multiple PRN opioids administered in previous 24 hours to manage pain, calculate the total dose of B/T opioids over previous 24 hours and add to syringe driver opioid dose or titrate up the opioid dose administered by syringe driver in previous 24 hours by 30%	Als de huidige pijn oncomfortabel is voor de patiënt en/of er meerdere “zo nodig” dosissen werden toegediend in de laatste 24u om de pijn te controleren, bereken dan de totale dosis van deze medicatie om naar de subcutane dosis en voeg deze toe aan de spuitdrijver, of titreer de totale dosis van de spuitdrijver van de voorbije 24u op met 30%.	Als de huidige pijn onvoldoende onder controle is voor de bewoner en/of er meer dan drie “zo nodig” dosissen werden toegediend in de laatste 24u in kader van pijncontrole contacteer dan de huisarts om: a. De totale dosis van deze medicatie om te rekenen naar de subcutane dosis, en deze toe te voegen aan de spuitdrijver,	Two recommendations were merged to improve clarity, without altering their content.

16	If greater than 3 doses of PRN opioids required for B/T pain over 24 hour period, request MO/NP review to consider changes to medication and syringe driver order	STAP 3: Als meer dan 3 dosissen van de “zo nodig” opioïden nodig zijn voor doorbraakpijn binnen de 24u, moet de huisarts worden gecontacteerd voor het aanpassen van de medicatie of spuitdrijver.	b. of de huidige 24uurs dosis van de subcutane spuitdrijver op te drijven met 30%.	
Which pharmacological treatment can be recommended for the management of dyspnea in dying nursing home residents? (clinical question 2)				
17	Dyspnoea is a common symptom experienced in advanced disease irrespective of diagnosis. The prevalence and severity can increase over time particularly in the last days of life.	Ongeacht de diagnose is dyspneu een frequent voorkomend symptoom bij gevorderde ziekte. De ernst en frequentie nemen toe over tijd, vooral in de laatste levensdagen.	Ongeacht de diagnose is dyspneu een frequent voorkomend symptoom bij gevorderde ziekte. De ernst en frequentie nemen toe over tijd, vooral in de laatste levensdagen.	Unchanged
18	There is limited evidence to support the use of oxygen to manage dyspnea at end of life. Oxygen has not been shown to relieve dyspnea in non-hypoxic patients.	Bij niet hypoxische patiënten heeft zuurstof toediening geen bewezen effect op de verbetering van dyspneu.	Er is beperkte evidentie rondom het gebruik van zuurstof bij de behandeling van dyspneu in de laatste levensdagen. Bij niet-hypoxische bewoner heeft zuurstoftoediening geen bewezen effect op de verbetering van dyspneu.	We stipulated the lack of evidence on the use of oxygen during the last days of life. No other changes were made.
19	If a resident is hypoxic, oxygen is recommended for provision of short term relief.	Wanneer een patiënt hypoxisch is wordt zuurstoftoediening wel aangeraden om een snelle verbetering van dyspneu te bekomen.	Er is beperkte evidentie rondom het gebruik van zuurstof bij de behandeling van dyspneu in de laatste levensdagen. Wanneer een bewoner hypoxisch is wordt zuurstoftoediening wel aangeraden om een snelle verbetering van dyspneu te bekomen.	
20	Oxygen should be continued for residents who have required long term use for the management of breathlessness in chronic respiratory illnesses.	Bij patiënten die reeds langdurig zuurstoftherapie krijgen voor een chronische longziekte wordt dit best verdergezet.	Er is beperkte evidentie rondom het gebruik van zuurstof bij de behandeling van dyspneu in de laatste levensdagen. Bij bewoners die reeds langdurig zuurstoftherapie krijgen voor een chronische longziekte wordt dit best verdergezet.	
21	Systemic opioids administered in appropriate doses are safe and effective in managing dyspnea.	Systemische toediening van opioïden, aan een correcte dosis, is veilig en effectief in de behandeling van dyspneu.	Systemische toediening van opioïden, aan een correcte dosis, is veilig en effectief in de behandeling van dyspneu.	Unchanged
22	Opioid naïve residents requiring opioids to manage symptoms should be commenced on the lowest opioid dose possible. Careful upward titration minimises the risk of toxicity.	Bij het voorschrijven van opioïden bij opioïd-naïeve bewoners moet er gestart worden met de laagst mogelijke dosis. Verder op te drijven door voorzichtig te titreren om zo het risico op toxiciteit zo laag mogelijk te houden.	Bij het voorschrijven van opioïden bij opioïd-naïeve bewoners moet er gestart worden met de laagst mogelijke dosis (bv. morfine 1.5mg subcutaan). Verder opdrijven gebeurt door voorzichtig optitreren om het risico op toxiciteit zo laag mogelijk te houden.	The dosage recommendation from the dyspnea flowchart was included.
23	Morphine should be avoided in residents with severe renal failure	Morfine wordt best vermeden bij bewoners met ernstige nierinsufficiëntie	Morfine wordt best vermeden bij bewoners met ernstige nierinsufficiëntie	In line with Recommendation 4, fentanyl was not considered as it

	(eGFR<30) due to the build-up of toxic metabolites. Fentanyl has no active metabolites of relevance and has been identified as an opioid that is least likely to cause harm in residents with severe renal impairment when used appropriately.	(eGFR<30ml/min) vanwege de opstapeling van toxische metabolieten. Fentanyl heeft geen relevante actieve metabolieten en is gekend als het opioïd met de minste kans om schade te veroorzaken in bewoners met ernstige nierinsufficiëntie, indien het gepast wordt gebruik.	(eGFR<30ml/min) vanwege het risico op opstapeling van toxische metabolieten.	is not available in Belgian nursing homes.
24			Achtergrond: Bij patiënten met ernstig renaal falen [eGFR<30ml/min], wordt beschreven in de literatuur dat morfine dient vermeden te worden o.w.v. stapeling van toxische metabolieten. In andere landen gaat daarom de voorkeur uit naar fentanyl en oxycodon subcutaan, die minder relevante toxische metabolieten hebben. Fentanyl en oxycodon subcutaan zijn in België echter niet beschikbaar of heel duur buiten de ziekenhuissetting. <u>Stelling:</u> De dosering van morfine bij ernstig renaal falen [eGFR<30ml/min] bij woonzorgcentrumbewoners in de laatste levensdagen dient niet aangepast te worden aan de nierfunctie (deze stelling werd toegevoegd door de onderzoekers).	Additionally, Recommendation 23 was included to assess whether experts would avoid using morphine entirely in residents with severe renal failure.
25	Anxiety is often associated with shortness of breath and benzodiazepines are effective in managing this symptom.	Angst is frequent geassocieerd met dyspneu en benzodiazepines zijn effectief in de behandeling van dit symptoom.	Angst is frequent geassocieerd met dyspneu, waardoor benzodiazepines effectief zijn in de behandeling hiervan.	Unchanged
Flowchart 3: Pharmacological Management of dyspnea for Residents on the Residential Aged Care End of Life Care Pathway (RAC EoLCP)				
26	Are opioids prescribed for any reason? YES: Ensure order written for PRN dose. PRN order = 1/12 q 2hrly of total daily subcut dose NO: Write/request opioid order for shortness of breath. Consider: Morphine 1.5 to 2.5 mg subcut PRN q 2hrly OR Fentanyl 25 to 50 mcg subcut PRN q 2hrly OR Hydromorphone 0.25 to 0.5 mg subcut PRN q 2hrly	Maken opioïden deel uit van de vaste of de “zo nodig” medicatie? JA: Zorg voor een voorschrift voor “zo nodig” medicatie, 1/12 van de totale subcutane dosis per 2 uur. NEE: voorzie voorschrift voor opioïden: bv. 1.5-2.5mg morfine “zo nodig” elke 2u.	Voorzie een “zo nodig” voorschrift opioïden zelfs als dyspneu niet aanwezig is: 1. Morfine 1.5-2.5mg subcutaan die elke 2u kan worden toegediend voor opioïd-naïeve bewoners. 2. 1/12 van de 24-uurs dosering die iedere 2u kan worden toegediend voor niet-opioïd-naïeve bewoners	See Recommendation 11.
27	Administer appropriate medication(s) as currently charted for respiratory distress.	Wanneer dyspneu aanwezig is dient de huidige medicatie die voorgeschreven is voor dyspneu te worden toegediend.	Wanneer dyspneu aanwezig is, dient de voorgeschreven medicatie voor (doorbraak) dyspneu te worden toegediend.	Content unchanged, textual changes to improve clarity.
28	If resident is very distressed and/or requiring multiple PRN opioids to manage breathlessness, may need	Bij ernstig discomfort en/of nood aan meerdere “zo nodig” dosissen om dyspneu onder controle te houden kan een hogere	Bij ernstig discomfort en/of nood aan drie of meer “zo nodig” dosissen om dyspneu onder controle te houden kan een hogere dosering in	Unchanged

	higher dose in syringe driver but generally advised not to titrate above 30% of previous daily requirements	dosering in de spuitdrijver nodig zijn, maar in het algemeen wordt geadviseerd om niet meer dan 30% boven de voorgaande dagdosisdosis te titreren.	de spuitdrijver nodig zijn, maar in het algemeen wordt geadviseerd om niet meer dan 30% boven de voorgaande dagdosis te titreren.	
29	If resident using regular opioids and unable to swallow consider converting regular oral opioids to appropriate subcut dose administered by CSCI using syringe driver over 24 hours.	Als de bewoner regelmatig opioïden gebruikt en niet meer kan slikken, zet dan de orale dosering opioïden om naar de overeenkomende subcutane dosis die via een subcutane spuitdrijver kan worden toegediend over 24u.	Als de bewoner regelmatig opioïden gebruikt en niet meer kan slikken, zet dan de orale dosering opioïden om naar de overeenkomende subcutane dosis die via een subcutane spuitdrijver kan worden toegediend over 24u (Zie omrekeningstabel BCFI (die wordt toegevoegd in de richtlijn) of gebruik de Orthodose app).	Potential conversion tools were added.
30	If greater than 3 doses of any PRN medication required over 24 hour period, request MO/NP review to consider changes to medications and syringe driver orders	Als meer dan 3 dosissen van de “zo nodig” medicatie nodig zijn voor binnen de 24u, moet de huisarts worden gecontacteerd voor het aanpassen van de medicatie of subcutane spuitdrijver.	Als meer dan 3 dosissen van de “zo nodig” medicatie nodig zijn binnen de 24u, moet de huisarts worden gecontacteerd voor het aanpassen van de medicatie.	Content unchanged, textual changes to improve clarity.
Which pharmacological treatment can be recommended for management of nausea and vomiting in dying nursing home residents? (clinical question 3)				
31	Factors contributing to nausea and vomiting in a resident with a life-limiting illness may include but are not limited to: drug toxicity, urinary tract infection, constipation, diseases of the gastrointestinal tract, metabolic and biochemical disturbance and organ failure. Cause(s) of nausea and vomiting in the last days of life may be unidentifiable and multi-factorial.	Factoren die nausea en braken kunnen veroorzaken bij bewoners met een levensbeperkende prognose omvatten, maar zijn niet beperkt tot: medicatie toxiciteit, urineweginfecties, constipatie, gastro-intestinale aandoeningen, metabole en biochemische afwijkingen en orgaan falen. Oorzaken van nausea en braken in de laatste levensdagen kunnen onbekend en multifactorieel zijn.	Factoren die nausea en braken kunnen veroorzaken bij bewoners met een levensbedreigende ziekte zijn onder andere: toxiciteit van medicatie, urineweginfecties, constipatie, gastro-intestinale aandoeningen, metabole en biochemische ontregeling en orgaanfalen. Oorzaken van nausea en braken in de laatste levensdagen kunnen ook onbekend of multifactorieel zijn. Het achterhalen van deze oorzaak is niet altijd aangewezen in de laatste levensdagen.	The last sentence was added to avoid unnecessary investigations for the resident.
32	Nausea is often under recognised and under treated.	Nausea wordt vaak niet herkend en onderbehandeld.	Nausea wordt vaak niet herkend en wordt hierdoor niet adequaat behandeld.	Content unchanged, textual changes to improve clarity.
33	There is limited evidence to guide the use of antiemetic therapy in the elderly.	Er is beperkte evidentie omtrent het gebruik van anti-emetica bij bewoners.	Er is beperkte evidentie omtrent het gebruik van anti-emetica bij bewoners.	Unchanged
34	Opioids commonly cause nausea and vomiting. Metoclopramide has been shown to be effective in the management of nausea and vomiting in patients with cancer who are on opioid therapy.	Opioïden zijn een frequente oorzaak van nausea en braken. Metoclopramide is effectief in de behandeling van nausea en braken in patiënten met kanker die behandeld worden met opioïden.	Opioïden zijn een frequente oorzaak van nausea en braken. Metoclopramide is effectief in de behandeling van nausea en braken bij patiënten met kanker die behandeld worden met opioïden.	Unchanged

35	Haloperidol can be trialed to manage nausea and vomiting if metoclopramide is ineffective.	Haloperidol kan op proef worden toegediend indien metoclopramide ineffectief is in de behandeling van nausea en braken.	Haloperidol kan op proef worden toegediend indien metoclopramide ineffectief is in de behandeling van nausea en braken.	Unchanged
36	Metoclopramide or haloperidol can cause extrapyramidal side effects. These medications need to be avoided or used with caution in residents with neurodegenerative disorders such as Parkinson's disease.	Metoclopramide of haloperidol kan extrapyramidale nevenwerkingen veroorzaken. Deze medicatie moet vermeden worden of met de nodige voorzichtigheid worden gebruikt bij bewoners met neurodegeneratieve aandoeningen zoals de ziekte van Parkinson.	Men moet bedacht zijn op extrapyramidale nevenwerkingen bij het gebruik van metoclopramide en haloperidol. Deze medicatie moet vermeden worden of met de nodige voorzichtigheid worden gebruikt bij bewoners met neurodegeneratieve aandoeningen zoals de ziekte van Parkinson.	Content unchanged, textual changes to improve clarity.
37	Subcutaneous infusion of antiemetics delivered via a syringe driver has been shown to be effective in managing persistent symptoms of nausea and vomiting.	Subcutane toediening van anti-emetica via een spuitdrijver heeft bewezen effectiviteit in de behandeling van aanhoudende nausea en braken.	Continue subcutane toediening van anti-emetica is effectief in de behandeling van aanhoudende nausea en braken.	Content unchanged, textual changes to improve clarity.
Flowchart 1: Pharmacological Management of nausea and vomiting for Residents on the Residential Aged Care End of Life Care Pathway (RAC EoLCP)				
38	Write/request medication order for metoclopramide 10 mg subcut PRN q tds	Voorzie een voorschrift voor metoclopramide 10mg subcutaan zo nodig tot 3 keer per dag.	Voorzie een "zo nodig" voorschrift voor anti-emetica zelfs indien nausea en braken afwezig zijn. Overweeg metoclopramide 10mg subcutaan tot drie keer per dag zo nodig.	Content unchanged, textual changes to improve clarity.
39	Review regularly for symptoms of nausea and vomiting. If symptom present: Administer appropriate medication as currently charted for nausea and vomiting.	Wanneer Nausea en braken aanwezig is, dient het voorgeschreven anti-emeticum te worden toegediend.	Wanneer nausea en/of braken aanwezig is/zijn, dient de voorgeschreven medicatie voor (doorbraak) nausea en braken te worden toegediend.	See Recommendation 11.
40	If nausea and vomiting persist, or if resident using regular oral antiemetic and unable to swallow, consider converting to metoclopramide 20 to 30 mg administered by CSCI using a syringe driver over 24 hours	Bij aanhoudende nausea en braken of bij regelmatige nood aan orale anti-emetica en wanneer het slikken niet meer lukt, overweeg dan om over te gaan op metoclopramide 20-30mg, toegediend over 24u via een subcutane spuitdrijver.	Overweeg continue subcutane toediening van metoclopramide 20-30mg over 24u met "zo nodig" dosissen van 10mg tot drie keer per dag indien: a. Nausea of braken aanwezig blijft ondanks toediening van "zo nodig" anti-emetica b. De bewoner niet meer kan slikken en er orale anti-emetica zijn voorgeschreven	Content unchanged, textual changes to improve clarity.
41	If nausea and vomiting persist consider trial of haloperidol 0.5 to 1.5 mg subcut PRN q bd	Bij persisterend nausea en braken, overweeg dan haloperidol 0.5-1.5mg subcutaan "zo nodig", tot twee keer per dag.	Bij aanhoudende nausea en/of braken, ondanks systematische toediening van metoclopramide, overweeg een "zo nodig" voorschrift haloperidol 0.5-1.5mg subcutaan, tot twee keer per dag.	Content unchanged, textual changes to improve clarity.

42	If haloperidol appears to be more effective than metoclopramide in managing nausea and vomiting consider changing to CSCI of haloperidol using a syringe driver over 24 hours.	Indien haloperidol effectiever is dan metoclopramide in de behandeling van nausea en braken, overweeg dan om over te schakelen naar haloperidol dat via een spuitdrijver kan worden toegediend over 24 uur.	Indien de behandeling van nausea en/of braken met haloperidol effectiever is dan met metoclopramide, overweeg dan overschakelen naar haloperidol via continue subcutane toediening.	Content unchanged, textual changes to improve clarity.
43	If greater than 3 doses of PRN metoclopramide 10 mg subcut required over 24 hour period, or if prescribed haloperidol dose ineffective over 24 hour period, request MO/NP review to consider changes in medication and syringe driver orders.	Als meer dan 3 dosissen van metoclopramide 10mg subcutaan “zo nodig” zijn toegediend binnen de 24u, of als de voorgeschreven dosis haloperidol ineffectief is over 24-uur moet de huisarts worden gecontacteerd voor het aanpassen van de medicatie of spuitdrijver.	Als meer dan 3 dosissen van metoclopramide 10mg subcutaan “zo nodig” zijn toegediend binnen de 24u, of als de voorgeschreven dosis haloperidol ineffectief is over 24u moet de huisarts worden gecontacteerd om een gespecialiseerd palliatief team in te schakelen.	We specified that the general practitioner should contact a specialized palliative care team if nausea persists, as it can cause significant distress and may be challenging to manage effectively. During the Delphi procedure the involvement of a specialized palliative care team was made optional.
Which pharmacological treatment can be recommended for management of agitation and restlessness in dying nursing home residents? (clinical question 6a)				
44	Restlessness and agitation occur commonly at end of life and can often be attributed to multiple causes. Investigating the underlying cause may not be appropriate in the last days of life.	Rusteloosheid en agitatie komen frequent samen voor op het einde van het leven, dit kan meerdere oorzaken hebben. Deze onderliggende oorzaak achterhalen is niet altijd aangewezen in de laatste levensdagen.	Rusteloosheid en agitatie komen frequent samen voor aan het levenseinde en kunnen meerdere oorzaken hebben. Deze onderliggende oorzaak achterhalen is niet altijd aangewezen in de laatste levensdagen.	Unchanged
45	It is important to assess and manage factors which contribute to restlessness and agitation such as pain, urinary retention, rectal impaction, hypoxia, environmental factors, psychological and spiritual distress.	Wel is het belangrijk om uitlokkende factoren van rusteloosheid en agitatie te behandelen, zoals o.a. pijn, urineretentie, faecale impactie, hypoxie, omgevingsfactoren, psychologische en spirituele nood.	Wel is het belangrijk om uitlokkende factoren van rusteloosheid en agitatie te behandelen, zoals o.a. pijn, urineretentie, faecale impactie, hypoxie, omgevingsfactoren, psychologische en spirituele nood.	Unchanged
Which pharmacological treatment can be recommended for the management of confusion and delirium in dying nursing home residents? (clinical question 6b)				
46	Low dose haloperidol is effective in managing restlessness and agitation associated with delirium.	Lage dosis haloperidol is effectief in de behandeling van rusteloosheid en agitatie veroorzaakt door delier.	Overweeg een lage dosis haloperidol voor de behandeling van agitatie of motorische onrust veroorzaakt door delier bij woonzorgcentrumbewoners in de laatste levensdagen.	Content unchanged, textual changes to improve clarity.
47	Extrapyramidal side effects (dystonia and akathisia) occur more commonly in doses of haloperidol above 4.5 mg per day.	Extrapiramidale nevenwerkingen (dystonie en acathisie) komen frequenter voor bij haloperidol dosissen boven 4.5 mg per dag.	Wees aandachtig voor extrapyramidale bijwerkingen bij het gebruik van antipsychotica.	The maximum dose was removed because the systematic review that informed this dose has been updated, and no new maximum dose has been established.

Flowchart 4: Pharmacological Management of Restlessness and Agitation for Residents on the Residential Aged Care End of Life Care Pathway (RAC EoLCP)				
48	Write/request antipsychotic order for persistent restlessness and agitation. Consider: Haloperidol 0.5 to 1 mg subcut PRN q bd	Voorzie een “zo nodig” voorschrift antipsychotica voor doorbraak rusteloosheid of agitatie. Overweeg: Haloperidol 0.5-1mg subcutaan zo nodig, tot 2 keer per dag	Voorzie een “zo nodig” voorschrift voor antipsychotica zelfs als er geen agitatie of rusteloosheid aanwezig zijn. Overweeg haloperidol 0.5-1mg subcutaan tot 2 keer per dag.	Unchanged
49	Review regularly for symptoms of restlessness and agitation If symptom present: Administer appropriate medication as currently charted for restlessness and agitation. Request MO/NP to review immediately current drugs, both regular and PRN orders	Evalueer regelmatig of rusteloosheid en agitatie aanwezig zijn. Indien aanwezig: Wanneer rusteloosheid en agitatie aanwezig zijn, dient de op dat moment voorgeschreven medicatie voor (doorbraak) rusteloosheid en agitatie te worden toegediend.	Wanneer een delier aanwezig is, dient de voorgeschreven medicatie te worden toegediend.	See Recommendation 11.
50	If resident only has a PRN oral or wafer antipsychotic convert to subcut dose	Als de bewoner behandeld wordt met zo nodig antipsychotica, schakel dan over naar subcutane toediening.	Indien de bewoner enkel behandeld wordt met “zo nodig” antipsychotica, overweeg dan om dit om te zetten naar een subcutane toediening.	This recommendation was changed to "consider" because the guideline development group determined that the conversion is not always necessary.
51	If resident on long term antipsychotic and unable to swallow, requires conversion to CSCI using syringe driver over 24 hours. Consider: Haloperidol commencement dose depends upon previous dose and severity of symptoms	Indien de bewoner niet meer kan slikken, maar langdurig behandeld werd met antipsychotica, zet dit dan om naar continue subcutane toediening via spuitdrijver over 24u. Overweeg: Haloperidol, startdosis hangt af van initiële antipsychotica dosering en de ernst van symptomen.	Indien de bewoner slikproblemen ontwikkelt, maar chronisch behandeld wordt met een antipsychoticum, zet dit dan om naar een continue subcutane toediening via spuitdrijver over 24u, bv. haloperidol. Startdosis hangt af van initiële antipsychotica dosering en de ernst van symptomen. (voor omrekening zie www.psychiatrienet.nl/switchtabel/)	A link to a conversion tool was added.
52	If greater than 3 doses of any PRN medication required for B/T restlessness/agitation over 24 hour period, request MO/NP review to consider changes to medication and syringe driver orders	Als meer dan 3 dosissen “zo nodig” medicatie nodig zijn voor doorbraak van rusteloosheid of agitatie binnen de 24u, moet de huisarts worden gecontacteerd voor het aanpassen van de medicatie of spuitdrijver.	Als meer dan 3 dosissen “zo nodig” medicatie nodig zijn voor doorbraak van rusteloosheid of agitatie binnen de 24u, moet de huisarts gecontacteerd worden om een gespecialiseerd palliatief zorgteam in te schakelen.	We specified that the general practitioner should contact a specialized palliative care team if restlessness/agitation persists, as it can cause significant distress and may be challenging to manage effectively. During the Delphi procedure the involvement of a specialized palliative care team was made optional.

Which pharmacological treatment can be recommended for management of anxiety in dying nursing home residents? (clinical question 6c)				
53	Anxiety is often associated with shortness of breath and benzodiazepines are effective in managing this symptom.	Angst is frequent geassocieerd met dyspneu en benzodiazepines zijn effectief in de behandeling van dit symptoom.	Angst is frequent geassocieerd met dyspneu en benzodiazepines zijn effectief in de behandeling hiervan.	Content unchanged, textual changes to improve clarity.
54	Restlessness and agitation at end of life can be caused by anxiety and distress. The addition of a low dose benzodiazepine can be effective in managing these symptoms.	Rusteloosheid en agitatie op het einde van het leven kunnen veroorzaakt worden door angst en onrust. Het toedienen van een lage dosis benzodiazepines kan effectief zijn in het behandelen van deze symptomen.	Rusteloosheid en agitatie op het einde van het leven kunnen veroorzaakt worden door angst. Het toedienen van een lage dosis benzodiazepines kan effectief zijn in de behandeling hiervan.	Content unchanged, textual changes to improve clarity.
Flowchart 4: Pharmacological Management of Restlessness and Agitation for Residents on the Residential Aged Care End of Life Care Pathway (RAC EoLCP)				
55	Write or request order for benzodiazepine to manage restlessness and agitation. Consider: Midazolam 2.5 to 5mg subcut PRN q 2hrly OR Clonazepam 0.25 to 0.5 mg oral drops or subcut PRN q 4hrly	Zorg voor een voorschrift voor “zo nodig” medicatie, overweeg subcutane midazolam of clonazepam per os. Overweeg: Midazolam 2.5-5mg subcutaan, zo nodig iedere 2u Overweeg: Clonazepam 0.25mg-0.5mg orale siroop	Voorzie een “zo nodig” voorschrift voor benzodiazepines zelfs als er geen rusteloosheid en agitatie aanwezig is. Overweeg midazolam 2.5-5mg elke 2u ofwel clonazepam orale druppels 0.25-0.5mg elke 4u, bij bewoners waar nog geen benzodiazepines zijn voorgeschreven.	Content unchanged, textual changes to improve clarity.
56	Review regularly for symptoms of restlessness and agitation If symptom present: Administer appropriate medication as currently charted for restlessness and agitation. Request MO/NP to review immediately current drugs, both regular and PRN orders	Evalueer regelmatig of rusteloosheid en agitatie aanwezig zijn. Indien aanwezig: Wanneer rusteloosheid en agitatie aanwezig zijn, dient de op dat moment voorgeschreven medicatie voor (doorbraak) rusteloosheid en agitatie te worden toegediend. Na toediening van de voorgeschreven medicatie, wordt de huisarts gecontacteerd om de voorgeschreven medicatie voor rusteloosheid en agitatie (zowel de standaard medicatie als de “zo nodig” medicatie) te herbekijken.	Wanneer angst aanwezig is, dient de voorgeschreven medicatie te worden toegediend.	See Recommendation 11.

57	If resident on long term regular benzodiazepine and unable to swallow, requires conversion to CSCI using syringe driver over 24 hours. Consider: Midazolam, usual commencement dose 5 to 10 mg over 24 hours but may need higher dose depending on previous 24 hour dose OR Clonazepam, usual commencement dose 1 to 2 mg over 24 hours but may need higher dose depending on previous 24 hour dose	Indien de bewoner niet meer kan slikken, maar langdurig behandeld werd met benzodiazepines, zet dit dan om naar continue subcutane toediening via spuitdrijver over 24u. Overweeg: Midazolam – start met 5-10mg over 24u, maar mogelijks nood aan hogere doseringen afhankelijk van de vorige 24uurs dosering. Ofwel Clonazepam, start met 1-2mg over 24u, maar mogelijks nood aan hogere doseringen afhankelijk van de vorige 24uurs dosering.	Indien de bewoner regelmatig nood heeft aan “zo nodig” dosissen benzodiazepines of chronisch behandeld wordt met orale benzodiazepines en slikmoeilijkheden ontwikkelt, zet dit dan om naar midazolam via continue subcutane toediening. Start minimaal met midazolam 5-10mg over 24u, maar hogere doseringen kunnen nodig zijn.	Since clonazepam is not available for subcutaneous injection in Belgium, this option was removed.
58	If resident only has PRN oral benzodiazepine tablet order convert to oral drops or subcut dose	Als de bewoner enkel behandeld wordt met zo nodig orale benzodiazepine tabletten, schakel dan over naar orale druppels of subcutane toediening.	Indien de bewoner enkel behandeld wordt met “zo nodig” orale benzodiazepines, overweeg dan om dit om te zetten naar midazolam via subcutane toediening.	Since clonazepam is not commonly used in this setting in Belgium, this option was removed. However, during the Delphi study, it was reintroduced.
59	If greater than 3 doses of any PRN medication required for B/T restlessness/agitation over 24 hour period, request MO/NP review to consider changes to medication and syringe driver order	Als meer dan 3 dosissen “zo nodig” medicatie nodig zijn voor doorbraak van rusteloosheid of agitatie binnen de 24u, moet de huisarts worden gecontacteerd voor het aanpassen van de medicatie of spuitdrijver.	Als meer dan 3 dosissen van de “zo nodig” medicatie nodig zijn binnen de 24u, moet de huisarts worden gecontacteerd voor het aanpassen van de medicatie.	Content unchanged, textual changes to improve clarity.
60	If symptoms persist consider use of antipsychotic medication.	Bij persisterende symptomen moet het gebruik van antipsychotica overwogen worden.	Bij persisterende angst of rusteloosheid ondanks systematische toediening van benzodiazepines, overweeg associatie van antipsychotica.	Content unchanged, textual changes to improve clarity.
Key principles and key points translated from source guideline				
61	Medications are prescribed, obtained, charted and administered according to the Australian National Medicines Policy and in accordance with regional jurisdictional requirements and local facility policies and procedures.	Medicatie wordt voorgeschreven, verkregen, genoteerd en toegediend volgens het lokale beleid en in overeenstemming met...	Medicatie wordt voorgeschreven door de behandelende huisarts, die een coördinerende rol heeft.	This recommendation was adapted to reflect Belgian practices.
62	Knowledge by the resident, or their substitute decision maker if appropriate, that the dying process is occurring and that medication administration may improve the quality of death.	De bewoner en/of hun plaatsvervangende beslissers moeten op de hoogte gebracht worden dat het stervensproces plaats vindt en dat medicatietoediening de kwaliteit van sterven kan verbeteren.	De bewoners en/of hun vertegenwoordiger moeten op de hoogte gebracht worden dat het stervensproces is gestart en dat medicatietoediening de kwaliteit van sterven kan verbeteren.	Content unchanged, textual changes to improve clarity.

63	Residents on the RAC EoLCP require 2 hourly symptom assessment. This allows for emergent symptoms to be detected quickly and treated pharmacologically if required. Efficacy of administered medications should be evaluated and documented.	Bewoners die in dit traject worden geïnccludeerd vereisen symptoom assessment om de 2u. Dit zorgt ervoor dat nieuwe symptomen snel worden gedetecteerd en farmacologisch kunnen worden behandeld als dat nodig is.	Het traject vereist dat symptomen om de 2u worden beoordeeld bij bewoners die in dit traject worden geïnccludeerd. Dit zorgt ervoor dat nieuwe symptomen snel worden gedetecteerd en waar nodig medicamenteus kunnen worden behandeld.	Content unchanged, textual changes to improve clarity.
64	If a medication is considered necessary, the most appropriate medicine is chosen and used safely and effectively	Als medicamenteuze behandeling nodig wordt geacht, moet het meest geschikte geneesmiddel worden gekozen dit dient dan veilig en effectief te worden gebruikt.	Als medicamenteuze behandeling nodig wordt geacht, moet het meest geschikte geneesmiddel worden geselecteerd, dat op een veilige en effectieve manier wordt toegediend	Content unchanged, textual changes to improve clarity.
65	Medications and doses prescribed should be based on careful assessment of the dying resident's condition and symptoms.	De voorgeschreven medicatiedoseringen zijn gebaseerd op regelmatige assessment en ernst van de symptomen bij bewoners.	De voorgeschreven medicatie en doseringen moeten gebaseerd zijn op nauwkeurige beoordeling van de bewoner zijn toestand en (ernst van de) symptomen.	Content unchanged, textual changes to improve clarity.
66	The burden of how medications are given and of potential side effects should be minimised. Palliative care medications at the end of life are usually given via the subcutaneous route, which is generally the least invasive and most reliable route in the dying resident.	De belasting veroorzaakt door medicatietoediening of bijwerkingen moet zo klein mogelijk zijn. Medicatie in de laatste levensdagen wordt meestal subcutaan toegediend, deze toedieningsweg is over het algemeen het minst invasief en meest betrouwbaar bij een stervende bewoner.	De belasting veroorzaakt door medicatietoediening of bijwerkingen moet zo klein mogelijk zijn. Medicatie in de laatste levensdagen wordt meestal subcutaan toegediend, deze toedieningsweg is over het algemeen het minst invasief en meest betrouwbaar bij een stervende bewoner.	Unchanged
67	Medications that have minimal therapeutic benefit in the terminal phase of life should be ceased.	Medicatie met een beperkt therapeutisch voordeel in de terminale levensfase moet worden gestopt.	Medicatie met een beperkt therapeutisch voordeel in de terminale levensfase moet worden gestopt.	Unchanged
68	Persistent symptoms require regular rather than PRN (as needed) orders.	Persisterende symptomen vereisen onderhoudsmedicatie op vaste tijdstippen eerder dan "zo nodig" medicatie.	Persisterende symptomen vereisen onderhoudsmedicatie op vaste tijdstippen eerder dan "zo nodig" medicatie.	Unchanged
69	Use of regular medications to manage symptoms does not preclude the need for appropriate breakthrough dose orders. PRN orders should be written for intermittent symptoms and to cover possible breakthrough events for persistent symptoms.	Deze onderhoudsmedicatie sluit de nood aan "zo nodig" medicatie niet uit. Een "zo nodig" medicatievoorschrift moet worden voorgeschreven om doorbraaksymptomen en symptomen die slechts tijdelijk aanwezig zijn snel en optimaal te kunnen behandelen	Deze onderhoudsmedicatie sluit de nood aan "zo nodig" medicatie niet uit. Een "zo nodig" medicatievoorschrift moet worden voorgeschreven om doorbraaksymptomen en symptomen die slechts tijdelijk aanwezig zijn snel en optimaal te kunnen behandelen.	Unchanged
70	Medications are immediately available to ensure optimal symptom control.	Geneesmiddelen zijn onmiddellijk beschikbaar om een optimale symptoomcontrole te verzekeren.	Geneesmiddelen zijn onmiddellijk beschikbaar om een optimale symptoomcontrole te verzekeren.	Unchanged

71	Charted medication doses are based on frequent assessment of the resident and are appropriate to the severity of the symptom(s). Persistent symptoms are treated with regular doses of medication while as needed doses of medication are charted to cover 'break through' symptoms. Medications are administered by the most reliable route.	De vermelde medicatiedoseringen zijn gebaseerd op frequente symptoomassessment bij de bewoner, en zijn geschikt voor de ernst van symptomen. Persisterende symptomen worden behandeld met regelmatige medicatie toediening, terwijl zo nodig medicatie wordt gebruikt voor doorbraak symptomen. Medicatie wordt toegediend volgens de meest betrouwbare toedieningsweg.	Effectiviteit van de toegediende medicatie dient te worden geëvalueerd en gedocumenteerd.	This recommendation was simplified to avoid misinterpretations. <i>"Persisterende symptomen worden behandeld met regelmatige medicatie toediening, terwijl zo nodig medicatie wordt gebruikt voor doorbraak symptomen. Medicatie wordt toegediend volgens de meest betrouwbare toedieningsweg".</i> Was deleted as this is already mentioned in Recommendations 66 and 69.
72	Always consider non-pharmacological interventions in addition to the pharmacological management of end of life (terminal) symptoms.	Naast de farmacologische behandeling, dient ook niet-farmacologische behandeling overwogen te worden.	Denk naast de farmacologische behandeling, ook steeds aan een niet-farmacologische aanpak van de symptomen.	Content unchanged, textual changes to improve clarity.
73	Registered and enrolled nurses are responsible for: regularly assessing symptoms;	Verpleegkundigen zijn verantwoordelijk voor: Regelmatig assessment van symptomen	De verpleegkundige is verantwoordelijk voor: a. Regelmatige beoordeling van symptomen	Content unchanged, textual changes to improve clarity.
74	Registered and enrolled nurses are responsible for: administering PRN medications when required	Verpleegkundigen zijn verantwoordelijk voor: Toedienen van "zo nodig medicatie" waar nodig	De verpleegkundige is verantwoordelijk voor: b. Toediening van "zo nodig" medicatie wanneer nodig	Content unchanged, textual changes to improve clarity.
75	Registered and enrolled nurses are responsible for: regularly monitoring and documenting the effectiveness of prescribed medication(s)	Verpleegkundigen zijn verantwoordelijk voor: Regelmatig monitoren en documenteren van de effectiviteit van voorgeschreven medicatie	De verpleegkundige is verantwoordelijk voor: c. Regelmatige monitoring en documentatie van de effectiviteit van voorgeschreven medicatie	Content unchanged, textual changes to improve clarity.
76	Registered and enrolled nurses are responsible for: identifying and reporting side effects/adverse reactions caused by prescribed medication.	Verpleegkundigen zijn verantwoordelijk voor: Identificeren en rapporteren van nevenwerkingen/bijwerkingen veroorzaakt door de voorgeschreven medicatie.	De verpleegkundige is verantwoordelijk voor: d. Identificatie en rapportering van nevenwerkingen/bijwerkingen veroorzaakt door de voorgeschreven medicatie.	Content unchanged, textual changes to improve clarity.
77	Responses to administered medications are charted and adverse reactions noted and notified. The Therapeutic Goods Administration encourages reporting of all suspected adverse reactions to prescription, over-the-counter and complementary medicines. Information on how to	Het effect van de toegediende medicatie wordt geregistreerd, bijwerkingen worden genoteerd en gemeld aan de behandelende arts. Het XXX moedigt de rapportage van alle bijwerkingen ten sterkste aan. Informatie over rapportage van bijwerkingen kan online gevonden worden op XX	Het effect van de toegediende medicatie wordt geregistreerd, bijwerkingen worden genoteerd en gemeld aan de huisarts. Het Federaal Agentschap voor Geneesmiddelen en Gezondheidsproducten moedigt de rapportage van alle bijwerkingen ten sterkste aan. Informatie over melden van bijwerkingen	The Belgian federal agency and site were added in place of the Australian one.

	lodge a report together with the 'blue card' adverse reaction reporting form are available online at http://www.tga.gov.au/reporting-medicine-and-vaccine-adverse-events		kan online gevonden worden op https://www.fagg.be/nl/bijwerking .	
78	Action is taken in the event of a medication error occurring - e.g. under- or over-dosing according to local policy and procedure documentation	Toepasselijke acties worden ondernomen in het geval van een medicatiefout (zijnde onder- of overdosering) volgens lokale procedures.	Een medicatiefout (bv. onder- of overdosering) dient gemeld te worden aan de behandelende huisarts.	This recommendation was adapted to reflect Belgian practices.

Chapter 11: Recommendations before consensus procedure

In the following step the levels of evidence of the translated recommendations were reevaluated in context of:

- Literature found through the systematic reviews (39,40). Additional references are added in **bold**.
- The two NICE guidelines.
- Reevaluation of the initial references from the Australian guideline. All references from the original Australian guideline were reevaluated on their methodological values and relevance to our guideline (Appendix G).

Limited evidence was found in the systematic reviews. The two NICE guidelines give broad recommendations without any specific advice on how to treat the different symptoms and the references from the original Australian guidelines do not focus on our goal population (people dying in nursing homes). This results in low levels of evidence, whereby the initial levels of evidence in the Australian guideline were frequently downgraded.

Table 10: List of all recommendations scored in the first Delphi round			
References	Recommendation	Source guideline	After review of the literature
Which pharmacological treatment can be recommended for management of pain in dying nursing home residents? (clinical question 1)			
(10) Mercadante & Acuri 2007	Pijn is een belangrijk probleem bij woonzorgcentrumbewoners, de prevalentie van aanhoudende pijn in deze populatie bedraagt 49-80%.	III-2	III-2
(11) Pergolizzi et al. 2008 (29) Fürst et al. 2020	Opioïden zijn effectief en worden meestal goed getolereerd door ouderen.	I /	V III-3
(11) Pergolizzi et al. 2008 (12) Elliott 2006	Bij bewoners die vooraf geen opioïden kregen, hierna opioïd-naïeve bewoners genoemd, gebruikt men de laagste dosis voor pijn management. Het voorzichtig ophogen van de dosis vermindert het risico op toxiciteit/ongewenste nevenwerkingen.	I V	V V
(13) King et al. 2011	Achtergrond: Bij patiënten met ernstig renaal falen [eGFR<30], wordt beschreven in de literatuur dat morfine dient vermeden te worden o.w.v. stapeling van toxische metabolieten. In andere landen gaat daarom de voorkeur uit naar fentanyl en oxycodon subcutaan, die minder relevante toxische metabolieten hebben. Fentanyl en oxycodon subcutaan zijn in België echter niet beschikbaar of heel duur buiten de ziekenhuissetting. Stelling: De dosering van morfine bij ernstig renaal falen [eGFR<30] bij woonzorgcentrumbewoners in de laatste levensdagen dient niet aangepast te worden aan de nierfunctie. (Stelling opgesteld door de onderzoekers)	I	V
(14) Farrington et al. 2010	Voor de controle van aanhoudende pijn, is 24u/24u toediening van opioïden noodzakelijk, rekening houdend met de werkingsduur van het voorgeschreven opioïde.	V	V
(15) Portenoy et al. 2006 (14) Farrington et al. 2010	Doorbraakpijn komt regelmatig voor bij bewoners met aanhoudende pijn. Naast de onderhoudsmedicatie, moet een "zo nodig" voorschrift worden voorzien voor doorbraakpijn. De dosering hiervan bedraagt 1/12 tot 1/6 van de 24-uurs dosering en kan elke 2u worden toegediend.	III-2 V	III-2 V
(16) Rossi et al. 2013	Transdermale opioïdpleisters (buprenorfine en fentanyl) zijn niet geschikt om te starten in de laatste levensdagen. Het pijnstillend effect laat namelijk even op zich wachten, waardoor geen snelle en veilige dosistitratie kan gebeuren.	I	I

(14) Farrington et al. 2010	Wanneer opioïden worden gestart in de laatste levensdagen of wanneer perorale intake niet langer mogelijk is, heeft continue subcutane toediening de voorkeur qua toedieningsweg.	V	V
(14) Farrington et al. 2010	Vaak voorkomende bijwerkingen van opioïden zijn obstipatie, misselijkheid en braken, duizeligheid en sufheid. De meeste bijwerkingen verminderen bij doorgezet gebruik behalve obstipatie, die aanwezig blijft. Voorschrift voor een laxativum “zo nodig” verkleint de kans op obstipatie.	V	V
Flowchart 2 Australian guideline	Voorzie een “zo nodig” voorschrift opioïden zelfs als er geen pijn aanwezig is: a. Morfine 2.5-5mg subcutaan dat elke 2u kan worden toegediend voor opioïd-naïeve bewoners. b. 1/12 tot 1/6 van de 24-uurs dosering dat iedere 2u kan worden toegediend voor niet opioïd-naïeve bewoners.	/	V
Flowchart 2 Australian guideline	Wanneer pijn aanwezig is, dient de voorgeschreven medicatie voor (doorbraak)pijn te worden toegediend.	/	V
Flowchart 2 Australian guideline	Indien de bewoner niet meer kan slikken, zet dan de orale dosering opioïden om naar de overeenkomende subcutane dosis, die via een spuitdrijver kan worden toegediend over 24u (Zie omreken tabel BCFI (die wordt toegevoegd in de richtlijn) of gebruik de Orthodose app).	/	V
Flowchart 2 Australian guideline	Als de huidige pijn onvoldoende onder controle is voor de bewoner en/of er meer dan drie “zo nodig” dosissen werden toegediend in de laatste 24u in kader van pijncontrole contacteer dan de huisarts om: a. De totale dosis van deze medicatie om te rekenen naar de subcutane dosis, en deze toe te voegen aan de spuitdrijver, b. of de huidige 24uurs dosis van de subcutane spuitdrijver op te drijven met 30%.	/	V
Flowchart 2 Australian guideline	Als een transdermale opioïdpleister in situ is, kan gekozen worden voor: a. Doorzetten van deze pleister in combinatie met subcutane “zo nodig” dosissen, eventueel kan morfine geassocieerd worden via een subcutane spuitdrijver b. of omzetten van de dosering van de pleister naar de overeenkomende subcutane dosis via spuitdrijver over 24u in combinatie met subcutane “zo nodig” dosissen.	/	V
Which pharmacological treatment can be recommended for management of dyspnea in dying nursing home residents? (clinical question 2)			
(17) Uptodate (Dudgeon) 2021 (18) Currow et al. 2010	Ongeacht de diagnose is dyspneu een frequent voorkomend symptoom bij gevorderde ziekte. De ernst en frequentie nemen toe over tijd, vooral in de laatste levensdagen.	I III-2	V (no longer available) III-2
(17) Uptodate (Dudgeon) 2021	Er is beperkte evidentie rondom het gebruik van zuurstof bij de behandeling van dyspneu in de laatste levensdagen. Bij niet-hypoxische bewoner heeft zuurstoftoediening geen bewezen effect op de verbetering van dyspneu.	I	V (no longer available)
(19) Booth et al. 2004	Er is beperkte evidentie rondom het gebruik van zuurstof bij de behandeling van dyspneu in de laatste levensdagen. Wanneer een bewoner hypoxisch is wordt zuurstoftoediening wel aangeraden om een snelle verbetering van dyspneu te bekomen.	I	II
(19) Booth et al. 2004	Er is beperkte evidentie rondom het gebruik van zuurstof bij de behandeling van dyspneu in de laatste levensdagen. Bij bewoners die reeds langdurig zuurstoftherapie krijgen voor een chronische longziekte wordt dit best verdergezet.	I	II
(20) Jennings et al. 2001	Systemische toediening van opioïden, aan een correcte dosis, is veilig en effectief in de behandeling van dyspneu.	I	V (Review retracted)

(11) Pergolizzi et al. 2008 (12) Elliott 2006	Bij het voorschrijven van opioïden bij opioïd-naïeve bewoners moet er gestart worden met de laagst mogelijke dosis (bv. morfine 1.5mg subcutaan). Verder opdrijven gebeurt door voorzichtig optitreren om het risico op toxiciteit zo laag mogelijk te houden.	I V	V V
(13) King et al. 2011	Morfine wordt best vermeden bij bewoners met ernstige nierinsufficiëntie (eGFR<30ml/min) vanwege het risico op opstapeling van toxische metabolieten.	I	V
(13) King et al. 2011	Achtergrond: Bij patiënten met ernstig renaal falen [eGFR<30], wordt beschreven in de literatuur dat morfine dient vermeden te worden o.w.v. stapeling van toxische metabolieten. In andere landen gaat daarom de voorkeur uit naar fentanyl en oxycodon subcutaan, die minder relevante toxische metabolieten hebben. Fentanyl en oxycodon subcutaan zijn in België echter niet beschikbaar of heel duur buiten de ziekenhuissetting. Stelling: De dosering van morfine bij ernstig renaal falen [eGFR<30] bij woonzorgcentrumbewoners in de laatste levensdagen dient niet aangepast te worden aan de nierfunctie. (deze stelling werd toegevoegd door de onderzoekers).	I	V
Developed by the GDG	Angst is frequent geassocieerd met dyspneu, waardoor benzodiazepines effectief zijn in de behandeling hiervan.	/	V
Flowchart 3 Australian guideline	Voorzie een “zo nodig” voorschrift opioïden zelfs als dyspneu niet aanwezig is: 1. Morfine 1.5-2.5mg subcutaan die elke 2u kan worden toegediend voor opioïd-naïeve bewoners. 2. 1/12 van de 24-uurs dosering die iedere 2u kan worden toegediend voor niet-opioïd-naïeve bewoners	/	V
Flowchart 3 Australian guideline	Wanneer dyspneu aanwezig is, dient de voorgeschreven medicatie voor (doorbraak) dyspneu te worden toegediend.	/	V
Flowchart 3 Australian guideline	Bij ernstig discomfort en/of nood aan drie of meer “zo nodig” dosissen om dyspneu onder controle te houden kan een hogere dosering in de spuitdrijver nodig zijn, maar in het algemeen wordt geadviseerd om niet meer dan 30% boven de voorgaande dagdosis te titreren.	/	V
Flowchart 3 Australian guideline	Als de bewoner regelmatig opioïden gebruikt en niet meer kan slikken, zet dan de orale dosering opioïden om naar de overeenkomende subcutane dosis die via een subcutane spuitdrijver kan worden toegediend over 24u (Zie omreken tabel BCFI (die wordt toegevoegd in de richtlijn) of gebruik de Orthodose app).	/	V
Flowchart 3 Australian guideline	Als meer dan 3 dosissen van de “zo nodig” medicatie nodig zijn binnen de 24u, moet de huisarts worden gecontacteerd voor het aanpassen van de medicatie.	/	V
Which pharmacological treatment can be recommended for management of Nausea and vomiting in dying nursing home residents? (clinical question 3)			
(21) Glare et al. 2011	Factoren die nausea en braken kunnen veroorzaken bij bewoners met een levensbedreigende ziekte zijn onder andere: toxiciteit van medicatie, urineweginfecties, constipatie, gastro-intestinale aandoeningen, metabole en biochemische ontregeling en orgaanfalen. Oorzaken van nausea en braken in de laatste levensdagen kunnen ook onbekend of multifactorieel zijn. Het achterhalen van deze oorzaak is niet altijd aangewezen in de laatste levensdagen.	V	V
(22) Hardy et al. 2002	Nausea wordt vaak niet herkend en wordt hierdoor niet adequaat behandeld.	I	II
(21) Glare et al. 2011	Er is beperkte evidentie omtrent het gebruik van anti-emetica bij bewoners.	V	V
(21) Glare et al. 2011 (22) Hardy et al 2002 (23) Porreca & Ossipov 2009	Opioïden zijn een frequente oorzaak van nausea en braken. Metoclopramide is effectief in de behandeling van nausea en braken bij patiënten met kanker die behandeld worden met opioïden.	V I V	V II V

(21) Glare et al. 2011 (22) Hardy et al 2002	Haloperidol kan op proef worden toegediend indien metoclopramide ineffectief is in de behandeling van nausea en braken.	V I	V II
(51) Lee et al. 2010	Men moet bedacht zijn op extrapiramidale nevenwerkingen bij het gebruik van metoclopramide en haloperidol. Deze medicatie moet vermeden worden of met de nodige voorzichtigheid worden gebruikt bij bewoners met neurodegeneratieve aandoeningen zoals de ziekte van Parkinson.	V	V
Developed by the GDG	Continue subcutane toediening van anti-emetica is effectief in de behandeling van aanhoudende nausea en braken.	V	V
Flowchart 1 Australian guideline	Voorzie een "zo nodig" voorschrift voor anti-emetica zelfs indien nausea en braken afwezig zijn. Overweeg metoclopramide 10mg subcutaan tot drie keer per dag zo nodig.	/	V
Flowchart 1 Australian guideline	Wanneer nausea en/of braken aanwezig is/zijn, dient de voorgeschreven medicatie voor (doorbraak) nausea en braken te worden toegediend.	/	V
Flowchart 1 Australian guideline	Overweeg continue subcutane toediening van metoclopramide 20-30mg over 24u met "zo nodig" dosissen van 10mg tot drie keer per dag indien: a. Nausea of braken aanwezig blijft ondanks toediening van "zo nodig" anti-emetica b. De bewoner niet meer kan slikken en er orale anti-emetica zijn voorgeschreven	/	V
Flowchart 1 Australian guideline	Bij aanhoudende nausea en/of braken, ondanks systematische toediening van metoclopramide, overweeg een "zo nodig" voorschrift haloperidol 0.5-1.5mg subcutaan, tot twee keer per dag.	/	V
Flowchart 1 Australian guideline	Indien de behandeling van nausea en/of braken met haloperidol effectiever is dan met metoclopramide, overweeg dan overschakelen naar haloperidol via continue subcutane toediening.	/	V
Flowchart 1 Australian guideline	Als meer dan 3 dosissen van metoclopramide 10mg subcutaan "zo nodig" zijn toegediend binnen de 24u, of als de voorgeschreven dosis haloperidol ineffectief is over 24u moet de huisarts worden gecontacteerd om een gespecialiseerd palliatief team in te schakelen.	/	V
Which pharmacological treatment can be recommended for management of death rattle in dying nursing home residents? (clinical question 4)			
(42) Wildiers et al. 2009	Doodsreutel is het geluid dat ontstaat doordat secreties in de luchtpijp of keel met de ademhaling meetrillen bij stervende bewoners die deze niet meer kunnen ophoesten.	/	II
Developed by the GDG	Doodsreutel treedt meestal op als de bewoner niet meer bij bewustzijn is. Het is dan ook onduidelijk of de bewoner hieronder lijdt. Voor de omgeving kan het echter zeer belastend zijn.	/	V
Developed by the GDG	Om doodsreutel minder belastend te maken voor de familie is een duidelijke uitleg over doodsreutel belangrijk.	/	V
Developed by the GDG	Positionering in zijligging kan helpen om opgestapelde secreties te draineren. Enkel uit te voeren indien deze positie comfortabel is.	/	V
Developed by the GDG	Beperken of stoppen van vochttoediening, ook indien doodsreutel nog niet aanwezig is, is belangrijk.	/	V
Developed by the GDG	Vooraleer medicatie wordt opgestart moeten andere oorzaken van een reutelende ademhaling door een arts worden uitgesloten: longoedeem, pneumonie, aspiratie van vocht/voedsel.	/	V
Developed by the GDG	Er is geen harde evidentie dat het gebruik van anticholinergica voor de behandeling van doodsreutel beter is dan het niet (medicamenteus) behandelen.	/	V

(42) Wildiers et al. 2009	Indien medicatie wordt opgestart, wordt dit best zo snel mogelijk gedaan bij het ontstaan van doodstreutel.	/	II
(42) Wildiers et al. 2009	Indien medicatie wordt opgestart voor doodstreutel wordt deze best verdergezet tot het moment van overlijden.	/	II
(45) Heisler et al. 2013 (43) Van Esch et al. 2021	Bij het gebruik van anticholinergica moet men aandachtig blijven voor bijwerkingen zoals: droge mond, urineretentie, onrust, agitatie, delier, tachycardie...	/	II II
(42) Wildiers et al. 2009	Buscopan® (hyoscine butylbromide) vermindert luchtwegsecreties. Het passeert de bloedhersenbarrière niet en kan zo geen duizeligheid of delier veroorzaken.	/	II
(48) Van Esch et al. 2021 (47) Mercadante et al. 2018	Profylactische toediening van Buscopan® (hyoscine butylbromide) vermindert de kans op het ontwikkelen van doodstreutel.	/	II II
(42) Wildiers et al. 2009	Buscopan® (Hyoscine butylbromide), Scopolamine® (hyoscine hydrobromide) en atropine hebben geen meetbaar verschil in bijwerkingen bij de behandeling van doodstreutel.	/	II
(46) Hugel et al. 2006	Robinul® (Glycopyrronium) en Scopolamine® (hyoscine hydrobromide) in continue toediening zijn even effectief in de behandeling van doodstreutel.	/	III-3
(45) Heisler et al. 2013	Toediening van 2 druppels atropine 1% oplossing, sublinguaal, heeft geen voordeel in de behandeling van doodstreutel in vergelijking met placebo.	/	II
Developed by the GDG	Voorzie een "zo nodig" voorschrift, zelfs als doodstreutel niet aanwezig is.	/	V
Developed by the GDG	Wanneer doodstreutel aanwezig is, dient de op dat moment voorgeschreven medicatie te worden toegediend.	/	V
(43) Van Esch et al. 2021 (47) Mercadante et al. 2018	Voor de profylactische behandeling van doodstreutel kan Buscopan® (hyoscine butylbromide) met een oplaaddosis van 20mg subcutaan, gevolgd door 60mg/24u via continue subcutane toediening of 4x20mg/d subcutaan gebruikt worden.	/	II II
(42) Wildiers et al. 2009	Bij de symptomatische behandeling van doodstreutel kan Buscopan® (hyoscine butylbromide) veilig gebruikt worden: a. 20mg subcutaan "zo nodig", dit mag iedere 2-4u toegediend worden b. Oplaaddosis van 20mg subcutaan, gevolgd door een continue toediening aan 60mg/24u of 10mg iedere 4u.	/	II
(46) Hugel et al. 2006 Clark et al. 2009 (42) Wildiers et al. 2009	Bij de symptomatische behandeling van doodstreutel kan Scopolamine® (hyoscine hydrobromide) veilig gebruikt worden. a. Oplaaddosis van 0.25mg subcutaan, gevolgd door een continue toediening van 1.5mg/24u of 0.25mg iedere 4u.	/	III-3 III-1 II
(42) Wildiers et al. 2009 (45) Heisler et al. 2013	Bij de symptomatische behandeling van doodstreutel kan atropine veilig gebruikt worden: a. Oplaaddosis van 0.5mg subcutaan, gevolgd door een continue toediening van 3mg/24u. b. 0,5 mg tot 3 mg/dag subcutaan c. Oogdruppels atropine 1% sublinguaal toe gediend iedere 4 uur 2-3 druppels	/	II II
(46) Hugel et al. 2006	Bij de symptomatische behandeling van doodstreutel kan Robinul® (glycopyrronium) veilig gebruikt worden: a. Oplaaddosis van 0.2mg gevolgd door een continue toediening aan 0.6-1.2mg/24u.	/	III-3

Which pharmacological treatment can be recommended for management of fever in dying nursing home residents? (clinical question 5)			
(49) Ouderenformularium BCFI	Terminale koorts (lichaamstemperatuursverhoging bij een stervende persoon) wordt meestal niet behandeld, tenzij de kans op stuipen reëel is. De ongewenste effecten eigen aan deze behandeling kunnen voor de bewoner meer belastend zijn dan de koorts zelf.	/	V
Developed by the GDG	Het is geen doel om de oorzaak van de koorts te achterhalen.	/	V
(49) Ouderenformularium BCFI	Terminale koorts wijkt meestal niet met koortswerende middelen.	/	V
Developed by the GDG	Voorzie een “zo nodig” voorschrift antipyretica onafhankelijk of koorts aanwezig is.	/	V
(49) Ouderenformularium BCFI (50) Palliatieve Zorg Vlaanderen	Als de behandeling van koorts gewenst is, kan gebruik gemaakt worden van Paracetamol 1g 2-4 keer per dag per os of rectaal.	/	V V
(50) Palliatieve Zorg Vlaanderen	82. Als de behandeling van tumorkoorts gewenst is, kan gebruik gemaakt worden van Naproxen 250mg twee tot drie keer per dag per os of rectaal.	/	V
(50) Palliatieve Zorg Vlaanderen	83. Als de behandeling van tumorkoorts gewenst is, kan gebruik gemaakt worden van Diclofenac 25mg twee tot drie keer per dag per os of rectaal.	/	V
(50) Palliatieve Zorg Vlaanderen (49) Ouderenformularium BCFI	Als de behandeling van tumorkoorts gewenst is, kan gebruik gemaakt worden van Ibuprofen 200-400mg twee tot drie keer per dag per os.	/	V V
(49) Ouderenformularium BCFI	Als de behandeling van tumorkoorts gewenst is, kan gebruik gemaakt worden van Dexamethason 4-8 mg 1 keer per dag, per os of subcutaan.	/	V
Which pharmacological treatment can be recommended for the management of confusion and delirium in dying nursing home residents? (clinical question 6b)			
(14) Palliative care expert group, 2010	Rusteloosheid en agitatie komen frequent samen voor aan het levenseinde en kunnen meerdere oorzaken hebben. Deze onderliggende oorzaak achterhalen is niet altijd aangewezen in de laatste levensdagen.	V	V
(14) Palliative care expert group, 2010	Wel is het belangrijk om uitlokkende factoren van rusteloosheid en agitatie te behandelen, zoals o.a. pijn, urineretentie, faecale impactie, hypoxie, omgevingsfactoren, psychologische en spirituele nood.	V	V
(52) Burry et al, 2018	Overweeg een lage dosis haloperidol voor de behandeling van agitatie of motorische onrust veroorzaakt door delier bij woonzorgcentrumbewoners in de laatste levensdagen.	I	II
(52) Burry et al, 2018	Wees aandachtig voor extrapiramidale bijwerkingen bij het gebruik van antipsychotica.	I	II
Developed by the GDG	De voorkeursbehandeling voor delier bestaat uit antipsychotica, behalve bij alcohol- of benzodiazepineonttrekkingsdelier.	/	V
Developed by the GDG	Benzodiazepines als monotherapie voor de behandeling van delier is enkel aangewezen bij alcohol- of benzodiazepineonttrekkingsdelier.	/	V
Flowchart 4 Australian guideline	Voorzie een “zo nodig” voorschrift voor antipsychotica zelfs als er geen agitatie of rusteloosheid aanwezig zijn. Overweeg haloperidol 0.5-1mg subcutaan tot 2 keer per dag.	/	V
Flowchart 4 Australian guideline	Wanneer een delier aanwezig is, dient de voorgeschreven medicatie te worden toegediend.	/	V

Flowchart 4 Australian guideline	Indien de bewoner enkel behandeld wordt met “zo nodig” antipsychotica, overweeg dan om dit om te zetten naar een subcutane toediening.	/	V
Flowchart 4 Australian guideline	Indien de bewoner slikproblemen ontwikkelt, maar chronisch behandeld wordt met een antipsychoticum, zet dit dan om naar een continue subcutane toediening via spuitdrijver over 24u, bv. haloperidol. Startdosis hangt af van initiële antipsychotica dosering en de ernst van symptomen. (voor omreken tabel zie https://www.psychiatrienet.nl/switchtabel/)	/	V
Flowchart 4 Australian guideline	Als meer dan 3 dosissen “zo nodig” medicatie nodig zijn voor doorbraak van rusteloosheid of agitatie binnen de 24u, moet de huisarts gecontacteerd worden om een gespecialiseerd palliatief zorgteam in te schakelen.	/	V
Which pharmacological treatment can be recommended for management of anxiety in dying nursing home residents? (clinical question 6c)			
(17) Dudgeon & Shadd, 2012 (28) Navigante et al, 2006	Angst is frequent geassocieerd met dyspneu en benzodiazepines zijn effectief in de behandeling hiervan.	V I	V II
(25) Kehl, 2004 (26) Breura et al, 2012	Rusteloosheid en agitatie op het einde van het leven kunnen veroorzaakt worden door angst. Het toedienen van een lage dosis benzodiazepines kan effectief zijn in de behandeling hiervan.	I V	II V
Flowchart 4 Australian guideline	Voorzie een “zo nodig” voorschrift voor benzodiazepines zelfs als er geen rusteloosheid en agitatie aanwezig is. Overweeg midazolam 2.5-5mg elke 2u ofwel clonazepam orale druppels 0.25-0.5mg elke 4u, bij bewoners waar nog geen benzodiazepines zijn voorgeschreven.	/	V
Flowchart 4 Australian guideline	Wanneer angst aanwezig is, dient de voorgeschreven medicatie te worden toegediend.	/	V
Flowchart 4 Australian guideline	Indien de bewoner regelmatig nood heeft aan “zo nodig” dosissen benzodiazepines of chronisch behandeld wordt met orale benzodiazepines en slikmoeilijkheden ontwikkelt, zet dit dan om naar midazolam via continue subcutane toediening. Start minimaal met midazolam 5-10mg over 24u, maar hogere doseringen kunnen nodig zijn.	/	V
Flowchart 4 Australian guideline	Indien de bewoner enkel behandeld wordt met “zo nodig” orale benzodiazepines, overweeg dan om dit om te zetten naar midazolam via subcutane toediening.	/	V
Flowchart 4 Australian guideline	Als meer dan 3 dosissen van de “zo nodig” medicatie nodig zijn binnen de 24u, moet de huisarts worden gecontacteerd voor het aanpassen van de medicatie.	/	V
Flowchart 4 Australian guideline	Bij persisterende angst of rusteloosheid ondanks systematische toediening van benzodiazepines, overweeg associatie van antipsychotica.	/	V

Key principles and key points translated from source guideline			
Australian guideline	Medicatie wordt voorgeschreven door de behandelende huisarts, die een coördinerende rol heeft.	/	V
Australian guideline	De bewoners en/of hun vertegenwoordiger moeten op de hoogte gebracht worden dat het stervensproces is gestart en dat medicatietoediening de kwaliteit van sterven kan verbeteren.	/	V
Australian guideline	Het traject vereist dat symptomen om de 2u worden beoordeeld bij bewoners die in dit traject worden geïncludeerd. Dit zorgt ervoor dat nieuwe symptomen snel worden gedetecteerd en waar nodig medicamenteus kunnen worden behandeld.	/	V
Australian guideline	Als medicamenteuze behandeling nodig wordt geacht, moet het meest geschikte geneesmiddel worden geselecteerd, dat op een veilige en effectieve manier wordt toegediend	/	V
Australian guideline	De voorgeschreven medicatie en doseringen moeten gebaseerd zijn op nauwkeurige beoordeling van de bewoner zijn toestand en (ernst van de) symptomen.	/	V
Australian guideline	De belasting veroorzaakt door medicatietoediening of -bijwerkingen moet zo klein mogelijk zijn. Medicatie in de laatste levensdagen wordt meestal subcutaan toegediend, deze toedieningsweg is over het algemeen het minst invasief en meest betrouwbaar bij een stervende bewoner.	/	V
Australian guideline	Medicatie met een beperkt therapeutisch voordeel in de terminale levensfase moet worden gestopt.	/	V
Australian guideline	Persisterende symptomen vereisen onderhoudsmedicatie op vaste tijdstippen eerder dan “zo nodig” medicatie.	/	V
Australian guideline	Deze onderhoudsmedicatie sluit de nood aan “zo nodig” medicatie niet uit. Een “zo nodig” medicatievoorschrift moet worden voorgeschreven om doorbraaksymptomen en symptomen die slechts tijdelijk aanwezig zijn snel en optimaal te kunnen behandelen.	/	V
Australian guideline	Geneesmiddelen zijn onmiddellijk beschikbaar om een optimale symptoomcontrole te verzekeren.	/	V
Australian guideline	Effectiviteit van de toegediende medicatie dient te worden geëvalueerd en gedocumenteerd.	/	V
Australian guideline	Denk naast de farmacologische behandeling, ook steeds aan een niet-farmacologische aanpak van de symptomen.	/	V
Australian guideline	De verpleegkundige is verantwoordelijk voor: a. Regelmatige beoordeling van symptomen	/	V
Australian guideline	De verpleegkundige is verantwoordelijk voor: b. Toediening van “zo nodig” medicatie wanneer nodig	/	V
Australian guideline	De verpleegkundige is verantwoordelijk voor: c. Regelmatige monitoring en documentatie van de effectiviteit van voorgeschreven medicatie	/	V
Australian guideline	De verpleegkundige is verantwoordelijk voor: d. Identificatie en rapportering van nevenwerkingen/bijwerkingen veroorzaakt door de voorgeschreven medicatie.	/	V

Australian guideline	De verpleegkundige is verantwoordelijk voor: e. contact met de huisarts om de voorgeschreven medicatie (zowel de standaard als de “zo nodig” medicatie) te herbekijken, wanneer de voorgeschreven medicatie werd toegediend.	/	v
Australian guideline	Het effect van de toegediende medicatie wordt geregistreerd, bijwerkingen worden genoteerd en gemeld aan de huisarts. Het Federaal Agentschap voor Geneesmiddelen en Gezondheidsproducten moedigt de rapportage van alle bijwerkingen ten sterkste aan. Informatie over melden van bijwerkingen kan online gevonden worden op https://www.fagg.be/nl/bijwerking	/	v
Australian guideline	Een medicatiefout (bv. onder- of overdosering) dient gemeld te worden aan de behandelende huisarts.	/	v

Chapter 12: Consensus procedure – Delphi procedure

A Delphi study was conducted as consensus procedure. The expert panel consisted of members of the PaGe working group⁷, a working group focused on palliative care in geriatrics, part of the Federation of Palliative Care Flanders. This working group consists of geriatricians and general practitioners. Besides these group we also included general practitioners who work in a nursing home and nurses specialized in palliative care in nursing homes in the expert group.

All recommendations from Chapter 11 were scored in the first round of the Delphi procedure. Where needed additional information and scientific background information was provided to the experts. The experts were asked to rate each recommendation on a five point Likert scale (strongly disagree, partially disagree, neutral, partially agree, strongly agree). Experts were always able to comment on the recommendations, comments were mandatory when experts disagreed with a recommendation. Recommendations where $\geq 78\%$ of experts strongly or partially agreed reached consensus and were added to the guideline. If recommendations did not reach this $\geq 78\%$ threshold for consensus, they were rephrased or combined and rescored in the following round. Some recommendations were deleted based on scores and comments of the experts.

A total of 15 experts participated in round one, 10 in round two and 8 in round three. After round three almost all recommendations had reached consensus ($\geq 78\%$ agreed). Six recommendations did not reach consensus in the third and final round of the Delphi study, these recommendations are discussed separately in Table 11.

Table 11: Recommendations that did not reach consensus in round 3 of the Delphi procedure		
Recommendation	Expert scores round 3	Additional information
Overweeg een “zo nodig” voorschrift voor nausea en braken bij het herkennen van de terminale fase.	63% agreed 25% neutral 13% partially disagreed	In round two of the Delphi procedure 50% of experts voted for an as needed prescription for nausea and vomiting. Therefore the recommendation to provide a as needed prescription was adapted to consider (“overweeg”) prescribing as needed medication for the treatment of nausea and vomiting once the last days of life are recognized. Ensuring that the treating physician is aware to consider medication. As this is already a weakened recommendation and the prescription of as needed medication for nausea of vomiting can contribute to patient comfort, the decision was made to keep this recommendation in the guideline despite not reaching consensus.
Een “zo nodig” voorschrift voor de behandeling van pijn in de laatste levensdagen bij opioïd-naïeve patiënten bestaat uit morphine 2.5 tot 5 mg subcutaan. Met voorkeur voor 2.5mg bij uitgesproken frailty, COPD, cachexie of nierinsufficiëntie [eGFR<30ml/min].	51% agreed 25% neutral 25% partially disagreed	In round two of the Delphi procedure we saw that our expert panel used varying doses of morphine in this setting, ranging from 1mg to 10mg, depending on the expert. As no evidence exists on what dose is recommended in this setting we favored the use of the most frequently used doses, 2.5 to 5mg, while adding rationale why a lower dose could be used.
Een “zo nodig” voorschrift voor de behandeling van pijn in de laatste levensdagen bij bewoners die reeds opioïden nemen bestaat uit morphine 1/6 van de 24-uurs dosering opioïden. Bij nierinsufficiëntie [eGFR<30ml/min] kan een lagere dosering bv. 1/12 worden overwogen.	51% agreed 25% neutral 25% partially disagreed	However, some experts find 2.5mg or 1/12 of 24h dose to low. There is however no evidence available to guide us when determining these dosages. Therefore we decided to keep these recommendations in the guideline despite not reaching consensus.

Overweeg een “zo nodig” voorschrift voor doodsreutel bij het herkennen van de terminale fase.	63% agreed 25% neutral 25% completely disagreed	Over the last few years, treatment of death rattle has been a topic of discussion in scientific literature(48). So far it remains unclear whether or not death rattle causes discomfort to the patient, though most experts agree that it does cause distress in relatives and health care workers (48). But, whether or not relatives’ distress is sufficient to start possibly harmful medication in patients has not reached consensus in scientific communities. Especially as no studies have proven pharmacological treatment to be superior to placebo in relieving death rattle once it occurs. Two recent randomized controlled trails did show a positive effect of prophylactic treatment with hyoscine butylbromide (Buscopan®) on the development of death rattle (43,47). But, whether or not prophylactic treatment for a symptom that might not be harmful is indicated is up for debate. It is therefore logical that we see the division whether or not treatment is necessary among our experts. In respect to the guideline we decided to keep these recommendations that did not reach consensus, but a statement will be added in the guideline, explaining current evidence on the uncertainty of death rattle treatment. Informing the treating physician so he/she can make an informed decision.
De profylactische behandeling van doodsreutel is aangewezen wanneer de kans op het ontwikkelen van doodsreutel groot is of wanneer dit zeer belastend zal zijn voor de familie.	38% agreed 13% neutral 38% partially disagreed	
De behandeling van doodsreutel kan overwogen worden als deze zeer uitgesproken is en/of zeer belastend is voor de familie.	63% agreed 13% neutral 25% partially disagreed	

12.1 Treatment suggested by experts

In the first round of the Delphi procedure experts suggested to add the following medication:

- Novalgine®(natriummetamizol) for fever

Although no evidence was found in the systematic review on the use of Novalgine® for the treatment of fever, it is common practice in one of the guideline development group’ hospitals. It was decided to incorporate a recommendation on the use of Novalgine® for the treatment of fever in the second round. Here however only 50% of the experts agreed with the use of Novalgine® in the treatment of fever, therefore the use of Novalgine® was not incorporated as a recommendation in the guideline.

- Litican ® (Alizapride) for nausea/vomiting

Once again no evidence for the use of Litican® (Alizapride) was available for the treatment of nausea and vomiting in this indication. However it is common practice in several hospitals. The decision was made to develop recommendations using Litican® as a first line treatment for nausea and vomiting, next to metoclopramide. These recommendations achieved consensus in the second round of our Delphi study, and were added in the guideline.

Chapter 13: Recommendations after Delphi procedure

All recommendations that reached consensus were added to the guideline, these are summarized in Table 12. The recommendations that were converted into flowcharts are highlighted in *italic*.

The levels of evidence used in the source guideline are designed by the National Health and Medical Research Council of Australia with the addition of a Level V, used for expert opinion only.

For each recommendation a strength of the recommendation was determined by the guideline development group. This is based on the GRADE system, whereby the strength of recommendation depends on a balance between all desirable and all undesirable effects of an intervention (i.e., net clinical benefit), quality of available evidence, values and preferences, and estimated cost (resource utilization) (53). For this guideline, no formal cost-effectiveness study was conducted.

A strong recommendation implies that most patients would want the recommended course of action. A weak recommendation implies that the majority of informed patients would want the intervention, but many would not. Specifically, a strong negative recommendation means the harms of the recommended approach clearly exceed the benefits whereas a weak negative recommendation implies that the majority of patients would not want the intervention, but many would.

Grade	Definition
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Strong	The desirable effects of an intervention clearly outweigh the undesirable effects (the intervention is to be put into practice), or the undesirable effects of an intervention clearly outweigh the desirable effects (the intervention is not to be put into practice)
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Weak	The desirable effects of an intervention probably outweigh the undesirable effects (the intervention probably is to be put into practice), or the undesirable effects of an intervention probably outweigh the desirable effects (the intervention probably is not to be put into practice)
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Table 12: Evidence to recommendation table after Delphi procedure

Which pharmacological treatment can be recommended for management of pain in dying nursing home residents? (clinical question 1)					
	Recommendation	Evidence to Decision	Health benefits (+)/Risks (-)	Level of evidence	Strength of recommendation
1	Pijn is een belangrijk probleem bij woonzorgcentrumbewoners, de prevalentie van aanhoudende pijn in deze populatie bedraagt 49-80%.	Prevalence as reported by Mercadante & Acuri (10)	(+) Enhancing the awareness of healthcare workers about pain prevalence can lead to faster recognition and management. (-) None	III-2	Weak
2	Opioiden zijn effectief en worden meestal goed getolereerd door ouderen.	This has been previously discussed by Pergolizzi et al. (11). Recent studies have shown the safe use of opioids during the last days of life in older people (39).	(+) Enhancing the awareness that opioid use is safe in the older population can lead to faster symptom management. (-) None	V III-3	Weak
3	Bij bewoners die vooraf geen opioiden kregen, hierna opioid-naïeve bewoners genoemd, gebruikt men de laagste dosis voor pijn management. Het voorzichtig ophogen van de dosis vermindert het risico op toxiciteit/ongewenste nevenwerkingen.	This has previously been discussed by Pergolizzi et al. (11).	(+) Slow titration decreases the risk of adverse events (-) Too slow titration might cause undertreatment of symptoms.	V	Strong
4	Bij patiënten met ernstig nierlijden [eGFR<30] wordt het gebruik van morfine afgeraden vanwege het risico op stapeling van toxische metabolieten [ref [6] – Level I]. Bij gebrek aan een parenterale toedieningsvorm van fentanyl kan morfine toch gebruikt worden in woonzorgcentra, mits extra aandacht wordt besteed voor het optreden van eventuele nevenwerkingen (misselijkheid en braken, duizeligheid, sufheid, myoclonus en obstipatie).	King et al. (13) discussed in their systematic review the use of opioids in patients with cancer pain and renal impairment. They advised against the use of morphine in those with severe renal impairment and recommended alternatives such as fentanyl, alfentanil, tramadol, and hydromorphone. However, these alternatives are either unavailable, restricted to hospital use, or very expensive in Belgium. Moreover, King et al. did not find hard evidence against the use of morphine in the last days of life, and their recommendations were based on evidence of low to very low quality. During our Delphi procedure, experts noted that nursing home personnel lack experience with these alternative opioids and that morphine can be used safely when initiated at a low dose and gradually increased.	(+) Morphine is a widely available and commonly used opioid with substantial user experience. (-) Risk of adverse events due to metabolite accumulation, especially in patients with renal failure.	V	Weak
5	Voor de controle van aanhoudende pijn, is 24u/24u toediening van opioiden noodzakelijk, rekening houdend met de werkingsduur van het voorgeschreven opioïde.	Continuous administration of medication for pain management is a cornerstone of palliative care, ensuring adequate symptom control, as stated by Farrington et al. (14).	(+) Continuous administration gives a more continuous way to treating symptoms, potentially preventing and minimizing breakthrough symptoms. (-) None, if the appropriate dose and regular symptom assessment are applied, with treatment adjustments if needed.	V	Strong

6	Transdermale opioïdpleisters (buprenorfine en fentanyl) zijn niet geschikt om te starten in de laatste levensdagen. Het pijnstillend effect laat namelijk even op zich wachten, waardoor geen snelle en veilige dosistitratie kan gebeuren.	Transdermal opioid medications are slowly absorbed, taking a few days to reach their maximum effect (16). This kind of delay is not desirable during the last days of life. However, during the Delphi procedure, experts noted that transdermal patches should be continued in patients who already received them. The reason why, is the unpredictable level of absorption in patients during their last days of life, who often have cachexia or edema. Therefore the experts in our Delphi procedure opted that the preferred approach is to continue the patch while administering additional subcutaneous morphine in case of additional opioid need.	(+) Helps treat symptoms quickly and efficiently while avoiding overtreatment. (-) May require more invasive treatment options, such as subcutaneous administration.	I	Strong
7	Als een transdermale opioïdpleister in situ is, gaat de voorkeur naar het doorzetten van deze pleister in combinatie met morfine via subcutane “zo nodig” dosissen of een continue subcutane spuitdrijver.		(+) Continuation avoids conversion mistakes leading to over- or underdosing. (-) Continuation might complicate calculation of dosage increases and the as needed dosage.	V	Weak
8	Wanneer opioïden worden gestart in de laatste levensdagen of wanneer perorale intake niet langer mogelijk is, heeft continue subcutane toediening de voorkeur qua toedieningsweg.	Continuous subcutaneous injection is considered the standard of care in end-of-life care (14).	(+) This ensures a continuous administration of opioids through a reliable administration route. (-) None, if the appropriate dose and regular symptom assessment are applied, with treatment adjustments if needed.	V	Strong
9	Indien opioïden worden toegediend is systematische toediening van een laxativum aanbevolen bij patiënten met een levensverwachting langer dan 1 week (bv. Laxoberon® 10 druppels om de 2 dagen).	In the general population, laxatives are recommended when starting opioid therapy. However, for residents in their last days of life, who often have reduced intake and may not experience constipation-related discomfort, this recommendation may not be as applicable. Therefore, the use of laxatives is still recommended, but should be evaluated on a case-by-case basis, considering the needs and characteristics of each resident (14).	(+) Prophylactic treatment helps to avoid constipation which might cause discomfort. (-) The unnecessary intake of laxatives in patients in their last days of life might cause discomfort.	V	Weak
10	<i>Voorzie standaard een “zo nodig” voorschrift voor pijn bij het herkennen van de terminale fase.</i>	Providing a prescription ‘as needed’ is standard practice in end-of-life care, ensuring rapid and effective management of pain. All experts in our Delphi procedure agreed on this recommendation.	(+) An "as needed" prescription allows nurses to promptly address symptoms as they arise, avoiding treatment delay. (-) None	V	Strong
11	<i>Wanneer pijn aanwezig is, dient de voorgeschreven medicatie voor (doorbraak)pijn te worden toegediend.</i>	All experts in our Delphi procedure agreed on this recommendation.	(+) Treatment of pain decreases patient discomfort. (-) None	V	Strong
12	<i>Indien de bewoner niet meer kan slikken, zet dan de orale dosering opioïden om naar de overeenkomende subcutane dosis, die via een spuitdrijver kan worden toegediend over 24u (Zie omreken tabel BCFI).</i>	All experts in our Delphi procedure agreed on this recommendation.	(+) This ensures the use of an appropriate administration route. (-) None	V	Strong
13	<i>Als de huidige pijn onvoldoende onder controle is voor de bewoner en/of er meer dan drie “zo nodig” dosissen werden toegediend in de laatste 24u in kader van pijncontrole contacteer dan de huisarts om: a. De totale dosis van deze medicatie om te</i>	Initially an augmentation of 30% was advised, but two experts in our Delphi procedure argued that a 50% increase was more appropriate, therefore this was adapted to 30-50%, and was accepted by all experts in the following Delphi round.	(+) Ensures a continuous adaptation of the treatment based on the patient’s needs. (-) None.	V	Strong

	rekenen naar de subcutane dosis, en deze toe te voegen aan de spuitdrijver, b. of de huidige 24uurs dosis van de subcutane spuitdrijver op te drijven met 30-50%.				
14	Een "zo nodig" voorschrift voor de behandeling van pijn in de laatste levensdagen bij opioïd-naïeve patiënten bestaat uit morfine 2.5 tot 5 mg subcutaan. Met voorkeur voor 2.5mg bij uitgesproken frailty, COPD, cachexie of nierinsufficiëntie [eGFR<30ml/min].	All Delphi experts agreed on the need for an "as needed" prescription, but there was debate about the exact dose to be used. Proposed dosages ranged from 1 to 10 mg, with the majority suggesting 2.5 to 5 mg. A preference was indicated for the lower dose in cases of frailty, COPD, cachexia, and kidney failure.	(+) Gradual titration of the opioid dose minimizes the risk of adverse events. (-) Potential undertreatment when the administered dosages are too low.	V	Weak
15	Een "zo nodig" voorschrift voor de behandeling van pijn in de laatste levensdagen bij bewoners die reeds opioïden nemen bestaat uit morphine 1/6 van de 24-uurs dosering opioïden. Bij nierinsufficiëntie [eGFR<30ml/min] kan een lagere dosering bv. 1/12 worden overwogen.	80% of Delphi experts agreed that 1/6th dose is adequate as an "as needed" prescription, the remaining experts argued 1/10th dose should be used.	(+) Gradual titration of the opioid dose minimizes the risk of adverse events. (-) Potential undertreatment when the administered dosages are too low.	V	Weak
16	De "zo nodig" dosis kan na 30 minuten opnieuw worden toegediend bij persisterende pijn. Indien deze tweede toediening onvoldoende symptoomcontrole geeft dient de huisarts te worden gecontacteerd.	a. Morphine's pharmacokinetic and pharmacodynamic profile shows peak blood concentrations 15 minutes after subcutaneous injection(54). b. The interval between doses was a topic of discussion in the Delphi study. Some participants suggested that a second dose should be administered as soon as the patient remains in pain 5 minutes after the first administration. Others argued for a dose interval of up to 4 hours after the first administration. The majority of experts recommended a 30-minute interval. Based on these two arguments, administering a second dose after 30 minutes seems appropriate and safe.	(+) Gradual titration of the opioid dose minimizes the risk of adverse events. (-) Potential undertreatment when the administered dosages are too low.	V	Weak
Which pharmacological treatment can be recommended for management of dyspnea in dying nursing home residents? (clinical question 2)					
	Recommendation	Evidence to Decision	Health benefits (+)/Risks (-)	Level of evidence	Strength of recommendation
17	Ongeacht de diagnose is dyspneu een frequent voorkomend symptoom bij gevorderde ziekte. De ernst en frequentie nemen toe over tijd, vooral in de laatste levensdagen.	This has previously been stated by Dudgeon et al (17) and Currow et al (18)	(+) Enhancing the awareness of healthcare workers about dyspnea prevalence can lead to faster recognition and management. (-) None	V (no longer available) III-2	Weak
18	Zuurstoftherapie opstarten wordt niet	The use of oxygen in the last days of life is controversial. While it may relieve dyspnea, it can also lead to mucus	(+) Avoids unnecessary start of oxygen, which might cause discomfort.	V (no longer	Weak

	aangeraden tijdens de laatste levensdagen .Er is beperkte evidentie rondom het gebruik van zuurstof bij de behandeling van dyspneu in de laatste levensdagen. Zuurstoftherapie op zich kan discomfort veroorzaken door het uitdrogen van de slijmvliezen en het creëren van drukpunten.	dehydration, cause discomfort from the oxygen mask, and potentially prolong the dying process, which is not the goal of treatment. Experts in our Delphi procedure advised against starting oxygen therapy for residents in their final days, even in cases of hypoxia, as it can lead to unnecessary discomfort. Instead, they recommend using morphine to relieve dyspnea.	(-) None	available)	
19	Bij bewoners die reeds langdurig zuurstoftherapie krijgen voor een chronische longziekte wordt dit enkel verdergezet zo dit bijdraagt aan het comfort van de bewoner.	In patients already receiving oxygen, discontinuing it can cause psychological distress, as it may be perceived as a sign that the medical team has given up on the patient. Additionally, abrupt withdrawal of oxygen might worsen dyspnea symptoms (19).	(+) Continuation of oxygen might be less confronting for the patient and/or family. (-) Continuation of oxygen might cause discomfort.	II	Weak
20	Systemische toediening van opioïden, aan een correcte dosis, is veilig en effectief in de behandeling van dyspneu.	All Delphi experts agreed on this recommendation. The four studies included in our systematic review on dyspnea used opioids and did not report any severe adverse events (39).	(+) Opioids are safe in the recommended dosages and help alleviate dyspnea. (-) None	V	Weak
21	Angst is frequent geassocieerd met dyspneu, waardoor benzodiazepines effectief zijn in de behandeling hiervan.	Dyspnea can often be a response to anxiety, and addressing the underlying cause is crucial for effective symptom management. All experts agreed on this recommendation.	(+) Highlighting the various causes of dyspnea and exploring different treatment methods enhances both the management of the condition and patient comfort. (-) None	V	Weak
22	<i>Voorzie standaard een “zo nodig” voorschrift voor dyspneu bij het herkennen van de terminale fase.</i>	Providing a prescription “as needed” is standard practice in end-of-life care, ensuring rapid and effective management of dyspnea. All but one expert from our Delphi procedure agreed.	(+) An "as needed" prescription allows nurses to promptly address symptoms as they arise, avoiding any delay in treatment. (-) None	V	Strong
23	<i>Wanneer dyspneu aanwezig is, dient de voorgeschreven medicatie voor (doorbraak) dyspneu te worden toegediend.</i>	All experts from our Delphi procedure agreed on this recommendation.	(+) Treatment of dyspnea once present decreases patient discomfort. (-) None	V	Strong
24	<i>Bij ernstig discomfort en/of nood aan drie of meer “zo nodig” dosissen om dyspneu onder controle te houden kan een hogere dosering in de spuitdrijver nodig zijn, maar in het algemeen wordt geadviseerd om niet meer dan 30-50% boven de voorgaande dagdosis te titreren.</i>	Initially an augmentation of 30% was advised, but two experts in our Delphi procedure argued that a 50% increase was more appropriate, therefore this was adapted to 30-50%, and was accepted by all experts in the following Delphi round.	(+) Ensures a continuous adaptation of the treatment based on the patient’s needs. (-) None.	V	Strong
25	<i>Als meer dan 3 dosissen van de “zo nodig” medicatie nodig zijn binnen de 24u, moet de huisarts worden gecontacteerd voor het aanpassen van de medicatie.</i>			V	Strong
26	<i>Als de bewoner regelmatig opioïden gebruikt en niet meer kan slikken, zet dan de orale dosering opioïden om naar de</i>	All experts from our Delphi procedure agreed on this recommendation.	(+) This ensures the use of an appropriate administration route. (-) None	V	Strong

	<i>overeenkomende subcutane dosis die via een subcutane spuitdrijver kan worden toegediend over 24u (Zie omrekening tabel BCFI).</i>				
27	<i>Voor de behandeling van dyspneu met morfine wordt dezelfde dosering gebruikt als voor de behandeling van pijn.</i>	Observational studies frequently indicate that lower opioid doses are used for dyspnea compared to pain, although research on this topic is limited (39). Given that we already advise starting with a low opioid dose for pain, it is reasonable to apply a similar approach for dyspnea. Consensus among the experts from our Delphi procedure reflects this: 80% recommend using the same dose as for pain, while 20% advocate for lower dosages.	(+) Using the same opioid doses for both pain and dyspnea simplifies treatment schedules, while still ensuring that low and safe dosages are maintained. (-) None	V	Weak
Which pharmacological treatment can be recommended for management of Nausea and vomiting in dying nursing home residents? (clinical question 3)					
	Recommendation	Evidence to Decision	Health benefits (+)/Risks (-)	Level of evidence	Strength of recommendation
28	Nausea is een vaak voorkomend symptoom bij patiënten die opioïden krijgen, maar is soms moeilijk te herkennen in de laatste levensdagen. Een verhoogde waakzaamheid is nodig om onderbehandeling te voorkomen.	All experts from our Delphi procedure agreed on this recommendation. This recommendation was based on literature on the causes of nausea in the last days of life (21).	(+) Increasing awareness of factors that can cause nausea and vomiting will help in identifying and addressing these symptoms more effectively. (-) None	V	Weak
29	Factoren die nausea en braken kunnen veroorzaken bij bewoners met een levensbedreigende ziekte zijn onder andere: constipatie, gastro-intestinale aandoeningen, toxiciteit van medicatie, orgaanfalen, metabole en biochemische ontregeling en (urineweg)infecties. Oorzaken van nausea en braken in de laatste levensdagen kunnen ook onbekend of multifactorieel zijn. Het achterhalen van deze oorzaak is niet altijd aangewezen in de laatste levensdagen.			V	Weak
30	Haloperidol kan op proef worden toegediend indien metoclopramide ineffectief is in de behandeling van nausea en braken.	Previous research proposed haloperidol as treatment for nausea or vomiting as a second line treatment (21)(22). All experts from our Delphi procedure agreed on this recommendation.	(+) Offering a variety of safe treatment options will improve the likelihood of effective symptom alleviation. (-) None	V II	Weak
31	Men moet bedacht zijn op extrapyramidale nevenwerkingen bij het gebruik van alizapride, metoclopramide en haloperidol. Deze medicatie moet vermeden worden of met de nodige voorzichtigheid worden gebruikt bij bewoners met neurodegeneratieve aandoeningen zoals de ziekte van Parkinson.	a. The development of extrapyramidal side effects is often mentioned as a reason to be hesitant for the use of antipsychotic medication. This warning seems to be based on one study with high attrition and was further discussed in a Cochrane review in 2007 (27). The updated version of this Cochrane review did not establish the same effect after pooling of the results (52).	(+) Raising awareness of potential adverse events helps recognizing them when they occur. (-) This might cause doctors to be too cautious in prescribing anti-emetic medication to patients prone to extrapyramidal side effects (e.g. patients	II	Weak

		b. In the treatment of symptoms during residents' final days, the experts from our Delphi procedure emphasized that the risk of developing extrapyramidal side effects is less important. The focus should be on maximizing patient comfort in cases of nausea or vomiting. The risk of adverse events is secondary as long as the benefit-risk ratio remains favorable.	with Parkinson's disease).		
32	Continue subcutane toediening van anti-emetica is effectief in de behandeling van aanhoudende nausea en braken.	Although no studies support this recommendation, all experts from our Delphi procedure agreed on this recommendation.	(+) Continuous medication administration will help to avoid breakthrough symptoms. (-) None, if the appropriate dose and regular symptom assessment are applied, with treatment adjustments if needed.	V	Weak
33	<i>Overweeg een “zo nodig” voorschrift voor nausea en braken bij het herkennen van de terminale fase.</i>	Only 50% of experts agreed that an “as needed” prescription should always be made for nausea and vomiting. Therefore the formulation of the recommendation was adjusted to “to consider”.	(+) An “as needed” prescription allows nurses to promptly address symptoms as they arise, avoiding any delay in treatment. (-) None	V	Weak
34	<i>Wanneer nausea en/of braken aanwezig is/zijn, dient de voorgeschreven medicatie voor (doorbraak) nausea en braken te worden toegediend.</i>	All experts from our Delphi procedure agreed on this recommendation.	(+) Treatment of nausea and vomiting once present decreases patient discomfort. (-) None	V	Strong
35	<i>Overweeg continue subcutane toediening van metoclopramide 20-30mg over 24u met “zo nodig” dosissen van 10mg tot drie keer per dag indien: a. Zo nodig medicatie voor Nausea of braken 2-3x per dag noodzakelijk is</i>	All experts from our Delphi procedure agreed on this recommendation.	(+) Safe dosage for treatment of nausea and vomiting with low risk of adverse events. (-) None	V	Weak
36	<i>Voor de behandeling van nausea en braken in de laatste levensdagen zijn metoclopramide (Primperan®) 10mg subcutaan, tot maximaal 30mg per dag, of alizapride (Litican®) 50mg subcutaan, tot 300mg per dag de eerste stap.</i>	All experts from our Delphi procedure agreed on this recommendation.	(+) Safe dosage for treatment of nausea and vomiting with low risk of adverse events. (-) Potential extrapyramidal side effects see recommendation 31.	V	Weak
37	<i>Als meer dan 3 dosissen van metoclopramide 10mg subcutaan “zo nodig” zijn toegediend binnen de 24u, of als de voorgeschreven dosis haloperidol ineffectief is over 24u moet de huisarts worden gecontacteerd om zo gewenst een gespecialiseerd palliatief team in te schakelen.</i>	In our Delphi procedure 93% of the experts agreed with this recommendation. Some expressed concerns about delaying contact with a doctor. Furthermore, many doctors are experienced in treating nausea and vomiting and are knowledgeable about second- and third-line treatment options, making involvement of a specialized palliative care team less necessary.	(+) This method ensures a feedback mechanism with the doctor, if treatment is insufficient. (-) None, if the appropriate dose and regular symptom assessment are applied, with treatment adjustments if needed.	V	Weak
38	<i>Voor de behandeling van nausea en braken in de laatste levensdagen is haloperidol (Haldol®) 1mg subcutaan, tot maximaal 3mg per dag, aangewezen bij onvoldoende effect</i>	In our Delphi procedure 90% of Delphi experts agreed that this is the standard second-line treatment. One Delphi expert noted that in agitated patients haloperidol might be a good first-line treatment option, as it can treat both	(+) Effective treatment of nausea and vomiting. (-) Potential extrapyramidal side effects see recommendation 31.	V	Weak

	van metoclopramide (Primperan®) en/of alizapride (Litican®).	symptoms simultaneously.			
39	Haldol is voorkeursbehandeling voor nausea en braken indien deze gepaard gaat met delier.		(+) Treatment of both symptoms with a single medication. (-) Potential extrapyramidal side effects see recommendation 31.	V	Weak
40	Indien de behandeling van nausea en/of braken met haloperidol effectiever is dan met metoclopramide, overweeg dan overschakelen naar haloperidol via continue subcutane toediening.	All experts from our Delphi procedure agreed on this recommendation.	(+) Avoiding unnecessary polypharmacy will help avoid adverse events. (-) None	V	Strong
41	Voor de behandeling van nausea en braken in de laatste levensdagen is Dexamethason (Acidexam®) 5-10mg subcutaan, 1 keer per dag, eerste keuze bij de aanwezigheid van hersenmetastasen, of kan dit gebruikt worden bij onvoldoende effect van de combinatie metoclopramide (Primperan®) of alizapride (Litican®) met haloperidol (Haldol®). Let op! Dexamethason mag niet toegevoegd worden aan de subcutane spuitdrijver.	This was proposed by one of the experts in the first Delphi round. This recommendation was scored in the second Delphi round, and approved by all experts.	(+) Effective treatment of nausea and vomiting in cases of brain metastases. (-) Dexamethasone might cause agitation or delirium in frail older people.	V	Weak

Which pharmacological treatment can be recommended for management of death rattle in dying nursing home residents? (clinical question 4)

	Recommendation	Evidence to Decision	Health benefits (+)/Risks (-)	Level of evidence	Strength of recommendation
42	Doodsreutel is het geluid dat ontstaat doordat secreties in de luchtpijp of keel met de ademhaling meetrillen bij stervende bewoners die deze niet meer kunnen ophoesten.	All experts from our Delphi procedure agreed on this recommendation based on a definition formulated in literature (42).	(+) Understanding the underlying causes of this symptom may help make the process less confronting for healthcare workers and relatives. (-) None	II	Strong
43	Doodsreutel treedt meestal op als de bewoner niet meer bij bewustzijn is. Het is dan ook onduidelijk of de bewoner hieronder lijdt. Voor de omgeving kan het echter zeer belastend zijn.	All experts from our Delphi procedure agreed on this recommendation.	(+) Understanding the underlying causes and consequences of this symptom may help make the process less confronting for healthcare workers and relatives. (-) None	V	Weak
44	Om doodsreutel minder belastend te maken voor de familie is een duidelijke uitleg over doodsreutel belangrijk.	Whether or not a patient experiences discomfort due to death rattle remains controversial among palliative care experts (48). Therefore informing relatives might be more important than the pharmacological treatment of death rattle. All experts from our Delphi procedure agreed on this recommendation.	(+) Understanding the underlying causes and consequences of this symptom may help make the process less confronting for relatives. (-) None	V	Strong

45	Positionering in zijligging kan helpen om opgestapelde secreties te draineren. Enkel uit te voeren indien deze positie comfortabel is.	As the effectiveness of pharmacological treatment for the death rattle remains uncertain, simple and non-discomforting non-pharmacological therapies may be appropriate (55). Although experts argued that the application of this posture needs to be evaluated on a case by case basis.	(+) May reduce the volume of the death rattle sound. (-) The true effectiveness of this position in managing death rattle is unclear.	V	Weak
46	Beperken of stoppen van vochttoediening, ook indien doodsreutel nog niet aanwezig is, is belangrijk.	This recommendation is based on expert opinion, with the assumption that additional fluid administration could increase respiratory secretions and subsequently exacerbate death rattle. Although no evidence exists to support this assumption. In addition, fluid administration might unnecessarily prolong the dying phase. Ninety-three percent of experts in our Delphi procedure agreed with this recommendation, while one expert neither agreed nor disagreed.	(+) Avoids unnecessary prolongation of the dying phase and might reduce death rattle occurrence. (-) Abruptly stopping fluid administration might cause distress in relatives.	V	Weak
47	Vooraleer medicatie wordt opgestart moeten andere oorzaken van een reutelende ademhaling door een arts worden uitgesloten: longoedeem, pneumonie, aspiratie van vocht/voedsel.	Experts agreed that preventing or reducing respiratory secretions to manage the death rattle is futile if other factors, such as pneumonia or pulmonary edema, are causing an excessive accumulation of secretions in the airways.	(+) Identifying the underlying cause of death rattle could help avoid futile pharmacological treatment. (-) None	V	Weak
48	Indien medicatie wordt opgestart, wordt dit best zo snel mogelijk gedaan bij het ontstaan van doodsreutel.	Wildiers et al. (42) indicated a greater likelihood of reducing death rattle severity when treatment is initiated earlier in the process. However, this study was not placebo-controlled. In our Delphi procedure one expert disagreed with this recommendation, arguing that treatment for the death rattle should never be started.	(+) Initiating treatment earlier may increase the likelihood of reducing the severity of the death rattle. (-). There may be potential side effects associated with pharmacological treatment.	II	Weak
49	Indien medicatie wordt opgestart voor doodsreutel wordt deze best verdergezet tot het moment van overlijden.	When anticholinergic medication is discontinued, it may theoretically lead to a rebound effect, increasing respiratory secretions, which can worsen the death rattle. Therefore, it is recommended to continue the medication until death, provided there are no severe side effects (42). All Delphi experts agreed.	(+) Continuing treatment can help prevent a rebound effect, which may otherwise lead to an increase in respiratory secretions. (-) There could be potential side effects associated with pharmacological treatment.	II	Weak
50	Bij het gebruik van anticholinergica moet men aandachtig blijven voor bijwerkingen zoals: droge mond, urineretentie, onrust, agitatie, delier, tachycardie...	Given the unclear effectiveness of death rattle treatment, it is crucial to minimize any discomfort associated with its treatment (43) (45) All experts from our Delphi procedure agreed with this recommendation. Some noted that the medication is typically initiated in unconscious patients, and a urinary catheter is often used to prevent urine retention.	(+) Monitoring for side effects will facilitate early detection and treatment. (-) None	II II	Strong
51	Buscopan® (hyoscine butylbromide) vermindert luchtwegsecreties. Het passeert de bloedhersenbarrière niet en kan zo geen duizeligheid of delier veroorzaken.	This recommendation states the working mechanism of buscopan, increasing awareness among health care workers (42). All experts in our Delphi procedure agreed with this recommendation	(+) Increasing awareness among health care workers of why medications are chosen will improve compliance. (-) None	II	Weak

52	<i>Overweeg een “zo nodig” voorschrift voor doodstreutel bij het herkennen van de terminale fase.</i>	Experts in our Delphi procedure disagree whether or not to treat death rattle with anticholinergics. This disagreement stems from uncertainty regarding the effectiveness of available pharmacological treatments and whether death rattle causes discomfort to the patient (48). Despite these uncertainties, death rattle can be highly distressing for relatives and healthcare workers, and pharmacological treatments are generally well tolerated. Consequently, some experts in our Delphi procedure support treatment, while others do not, making both approaches viable options. In our expert panel 50% agreed that an “as needed” prescription should always be provided for death rattle. Therefore the recommendation was adjusted to “to consider”.	(+) An "as needed" prescription allows nurses to promptly address symptoms as they arise, avoiding any delay in treatment. (-) None	V	Weak
53	<i>Wanneer doodstreutel aanwezig is, dient de op dat moment voorgeschreven medicatie te worden toegediend.</i>	If an “as needed” prescription is provided, it indicates that the goal is to treat death rattle as it occurs. Eighty-five percent of experts in our Delphi procedure agreed with this recommendation.	(+) This recommendations helps to promptly address the symptom once it occurs, avoiding any delay in treatment. (-) None	V	Weak
54	<i>De profylactische behandeling van doodstreutel is aangewezen wanneer de kans op het ontwikkelen van doodstreutel groot is of wanneer dit zeer belastend zal zijn voor de familie.</i>	Prophylactic treatment has been shown to reduce the likelihood of developing the death rattle and decrease its severity if it does occur (43). However, since it is still uncertain whether death rattle causes discomfort to the resident, it may not be advisable to treat all patients prophylactically. The expert panel in our Delphi procedure did not reach a consensus on this issue. Consequently, a weak recommendation was formulated: prophylactic treatment might be considered for patients with a high risk of developing death rattle or when a significant burden is expected for family members.	(+) Prophylactic treatment can avoid the development of death rattle, and decreases death rattle severity if it develops. (-) Prophylactic treatment might lead to side effects, which is particularly undesirable for patients who would not otherwise develop death rattle.	V	Weak
55	<i>De behandeling van doodstreutel kan overwogen worden als deze zeer uitgesproken is en/of zeer belastend is voor de familie.</i>	Experts in our Delphi procedure disagree whether or not to treat death rattle with anticholinergics. This disagreement stems from uncertainty regarding the effectiveness of available pharmacological treatments and whether death rattle causes discomfort to the patient (48). Despite these uncertainties, death rattle can be highly distressing for relatives and healthcare workers, and pharmacological treatments are generally well tolerated. Consequently, some experts in our Delphi procedure support treatment, while others do not, making both approaches viable options.	(+) Treatment may increase the likelihood of reducing the severity of the death rattle. (-) There could be potential side effects associated with pharmacological treatment.	V	Weak
56	<i>Indien doodstreutel zou behandeld worden wordt deze medicatie verkozen: Buscopan (hyoscine butylbromide) 20mg subcutaan tot 4 keer per dag, of 60mg per dag via continue toediening</i>	Buscopan is theoretically less likely to cause restlessness because it does not cross the blood-brain barrier and is equipotent to other anticholinergics (42). Additionally, it is the only medication tested in a prophylactic setting and has been shown to be effective in reducing the occurrence of the death rattle (43) (47). Eighty percent of experts in our	(+) Treatment may increase the likelihood of reducing the severity of the death rattle. (-) There could be potential side effects associated with pharmacological treatment.	II	Weak

		Delphi procedure agreed with the use of buscopan, while two other experts preferred the use of scopolamine or atropine.			
Which pharmacological treatment can be recommended for management of fever in dying nursing home residents? (clinical question 5)					
	Recommendation	Evidence to Decision	Health benefits (+)/Risks (-)	Level of evidence	Strength of recommendation
57	Terminale koorts (lichaamstemperatuursverhoging bij een stervende persoon) wordt behandeld indien dit discomfort veroorzaakt bij de bewoner, of wanneer de kans op stuipen reëel is. De ongewenste effecten van de behandeling kunnen voor de bewoner soms meer belastend zijn dan de koorts zelf.	There is very little evidence on how and when fever should be treated during the last days of life. Fever itself most often does not cause discomfort, but the treatment of fever might cause discomfort (due to side effects). The experts in our Delphi agreed on this recommendation (49).	(+) Avoids unnecessary treatment (-) None	V	Weak
58	Het is geen doel om de oorzaak van de koorts te achterhalen.	All experts from our Delphi procedure agreed on this recommendation.	(+) Avoids unnecessary interventions for the resident. (-) None.	V	Weak
59	Terminale koorts wijkt meestal niet met koortswerende middelen.	Experts in our Delphi procedure agreed, based on their experience, that the complete resolution of terminal fever is difficult. This is in line with other expert based recommendations (49).	(+) Avoids overtreatment by informing health care workers. (-) None	V	Weak
60	<i>Als de behandeling van koorts gewenst is, kan gebruik gemaakt worden van Paracetamol 1g 2-4 keer per dag per os of rectaal.</i>	Ninety percent of experts in our Delphi procedure preferred the use of paracetamol for treatment of fever during the last days of life. Paracetamol is also the preferred treatment option in several expert based guidelines (49)(50).	(+) Safe treatment option which is effective. (-) Treatment might cause discomfort due to the administration route or due to side effects (e.g. chills...).	V V	Weak
Which pharmacological treatment can be recommended for the management of confusion, delirium and anxiety in dying nursing home residents? (clinical question 6)					
	Recommendation	Evidence to Decision	Health benefits (+)/Risks (-)	Level of evidence	Strength of recommendation
61	Er is weinig evidentie over de behandeling van terminaal delier. De behandeling is daarom gebaseerd op de behandeling van delier in de niet-terminale fase.	In our Delphi procedure 80% of experts agreed on this recommendation.	(+) None (-) None	V	Weak
62	Rusteloosheid en agitatie komen frequent samen voor aan het levenseinde en kunnen meerdere oorzaken hebben. De exacte oorzaak achterhalen is niet altijd aangewezen in de laatste levensdagen. Wel is het belangrijk om uitlokkende factoren van rusteloosheid en agitatie te behandelen,	The initial recommendation stated that the underlying cause of agitation and restlessness should not be investigated (14). However, experts pointed out that some factors can easily be discovered and treated, without causing discomfort in the patient.	(+) Avoiding unnecessary investigations that may cause discomfort, while addressing easily identifiable causes, can enhance patient comfort and reduce the need for pharmacological management. (-) None.	V	Weak

	zoals o.a. medicatie, pijn, urineretentie, fecale impactie, hypoxie, omgevingsfactoren, psychologische en spirituele nood.				
63	Rusteloosheid en agitatie op het einde van het leven kunnen veroorzaakt worden door angst. Het toedienen van een lage dosis benzodiazepines kan effectief zijn in de behandeling hiervan.	Benzodiazepines have proven to be effective in the treatment of anxiety (25)(26). All experts in our Delphi procedure agreed with this recommendation.	(+) The most appropriate treatment will increase the likelihood of achieving symptom control. (-) Possible increase in somnolence due to benzodiazepine use.	II	Strong
64	Angst is frequent geassocieerd met dyspneu, waardoor benzodiazepines effectief zijn in de behandeling van angst geassocieerde dyspneu.			II	Strong
65	De voorkeursbehandeling voor delier bestaat uit antipsychotica, behalve bij alcohol- of benzodiazepineonttrekingsdelier.	In the non-end-of-life setting benzodiazepines are first-line treatment for alcohol and benzodiazepine withdrawal delirium. Seventy-eight percent of experts in our Delphi procedure agreed, other experts withheld due to a lack of experience.	(+) The most appropriate treatment will increase the likelihood of achieving symptom control. (-) None	V	Strong
66	Overweeg een lage dosis haloperidol voor de behandeling van agitatie of motorische onrust veroorzaakt door delier bij woonzorgcentrumbewoners in de laatste levensdagen.	Eighty-six percent of experts from our Delphi procedure agreed with this recommendation, one expert preferred the use of benzodiazepines. The Australian guideline referenced a review by Lonergan et al. from 2007 which discussed that haloperidol, risperidone, and olanzapine are equally effective in treating terminal delirium (27). This study took place in the general adult population, not during the last days of life. Our systematic review primarily suggested that haloperidol may be used as monotherapy for treatment of delirium during the last days of life of older people (40). Given its widespread use, availability in Belgium, and compatibility with various medications for subcutaneous injection, haloperidol was chosen as the main treatment for delirium. Some randomized controlled trials have studied the effects of haloperidol and benzodiazepine combination therapy with good results on agitation management(40). However, these studies took place in other populations.	(+) Haloperidol treatment increases the chance of delirium resolution. (-) There is a small risk of developing extrapyramidal symptoms, see recommendation 31.	II	Strong
67	Wees aandachtig voor extrapiramidale bijwerkingen bij het gebruik van antipsychotica.	Informative recommendation to make health care workers aware of possible side effects (27). This has been discussed more in detail for recommendation 31.	(+) Increases awareness of side effects, leading to faster detection and treatment. (-) None	II	Weak
68	Bij persisterende angst of rusteloosheid ondanks systematische toediening van benzodiazepines, overweeg associatie van antipsychotica.	Diagnosing the underlying cause of restlessness can be difficult. Therefore it is common practice to combine antipsychotics with benzodiazepines when single therapy is insufficient. All experts in our Delphi procedure agreed. Two randomized controlled trials, which were excluded from our systematic reviews because of their broad age criteria,	(+) Combination therapy might achieve better symptom control with lower dosages, reducing the risk of adverse events and increasing the chance of obtaining symptom control. (-) Polypharmacy might increase the risk of	II	Weak

		studied the combination effect of antipsychotics (e.g. haloperidol) and benzodiazepines (e.g. midazolam) during the last days of life in patients with cancer (56,57). Both studies compared the use of a low dose haloperidol versus the combination of low dose haloperidol and benzodiazepines (midazolam and lorazepam respectively). Both studies showed a reduced agitation scores in the combination group compared with the monotherapy group. However the study by Ferraz Gonçalves did show a higher sedation score in the combination group, which might not be unfavorable in this setting.	adverse events.		
69	<i>Voorzie standaard een “zo nodig” voorschrift voor agitatie en rusteloosheid bij het herkennen van de terminale fase.</i>	Providing a prescription “as needed” is standard practice in end-of-life care, ensuring rapid and effective management of agitation and restlessness. All experts agreed.	(+) An "as needed" prescription allows nurses to promptly address symptoms as they arise, avoiding any delay in treatment. (-) None	V	Strong
70	<i>Wanneer angst aanwezig is, dient de voorgeschreven medicatie te worden toegediend.</i>	All experts from our Delphi procedure agreed on this recommendation.	(+) Treatment of anxiety once present decreases patient discomfort. (-) None	V	Strong
71	<i>Indien de bewoner regelmatig nood heeft aan “zo nodig” dosissen benzodiazepines of chronisch behandeld wordt met orale benzodiazepines en slikmoeilijkheden ontwikkelt, zet dit dan om naar midazolam via continue subcutane toediening. Start minimaal met midazolam 5-10mg over 24u, maar hogere doseringen kunnen nodig zijn.</i>	The start of continuous subcutaneous injection when patients need regular medication injections is standard practice in end-of-life care. All experts in our Delphi procedure agreed.	(+) Continuous treatment helps to avoid breakthrough symptoms. (-) Benzodiazepine use might cause an increase in somnolence.	V	Strong
72	<i>Als meer dan 3 dosissen van de “zo nodig” medicatie nodig zijn binnen de 24u, moet de huisarts worden gecontacteerd voor het aanpassen van de medicatie.</i>	All experts from our Delphi procedure agreed on this recommendation.	(+) Ensuring a feedback loop to the doctor makes timely interventions possible. (-) None	V	Strong
73	<i>Wanneer een delier aanwezig is, dient de voorgeschreven medicatie te worden toegediend.</i>	All experts from our Delphi procedure agreed on this recommendation.	(+) Treatment of delirium once present decreases patient discomfort. (-) None	V	Strong
74	<i>Als meer dan 3 dosissen “zo nodig” medicatie nodig zijn voor doorbraak van rusteloosheid of agitatie binnen de 24u, moet de huisarts gecontacteerd worden om eventueel een gespecialiseerd palliatief zorgteam in te schakelen.</i>	All experts from our Delphi procedure agreed on this recommendation.	(+) Ensuring a feedback loop to the doctor makes timely interventions possible. (-) If the patient is still in uncomfortable and the doctor is unavailable, this might cause a delay in achieving adequate symptom control.	V	Strong
75	<i>Indien de bewoner slikproblemen ontwikkelt en blijvend nood heeft aan behandeling met een antipsychoticum, zet dit dan om naar een continue subcutane toediening via</i>	The start of continuous subcutaneous injection when patients need regular medication injections is standard practice in end-of-life care. Ninety-three percent of experts agreed with the recommendation. Some experts	(+) This ensures the use of an appropriate administration route. (-) None	V	Strong

	<i>sputdrijver over 24u, bv. haloperidol. Startdosis hangt af van initiële antipsychotica en/of ernst van het delier. (voor omreken tabel zie https://www.psychiatrienet.nl/switchtabel/)</i>	commented that if a patient is being treated with high doses of opioids and/or benzodiazepines in the last days of life, there might not be a need to continue or substitute their chronic antipsychotic medication.			
76	<i>Bij de behandeling van rusteloosheid en agitatie bij benzodiazepine-naïeve bewoners kan lorazepam sublinguaal (Temesta expedit®) 1mg tot 3 keer per dag worden toegediend indien de mond en slijmvliezen van de bewoner niet te droog zijn</i>	The use of lorazepam was suggested by the experts in round one of the Delphi procedure and agreed upon by 90% of experts in the second Delphi round. The remaining expert from our Delphi procedure noted that Temesta Expedit® might act too slow to achieve fast and adequate symptom control.	(+) Easily administrable if patient is still conscious. (-) Not recommended if the resident is unconscious, in which case a conversion to subcutaneous midazolam is often necessary. Risk of increasing somnolence.	V	Weak
77	<i>Bij de behandeling van rusteloosheid en agitatie bij benzodiazepine-naïeve bewoners kunnen clonazepam druppels 0.5mg per os tot 6 keer per dag worden toegediend, indien slikken nog mogelijk is. (0.5mg clonazepam is equivalente dosis van 1mg lorazepam)</i>	The Australian guideline recommended 0.25-0.5 mg per administration point. To simplify the conversion to subcutaneous midazolam, we aimed to provide equivalent dosage recommendations. Eighty percent of experts agreed with this recommendation, the remaining two experts had no experience with clonazepam drops.	(+) Easily administrable if patient is still conscious. (-) Not applicable if a continuous subcutaneous infusion is present or considered, in which case subcutaneous midazolam is a more appropriate option. (-) Risk of increasing somnolence.	V	Weak
78	<i>Bij de behandeling van rusteloosheid en agitatie bij benzodiazepine-naïeve bewoners kan midazolam 2.5mg subcutaan tot 6 keer per dag worden toegediend. (2.5mg midazolam is equivalente dosis van 1mg lorazepam)</i>	All experts in our Delphi procedure agreed with the recommendation. One expert mentioned that rapid dose titration might be necessary if adequate symptom control is not achieved after a single administration.	(+) Easy and safe to administer subcutaneously. (-) Risk of increasing somnolence.	V	Weak
79	<i>Bij de behandeling van rusteloosheid en agitatie in de laatste levensdagen bij een vermoeden van onderliggend delier, is de voorkeursbehandeling haloperidol: a. Haloperidol 10 druppels per os tot 2 keer per dag, indien slikken nog mogelijk is b. Haloperidol 0.5-1mg subcutaan tot 2 keer per dag. c. Haloperidol 2.5mg/24u subcutaan via continue subcutane spuitdrijver</i>	Eighty percent of experts in our Delphi procedure agreed, with some indicating to immediately start continuous subcutaneous injection and one expert suggesting to start with 3x5drops per day, increasing to 3x10drops per day if necessary.	(+) Treatment with haloperidol increases the chance of decreasing agitation and restlessness. (-) There is a small risk of developing extrapyramidal symptoms (as discussed in recommendation).	V	Weak

Key principles and key points translated from source guideline

	Recommendation	Level of evidence	Strength of recommendation
80	Medicatie wordt voorgeschreven door de behandelende huisarts, die een coördinerende rol heeft.	V	Strong
81	De bewoners en/of hun vertegenwoordiger moeten op de hoogte gebracht worden dat het stervensproces is gestart en dat medicatie om eventuele symptomen te bestrijden de kwaliteit van sterven kan verbeteren.	V	Strong
82	De bewoner en/of vertegenwoordiger worden geïnformeerd dat medicatie voor de behandeling van symptomen in de laatste levensdagen wordt gestart.	V	Strong

83	Onrust komt frequent voor aan het levenseinde en kan meerdere oorzaken hebben. De exacte oorzaak achterhalen is niet altijd aangewezen in de laatste levensdagen. Wel is het belangrijk om volgende oorzaken te herkennen en correct mee om te gaan: pijn, dyspneu, delier, angst, nausea en braken, urineretentie, faecale impactie, medicatie, psychologische, sociale en spirituele noden...	V	Strong
84	Het traject vereist dat symptomen elke 8u shift minstens éénmaal worden beoordeeld. Dit zorgt ervoor dat nieuwe symptomen snel worden gedetecteerd en waar nodig medicamenteus kunnen worden behandeld. Bij nieuwe symptomen en/of na medicamenteuze aanpassingen is herevaluatie om de 2u aangewezen.	V	Strong
85	Als medicamenteuze behandeling nodig wordt geacht, moet het meest geschikte geneesmiddel worden geselecteerd, dat op een veilige en effectieve manier wordt toegediend.	V	Strong
86	De voorgeschreven medicatie en doseringen moeten gebaseerd zijn op zorgvuldige evaluatie van de toestand van de bewoner en (ernst van de) symptomen.	V	Strong
87	De belasting veroorzaakt door medicatietoediening of -bijwerkingen moet zo klein mogelijk zijn. Medicatie in de laatste levensdagen wordt meestal subcutaan toegediend, deze toedieningsweg is over het algemeen het minst invasief en meest betrouwbaar bij een stervende bewoner.	V	Strong
88	Medicatie met een beperkt therapeutisch voordeel in de terminale levensfase moet worden gestopt.	V	Strong
89	Persisterende symptomen vereisen onderhoudsmedicatie op vaste tijdstippen eerder dan “zo nodig” medicatie.	V	Strong
90	Deze onderhoudsmedicatie sluit de nood aan “zo nodig” medicatie niet uit. Een “zo nodig” medicatievoorschrift moet worden voorgeschreven om doorbraaksymptomen en symptomen die slechts tijdelijk aanwezig zijn snel en optimaal te kunnen behandelen.	V	Strong
91	Effectiviteit van de toegediende medicatie dient te worden geëvalueerd en gedocumenteerd.	V	Strong
92	Voorzie de aanwezigheid van de meest courante “zo nodig” medicatie, of maak afspraken met de lokale apotheek zodat deze medicatie snel kan geleverd worden.	V	Strong
93	Denk naast de farmacologische behandeling, ook steeds aan een niet-farmacologische aanpak van de symptomen.	V	Strong
94	De verpleegkundige is verantwoordelijk voor: a. Regelmatige beoordeling van symptomen	V	Strong
95	De verpleegkundige is verantwoordelijk voor: b. Toediening van “zo nodig” medicatie wanneer nodig	V	Strong
96	De verpleegkundige is verantwoordelijk voor: c. Regelmatige monitoring en documentatie van de effectiviteit van voorgeschreven medicatie	V	Strong
97	De verpleegkundige is verantwoordelijk voor: d. Identificatie en rapportering van nevenwerkingen/bijwerkingen veroorzaakt door de voorgeschreven medicatie.	V	Strong
98	De verpleegkundige is verantwoordelijk voor: e. Het contacteren van de huisarts wanneer de toegediende medicatie niet het gewenste effect heeft, om de voorgeschreven medicatie (zowel de standaard als de “zo nodig” medicatie) te herbekijken.	V	Strong

99	Het effect van de toegediende medicatie wordt geregistreerd, bijwerkingen worden genoteerd en gemeld aan de behandelende huisarts. Het Federaal Agentschap voor Geneesmiddelen en Gezondheidsproducten moedigt de rapportage van alle bijwerkingen ten sterkste aan. Informatie over melden van bijwerkingen kan online gevonden worden op https://www.fagg.be/nl/bijwerking	V	Strong
100	Een medicatiefout (bv. onder- of overdosering) dient gemeld te worden aan de behandelende huisarts.	V	Strong
101	Bij onvoldoende symptoomcontrole na het doorlopen van de verschillende stappen van deze richtlijn kan de huisarts beroep doen op Multidisciplinaire BegeleidingsEquipes voor palliatieve verzorging (MBE) (https://www.zorg-engezondheid.be/multidisciplinaire-begeleidingsequipes-voor-palliatieveverzorging-mbe)	V	Strong

Chapter 14: Implementation

This clinical practice guideline on the pharmacological management of symptoms in the last days of life of nursing home residents is part of a larger project called “Development, implementation and evaluation of the Nursing Home Care Program for the Last Days of Life” and is accomplished by the End-Of-Life Care research group with funding by ‘Kom op tegen Kanker’ vzw. The aim of this project is to develop and implement a Nursing Home Care Program for the Last Days of Life, including an evidence-based clinical practice guideline for the pharmacological and non-pharmacological treatment of symptoms in the last days of life, a Care Guide for the Last Days of Life with recommendations for health care providers on recognizing dying, communication with family, regular assessment of symptoms, psychosocial and spiritual support, interprofessional and transmural communication and an implementation guide.

Implementation of our pharmacological management guideline will be done as a part of this larger project. This project will be enrolled in eight nursing homes through participatory action research, where the staff of the participating nursing homes will actively shape the implementation and put the care program into practice, adjusting it as necessary.

Chapter 15: Peer review

The guideline together with this methodological report will be submitted to CEBAM for review.

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APPENDICES

Appendix A: List of associations and stakeholders in the development of this guideline

Guideline development group

Tim Biesbrouck (MD)^{1,2,3}, Maaïke De Roo (MD, PhD)^{1,2}, Dine AD Jennes (MD)⁴, Nele Van Den Noortgate (MD, PhD)^{3,5}

1 Department of Geriatric Medicine, University Hospital Leuven, Leuven, Belgium

2 Department of Public Health and Primary Care, Gerontology and Geriatrics, KU Leuven, Leuven, Belgium

3 Department of Geriatric Medicine, Ghent University Hospital, Ghent, Belgium

4 Department of Geriatric Medicine, Antwerp University Hospital, Edegem, Belgium

5 End-of-life Care Research Group, Vrije Universiteit Brussel (VUB) and Ghent University, Laarbeeklaan, Brussels, Belgium

In Chapter 2.1 (page 6) of this guideline, you'll find a more detailed description of the guideline development group.

List of participating experts in the Delphi procedure

A total of fifteen experts participated:

- Three geriatricians who are a member of the “Palliatieve zorg en GERiatrie” workgroup of Palliative Care Flanders;
- Three general practitioners who are also CRA, and member of the “Palliatieve zorg en GERiatrie” of Palliative Care Flanders
- Five additional general practitioners who are CRA;
- Four nurses with special expertise in palliative care (in nursing homes).

As this was an anonymous Delphi procedure, the identities of the experts cannot be disclosed.

Others involved during the development

Prof. Lieve Van den Block, End-of-Life Care Research Group, Department of Family Medicine and Chronic Care, Vrije Universiteit Brussel & Ghent University

Prof. Tinne Smets, End-of-Life Care Research Group, Department of Family Medicine and Chronic Care, Vrije Universiteit Brussel & Ghent University

Fien Van Sint Jan (helped in the initial phases of the guideline development)

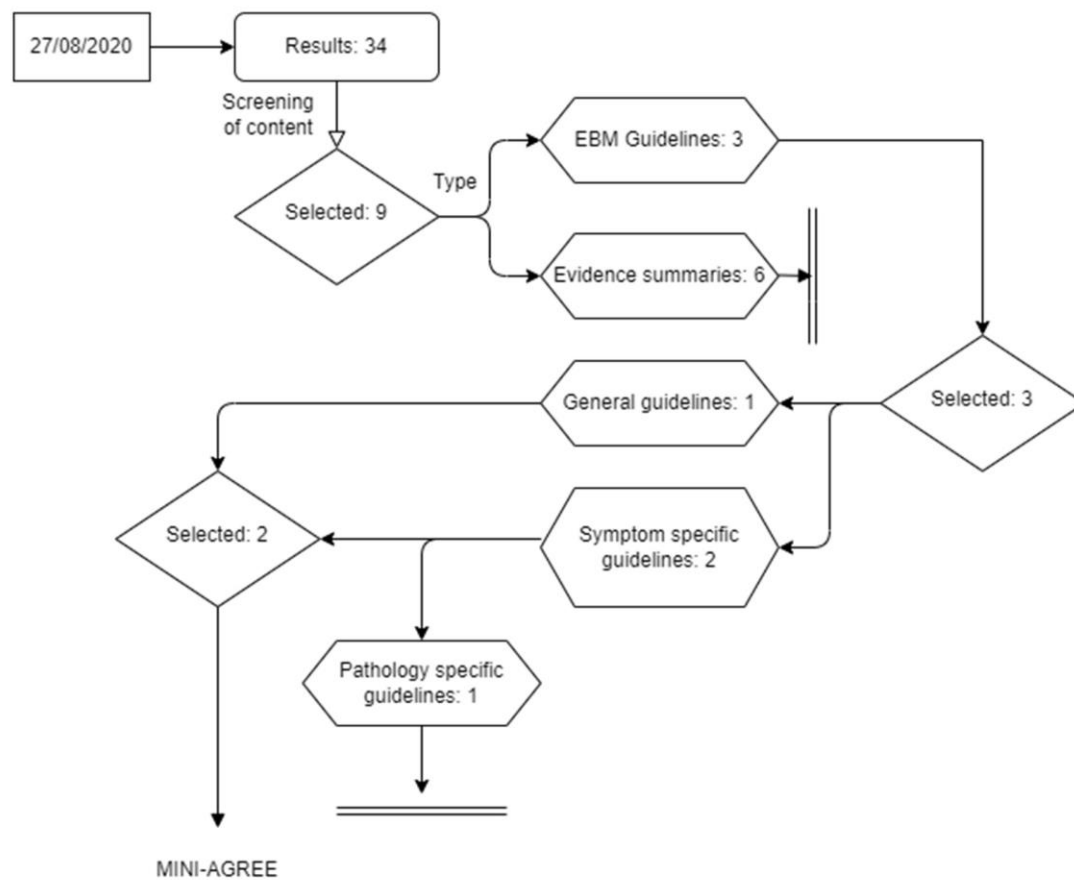
Appendix B: Summary of the guideline database search strategy and search results

Ebpracticenet: www.ebpnet.be	Search date: 10/08/2020	Search string: "palliatieve zorg" OR "palliative care" OR "terminal care" OR "terminale zorg"
		Results: 28
		Selected: 7
	Search date: 21/11/2020	Search string: "palliatieve zorg" OR "palliative care" OR "terminal care" OR "terminale zorg"
		Results: 30 (2 additional guidelines were detected in comparison with search dd 10/08/2020)
		Selected: 1 leading to 8 guidelines in total
Duodecim: www.ebm-guidelines.com	Search date: 27/08/2020	Search string: 'palliative care', giving the same results as 'palliative treatment'
		Results: 34
		Selected: 9
Guidelines International Network (G.I.N): www.g-i-n.net	Search date: 15/08/2020	Search string: "palliative care" OR "palliative treatment" OR "palliative therapie" OR "terminal care"
		Results: 89
		Selected: 28
Clinical Knowledge Summaries (CKS) from the National Institute for Health and Care Excellence (NICE): (https://www.nice.org.uk/about/what-we-do/evidence-services/clinical-knowledge-summaries ; http://www.evidence.nhs.uk)	Search date: 17/08/2020, 25/08/2020, 27/08/2020	Search strings: palliative care with filters for document type: guidance (guidance category: NICE guidelines and Clinical guidelines) AND quality standards
		Results: 45
		Selected: 3
Scottish Intercollegiate Guidelines Network (SIGN): (SIGN; http://www.sign.ac.uk)	Search date: 22/12/2020	Manual selection of 'current' and 'archived' guidelines next to all guidelines in development
		Results: 0
		Selected: 0
Nederlands Huisartsen Genootschap (NHG; http://www.artsennet.nl/Richtlijnen.htm)	Search date: 21/10/2020	Search string: terminale zorg
		Results: 11
		Selected: 2
Richtlijnen NVKG	Search date: 29/10/2020	Search string: palliatieve zorg
		Results: 9
		Selected: 2
Nederlandse richtlijnen database: IKNL using pallialine.nl and pallialine.be	Search date: 23/11/2020, 30/11/2020, 01/12/2020	Search string: Manual selection of all available guidelines
		Results: 56 (59 results minus 3 patient information folders)
		Selected: 17
Guidelines SumSearch	Search date: 29/10/2020	Search string: "end of life" AND "guideline", restriction: 'Guidelines'; focus: intervention; age: adult; human only; require abstracts; max. iterations: 6; target of original studies: 50
		Results: 29
		Selected: 2
Guideline database Federatie van Medisch specialisten (searched because of a referenced in the guideline delirium on pallialine.nl)	Search date: 30/11/2020	Search string: Palliatieve zorg.
		Results: 34
		Selected: 3
Caresearch and (https://www.clinicalguidelines.gov.au/)	Search date: 08/12/2020	Search string: Manual search all available guidelines
		Results: 1
		Selected: 1

Appendix C: Guideline screening and selection

EBM Guidelines

Finish collection of evidence-based guidelines for use in primary care

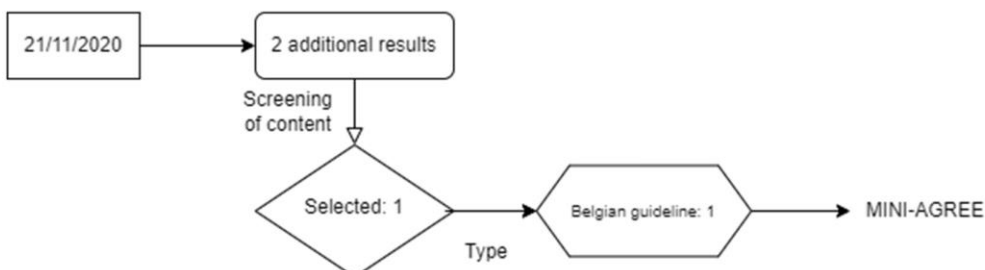
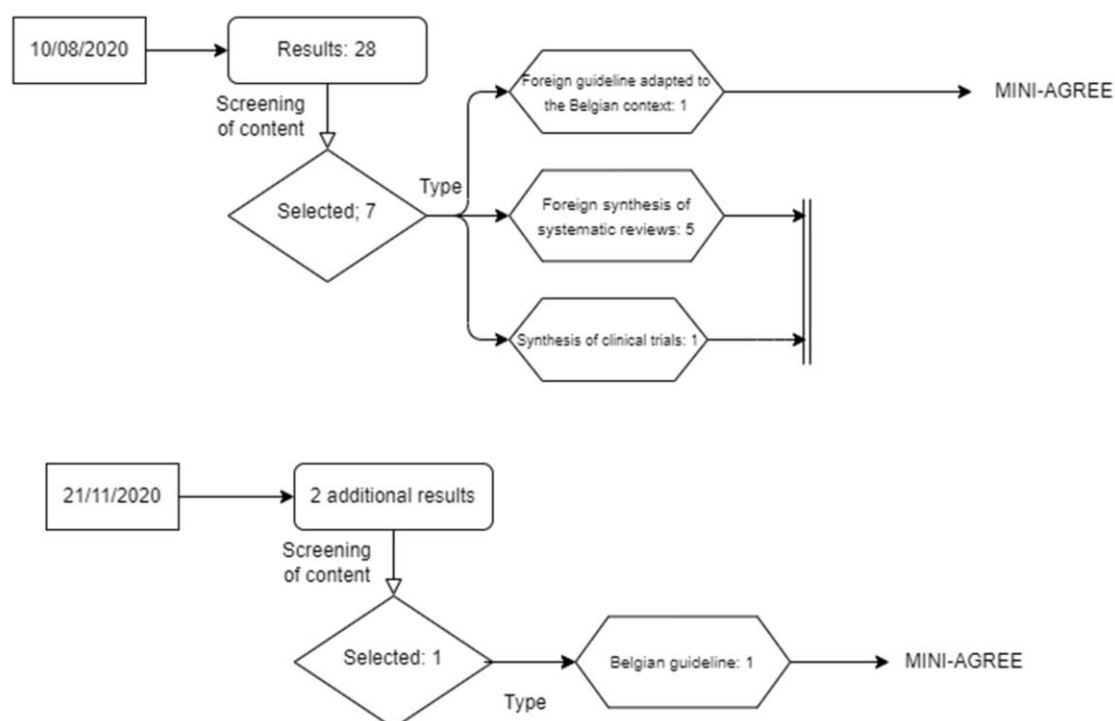


⁸ Selection based on a first screening of content (based on title, or content summary or abstract)

⁹ Additional search because of a new published guideline which was encountered during methodological review of other guidelines during the MINI-AGREE phase.

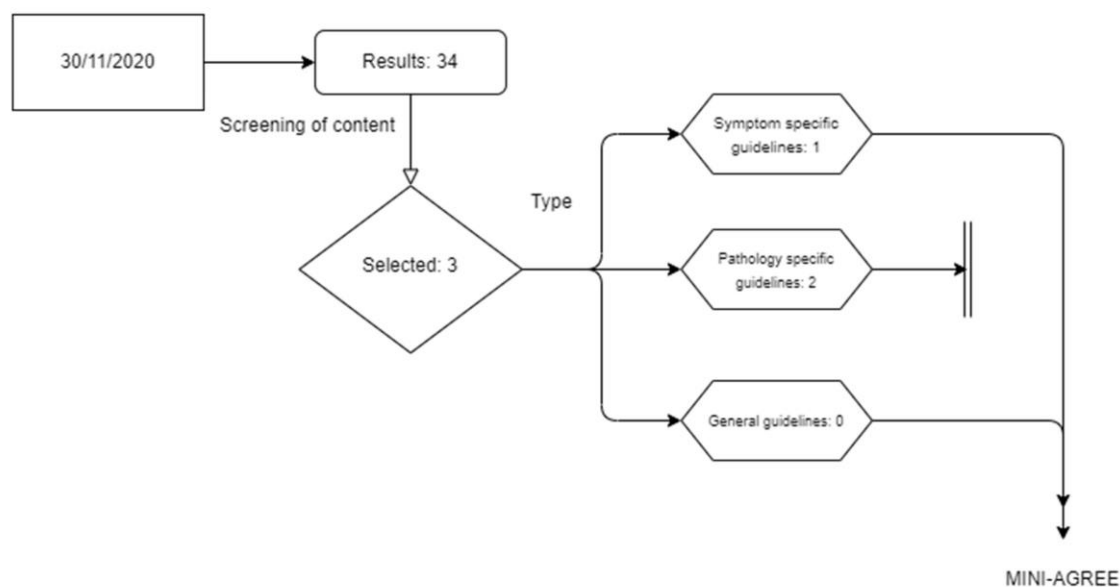
EBpracticenet

Reference centre for evidence based practice for Belgian health care workers



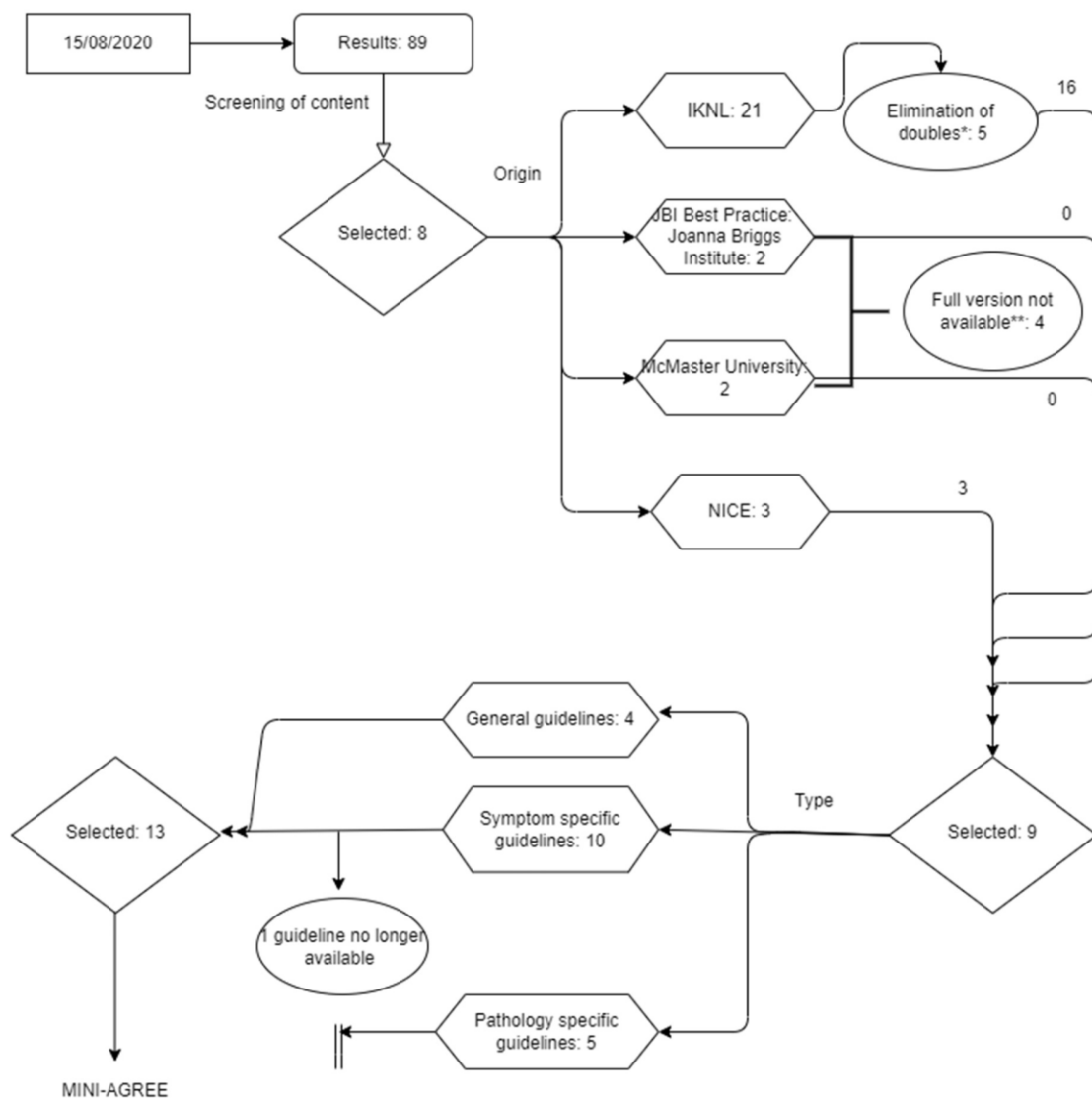
FMS: Federatie Medisch Specialisten

Dutch society of 33 medical specialties.



GIN

Guidelines International Network, a Scottish non-profit organisation that was founded in light of the European AGREE project in 2002 with the goal of creating a guideline library.

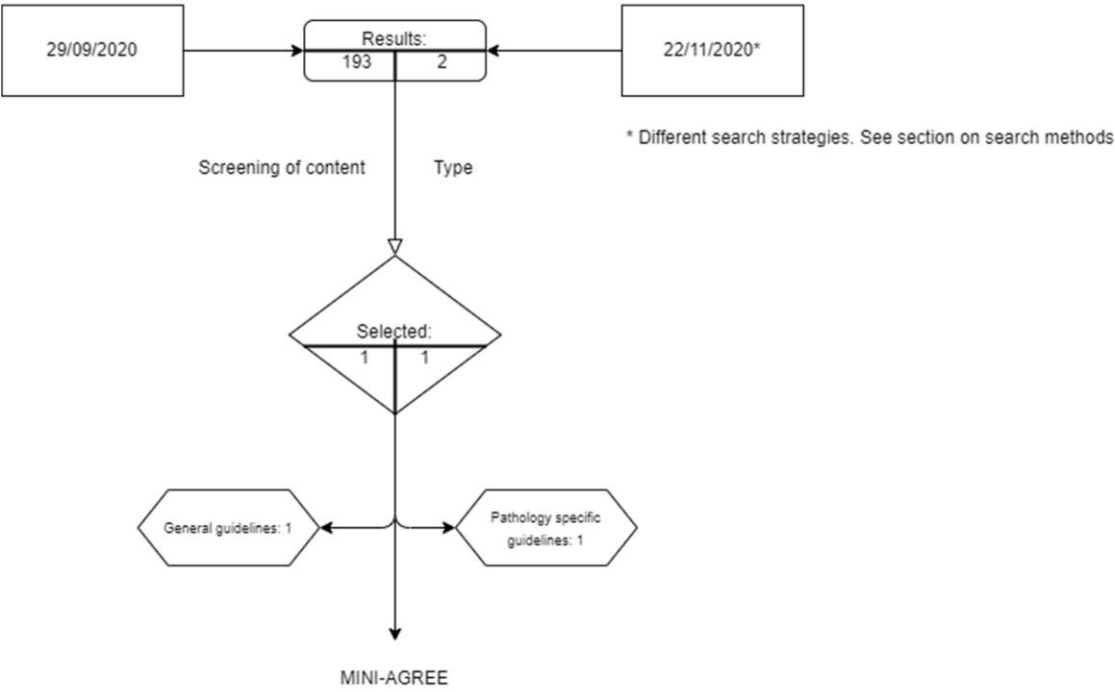


* English and Dutch version, identical content.

** Full version not available after request with publisher

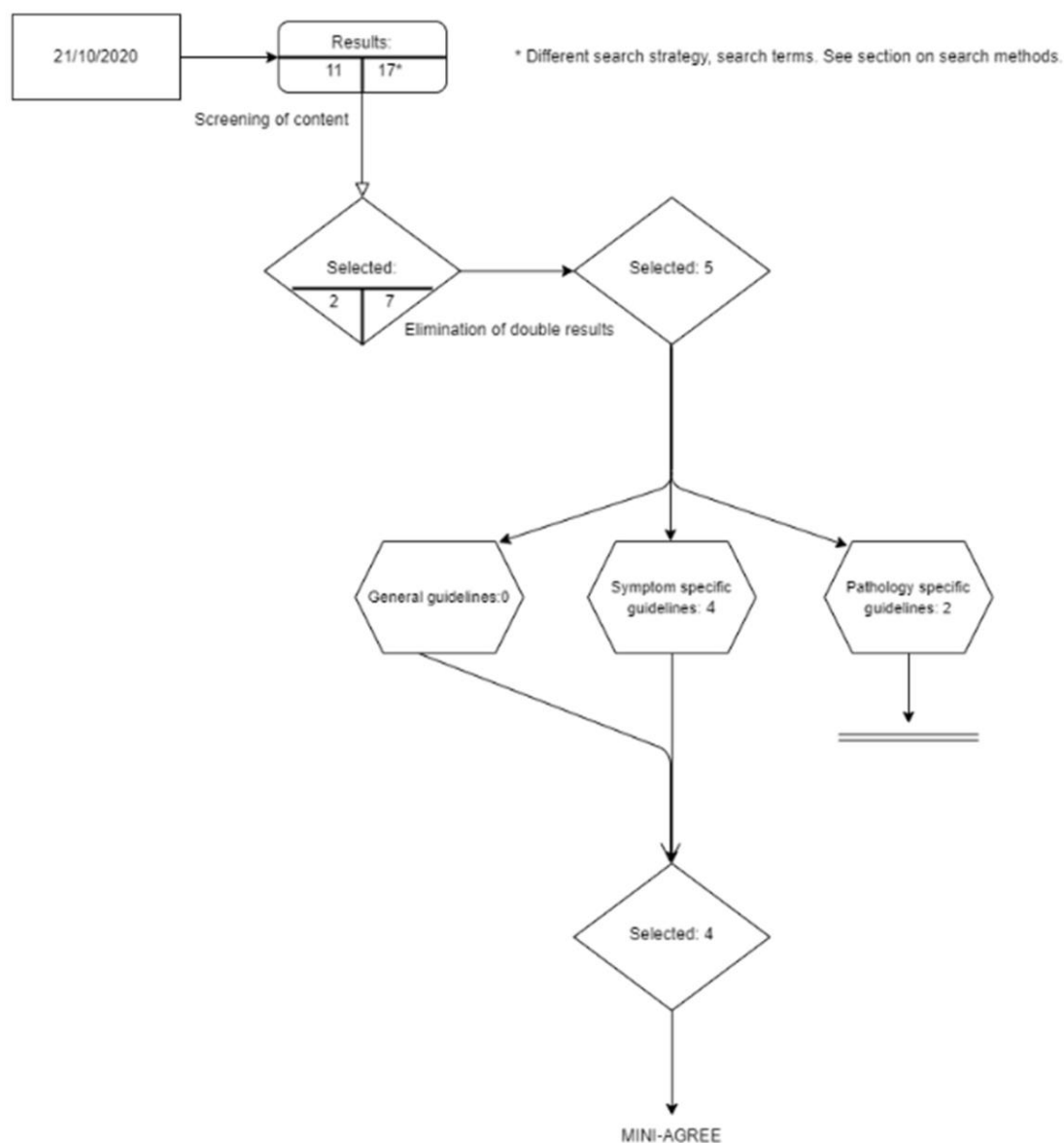
Guideline Central

American database



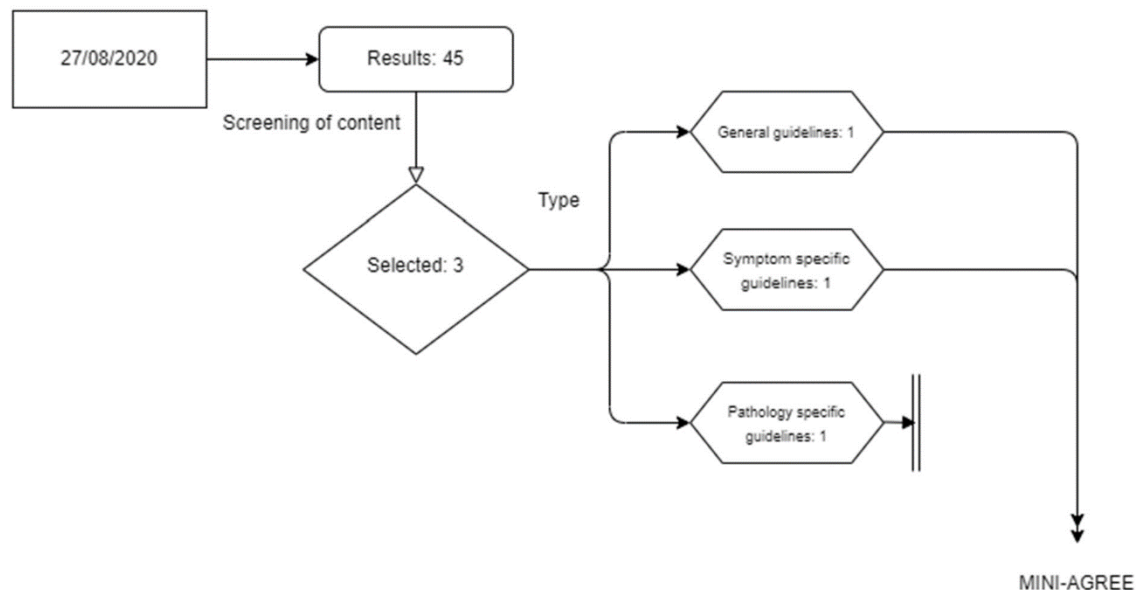
NHG

The Dutch College of General Practitioners, the scientific association of general practitioners in the Netherlands.



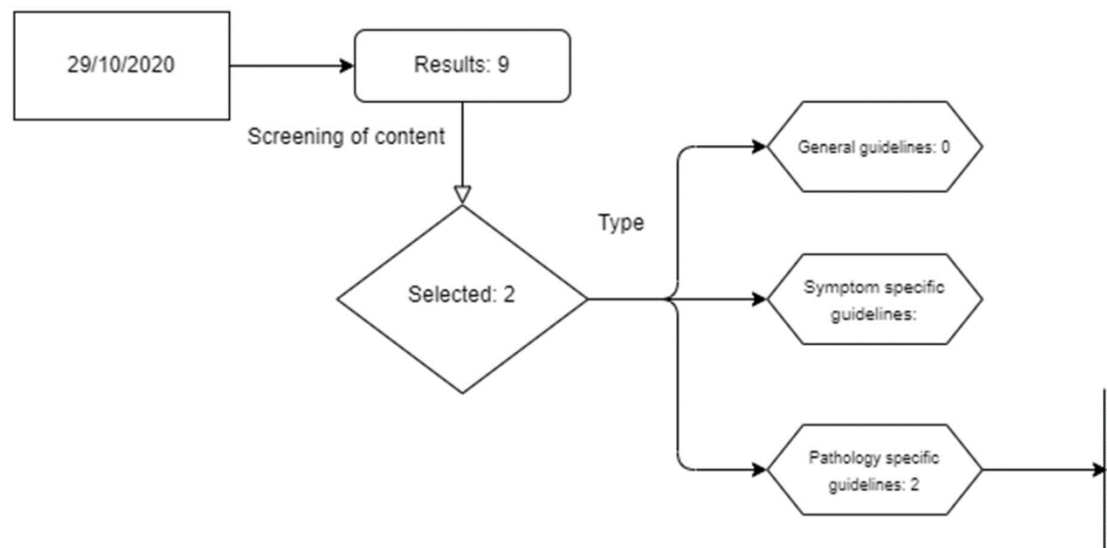
NICE

National Institute for Health and Care Excellence: database and guideline development centre for use in the UK's NHS.



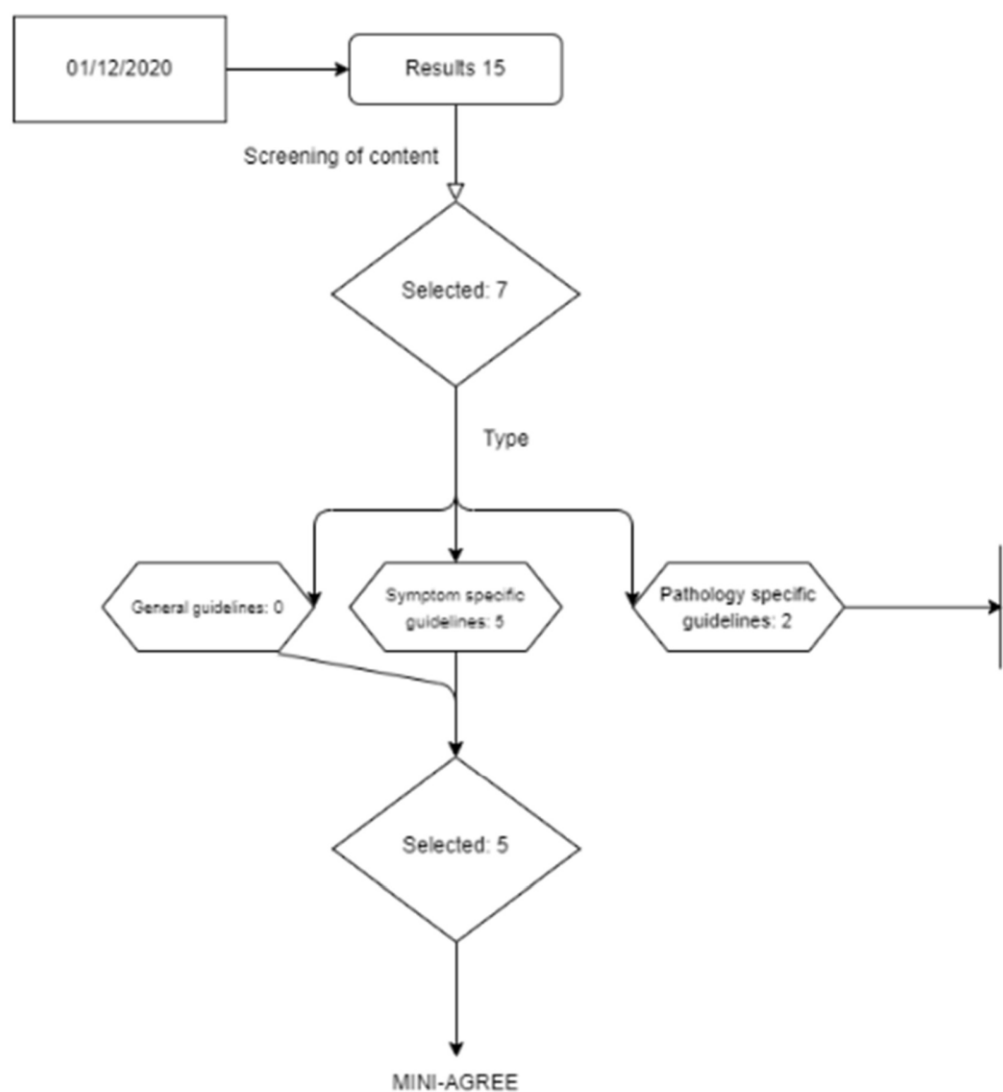
NVKG

Nederlandse vereniging voor klinische geriatrie.



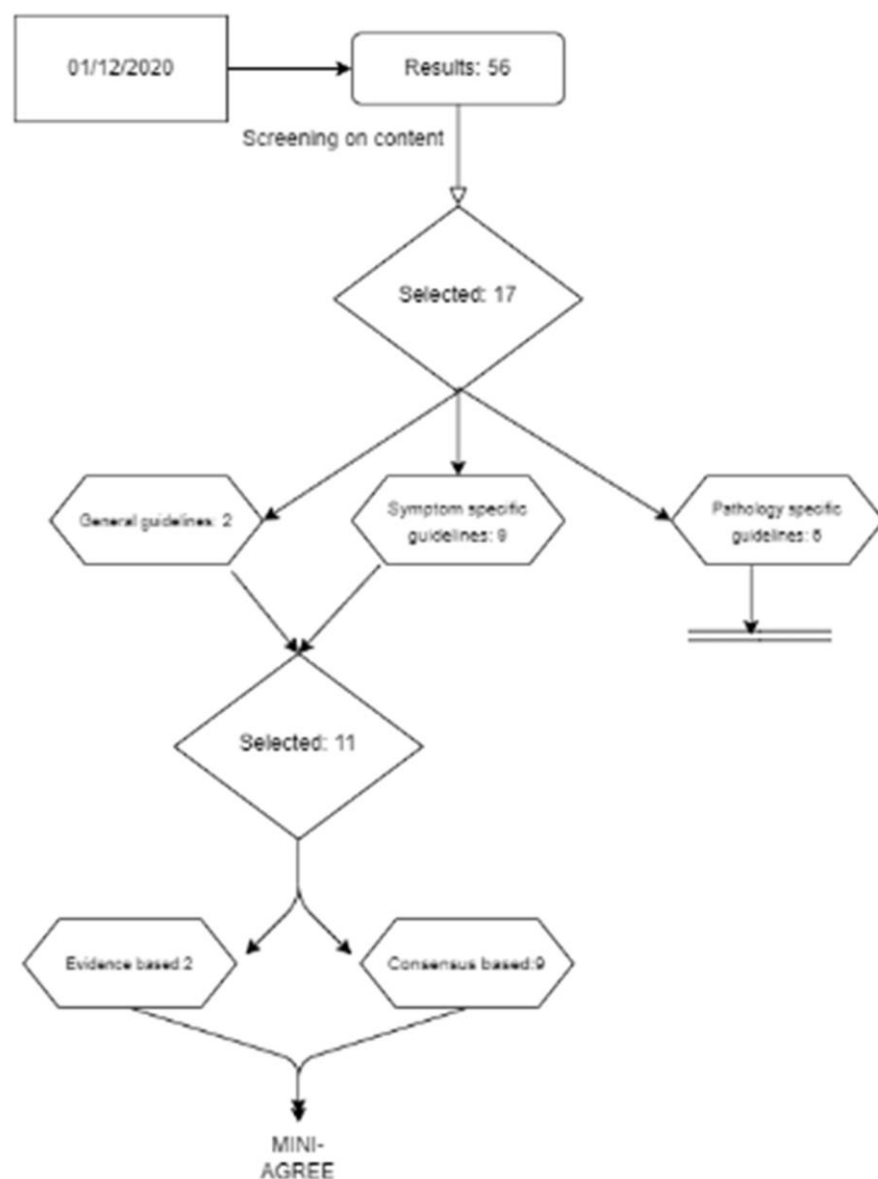
Pallialine.be

An initiative of the Federatie Palliatieve Zorg Vlaanderen vzw (now Palliatieve Zorg Vlaanderen) and based on the Dutch website: www.pallialine.nl of the Netherlands Comprehensive Cancer Organisation (IKNL)



Pallialine.nl - IKNL

Pallialine is based on the website oncoline, also a product of the IKNL (the Netherlands Comprehensive Cancer Organisation)



Appendix D: MINI-AGREE RESULTS

	ITEM 7		ITEM 9		ITEM 23		RELEVANCE BASED ON CONTENT		CONCLUSION
RESEARCHER:	1	2	1	2	1	2	1	2	
Palliatieve behandelingen (Duodecim, 2019)	2	2	1	1	1	1	4	5	Excluded
Palliatieve zorg (symptoomcontrole), KUL 2020	4	5	6	6	6	7	6	6	Accepted for further evaluation
Palliative Care/treatment (Duodecim 2019)	1	2	4	2	2	1	5	5	Excluded
Delirium in the elderly (EBM, 2018)	1	3	4	3	3	1	2	4	Excluded
Angst (GIN/ IKNL, 2009)	<u>1</u>	<u>2</u>	<u>3</u>	<u>2</u>	<u>1</u>	<u>4</u>	<u>4</u>	<u>4</u>	Excluded
Obstipatie (GIN / IKNL, 2010)	1	2	3	3	1	4	4	4	Excluded
Algemene inleiding richtlijn palliatieve zorg. National consensus based richtlijn.	1	3	1	3	1	4	5	2	Excluded
Delier (GIN/ IKNL, 2010)	1	2	3	2	1	4	4	4	Excluded
Diarree (GIN/ IKNL, 2010)	1	2	3	2	1	4	4	4	Excluded
Dyspnee (GIN, IKNL, 2010)	6	7	7	6	5	4	4	4	Accepted for further evaluation
Hoesten (GIN / IKNL, 2010)	1	2	3	3	1	1	4	4	Excluded
Jeuk (GIN/ IKNL, 2010)	1	2	3	3	1	1	4	4	Excluded
Misselijkheid en braken (GIN/ IKNL, 2010)	4	3	2	1	4	6	4	4	Excluded
Palliatieve zorg voor mensen met een verstandelijke beperking (GIN/ IKNL, 2010)	2	1	2	1	5	1	3	4	Excluded
Zorg in de stervensfase (GIN / IKNL, 2010)	1	2	3	3	1	1	2	4	Excluded
Palliative care for adults: strong opioids for pain relief (GIN/ NICE, 2012)	7	7	6	6	4	5	3	5	Accepted for further evaluation
Care of dying adults in the last days of life (GIN/ NICE, 2015)	7	7	6	7	6	7	6	6	Accepted for further evaluation

Delier (NHG,2014)	3	3	2	5	6	4	2	4	Excluded
Pijn (NHG, 2018)	4	4	2	5	5	7	1	4	Excluded
Handreiking bij risico op een acute massieve longbloeding. Landelijke richtlijn, Versie 1.0	1	2	1	2	1	2	5	4	Excluded
Delier bij volwassenen	5	5	6	6	7	7	5	4	Accepted for further evaluation
Pijnbestrijding bij palliatieve patiënten. Richtlijn, Versie 1.0.	3	3	5	2	1	2	3	5	Excluded
Delirium in de palliatieve zorg - een richtlijn, Versie 1.0.	1	3	4	2	1	2	3	5	Excluded
Dyspnoe in de palliatieve zorg - een richtlijn.	1	3	4	2	1	2	3	5	Excluded
Hoesten in de palliatieve zorg - een richtlijn.	1	3	4	2	1	2	3	5	Excluded
Misselijkheid en braken in de palliatieve zorg - een richtlijn.	1	3	4	2	1	2	3	5	Excluded
Guide to the Pharmacological Management of End of Life (Terminal) Symptoms in Residential Aged Care Residents.							7	7	Accepted for further evaluation

Appendix E: AGREE-II scores

AGREE-II scores of the selected guidelines after the mini-AGREE procedure.

Palliative care (symptom control), KUL (2020)

Item	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Score reviewer 1	6	6	5	3	2	6	3	1	5	4	6	6	5	7	6	6	7	5	7	4	2	7	7
Score reviewer 2	7	7	7	4	3	6	3	1	5	4	6	6	5	7	7	6	7	4	6	4	2	7	7

Dyspnoea/ dyspnee, GIN/IKNL (2010)

Item	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Score reviewer 1	6	7	6	5	7	7	6	5	6	5	7	7	2	3	7	7	7	5	6	5	2	6	5
Score reviewer 2	7	7	7	7	7	7	6	6	7	6	7	7	4	5	7	7	7	3	4	4	2	6	5

Palliative care for adults: strong opioids for pain relief, GIN/NICE (2012)

Item	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Score reviewer 1	6	7	7	7	7	7	7	6	7	5	7	5	6	5	4	4	4	6	4	5	7	6	6
Score reviewer 2	6	7	7	7	6	7	7	6	7	5	7	6	7	6	7	7	7	7	4	5	7	5	5

Care of dying adults in the last days of life, NICE (2015)

Item	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Score reviewer 1	6	7	7	7	6	6	7	6	7	7	7	7	7	5	4	5	7	4	4	7	4	6	7
Score reviewer 2	6	6	7	6	5	6	7	6	7	7	7	7	7	5	5	7	7	7	6	7	7	6	7

Appendix F: Interview with Guideline developer about methodology of the Guide to the pharmacological management of end-of-life (terminal) symptoms in residential aged care residents (Brisbane South Palliative Care Collaborative)

Call with Elizabeth Reymond – 24/05/2021

Australian palliative care medical specialist

- Guidelines were developed as a part of the palliative care toolkit (10 years ago); which are available to every RACF (in Australia or Queensland?)
- Funding: Australian government, Health & Ageing department → to improve palliative care in residential aged care facilities (RACF)
 - Because of the different funding arrangements in Australia, there is a barrier to provide proper palliative care to RACF residents.
 - Partners in the project: specialists in palliative care, royal college of general practice, Australian & New Zealand society for palliative care medicine, RACF associations of au (profit organizations), Queensland pharmacists, university Queensland.
- Steering committee: partners
- Working group: partners + stakeholders like GP's, nurses, ...
- Literature review by the university colleagues of which there is no existing methodological report
- Selection criteria for medicine:
 - Easily available
 - Applicable for different symptoms
 - Refunded
 - Easily stored and administered
 - Phobia to store opioid on campus (note: two nurses need to be present in order to administer an opioid; problem of not having enough staff available)
- Then discussed in a Delphi procedure to determine the list of medications, doses (for example, if a resident is naïve or not)
- Collect results and send around for feedback
- Workshops on the toolkits are organized; 1600/3000 nursing homes in Australia use the guidelines and toolkit.
 - Staff is very keen to learn about this subject!
 - National funding for the workshops
- National prescribing service: PalliMeds → app developed from guidelines (Android and iPhone)
- See also study from Wales that used AU Guidelines for a home-based study
- Important to look at ACP!

Appendix G: Evidence tables of the Guide to the pharmacological management of end-of-life (terminal) symptoms in residential aged care residents (Brisbane South Palliative Care Collaborative)

Study (source)	Commonwealth of Australia (2012)
Study type	Guiding principle for medication management in RACF
Comments/notes	Not relevant for our research

Study	Hanson et al (2008)
Study type	Structured after-death interviews with staff and family caregivers
Number of participants	792 participants: - 677 staff members; 3 incomplete → 674 - 451 family members; 5 incomplete → 446 included 331 paired interviews
Lost to follow-up	/
Setting and country	230 LTC sites in Florida, New Jersey, North Carolina and Maryland
Duration of study	July 2002 – December 2004
Population characteristics	Primary family caregivers of residents who had lived at least 15 days of their final month in the LTC and died there, or less than 3 days after transfer to another facility Staff caregivers Mean age 85.4y, 89% white, 77% cognitive impaired, 65% unable to transfer, 19% cancer patients 85% died in long term care facility NH residents were more likely to have problems with cleanliness and with intake of food and water during their final month of life. Dyspnoea and pain symptoms similar between nursing home and residential care/ assisted living.
Inclusion criteria	- Primary family caregivers and staff caregivers of residents who had lived at least 15 days of their final month in the LTC and died there, or less than 3 days after transfer to another facility
Exclusion criteria	/
Risk of bias	- Recall bias or grief may affect after death-interviews - Potential bias by nonresponse: lower response rate for family caregivers and the under-representation of minority populations - Observation might underestimate symptomatology especially in cognitive impaired residents --> low/moderate bias
Imprecision	No confidence intervals given --> NVT High sample size --> low bias
Inconsistency	Other studies show that healthcare providers may underestimate patient pain and family caregivers may perceive symptoms as more severe --> moderate bias
Indirectness	No indirectness found --> very low bias
Publication bias	No studies are missing --> very low bias
Funding	Grants from the National Institute on Aging The sponsor had no role in any aspect of the study
Outcomes	- Frequency and severity of physical symptoms - Effectiveness of treatment - Recommendations to improve care - Staff report of assessment
Results	Symptoms: - Pain 47% - Dyspnoea 48% - Problems with cleanliness 90% - Symptoms affecting intake 72% Symptom burden is somewhat greater for residents of NHs than for residents in RC/AL. Staff and family respondents provided similar overall estimates of the quality of care for symptoms, although they did not often agree on symptom experience of an individual resident who was dying. Both groups recommended improved application of treatment and increased staffing to improve care.
Comments/notes	/

Study	Caprio et al (2008)
Study type	After-death interviews
Number of participants	336 paired interviews from a total population of 677 staff caregivers and 451 family caregivers, from 114 facilities. 11 pairs were excluded because of missing data --> final sample 325 paired interviews about the same decedent from 111 facilities
Lost to follow-up	/
Setting and country	111 NHs and RC/AL facilities in Florida, New Jersey, North Carolina and Maryland
Duration of study	July 2002-January 2005
Population characteristics	Primary family caregivers of residents who had lived at least 15 days of their final month in the LTC and died there, or less than 3 days after transfer to another facility Staff caregivers
Inclusion criteria	- Primary family caregivers and staff caregivers of residents who had lived at least 15 days of their final month in the LTC and died there, or less than 3 days after transfer to another facility
Exclusion criteria	/
Risk of bias	- Recall bias or grief may affect after death-interviews - Staff reports of pain and dyspnoea may differ from those of cognitively intact residents - QOD-LTC: doesn't capture all aspects of dying experience, has limited validation and sensitivity to change hasn't been determined --> moderate bias
Imprecision	- High sample size - Large confidence intervals --> moderate bias
Inconsistency	/
Indirectness	No indirectness --> very low bias
Publication bias	No missing --> very low
Funding	Grants from the National Institute on Aging The sponsor had no role in any aspect of the study
Outcomes	Staff caregivers: - Reported the presence, frequency and severity of pain and dyspnoea Family caregivers: - Quality of Dying in Long-Term Care (QOD-LTC)
Results	The experiences of pain and dyspnoea are not strongly related to family perceptions of the quality of dying in LTC.
Comments/notes	/

Study (source)	Vandervoort et al (2013)
Study type	Post-mortem study using random cluster sampling Retrospective cross-sectional study
Number of participants	69 nursing homes 198 deaths
Lost to follow-up	/
Setting and country	Nursing homes in Flanders, Belgium
Duration of study	May-October 2010 (?)
Population characteristics	- Nurses most involved in care - Family physician - Nursing home administrator
Inclusion criteria	Residents dying in nursing homes or elsewhere and meeting one of the next criteria used by the Belgian health insurance system to allocate financial resources: - Category C dementia - Disorientation in time and space >3 Additional: FP or nurse indicates that residents had dementia or was diagnosed with dementia
Exclusion criteria	
Risk of bias	Some study limitations
Imprecision	Some imprecision is possible
Inconsistency	/
Indirectness	Ok
Publication bias	/
Funding	Major grant from Vrije Universiteit Brussel

Outcomes	- Health status in the last month of life - Clinical complications in last month of life - Distressing symptoms in last month of life - Clinical characteristics and outcomes in last week of life
Results	Health status in the last month of life - 50% AD - 83,3% able to (partly) express their wishes at the time of admission - 60,7% (partly) competent for medical decision making - 70,3% had at least one co-existing condition at time of death; cardiovascular diseases were reported most frequently Clinical complications in last month of life - 95,5% had one or more sentinel events - 65,7% eating or drinking problems - 42,9% febrile episodes - 32,3% pneumonia - Prevalence of eating/drinking problems in the last month of life increased with severity of dementia Distressing symptoms in last month of life - Pain, fear, anxiety, resistance to care and agitation were most frequently reported using the SM-EOLD scale - Presence of symptoms did not differ significantly in stages of dementia Clinical characteristics and outcomes in last week of life - Low levels of comfort were reported for difficulty swallowing, pain, lack of serenity, lack of peace, lack of calm. - QoL was 28,9 (QUALID scale) - 26,9% pressure sores - 89,2% urine incontinence - 85,6% faeces incontinence - 45,8% (severe) cachexia - 38,6% (severe) dehydration - 21,4% restraints used - 50,3% death was expected - 8,9% deaths in hospital - Statistically significant differences between stages of dementia, all being more common with more severe dementia: choking, swallowing, incontinence, dehydration
Comments/notes	- Background info

Study (source)	Konetzka et al (2008)
Study type	Literature review
Number of participants	/
Lost to follow-up	/
Setting and country	/
Duration of study	/
Population characteristics	55 articles
Inclusion criteria	- Peer reviewed articles - Between 1990 and 2005 - Articles that empirically assess the determinants of inpatients admissions from LTC settings or test the effectiveness of particular interventions in reducing hospital admissions and readmission rates, length of stay, emergency room visits or hospital costs among elderly LTC recipients - Mean age of study population at least 65 - Studies conducted in formal settings Inclusion criteria for experimental and quasi-experimental (observational) studies: - Sample size of minimum 20 individuals - Attempt to control for confounding and selection bias using at least one of the following: (1) randomization to treatment and control groups, (2) inclusion of a non- random control group with multivariate adjustment or matching for differences in risk and (3) inclusion of a non-random control group with pre/post measurements on both groups to account for secular trends (change or "difference in differences" model).
Exclusion criteria	- Studies that focused primarily on the adult disabled or children - Studies where the intervention was not explicit
Risk of bias	- Number of studies on this topic is small - Selection bias could not be ruled out in a lot of the selected studies - Studies of the benefit from higher RN staffing; while fairly consistent in direction and significance are almost entirely cross-sectional and observational

Imprecision	
Inconsistency	
Indirectness	
Publication bias	No missing --> very low bias
Funding	Not mentioned
Outcomes	Assessment of LTC interventions with potential to reduce hospitalizations among the elderly regardless of original setting.
Results	Almost all studies showed a significant decrease in hospitalisations. Following interventions hold the most promise: - Use of nurse practitioners and physician assistants: evidence points towards their use as a cost-effective strategy to reduce hospitalisations - Increasing RN staffing in LTC facilities: consistent relationship between more RN staffing and better outcomes, including lower hospitalizations rates, though the exact nature of that relationship is unclear - Improving the hospital-to-home transition - Substituting home health care for hospital days - Aligning financial incentives: NHs have a financial incentive to hospitalise residents who could be treated in the facility
Comments/notes	Not a systematic review or meta-analysis. No attempt was made to standardise findings across studies.

Study (source)	Reymond et al (2011)
Study type	Development of the RAC EoLCP: - Iterative process requiring numerous drafts to achieve a user-friendly final document. Evaluation based on outcome and process measures concerning dying residents, staff and bereaved relatives --> pre-post audit
Number of participants	299 deaths (46 pre-implementation; 253 during implementation) Staff: - Pre-implementation: 170 - After 3 months: 88 - After 9 months: 114 - After 18 months: 96 Bereaved relatives: - Pre-implementation: 21 - Implementation: 66
Lost to follow-up	/
Setting and country	RACF Brisbane, Australia
Duration of study	June 2007-December 2008
Population characteristics	
Inclusion criteria	- Large number of high care beds - Management knowledge of the Department of Health and Ageing document 'Guidelines for a palliative approach in residential aged care facilities' - Management support for the project
Exclusion criteria	
Risk of bias	Some limitations
Imprecision	Moderate
Inconsistency	/
Indirectness	Ok
Publication bias	Ok
Funding	Department of Health and Ageing under the National Palliative Care Strategy.
Outcomes	Primary outcome: - Transfer to hospital Secondary measures: - Staff perceived changes in quality of palliative care provided and family satisfaction with care
Results	The pathway, delivered with a care framework that guides provision of palliative care, resulted in improved resident outcomes and decreased inappropriate transfers to acute care settings. Primary outcome: significant difference in place of death Secondary outcomes: - RACF staff surveys: increase in almost all items - Bereaved relatives: no significant difference in relative's rating (satisfaction was already high)
Comments/notes	- Nothing about pharmacological management of dying symptoms

Study (source)	Commonwealth of Australia (2006) (only pages 161 to 181 are reviewed)
Study type	Guidelines for a palliative approach in RACF Chapter: End-of-life (terminal) care
Number of participants	/
Lost to follow-up	/
Setting and country	Australia
Duration of study	/
Population characteristics	
Inclusion criteria	
Exclusion criteria	
Risk of bias	
Imprecision	
Inconsistency	
Indirectness	
Publication bias	
Funding	
Outcomes	
Results	- Pain: opioids (morphine) or benzodiazepines are recommended - Delirium and agitation: haloperidol and clonazepam, sedation if symptoms not controlled - Respiratory secretions and death rattle: hyoscine hydrobromide and atropine sulphate (suction possible in case of viscous, sticky secretions in the upper airways but not routinely)
Comments/notes	General guidelines palliative care

Study (source)	Commonwealth of Australia (1999)
Comments/notes	Not available

Study (source)	Palliative Expert Group (2010)
Comments/notes	Book: updated in 4th version

Study (source)	Commonwealth of Australia (2002)
Study type	Australia's National Strategy for Quality Use of Medicine
Outcomes	Objectives: - Improve QUM by healthcare consumers - Improve QUM by healthcare practitioners, healthcare providers and health educators - Gain the commitment of the medicines industry (including manufacturers and distributors) to QUM - Gain the commitment of governments to QUM - Improve the commitment of healthcare consumers, health practitioners and educators, the medicines industries, the media, healthcare facilities, funders and purchasers and governments (commonwealth, state and territory) to working in partnership to achieve QUM.
Results	Levels of QUM pyramid: - Awareness level - Knowledge and skills - Action and evaluation
Comments/notes	- Describes Australia's National Strategy for Quality Use of Medicine - Not specific for our context or population

Study (source)	WA Cancer and Palliative Care Network (2011)
Study type	Evidence based clinical guidelines for adults in the terminal phase

Results	Overall: call for assistance from a palliative care specialist if there is an inadequate response to the recommendations Management of dyspnoea: <i>If patient is opioid naive:</i> - Prescribe and have available pre-emptively: morphine 2.5mg hourly SC prn - Re-assessment after 24h <i>If patient takes opioid – not regular:</i> - Convert dose of opioid to regular oral dose - OR give dose SC (with appropriate dose of prn opioid) - Re-assessment after 24h - Titrate dose of regular and prn opioids - If patient is anxious: provide reassurance lorazepam 0.5mg sublingual hourly prn <i>If patient takes opioid – regular:</i> - If tolerating current dose, increase regular opioid dose by 25% (increase prn dose accordingly) - Re-assessment after 24h - Titrate dose of regular and prn opioids - If patient is anxious: provide reassurance lorazepam 0.5mg sublingual hourly prn Summary of evidence for dyspnoea: - Dyspnoea is a subjective experience - Opioids are beneficial in reducing the symptom of dyspnoea - SC or oral morphine improve the symptom of dyspnoea without detrimental effect on respiratory function - Low dose morphine is effective in relieving dyspnoea - There is insufficient evidence to recommend the use of inhaled opioids - Dyspnoea is significantly associated with anxiety - The addition of benzo's (midazolam) to a baseline of regular morphine improves sensation of dyspnoea in presence of anxiety - Oxygen saturation does not correlate with sensation of dyspnoea - Oxygen (and air) decrease the sensation of dyspnoea at rest regardless of whether hypoxia is present - Some patients with cancer feel better during oxygen inhalation - Air flow over the face and nasal mucosa lessens the sensation of dyspnoea Management of nausea and vomiting: When symptoms are <u>present</u>: <i>*Patient is taking regular antiemetic:</i> - In case of SC: increase dose of current antiemetic OR change to promethazine 12.5-25mg over 24h by SC infusion - In case of no SC administration: change to SC (ensure SC prn antiemetic also available) <i>*Patient is not taking regular antiemetic:</i> - Initiate regular metoclopramide 30mg over 24h by SC infusion - Reassess after 24h When symptoms are <u>not present</u>: <i>*Patient is taking antiemetic:</i> - In case of SC: continue to assess presence or absence of symptoms - In case of no SC administration: consider changing total dose of antiemetic used in last 24h to a SC infusion (with prn antiemetic available) - Continue to assess presence or absence of symptoms <i>*Patient is not taking antiemetic:</i> - Prescribe and have available pre-emptively: metoclopramide 10mg SC tds prn - Continue to assess presence or absence of symptoms Summary of evidence for nausea and vomiting: - The rate of both nausea and vomiting generally decreases as patients enter the terminal phase of disease - The appropriate medication is usually selected dependent upon its receptor and neurotransmitter affinity, after determining the most likely cause of emesis - This mechanistic approach may not be so relevant in eol symptom management - To provide continuous relief of nausea or vomiting, medications should be given regularly (that is, at regular fixed intervals or by infusion) - If unable to swallow or vomiting, the SC route of administration is recommended - An appropriate first line antiemetic in the terminal phase is metoclopramide due to its prokinetic and dopamine antagonist effects - Metoclopramide is effective in the management of nausea and vomiting in patients with advanced cancer - Promethazine is an appropriate second line antiemetic in the terminal phase as it exerts its effect on histamine receptors in the vomiting centre Management of pain: Patient is in pain: assess and review cause of pain; review analgesic medication <i>*Taking regular opioid:</i>
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	- Increase background dose of opioid by total prn dosage used in the last 24h (max. increase 30%); convert to equivalent dose of SC infusion (with appropriate dose of prn opioid) Not taking regular opioid: - Initiate morphine 2.5-5mg SC hourly prn - Reassess after 24h: if ≥ 3 doses were required: initiate SC infusion morphine 10-15mg over 24h (with appropriate dose of prn opioid) Pain is controlled: *Taking an opioid: - If regular, oral opioid: consider pre-emptively converting to equivalent dose of morphine by subcutaneous infusion. Prescribe and have available pre-emptively an appropriate dose of morphine SC hourly prn - Continue to assess presence or absence of symptoms Summary of evidence for pain: - Morphine is the opioid of choice and is as effective as other opioids. If morphine is not tolerated, contact a palliative care specialist for advice - Analgesia should be by the oral route of administration if possible. It is anticipated that patients will be increasingly less able to swallow as their condition deteriorates - If unable to swallow, the SC route of administration is recommended; patients prefer administration of opioids SC compared to IM; when changing from oral to SC morphine start with 1/3rd of oral dose to allow for the decreased oral bioavailability - To provide continuous pain relief, analgesia should be given "by the clock" (that is, at regular fixed intervals or by infusion), this is the background dose; SC infusions of opioids are as effective as IV infusions - Adequate dose of breakthrough opioid should be available. The dose is calculated as 1/6th or 1/12th of the daily dose - Breakthrough doses of opioid should be prescribed 1 hourly 'when required' (prn) - Morphine should be prescribed with caution in renal dysfunction due to accumulation of neurotoxic metabolites - Pethidine should not be used because of risk of accumulation of its toxic metabolite norpethidine - Transdermal fentanyl is not appropriate to initiate in the terminal phase Management of respiratory tract secretions: Symptom is present: - Reassure and explain symptom to patient, family and carers. - Position patient to promote drainage of secretions - Consider suction if available and tolerated by patient - Prescribe and administer stat SC: hyoscine hydrobromide 400mg OR hyoscine butylbromide 20mg OR glycopyrrolate 200-400mg - Reassess after 1h - If symptom is resolved: continue to assess presence or absence of symptom - If symptom is not resolved: repeat dose of previously administered anticholinergic medication and reassess after 1h - If ≥ 3 doses are administered within 24h then consider initiating SC infusion: hyoscine hydrobromide 800-1600 mg/24h OR hyoscine butylbromide 80-120 mg/24h OR glycopyrrolate 600-1200 mg/24h Symptom is not present: - Consider risk factors for development of secretions - Prescribe and have available pre-emptively: preferred/obtainable anticholinergic medication - Continue to assess presence or absence of symptoms Summary of evidence for respiratory tract secretions: - Reported incidence of terminal respiratory tract secretions ranges from 23% to 92% (most studies reported rates between 30-56%) - Incidence increases as patients are closer to death - Management is effective in approximately 40-70% of cases - There is no consistent measure for terminal secretions, which may explain the variation in reported incidence and rates of effective management - Dehydration does not prevent the symptom of death rattle - Family and carers distress about this symptom can be relieved with explanation of the reason for noisy breathing and reassurance - Positioning to promote drainage of secretions if effective. Positioning with occasional suctioning improved symptoms without the use of medications in 31% of patients. - Medications do not dry existing secretions, hence administering medication as early as possible is advantageous - High incidence of symptom and the need for prompt treatment supports anticipatory prescribing to avoid delays in controlling symptom - There is no conclusive evidence of significant difference in efficacy of anticholinergic medications and therefore no particular medication can be recommended - Medication choice to be based on ease of access, cost and differing pharmacological profiles Management of terminal restlessness/agitation: When symptom is present: - Consider: calm and quiet environment; familiar people in attendance; room lighting appropriate to time of day - Initiate haloperidol 1 mg SC hourly prn - Reassess after 24h - If ≥ 3 doses have been required, initiate haloperidol 2.5 mg/24h by SC infusion + 1mg hourly prn WITH clonazepam 0.5 mg sublingual bd + hourly prn OR lorazepam 0.5 mg sublingual bd + hourly prn OR midazolam 5 mg/24h by SC infusion + 1-2 mg hourly When symptom is absent:
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	- Consider risk factors for development of terminal restlessness - Prescribe and have available pre-emptively: haloperidol 1 mg SC hourly prn AND clonazepam sublingual 0.5mg hourly prn - Continue to assess presence or absence of symptoms Summary of evidence for terminal restlessness/agitation: - Terminal restlessness is an agitated delirium that occurs in some patients during the last few days of life - Reported incidence of terminal restlessness is 62%-88% - Family members find terminal restlessness very distressing - The development of delirium leads to increased caregivers anxiety - Non-pharmacological measures can be helpful in managing delirium - Where possible, any potentially reversible causes should be treated - Prompt pharmacological treatment is required to reduce the possibility of harm for the patient and distress for the family - There is limited evidence from clinical trials on the role of drug therapy for the treatment of delirium in terminally ill patients - Haloperidol is the drug of choice for the treatment of patients with delirium near the end of life - Low dose haloperidol (<3mg per day) is effective in treating delirium with few adverse effects - Sedation may be necessary to complement the effect of haloperidol, and benzodiazepines are effective in this role - Benzodiazepines should not be used alone as they may worsen the delirium.
Comments/notes	Not specified for nursing home residents Useful flowcharts and evidence based clinical guidelines

Study (source)	Glare et al (2011)
Study type	Literature review
Number of participants	/
Lost to follow-up	/
Setting and country	/
Duration of study	/
Population characteristics	Patients with nausea and vomiting in palliative care
Inclusion criteria	/
Exclusion criteria	/
Risk of bias	/
Imprecision	/
Inconsistency	Some inconsistencies are found
Indirectness	Not always similar to population
Publication bias	Ok
Funding	Authors have no proprietary or financial interests in any of the drugs mentioned; none has a financial arrangement with the companies that make of any of these products.
Outcomes	

Results	Approaches to pharmacological treatment of nausea 1) Empirical approach: leaves antiemetic drug selection to physician preference 2) Mechanistic approach: applies current understanding of the neuropharmacology of the 'emetic pathway' to the selection of antiemetic agents; antiemetic selection is based on knowing which drugs block the receptors found in the structure where the cause of nausea and vomiting is acting on. The selected agent should be titrated to effect. An empiric approach is justifiable in that the aetiology of chronic nausea in advanced disease is unidentifiable or multifactorial, drugs act on multiple receptors and other pathways may also be operating. Treatment should be initiated with a dopamine antagonist; for those not responding to a dopamine antagonist, agents from different classes are combined. In patients with refractory nausea and vomiting, a broad spectrum agent which blocks multiple neurotransmitters should be added such as olanzapine or methotrimeprazine. Data on the effectiveness of the palliative care approach to choosing which antiemetic to choose is limited. Drugs for treating nausea and vomiting Prokinetic agents: - Stimulate the motility of the upper gastro-intestinal tract - Should not be used when stimulation of muscular contractions might adversely affect the gut - Prokinetic effects are blocked by antimuscarinic agents (including antihistamine antiemetics). - Metoclopramide: 10 mg 3-4 times a day; side effects: restlessness, drowsiness and fatigue; administration beyond 12 weeks is not recommended (FDA) - Mirtazapine - Erythromycin; no data for use in palliative care; 250mg 3x a day po or 250-500mg IV; side effects: impaired hepatic function and prolongation of QTc-interval - Domperidone and cisapride were formerly available, but are removed or restricted due to their potential for cardiac toxicity associated with QTc-prolongation - Domperidone only acts at peripheral dopamine receptors; extrapyramidal symptoms are much less likely - Cisapride: prone to many drug interactions via CYP 3A4, which increases its blood levels and aggravates its arrhythmogenic effect Dopamine receptor antagonists: - Phenothiazines and other antipsychotic agents such as haloperidol and olanzapine - All (but haloperidol) has a broad spectrum of activity, also blocking histaminic, muscarinic, serotonergic, and/or alpha-adrenergic receptors; may also produce vagal blockade in the gastrointestinal tract, giving them some prokinetic activity; - Many side effects: sedation, hypotension, anticholinergic side effects, dystonia, extrapyramidal symptoms, may prolong QTc interval, lower the white blood cell count and lower seizure thresholds. - Not ideal to use in the elderly; carry a warning about precipitation of psychosis in demented elderly patients; increased risk of death compared to placebo - Prochlorperazine: phenothiazine commonly used as antiemetic; dose: 5-10mg 3-4x a day po, 25mg 2-3 a day rectally or 5-10 mg every 3-4h IM (to max 40mg daily); initial geriatric dose is 2,5-5mg 1-2 daily, increased once or twice a week by 2,5-5 mg/day, daily max dose should not exceed 75mg; should be avoided in patients with an absolute neutrophil count of <1000 cells/μL - Chlorpromazine: phenothiazine commonly used as antiemetic; more sedating than prochlorperazine - Haloperidol: a butyrophenone; a more potent and pure D2 receptor blocker than the other agents; antiemetic doses: 1,5-5mg 2-3x a day po, 0,5-2mg IV every 8h (lower than usual antipsychotic doses); side effects are similar to phenothiazines, except causes less sedation and hypotension, more extrapyramidal symptoms, should be avoided in Parkinson's disease; can exacerbate narrow-angle glaucoma; coadministration with carbamazepine, phenytoin, phenobarbital, rifampicin or quinidine may alter the clinical pharmacokinetics of haloperidol - Levomepromazine: used as second or third line therapy for refractory nausea and vomiting in palliative care; dose: 6,25-25mg 2x a day, 25-50mg/day continuous SC; same potential problems in the elderly as the other dopamine antagonists (anticholinergic effects, confusion, hallucinations, dystonic reactions); more sedative and more likely to cause postural hypotension than chlorpromazine, may prolong QT-interval - Olanzapine: atypical antipsychotic; dose: 5-10mg/day po, a lower starting dose (2,5-5 mg/day) is recommended for the elderly, which may be increased as tolerated with close monitoring of orthostatic blood pressure; causes fewer extrapyramidal symptoms, does not usually cause QTc prolongation; reported side effects: dry mouth, constipation, increased appetite, agitation, hyperglycaemia and oedema; drug interactions are not usually an issue; precautions to consider in an elderly palliative care population: severe cardiac disease, hemodynamic instability, prior myocardial infarction, ischemic heart disease, hypercholesterolemia, Parkinson's disease and renal hepatic impairment. - Evidence to support dopamine antagonists in palliative care is scant Antihistaminic agents: - Promethazine: little anticholinergic activity; dose: 25mg po or IV every 4-6h (mx. 100mg/day), lowest doses should be used to avoid side effects; main side effect is sedation; recommended to lower dose in elderly population and to discontinue when not effective. - Other H1 antagonists mentioned in palliative care context: diphenhydramine, dimenhydrinate, hydroxyzine, meclizine and cyclizine; all are problematic in geriatric populations, with diphenhydramine and hydroxyzine having the highest risk for a severe adverse outcome - There is few data for H1 blockers as antiemetics in palliative care Selective 5HT3 receptor antagonists: - Primarily used for chemotherapy induced emesis and have a limited role in palliative care, where they are generally used as third line agents for refractory cases; they have been shown to be effective in bowel obstruction and renal failure - Ondansetron: dose: 4-8mg 1-2x a day - Granisetron: 1mg 2x a day - Dolasetron: 200mg once a day
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	- Palonosetron: 0,25mg IV once a day - Tropisetron - Side effects: usually mild and transient; constipation is biggest potential problem for palliative care patients - Evidence of benefit for refractory nausea in palliative and supportive care is less strong Corticosteroids: - Mainly studies as antiemetics in chemotherapy induced emesis, in the management of malignant bowel obstruction and raised intracranial pressure. Also used as second line therapy in chronic nausea of advanced cancer. - Dexamethasone: 4-8mg/day for chronic nausea, up to 16mg/day for malignant bowel obstruction or raised intracranial pressure; lowest possible dose should be used for the briefest period - Limited and conflicting data on the benefit of steroids in chronic nausea due to non-specific causes in terminal cancer Benzodiazepines: - Generally to be avoided in elderly patients; lorazepam is preferred because it is relatively short-acting Hyoscine: - Also useful in the medical management of terminal bowel obstruction - Hyoscine butylbromide: 80-120mg daily via continuous SC infusion - Hyoscine hydrobromide po, 0,1-0,4µg every four hours, SC or IV, 0,6-2,4mg daily via continuous infusion or 1-3mg transdermal changed every third day - Anticholinergic effects are a potential problem in the elderly Ocreotide: - Used for palliation of malignant bowel obstruction, especially when there is high output vomiting not responding to other measures. - Shown to produce analgesic effects - Dose: 100µg SC 3x a day or 100-600µg per day via IV or continuous SC infusion - Side effects: local skin reactions (pain, stinging, burning), gastrointestinal effects (cramps, nausea, vomiting, diarrhoea or constipation), gallstones, headache, hypothyroidism, cardiac toxicity Cannabinoids: - Few clinical evidence for efficacy in palliative care - Short duration of action and side effects are a major problem of current commercial formulations; especially in the elderly - Dose should be titrated slowly, with close monitoring for adverse effects. Other modes of nausea control: Avoidance of environmental stimuli (lights, sounds, smells). Behavioural approaches (relaxation and distraction) may decrease psychological arousal and distress and increase the patient's feelings of control over the symptoms. Massage, acupuncture and ginger has been reported to be effective, but no studies are available for the symptom management at the end-of-life Nonsurgical procedures in palliation of cancer-related nausea and vomiting PEG and other techniques
Comments/notes	- Not a systematic review or meta-analysis - Found no citations in Medline for "nausea and vomiting" or "palliative care" when limited to human subjects, English language and age over 65 years; 19 trials were found in the clinical trial database maintained by the US National Library of Medicine, but none evaluating pharmacological therapies in this population, and none being specifically for the elderly.
	- Lots of background info: epidemiology, definitions, causes, assessment - Interesting article for information on doses, interactions, precautions and side effects

Study	Hardy et al (2002)
Study type	Double blind, randomised, parallel group, multinational, multicentre study
Number of participants	92: - 30 placebo, p.o. tds. - 29 ondansetron, 24mg p.o. once daily - 22 metoclopramide, 10mg p.o. tds. 94 participants received study medication
Lost to follow-up	2 were excluded because of serious non-compliance with the protocol
Setting and country	Recruitment in 10 different countries, Majority recruited in Greece
Duration of study	Patient screening: April 1998-1996 Actual study: ? Duration of study: 48h per patient

Population characteristics	100% Caucasian Average age: 62 (+/-11), age range: 39-89 Recruitment in 10 different countries Majority recruited in Greece 34% incidence reported in this study
Inclusion criteria	- At least 18 years old - Diagnosis of cancer - Receiving a full opioid agonist for pain control (either opioid-naïve, received no opioids within 1 month prior to the study; required a clinically relevant escalation of their current dose of opioid; or required a change of opioid dosing regimen, eg. route or formulation)
Exclusion criteria	- Having other known reasons for nausea and/or emesis - Experiencing nausea or emesis not due to opioid 24h before the initiation of opioid treatment - Experienced chemotherapy or radiotherapy induced nausea and/or emesis in the 7 days before study entry - Concurrent medical conditions likely to be associated with nausea or vomiting - Severe concurrent illness - Planned medication with known or potential anti-emetic activity 24h before or during opioid treatment or during the study treatment period - Surgical procedures or non-invasive procedures under general anaesthesia within 3 days of the start of the study or during the study period - Having conditions likely to limit the validity of consent and contraindications or hypersensitivity to opioids, ondansetron or metoclopramide were also excluded - Pregnant or breast-feeding patients
Risk of bias	Recruitment to study was terminated early
Imprecision	Small sample size + low incidence of OIE No confidence intervals given --> moderate risk of bias
Inconsistency	/
Indirectness	
Publication bias	
Funding	Funding provided by Glaxo Wellcome Research and Development.
Outcomes	Primary measure of efficacy: based on the comparison between the proportion of treated patients who experienced complete control of emesis in the 24h study treatment period. Second measure of efficacy: based on the comparison between the proportion of treated patients who experienced complete control of emesis in the first 8h following administration of the first dose of study drug Compare the anti-emetic efficacy, safety and tolerability of oral (p.o.) ondansetron 24mg once daily (o.d.) with placebo p.o. and metoclopramide 10mg p.o. three times daily (tds) in the treatment of OIE in cancer patients receiving a full opioid agonist for pain control. Emetic episodes, severity of nausea and pain, and patient satisfaction with the study drug were recorded on diary cards by each patient for the 24h period following study drug administration. Nausea and pain were assessed on an increasing numerical scale from 0 to 10, where 0 represented 'no nausea/pain'
Results	Emesis and nausea were completely controlled at 24h in about one half and one third of patients who developed OIE. Neither ondansetron 24mg once daily nor metoclopramide 10mg tds given orally was significantly more effective than placebo in the control of OIE in cancer patients. There was no statistically significant difference between the groups in proportion achieving complete control of emesis: - 33% of pts on placebo - 48% on ondansetron - 52% on metoclopramide There was no statistically significant difference between the groups in proportion achieving complete control of nausea: - 23% of pts on placebo - 17% on ondansetron - 36% on metoclopramide Rescue anti-emetics (= receiving other anti-emetic medications than the study medications) were required in: - 8 of 33 patients on metoclopramide - 4 of 29 on ondansetron - 3 of 30 on placebo --> incidence of adverse events was very low and similar in all treatment groups Questionnaire showed a comparable proportion of patients were satisfied or very satisfied with their treatment in each of the three treatment groups.
Comments/notes	- Different population and setting than the guideline: younger and cancer patients, not in last days of life - Not clear what countries

Study	Porreca & Ossipov (2009)
Study type	Review of data

Number of participants	/
Lost to follow-up	/
Setting and country	Not stated
Duration of study	Not stated
Population characteristics	Not relevant
Inclusion criteria	Not stated
Exclusion criteria	Not stated
Risk of bias	/
Imprecision	/
Inconsistency	/
Indirectness	/
Publication bias	/
Funding	Not stated
Outcomes	Examining the mechanisms that produce these side effects, their impact on treatment outcomes in chronic pain patients and counteractive strategies
Results	Mechanisms by which opioids produce nausea and vomiting: - Central and peripheral sites - Vomiting centre - Chemoreceptor trigger zones - Cerebral cortex - Vestibular apparatus of the brain - GI tract Nausea and vomiting have a negative impact on treatment efficacy and successful patient management, because: - They limit the effective analgesic dosage that can be achieved - Frequently reported as the reason for discontinuation of opioid pain medication or missed doses Management: - Various strategies as antiemetic agents or opioid switching can be employed to control these side effects, but neither is ideal because they are not always effective and incur additional costs and inconvenience - Opioid sparing analgesic agents may provide a further alternative
Comments/notes	- Not systematic review - Not specified to our population or setting - Limited methodological information

Study (source)	CareSearch (2013)
VERSION NO LONGER AVAILABLE	

Study (source)	Won et al (2004)
Study type	Cross-sectional study
Number of participants	21 380 nursing home residents
Lost to follow-up	
Setting and country	Nursing homes from 10 US states (Alaska, Illinois, Indiana, Kansas, Maine, Mississippi, Ohio, Texas, South-Dakouta, West-Virginia)
Duration of study	
Population characteristics	Nursing home residents, aged 65 and older with persistent pain
Inclusion criteria	- Residents from US states that had complete pain and drug data - Aged 65 and older, admitted to Medicare/Medicaid-certified nursing home in 1998 (June to December), 1999 and 2000 - Persistent pain
Exclusion criteria	- Moderate to severe cognitive impairment (MMSE < 19) - Moderate to severe communication difficulties - Residents with cancer or a terminal diagnosis
Risk of bias	Some limitations, moderate ROB
Imprecision	Big sample size (no confidence intervals) Low risk of imprecision
Inconsistency	Moderate risk
Indirectness	Low risk of inconsistency
Publication bias	Low risk of publication bias

Funding	Grant of Janssen Pharmaceutical Products LP to the Hebrew Rehabilitation Centre for Aged Grants the National Institute on Aging, Bethesda, Maryland
Outcomes	- The MDS pain evaluation, third assessment (Minimum Data Set, a resident assessment instrument used in all US Medicare and Medicaid LTCFs) - Analgesic use - MDS CPS: categorical measure of cognition - Evaluation of physical function using MDS (based on an ADL evaluation classification system) - Mood was measured with the MDS Depression Rating Scale - Index of Social engagement (assesses the resident's capacity to respond to social overtures and initiate social interactions) - Diagnosis of interest: falls within the past 6 months, fractures within the past 6 months and surgery. Constipation, dysphagia, unsteady gait, falls, delirium, depression, gastrointestinal bleed, congestive heart failure, oedema and renal failure were examined.
Results	Prevalence of pain - 48,5% of the residents had persistent pain Analgesic use Of the population with persistent or recurring pain, at baseline: - 38,4% received opioids - 37,1% received nonopioid analgesics - 24,5% received no analgesics The use of short-acting opioids: - Alone: 4,0% - Short-acting opioid combination formulations: 13,8% - With a separate nonopioid prescription was 16,6% Use of long-acting opioids: - Alone: 1,1% - In conjunction with nonopioid analgesics: 1,0% - With short acting opioids: 1,7% Most frequent prescribed analgesics: - Acetaminophen (37,2%) - Propoxyphene (18,2%) - Hydrocodone (6,8%) - Ibuprofen (6,2%) - Tramadol (5,4%) - Celecoxib (3,1%) Discussion Undertreatment of pain in NHs is a significant problem: one quarter of residents experiencing persistent pain are not receiving any analgesics; special attention is needed to vulnerable populations
Comments/notes	- Terminal residents were excluded

Study (source)	Mercadante & Acuri (2007)
Study type	Literature review
Number of participants	/
Lost to follow-up	/
Setting and country	/
Duration of study	/
Population characteristics	Elderly cancer patients
Inclusion criteria	
Exclusion criteria	
Risk of bias	/
Imprecision	/
Inconsistency	/
Indirectness	/
Publication bias	/
Funding	No sources of funding were used
Outcomes	

Results	Recommendations for managing cancer pain in the elderly: 1) <i>Evaluation</i> - Concurrent medications and concomitant diseases - Characteristics of pain: causes, mechanisms, duration - Previous response to prescribed analgesics, if any - Stage of cancer disease and setting of care 2) <i>Treatment</i> - Careful drug titration according to need - Frequent pain and symptom intensity assessment - Addition of only one drug to a regimen - Re-evaluation for changes in dosages and drugs
	Existing non-pharmacological treatments for chronic pain in the elderly - Radiotherapy for bone metastases - Surgical correction of impending or completed pathological fractures - Ablative procedures using alcohol, radiofrequency, cryoprobes - Bone stabilisation with vertebroplasty and kyphoplasty - Cognitive, behavioural, biofeedback, diversion therapy - Psychological intervention for existential and spiritual pain - Nerve blocks, intrathecal therapies
Comments/notes	- Population semi-relevant: elderly cancer-patients, not terminal - Not a systematic review

Study (source)	Pergolizzi et al (2008)
Study type	Consensus statement and review of evidence
Funding	Unrestricted educational grant from Grünenthal GmbH, Aachen, Germany. Zaicom MMC Ltd, Horsham, UK for editorial support. Following authors received honoraria and travel expenses: Keith Budd, Albert Dahan, Hans-Georg Kress, Richard Langford, Joseph Pergolizzi, Robert Raffa, Paola Sacerdote, Raner Boeger
Outcomes	Aim of the meeting was to review and critically evaluate published evidence for the efficacy and tolerability of the 6 clinically most often used WHO step III opioids in the elderly population (buprenorphine, fentanyl, hydromorphone, methadone, morphine, oxycodone), in order to provide practical recommendations to physicians on the optimal use of these drugs in the target population, i.e., elderly patients with chronic severe pain requiring strong opioids.
Results	Recommendations Cancer-related pain: - High level data is available for the use of step III opioids in cancer related pain - No well-designed specific studies in the elderly cancer patients - Fentanyl and buprenorphine are available in transdermal formulation; fentanyl is the most investigated, but both seem to be effective with low toxicity and good tolerability profiles, especially at low doses Non-cancer related pain: - Evidence is growing that opioids are highly efficacious in non-cancer pain, but need individual dose titration and consideration of the respective tolerability profiles. - No specific studies in the elderly - The appropriate drug should be chosen based on safety and tolerability considerations Neuropathic pain: - Use of opioids in neuropathic pain has been in debate, but is more and more accepted. - Incorporation of opioids earlier on might be beneficial, at least in a number of patients - Buprenorphine shows a distinct benefit in improving neuropathic pain symptoms, which is considered a result of its specific pharmacological profile. Safety and tolerability Common adverse reactions: - Constipation, nausea and vomiting - Sedation - Impaired judgement - Impaired psychomotor function - Respiratory depression * <i>GI system</i>: buprenorphine and transdermal fentanyl produces less constipation than morphine and oxymorphone. * <i>Central nervous system effects</i>: opioid neurotoxicity is a significant issue in the elderly, presenting as hallucinations, confusion and loss of cognition. Most opioids are associated with this when given long-term at high doses. This effects have been demonstrated for all opioid, except for buprenorphine, but more data on buprenorphine in this patient group is needed.

	* Addiction: a review (Fishbain et al) showed that addictive behaviour was not common in the general chronic pain population. Many clinical studies have shown that long-term opioid therapy can be maintained without escalation of dose or tolerance to effect presenting. * Drug interactions: potential for drug-drug interactions and exacerbation of adverse events is high in the average nursing home resident. A number of drug interactions have to be taken into account for morphine, fentanyl and tramadol. * Recommendation for selecting opioids with regard to tolerability profile: agents should be used that have a good tolerability profile (especially regarding CNS and GI effects) and that are as safe as possible in overdose. Slow dose titration helps to reduce the incidence of adverse events. Sustained release preparations, including transdermal formulations, increase patient compliance. * Impaired hepatic and renal function: in elderly patients with impaired hepatic and renal function, it is important to be cognizant of the accumulation of metabolites from certain opioids, such as morphine. In practice, it is preferred to avoid such accumulation, by using compounds such as hydromorphone and buprenorphine. Doses should be reduced, a longer time interval be used between doses and creatinine clearance be monitored. Oxycodone, hydromorphone and buprenorphine appear to be a safe choice for opioid treatment in the elderly. * Respiratory depression: CNS depressants (benzo's, barbiturates, antidepressants, phenothiazine derivatives) and alcohol increase the risk of respiratory depression if taken with any opioid analgesic, this may progress to total apnoea. The different features of opioids regarding respiratory effects should be considered when treating patients at risk for respiratory problems. * Immunosuppression: several opioids possess intrinsic immunosuppressive activities. Morphine is the most immunosuppressive of the opioids; fentanyl has the same immune-pharmacological profile as morphine. Buprenorphine, hydromorphone and oxycodone have been reported to be less immunosuppressive than morphine. There is little evidence available to gauge the immunosuppressive effects of other opioids and even less in the elderly. Opioids with minimal immunosuppressive characteristics should be used in the elderly. The immunosuppressive effects of most opioids are poorly described and this is one of the problems in assessing the true effect of the opioid spectrum. Buprenorphine can be recommended, where morphine and fentanyl cannot. Few data is available for oxycodone and hydromorphone.
Comments/notes	Consensus statement, based on review

Study (source)	Elliott, Rohan A (2006).
Study type	Review of Australian literature
Number of participants	Not stated
Lost to follow-up	No information available
Setting and country	Australia
Duration of study	Different studies between 1997-2006
Population characteristics	Elderly, but heterogeneous population
Inclusion criteria	There is no description of inclusion criteria
Exclusion criteria	There is no description of exclusion criteria
Risk of bias	Bias is not described
Imprecision	Insufficient information presented to properly evaluate
Inconsistency	Insufficient information presented to properly evaluate
Indirectness	Insufficient information presented to properly evaluate
Publication bias	Insufficient information presented to properly evaluate
Funding	Insufficient information presented to properly evaluate
Outcomes	Patients' management of their medications is often suboptimal in different settings
Results	Strategies need to be developed to avoid overprescribing and under prescribing and improve medication management.
Comments/notes	Review with scope outside of our clinical research questions Research strategy is not described Very low quality evidence

Study (source)	King S, Hanks GW, Ferro CJ, Chambers EJ (2011)
Study type	Systematic review, narrative. No meta-analysis
Number of participants	- Selection of very heterogeneous studies with different number of participants ranging from 11 –to 300 - 8 prospective observational studies - 7 retrospective studies
Lost to follow-up	/
Setting and country	UK for the review Country was not an exclusion factor
Duration of study	Review up to 31/07/2009
Population characteristics	- Cancer-related pain in combination with renal impairment - Population relevant for palliative cancer care but not restricted to palliative cancer care
Inclusion criteria	- Studies with a clinical outcome relevant to the use of the following opioids: morphine, diamorphine, codeine, dextropropoxyphene, dihydrocodeine, oxycodone, hydromorphone, buprenorphine, tramadol, alfentanil, fentanyl, sufentanil, remifentanil, pethidine and methadone
Exclusion criteria	- Studies assessing the long-term efficacy

Risk of bias	- Risk of bias in the study methodology in all studies was identified by the reviewers - No randomised controlled trials were identified - Many potential confounding factors were identified
Imprecision	- Heterogeneous population throughout different studies
Inconsistency	- The clinical question didn't produce enough quality evidence for the formulation of recommendations - Recommendations were formulated outside of the scope of the initial clinical questions using alternative and less evidence based resources: pharmacokinetic modelling
Indirectness	- There are no studies giving a clear relative risk for one opioid compared to another although precisely this is what is found in the conclusion - Evidence from population groups outside of the initial clinical question was used for the formulation of the conclusion
Publication bias	Risk of publication bias reported by the reviewers
Funding	European Palliative Care Research Collaboration (EPCRC) within the EC 6th Framework Programme and by the Department of Palliative Medicine of the University of Bristol, UK
Outcomes	- 15 articles - Evidence was considered of very low quality by the reviewers - Very heterogeneous group of selected articles with very different outcome measures. - Insufficient evidence for the formulation of guideline - Suggestive for differences in risk between different types of opioids
Results	Fentanyl, alfentanil and methadone identified as least likely to cause harm. Morphine is associated with increased toxicity in patients with renal impairment Opioids for which there are thought to be no clinically significant metabolites include fentanyl, alfentanil and methadone. For fentanyl and alfentanil the assumption of no relevant activity is largely based on animal experiments or the absence of opioid receptor mediated effects
Comments/notes	Recommendations are based on clinical experience guided by relevant pharmacological studies and very few on the clinical data that were part of the scope of the study.

Study (source)	Portenoy RK, Bennett DS, Rauck R, Simon S, Taylor D, Brennan M, Shoemaker S (2006)
Study type	Prospective observational study without control group
Number of participants	228
Lost to follow-up	19
Setting and country	- 9 pain treatment programs - USA
Duration of study	February – April 2004
Population characteristics	- Patients with diverse types of chronic non-cancer pain - Receiving opioid analgesics - Adult population: 18-75 years. Median 47 y. - > 50% musculoskeletal pain
Inclusion criteria	- Chronic non-cancer pain patients - Already receiving opioid analgesics at baseline
Exclusion criteria	- Uncontrolled baseline pain control
Risk of bias	- Population bias: only patients in pain treatment programs - Population was overall white and female - Moderate risk of bias
Imprecision	- No description of calculated power
Inconsistency	- Inclusion criteria were not consistent throughout the study. - 4 patients were excepted outside of the inclusion criteria for age
Indirectness	- Different population, different age range, different underlying pathology
Publication bias	/
Funding	Cephalon, inc.
Outcomes	- Breakthrough pain in non-cancer pain is similar to that in cancer related breakthrough pain.
Results	- 19% of breakthrough pain occurred at the end of the dosing interval - Pharmacologic (80%) and non-pharmacologic (60%) interventions were used in breakthrough pain. - Only 35% of patients had a systematic adequate response to interventions for breakthrough pain (pharmacologic or non-pharmacologic)
Comments/notes	- Not in a palliative setting - Different and younger population than in our scope

Study (source)	Rossi S (ed) (2013)
Study type	Australian medicine handbook

Study (source)	Abraham J, Savarese D, Portenoy RK (2012)
VERSION NO LONGER AVAILABLE	

Study (source)	Dudgeon D, Shadd J (2012)
VERSION NO LONGER AVAILABLE	

Study (source)	Currow DC, Smith J, Davidson PM, Newton PJ, Agar MR, Abernethy AP (2010)
Study type	Cohort study
Number of participants	5862 consecutive patients
Lost to follow-up	None described
Setting and country	Single centre study Hospice Care centre
Duration of study	January 2004-december 2007
Population characteristics	Mean age 69.69, 42% female
Inclusion criteria	- Breathlessness as a symptom at any time between referral and death
Exclusion criteria	- Non stated
Risk of bias	- 92% of patients had cancer diagnosis. This makes the statistical power for conclusions about the noncancer group much smaller - Selection bias very plausible because of the limitation concerning setting
Imprecision	- Data were not collected at predefined moments - Number of evaluations differed between subjects and are not disclosed
Inconsistency	- Population probably at higher a priori risk of dyspnoea than in out-patient setting and in lower a priori risk of dyspnoea than in hospital setting
Indirectness	- Use of surrogate outcome measures. Data were not recorded specifically for the clinical question
Publication bias	/
Funding	Not disclosed
Outcomes	- Dyspnoea evaluation through numeric rating scale (NRS) - Onset and intensity of breathlessness over time as death approached, stratified by lung cancer (primary or secondary) and other diagnoses subsequently broken down into cardiac failure, documented respiratory failure, or no obvious cardiorespiratory cause
Results	- Breathlessness was scored significantly higher in non-cancer diagnoses. - Increase in breathlessness at days 10 and 3 prior to death for cancer patients
Comments/notes	- Descriptive, epidemiological study without intervention - Outside of the scope of our clinical questions concerning medical therapy - 25% was suffering from severe breathlessness at the time of death - Mean population younger than our target population - Different setting than our target setting

Study (source)	Booth S, Anderson H, Swannick M, Waded R, Kitee S, Johnson M (2004)
Study type	Review - No large RCTs - Small controlled trials - Single or double blind
Number of participants	22 Small trials with participants varying between 9 and 50
Lost to follow-up	Not stated
Setting and country	- Review from the UK - Studies included in different countries and different settings
Duration of study	Different study duration
Population characteristics	Very few studies on heart failure. Age not mentioned.
Inclusion criteria	- Different levels of hypoxemia
Exclusion criteria	- Not stated
Risk of bias	- Not all studies were double blind.
Imprecision	- The power of each study was very small because of limited number of participants.
Inconsistency	- Important inconsistency throughout the trials.

Indirectness	- Most trials measure QOL as opposed to breathlessness
Publication bias	Possible, although attempts were made to search grey literature. Broad search terms but the specific search strings were not disclosed.
Funding	Not declared
Outcomes	- No studies specifically looking at the effects of oxygen on reducing the sensation of breathlessness in severe heart failure - Secondary outcome measures from the trials were used as primary outcome measure in the review: reduction in breathlessness as measured with a specific tool or scale: VAS or Borg scale
Results	- Oxygen may be helpful in patients with breathlessness and cancer but there is little evidence to predict which patient will have a benefit - In patients with heart failure, no patients with severe dyspnoea were included. There is no evidence to support or to contradict the benefit of oxygen use.
Comments/notes	Very limited body of evidence, primary endpoints not breathlessness, different population than in our scope

Study (source)	Jennings AL, Davies AN, Higgins JP, Broadly K (2001).
REVIEW RETRACTED ON 11/07/2012	

Study (source)	Navigante et al (2006)
Study type	Single centre, single blind randomized trial
Number of participants	101 patients
Lost to follow-up	None
Setting and country	Argentina
Duration of study	48 hours
Population characteristics	- Mean Age: 57 y - Both sexes equally represented except for the midazolam group where there were 13 to 20 M/F - Types of underlying neoplasms equally divided within different groups. Midazolam group has fewer lung cancer patients than the other groups.
Inclusion criteria	- 18 years of age or older - documented diagnosis of terminal advanced cancer, - life expectancy less than a week - Mini-Mental Status Exam (MMSE) > 23/30 - severe dyspnoea at rest - performance status of 4 (Eastern Cooperative Oncology Group categorical scale) = completely disabled
Exclusion criteria	- Chronic obstructive pulmonary disease with hypercapnia - Non-compensated congestive heart failure - Severe renal or hepatic failure (clinically and/or biochemically detected) - Other uncontrolled (numerical rating scale > 3/10) symptoms (excepting anxiety associated with dyspnoea) that could require the use of opioids, benzodiazepines, glucocorticosteroids, phenothiazines, bronchodilators, or methylxantines
Risk of bias	- Concerns about bias due to lack of double blinding - Possible moderate impact on study outcome – attempt to reduce bias by having 2 physicians evaluating independently
Imprecision	- Small sample size (only 33 patients in each study group) - Larger confidence intervals for de midazolam only group, in combination with small sample size, resulting in possible erroneous interpretation
Inconsistency	- Pretty homogenous group
Indirectness	- Direct response for clinical question - Significant difference between their study group and our target group
Publication bias	/
Funding	Research funds from the Angel H. Roffo Cancer Institute, University of Buenos Aires
Outcomes	- Assessment of the role of midazolam as adjunct therapy to morphine in the alleviation of severe dyspnoea perception in terminally ill cancer patients - principal endpoints were : - dyspnoea intensity (modified Borg scale)12 - dyspnoea relief (yes-no) after the intervention - Additional endpoints: - episodes of breakthrough dyspnoea (BD) requiring rescue medication (episodes/day) - frequency and severity of medication-related side effects
Results	- Mo =morphine baseline, Mi = midazolam baseline, MM = morphine and midazolam baseline - Significant correlation between dyspnoea and anxiety at baseline (p=0.038), 24 hours (P = 0.040) and 48 hours P = 0.002) - Oxygen saturation did not differ significantly by intragroup comparison (t-test) at baseline, 24h nor 48h. - Dyspnoea relief: T24h: Mo 69% - Mi 46% - MM 92%. Larger confidence intervals in the Migroup. - No dyspnoea relief: Mo 12.5% - Mi 26% - MM 4% - Breakthrough dyspnoea episodes significantly lower in Mo versus Mi but not significantly for Mo vs MM. Percentage of patients with breakthrough dyspnoea did not differ significantly (Mo

	38% vs Mi 38.5% vs MM 24%) - MM group had more significant adverse events versus Mo group but more somnolence in Mo group - More patients with clinical relevant toxicity reported in the Mo group although definition of toxicity not stated
Comments/notes	- Population different from our population, especially regarding mean age and the exclusion of other underlying conditions - Small study, only single blinded - Somnolence in Mo group probably due to more frequent breakthrough dyspnoea and need for midazolam rescue doses - Less nausea in midazolam group (notion of anti-emetic effect)
Study (source)	Wildiers et al (2009)
Study type	Open label, prospective, randomized phase III trial
Number of participants	333
Lost to follow-up	- 81 patients were not lost to follow up but excluded for final analysis because of the lack of an informed consent - 18 patients lost to follow up for effectiveness score after 1 hour for primary end point (because of early death)
Setting and country	6 Flemish (Belgian) palliative care centres
Duration of study	120 hours
Population characteristics	- Mean age in all groups between 70 and 75 - Predominantly cancer patients - Rattle score of 2 was the most frequent score at baseline
Inclusion criteria	- Aged 18 or older - Terminally ill - Any audible death rattle (intensity score ≥ 1)
Exclusion criteria	- Secondary causes of rattle = pseudo death rattle - Respiratory infection - Food/fluid aspiration - Cardiac failure with pulmonary oedema
Risk of bias	- Absence of blinding - Positive result reported for treatment at lower doses is at high risk of bias because of the lack of a placebo arm. - For the other clinical questions and results there is only a small to moderate risk of bias because of this.
Imprecision	- Power study done, sample size sufficient.
Inconsistency	- Patient characteristics equally divided within different treatment arms - Under representation of terminally ill non cancer patients - Outcome definition is subjective appreciation by different subjects in different centres. Nurse administering therapy responsible for the evaluation. - Differences between the centres not described in the study. - No systematic recording of intravenous and oral fluid intake, which might have influenced the development and evolution of the rattle
Indirectness	- Slightly younger age range - Almost solely cancer patients - Different setting: hospice care
Publication bias	/
Funding	- Bart Van den Eynden (Chair of Palliative Care, University of Antwerp) for funding the data management. - Small unrestricted grant from Boehringer-Ingelheim of 500 euros. Firm not implicated in study protocol.
Outcomes	- Primary objective: Difference in effectiveness between the three anticholinergic drugs after 1 h. Effectiveness defined as lowered intensity to score 0 or 1 - Secondary end-points: - Differences in effectiveness at other time end points - Differences in side effects - Time interval between start treatment and death - Influence of underlying pathology on the effectiveness of the treatment

Results	- T1h : no significant difference in effectiveness among atropine (42%), hyoscine butylbromide (42%), and scopolamine (37%) (Chi-square χ^2 0.66, degrees of freedom [DF] χ^2 2, P χ^2 0.72) - T1h: 50% of patients had no response - No conclusions about effectiveness can be made after 48h because of the small number of survival. - Treatment effectiveness increases up to 24 h and stabilises thereafter with 30% of patients continuing to have rattlescore 2-3 - Treatment is started at the low rattle score intensity of 1, treatment is much more effective than when started at a rattle score of 2 or 3 (Chisquare χ^2 72, DF χ^2 2, P < 0.000001), especially in the initial hours - Primary lung tumour or lung metastases benefited least from anticholinergic therapy (32%) compared with patients without tumour in the lung (46%, Chisquare χ^2 6.82, DF χ^2 1, P χ^2 0.009 - Median survival was similar among the three treatment groups showing that death rattle is an objective sign of imminent death. Patients with higher baseline rattle scores die earlier - Consciousness decreased significantly more at 12 hours for scopolamine compared with the other two drugs (Chi-square χ^2 20.83, DF χ^2 8, P χ^2 0.0076), but this was not significant at 24 hours - None were reported to have increased confusion at 12 and 24 hours - Clinical signs of cardiac failure, ileus, and bladder retention were rarely reported; at 12 hours, there was one case of clinical cardiac failure, one case of ileus, and six cases of bladder retention
Comments/notes	- No placebo arm: impact of anticholinergic drugs regarding to rattle score is open for bias - Mean age slightly younger. Especially cancer patients - Decreased consciousness to be theoretically expected for scopolamine in regard to hyoscine bromide but not in regard to atropine - Conflicting results from other (smaller and lower quality studies) studies - Glycopyrrolate not included because of supply problems at that time - No data in regard to differences in effectiveness between IV and SC route of administration

Study (source)	Tabet & Howard (2009)
Study type	Review about non-pharmacological interventions for delirium
Comments/notes	Outside the scope of our review questions because we solely focus on pharmacological interventions

Study (source)	Kehl (2004)
Study type	Literature review
Number of participants	30+86+20+60+100
Lost to follow-up	Not described
Setting and country	- USA, UK, Canada, Australia - Mostly palliative care unit or hospice care. One study also outpatient treatment
Duration of study	Articles from 1966-2002
Population characteristics	Very little information about patient characteristics. Different underlying pathologies: HIV, cancer, ..
Inclusion criteria	- All available literature, in all languages, from 1966-2002 including randomized trials, prospective and retrospective studies, case studies, and clinical practice guidelines were included in the analysis
Exclusion criteria	- Studies regarding patients who did not have advanced cancer, HIV or a terminal prognosis - Review articles, expert opinion articles
Risk of bias	- Search strings limited. Risk
Imprecision	- Sample sizes very small:
Inconsistency	- Frequency of symptom varies widely among studies (25-88%) - Major heterogeneity. There are very large differences within the studied populations and within the interventions. - Very different methodologies between studies. - Very different outcome definitions.
Indirectness	- Indirect comparisons
Publication bias	- Moderate risk of publication bias - Small studies were included
Funding	- Grant from the National Institutes for Nursing Research
Outcomes	- Empiric evidence about the pharmacological treatment for terminal restlessness
Results	- Considerable variety in the recommendations for terminal restlessness or delirium at the end of life - Only one randomized double blind study concerning 30 HIV patients that favoured chlorpromazine and haloperidol over lorazepam. Lorazepam arm was unblinded during study because of number of side effects - Midazolam was found safe and effective in two studies. 44,46 In a third study, 24 midazolam was found effective for restlessness, but not delirium - One study found phenobarbitone safe

Comments/notes	- Chlorpromazine can be magistral prepared as suppositories but is not as a brand name marketed in Belgium, very little experience with this drug - No good clinical evidence regarding treatment of terminal restlessness for out target population
Study (source) VERSION NO LONGER AVAILABLE	Bruera et al (2012)
Study (source)	Lonergan et al (2007)
Study type	Systematic review
Number of participants	Han (2004) N24 + Hu (2004) N175 + Kalisvaart (2005) N430
Lost to follow-up	- No lost to follow up in study 1. Study 2 had 5 dropouts or 2.7%. In the 3th study there were 20 dropouts (9.4%) in haloperidol treated patients and 28 dropouts (12%) in placebo treated patients. - Dropouts were evaluated as non-drug related.
Setting and country	Not described
Duration of study	Han (2004) 7d + Hu (2004) 7d
Population characteristics	- Hospitalised patients - Adult patients - No patients with pre-existing psychotic disorders - All types of delirium except for antipsychotic medication induced delirium and alcohol-withdrawal induced delirium - Average age Hu (2004): 74 y old, male predominance, Kalisvaart (2005) average age 79 but all preoperative patients
Inclusion criteria	- Unconfounded, randomised trials with concealed allocation of subjects. - Trials had to have assessed patients pre- and post-treatment. Where cross-over studies are included, only data from the first part of the study were examined - For meta-analysis: Only 'intention to treat' data were included
Exclusion criteria	- Interrupted time series were excluded
Risk of bias	- Unclear concealment in all trials selected. No concealment trials were excluded. - Concealment impaired by different administration routes
Imprecision	- One small sized trial, other trials had acceptable sample sizes - Intervals that are given are not too wide
Inconsistency	- Large heterogeneity in intervention type and setting - Large methodological diversity - Different outcome definitions
Indirectness	- Recommendations are not for patients in palliative setting
Publication bias	- Inclusion of very small number of studies due to methodological requirements but non included studies were also reviewed.
Funding	Non declared
Outcomes	- Primary outcome measure: To compare the efficacy and incidence of adverse effects of haloperidol with risperidone, olanzapine, and quetiapine in the treatment of delirium. - Secondary outcome measures: - Compare efficacy of each drug with placebo - Incidence and types of adverse effects - Adverse drug effect confounding response evaluations - Whether response is influenced by, - o character: hypo- versus hyperactive - o contributing factors: underlying pathology, drugs and age - o presence of cognitive impairment - o dose of drug and duration of treatment
Results	- Meta-analysis was possible only in comparing the results of risperidone v haloperidol on delirium (Han 2004) and olanzapine v haloperidol on delirium (Hu 2004). These results (See Comparisons and data) showed no significant difference in the overall effect on delirium of olanzapine or risperidone, compared with haloperidol - Where placebo controls were present (Hu 2004; Kalisvaart 2005) patients receiving haloperidol or olanzapine showed a significant improvement in delirium scores compared with placebo controls - More adverse effects with haloperidol in high doses (mean 7.5 mg/day). - Incidence of adverse effects for low dose haloperidol (< 4.5 mg/d) was comparable to the incidence of adverse effects for the atypical antipsychotics that were reviewed - Non-superiority of atypical antipsychotics but also non-inferiority: so to be considered when high doses or increased likelihood of developing haloperidol related side effects. - Preliminary evidence supports the use of pre-operative haloperidol to diminish the duration and intensity of delirium in post-surgery patients, but the single study supporting this recommendation requires testing by further research
Comments/notes	- Not in palliative setting, largest study in pre-operative setting as preventative measure.

Appendix H: Search strings

Pain: clinical research question 1

MEDLINE/PubMed: (#1 AND #2 AND #3 AND #4) NOT #5

#1	("Nursing Homes"[MeSH Terms] OR "Nursing Home"[Title/Abstract] OR "Housing for the Elderly"[MeSH Terms] OR "Housing for the Elderly"[Title/Abstract] OR "Homes for the Aged"[MeSH Terms] OR "Homes for the Aged"[Title/Abstract] OR "Long-term care"[MeSH Terms] OR "Long-term care"[Title/Abstract] OR "Long term care"[Title/Abstract] OR "Aged care facilitat"[Title/Abstract] OR "Aged, 80 and over"[MeSH Terms] OR "Elderly"[Title/Abstract] OR "Geriatrics"[MeSH Terms] OR "Geriatr"[Title/Abstract] OR "Older patient"[Title/Abstract] OR "Older person"[Title/Abstract] OR "Older adult"[Title/Abstract] OR "Advanced age"[Title/Abstract] OR "Octogenarian"[Title/Abstract] OR "Nonagenarian"[Title/Abstract] OR "Centenarian"[Title/Abstract] OR "Frailty"[MeSH Terms] OR "Frail elderly"[Mesh] OR "Frail"[Title/Abstract] OR "Vulnerable"[Title/Abstract] OR "Dementia"[MeSH Terms] OR "Dement"[Title/Abstract] OR "Cognitive disease"[Title/Abstract] OR "Cognitive decline"[Title/Abstract] OR "Cognitive impairment"[Title/Abstract] OR "Multimorbidity"[MeSH Terms] OR "Multimorbidity"[Title/Abstract] OR "Polypatholog"[Title/Abstract] OR "Multipatholog"[Title/Abstract] OR "Cognitive dysfunction"[MeSH Terms] OR "Cognitive dysfunction"[Title/Abstract] OR "Cognitive disfunction"[Title/Abstract] OR "Cognitive disorder"[Title/Abstract] OR "Neurocognitive disorders"[MeSH Terms] OR "Neurocognitive disorder"[Title/Abstract] OR "Neurocognitive decline"[Title/Abstract] OR "Neurocognitive impairment"[Title/Abstract] OR "Neurocognitive disease"[Title/Abstract] OR "Aged"[MeSH Terms] OR "Aging"[Title/Abstract])
#2	("Terminal Care"[MeSH Terms] OR "Terminal care"[Title/Abstract] OR "Terminal Treatment"[Title/Abstract] OR "Terminal Condition"[Title/Abstract] OR "Terminal Disease"[Title/Abstract] OR "Terminal Illness"[Title/Abstract] OR "Terminally ill"[Title/Abstract] OR "Terminally Sick"[Title/Abstract] OR "Terminal Stadi"[Title/Abstract] OR "Terminal Stag"[Title/Abstract] OR "Terminal Phase"[Title/Abstract] OR "Death"[MeSH Terms] OR "Deathbed"[Title/Abstract] OR "Dying"[Title/Abstract] OR "Last Day"[Title/Abstract] OR "Last days"[Title/Abstract] OR "Final day"[Title/Abstract] OR "Final days"[Title/Abstract] OR "Last Minute"[Title/Abstract] OR "Final minute"[Title/Abstract] OR "Last Hour"[Title/Abstract] OR "Final hour"[Title/Abstract] OR "Last Week"[Title/Abstract] OR "Final week"[Title/Abstract] OR "Final moment"[Title/Abstract] OR "End of life"[Title/Abstract] OR "End-of-life"[Title/Abstract] OR "Palliative Medicine"[MeSH Terms] OR "Hospice and palliative care nursing"[MeSH Terms] OR "Palliative care"[MeSH Terms] OR "Palliative"[Title/Abstract] OR "Hospice Care"[MeSH Terms] OR "Hospice"[Title/Abstract] OR "Life Treathening"[Title/Abstract] OR "Palliation"[Title/Abstract] OR "Supportive Care"[Title/Abstract] OR "Supportive Treatment"[Title/Abstract] OR "Supportive Management"[Title/Abstract] OR "Patient comfort"[MeSH Terms] OR "Comfort"[Title/Abstract] OR "Symptom relief"[Title/Abstract])
#3	("Pharmacology"[MeSH Terms] OR "Pharmac"[Title/Abstract] OR "Drug therapy"[MeSH Terms] OR "Drug therap"[Title/Abstract] OR "Analgesics"[MeSH Terms] OR "Analgesic"[Title/Abstract] OR "Analgesia"[MeSH Terms] OR "Analgesia"[Title/Abstract] OR "Analgesics, Opioid"[MeSH Terms] OR "Opioid"[Title/Abstract] OR "Opiate alkaloids"[MeSH Terms] OR "Opiat"[Title/Abstract] OR "narcotic"[Title/Abstract] OR "Tramadol"[MeSH Terms] OR "Tramadol"[Title/Abstract] OR "Morphine"[MeSH Terms] OR "Morphin"[Title/Abstract] OR "Morfin"[Title/Abstract] OR "Oxycodone"[MeSH Terms] OR "Oxycodon"[Title/Abstract] OR "Oxicodon"[Title/Abstract] OR "Hydromorphone"[MeSH Terms] OR "Hydromorphin"[Title/Abstract] OR "Hydromorphon"[Title/Abstract] OR "Buprenorphine"[MeSH Terms] OR "Buprenorphin"[Title/Abstract] OR "Buprenorfin"[Title/Abstract] OR "Fentanyl"[MeSH Terms] OR "Fentanyl"[Title/Abstract] OR "Phentanyl"[Title/Abstract] OR "Fentanil"[Title/Abstract] OR "Pirinitramide"[MeSH Terms] OR "Piritramid"[Title/Abstract] OR "Dipidolor"[Title/Abstract] OR "Dipydolor"[Title/Abstract] OR "Naltrexone"[MeSH Terms] OR "Methylnaltrexon"[Title/Abstract] OR "Naltrexon"[Title/Abstract] OR "Methadone"[MeSH Terms] OR "Methadon"[Title/Abstract] OR "Anti-Inflammatory Agents, Non-Steroidal"[MeSH Terms] OR "Non-Steroidal Anti-inflammatory"[Title/Abstract] OR "NSAID"[Title/Abstract] OR "Acetaminophen"[MeSH Terms] OR "Acetaminophen"[Title/Abstract] OR "paracetamol"[Title/Abstract] OR "Aspirin"[MeSH Terms] OR "Aspirin"[Title/Abstract] OR "Dipyron"[MeSH Terms] OR "Dipyron"[Title/Abstract] OR "Metamizol"[Title/Abstract] OR "methamizol"[Title/Abstract] OR "Novalg"[Title/Abstract] OR "Dipyrionium"[Title/Abstract] OR "Biopyrin"[Title/Abstract] OR "Pyrargin"[Title/Abstract] OR "Novaminsulfon"[Title/Abstract] OR "Sulpyrin"[Title/Abstract] OR "Optalgin"[Title/Abstract] OR "Noramidopyrin"[Title/Abstract] OR "Novamid"[Title/Abstract] OR "Methampyrone"[Title/Abstract] OR "Algopyrin"[Title/Abstract] OR "Analgin"[Title/Abstract] OR "Naron"[Title/Abstract] OR "Nefopam"[MeSH Terms] OR "Nefopam"[Title/Abstract] OR "Acedclofenac" [Supplementary Concept] OR "acedclofenac"[Title/Abstract] OR "Diclofenac"[MeSH Terms] OR "Diclofenac"[Title/Abstract] OR "Ketorolac"[MeSH Terms] OR "Ketorolac"[Title/Abstract] OR "Ibuprofen"[MeSH Terms] OR "Ibuprofen"[Title/Abstract] OR "Naproxen"[MeSH Terms] OR "Naproxen"[Title/Abstract] OR "Meloxicam"[MeSH Terms] OR "Meloxicam"[Title/Abstract] OR "Piroxicam"[MeSH Terms] OR "Piroxicam"[Title/Abstract] OR "Tenoxicam" [Supplementary Concept] OR "Tenoxicam"[Title/Abstract] OR "Celecoxib"[MeSH Terms] OR "Celecoxib"[Title/Abstract] OR "Etoricoxib"[MeSH Terms] OR "Etoricoxib"[Title/Abstract] OR "Parecoxib" [Supplementary Concept] OR "Parecoxib"[Title/Abstract] OR "Glucocorticoids"[MeSH Terms] OR "Glucocorticoid"[Title/Abstract] OR "Corticosteroid"[Title/Abstract] OR "Betamethasone"[MeSH Terms] OR "Betamethasone"[Title/Abstract] OR "Dexamethasone"[MeSH Terms] OR "Dexamethasone"[Title/Abstract] OR "Methylprednisolone"[MeSH Terms] OR "Methylprednisolon"[Title/Abstract] OR "Prednisolone"[MeSH Terms] OR "Prednisolon"[Title/Abstract] OR "Prednisone"[MeSH Terms] OR "Prednisone"[Title/Abstract] OR "Hydrocortisone"[MeSH Terms] OR "Hydrocortison"[Title/Abstract] OR "Carbamazepine"[MeSH Terms] OR "Carbamazepin"[Title/Abstract] OR "Gabapentin"[MeSH Terms] OR "Gabapentin"[Title/Abstract] OR "Pregabalin"[MeSH Terms] OR "Pregabalin"[Title/Abstract] OR "Phenytoin"[MeSH Terms] OR "Phenytoin"[Title/Abstract] OR "Topiramate"[MeSH Terms] OR "Topiram"[Title/Abstract] OR "Ketamine"[MeSH Terms] OR "Ketamin"[Title/Abstract] OR "Clonidine"[MeSH Terms] OR "Clonidin"[Title/Abstract] OR "Lidocaine"[MeSH Terms] OR "Lidocain"[Title/Abstract] OR "Xylocain"[Title/Abstract] OR "Butylscopolammonium Bromide"[MeSH Terms] OR "Butylscopol"[Title/Abstract] OR "Butylhyoscine"[Title/Abstract] OR "Scopolamine"[MeSH Terms] OR "Scopolamin"[Title/Abstract])
#4	("Pain"[MeSH Terms] OR "Pain"[Title/Abstract] OR "Pain Management"[MeSH Terms] OR "Painfull"[Title/Abstract] OR "Breakthrough Pain" [MeSH Terms] OR "Discomfort"[Title/Abstract] OR "Dyscomfort"[Title/Abstract] OR "Suffering" [Title/Abstract] OR "Agon"[Title/Abstract] OR "Aching"[Title/Abstract] OR "Hurt"[Title/Abstract] OR "Polyneuropathies"[MeSH Terms] OR

	"Polyneuropath*" [Title/Abstract] OR "Neuropath*" [Title/Abstract] OR "Neuralgia" [MeSH Terms] OR "Neuralg*" [Title/Abstract] OR "Antalg*" [Title/Abstract] OR "Colic" [MeSH Terms] OR "Colic*" [Title/Abstract] OR "Cramp*" [Title/Abstract] OR "Cancer pain" [MeSH Terms] OR "Cancer pain" [Title/Abstract]
#5	("Veterinary" [Subheading] OR "Veterinary" [Title/Abstract] OR "Animals" [MeSH Terms: noexp] OR "Models, Animal" [MeSH Terms] OR "Rodentia" [MeSH Terms] OR "Animal Experimentation" [MeSH Terms] OR "Animals, laboratory" [MeSH Terms] OR "Animal*" [Title/Abstract] OR "Mice" [Title/Abstract] OR "Mouse" [Title/Abstract] OR "Rats" [Title/Abstract] OR "Rat" [Title/Abstract] OR "In Vitro Techniques" [MeSH Terms] OR "In vitro" [Title/Abstract] OR "Cells" [MeSH Terms] OR "Cell*" [Title] OR "Child" [MeSH Terms] OR "Infant" [MeSH Terms] OR "Infant*" [Title/Abstract] OR "Adolescent" [MeSH Terms] OR "Adolescent*" [Title/Abstract] OR "Juvenile*" [Title/Abstract] OR "Pediatrics" [MeSH Terms] OR "Pediatric*" [Title/Abstract] OR "Paediatric*" [Title/Abstract] OR "Substance-Related Disorders" [MeSH Terms] OR "Addiction Medicine" [MeSH Terms] OR "Addiction Medicine" [Title/Abstract] OR "Surgical procedures, Operative" [MeSH Terms] OR "Surgical procedure*" [Title/Abstract] OR "Operating Rooms" [MeSH Terms] OR "Operating room*" [Title/Abstract] OR "Specialties, surgical" [MeSH Terms] OR "Intensive care units" [MeSH Terms] OR "Intensive care" [Title/Abstract] OR "Antineoplastic Agents" [MeSH Terms] OR "Chemotherapy*" [Title/Abstract] OR "Psychiatric Department, Hospital" [MeSH Terms] OR "Psychiatric Department*" [Title/Abstract] OR "Psychiatric Service*" [Title/Abstract] OR "Psychiatric Ward*" [Title/Abstract] OR "Letter" [Publication Type] OR "Correspondence as topic" [MeSH Terms] OR "Editorial" [Publication Type] OR "Comment" [Publication Type] OR "Case Reports" [Publication Type] OR "Letter" [Title/Abstract] OR "Comment" [Title] OR "Case report*" [Title/Abstract] OR "Case" [Title] OR "Address" [Publication Type] OR "Clinical trial, veterinary" [Publication type])

EMBASE: (#1 AND #2 AND #3 AND #4) NOT #5

#1	('Nursing home'/exp OR 'Nursing home\$':ti,ab OR 'Institutionalized elderly'/exp OR 'Institutionalized NEAR/3 elderly OR 'Home for the aged'/exp OR 'home\$ NEAR/4 aged OR 'Long-term care' NEAR/3 facilit* OR 'Long term care' NEAR/3 facilit* OR 'Aged care facilit*':ti,ab OR 'Very elderly'/exp OR 'elderly':ti,ab OR 'Aged hospital patient'/exp OR 'Aged hospital patient':ti,ab OR 'Geriatric care'/exp OR 'Geriatrics'/de OR 'Geriatr*':ti,ab OR 'Older patient\$':ti,ab OR 'Older person\$':ti,ab OR 'Older adult\$':ti,ab OR 'Advanced age':ti,ab OR 'Octogenarian\$':ti,ab OR 'Nonagenarian\$':ti,ab OR 'Centenarian\$':ti,ab OR 'Frailty'/exp OR 'Frail elderly'/exp OR 'Frail':ti,ab OR 'Frailty':ti,ab OR 'Vulnerable population'/exp OR 'Dementia'/de OR 'Dement*':ti,ab OR 'Cognitive aging'/de OR 'Cognitive decline':ti,ab OR 'Cognitive impairment':ti,ab OR 'Cognitive defect'/de OR 'Cognitive dysfunction*':ti,ab OR 'Cognitive disfunction*':ti,ab OR 'Cognitive disorder\$':ti,ab OR 'Memory disorder'/de OR 'Neurocognitive disorder\$':ti,ab OR 'Neurocognitive decline':ti,ab OR 'Multiple chronic conditions'/exp OR 'Multimorbidity*':ti,ab OR 'Polypatholog*':ti,ab OR 'Multipatholog*':ti,ab OR 'Aged'/de OR 'Aging'/de OR 'Aging':ti,ab)
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#3	('morphin*':ti,ab OR 'morphin*':ti,ab OR 'morphine':ti,ab OR 'oxycodone'/exp OR 'oxycodone':ti,ab OR 'oxycodon*':ti,ab OR 'oxicodone':ti,ab OR 'hydromorphone':ti,ab OR 'hydromorphone':ti,ab OR 'buprenorphine':ti,ab OR 'buprenorphin*':ti,ab OR 'buprenorfine':ti,ab OR 'buprenorfin*':ti,ab OR 'fentanyl derivative'/exp OR 'fentanyl':ti,ab OR 'phentanyl':ti,ab OR 'fentanil':ti,ab OR 'Pirinitramide':ti,ab OR 'Pirinitramid*':ti,ab OR 'dipidolor':ti,ab OR 'dipydolor':ti,ab OR 'naltrexone'/exp OR 'naltrexon\$':ti,ab OR 'naltrexon*':ti,ab OR 'methylnaltrexon\$':ti,ab OR 'methylnaltrexon*':ti,ab OR 'methadone'/exp OR 'methadone':ti,ab OR 'methadon*':ti,ab OR 'antiinflammatory agent'/exp OR 'Non-Steroidal Anti-Inflammatory':ti,ab OR 'NSAID\$':ti,ab OR 'paracetamol'/exp OR 'paracetamol':ti,ab OR 'acetaminophen':ti,ab OR 'acetylsalicylic acid'/exp OR 'aspirin':ti,ab OR 'dipyrone'/exp OR 'dipyrone':ti,ab OR 'metamizol\$':ti,ab OR 'novalg*':ti,ab OR 'dipyrionium':ti,ab OR 'biopyrrin'/exp OR 'biopyrrin':ti,ab OR 'Pyralgin':ti,ab OR 'Novaminsulfon':ti,ab OR 'Sulpyrin\$':ti,ab OR 'optalgin'/exp OR 'optalgin':ti,ab OR 'Noramidopyrin\$':ti,ab OR 'Novamid*':ti,ab OR 'Methampyrone\$':ti,ab OR 'Algopyrin\$':ti,ab OR 'Analgin\$':ti,ab OR 'naron\$':ti,ab OR 'nefopam':ti,ab OR 'Aceclofenac':ti,ab OR 'diclofenac':ti,ab OR 'Ketorolac':ti,ab OR 'ibuprofen':ti,ab OR 'naproxen':ti,ab OR 'meloxicam':ti,ab OR 'piroxicam':ti,ab OR 'tenoxicam':ti,ab OR 'celecoxib':ti,ab OR 'Etoricoxib':ti,ab OR 'Parecoxib':ti,ab OR 'glucocorticoid'/exp OR 'glucocorticoid\$':ti,ab OR 'corticosteroid'/exp OR 'corticosteroid\$':ti,ab OR 'Betamethason\$':ti,ab OR 'Betamethason*':ti,ab OR 'Dexamethason\$':ti,ab OR 'Methylprednisolon\$':ti,ab OR 'Methylprednisolon*':ti,ab OR 'prednisolon\$':ti,ab OR 'prednisolon*':ti,ab OR 'Prednison\$':ti,ab OR 'Prednison*':ti,ab OR 'Hydrocortison\$':ti,ab OR 'Hydrocortison\$':ti,ab OR 'carbamazepine'/exp OR 'carbamazepine\$':ti,ab OR 'gabapentin'/exp OR 'gabapentin\$':ti,ab OR 'Pregabalin\$':ti,ab OR 'phenytoin'/exp OR 'phenytoin':ti,ab OR 'topiramate'/exp OR 'topiramate':ti,ab OR 'topiram*':ti,ab OR 'ketamine'/exp OR 'ketamin\$':ti,ab OR 'clonidine'/exp OR 'clonidine\$':ti,ab OR 'lidocaine'/exp OR 'lidocain\$':ti,ab OR 'Xylocain\$':ti,ab OR 'scopolamine butyl bromide'/exp OR 'butylscopol*':ti,ab OR 'Butylhyoscine':ti,ab OR 'Butylhyoscine*':ti,ab OR 'scopolamine'/exp OR 'scopolamine\$':ti,ab OR 'scopolamine*':ti,ab)
#4	('pain'/exp OR 'analgesia'/exp OR 'pain management':ti,ab OR 'Breakthrough Pain':ti,ab OR 'pain':ti,ab OR 'painful':ti,ab OR 'discomfort'/exp OR 'discomfort':ti,ab OR 'dyscomfort':ti,ab OR 'suffering'/exp OR 'suffering':ti,ab OR 'agony'/exp OR 'agony':ti,ab OR 'aching':ti,ab OR 'hurt':ti,ab OR 'hurt*':ti,ab OR 'polyneuropathy'/exp OR 'polyneuropathy':ti,ab OR 'polyneuropath*':ti,ab OR 'neuralgia'/exp OR 'neuralgia':ti,ab OR 'neuralg*':ti,ab OR 'analgesic agent'/exp OR 'antalgic':ti,ab OR 'colic'/exp OR 'colic*':ti,ab OR 'cramp\$':ti,ab OR 'cancer pain':ti,ab)
#5	('Veterinary medicine'/exp OR 'Veterinary study'/exp OR 'Veterinary':ti,ab OR 'Animal'/de OR 'Animal model'/exp OR 'Rodent'/exp OR 'Animal experiment'/exp OR 'Experimental animal'/exp OR 'Animal tissue, cells or cell components'/exp OR 'Animal\$':ti,ab OR 'Mouse'/exp OR 'Mouse':ti,ab OR 'Mice':ti,ab OR 'Rat'/exp OR 'Rat\$':ti,ab OR 'In vitro study'/exp OR 'In vitro':ti,ab OR 'Cells'/exp OR 'Cell*':ti,ab OR 'Child'/exp OR 'Infant'/exp OR 'Infant\$':ti,ab OR 'Adolescent'/exp OR 'Adolescent\$':ti,ab OR 'Juvenile'/exp OR 'Juvenile\$':ti,ab OR 'Young adult'/exp OR 'Pediatrics'/exp OR 'Pediatrics':ti,ab OR 'Paediatric\$':ti,ab OR 'Drug dependence':exp OR 'Addiction medicine'/exp OR 'Addiction Medicine':ti,ab OR 'Surgery'/exp OR 'Surgical procedure\$':ti,ab OR 'Operating room'/exp OR 'Operating room\$':ti,ab OR 'Intensive care'/exp OR 'Intensive care unit\$':ti,ab OR 'Cancer chemotherapy'/exp OR 'Palliative NEAR/2 chemotherapy OR 'Psychiatric department'/exp OR 'Psychiatric department\$':ti,ab OR 'Psychiatric service\$':ti,ab OR 'Psychiatric ward\$':ti,ab OR 'Letter'/exp OR 'Note'/exp OR 'Editorial'/exp OR 'Case report'/exp OR 'Letter\$':ti,ab OR 'Editorial':ti,ab OR 'Case':ti OR 'Case report':ti,ab)

Dyspnea: clinical research question 2

MEDLINE/PubMed: (#1 AND #2 AND #3 AND #4) NOT #5

#1	("Nursing Homes"[MeSH Terms] OR "Nursing Home"[Title/Abstract] OR "Housing for the Elderly"[MeSH Terms] OR "Housing for the Elderly"[Title/Abstract] OR "Homes for the Aged"[MeSH Terms] OR "Homes for the Aged"[Title/Abstract] OR "Long-term care"[MeSH Terms] OR "Long-term care"[Title/Abstract] OR "Long term care"[Title/Abstract] OR "Aged care facilitat"[Title/Abstract] OR "Aged, 80 and over"[MeSH Terms] OR "Elderly"[Title/Abstract] OR "Geriatrics"[MeSH Terms] OR "Geriatr"[Title/Abstract] OR "Older patient"[Title/Abstract] OR "Older person"[Title/Abstract] OR "Older adult"[Title/Abstract] OR "Advanced age"[Title/Abstract] OR "Octogenarian"[Title/Abstract] OR "Nonagenarian"[Title/Abstract] OR "Centenarian"[Title/Abstract] OR "Frailty"[MeSH Terms] OR "Frail elderly"[Mesh] OR "Frail"[Title/Abstract] OR "Vulnerable"[Title/Abstract] OR "Dementia"[MeSH Terms] OR "Dement"[Title/Abstract] OR "Cognitive disease"[Title/Abstract] OR "Cognitive decline"[Title/Abstract] OR "Cognitive impairment"[Title/Abstract] OR "Multimorbidity"[MeSH Terms] OR "Multimorbidit"[Title/Abstract] OR "Polypatholog"[Title/Abstract] OR "Multipatholog"[Title/Abstract] OR "Cognitive dysfunction"[MeSH Terms] OR "Cognitive dysfunction"[Title/Abstract] OR "Cognitive disfunction"[Title/Abstract] OR "Cognitive disorder"[Title/Abstract] OR "Neurocognitive disorders"[MeSH Terms] OR "Neurocognitive disorder"[Title/Abstract] OR "Neurocognitive decline"[Title/Abstract] OR "Neurocognitive impairment"[Title/Abstract] OR "Neurocognitive disease"[Title/Abstract] OR "Aged"[MeSH Terms] OR "Aging"[Title/Abstract])
#2	("Terminal Care"[MeSH Terms] OR "Terminal care"[Title/Abstract] OR "Terminal Treatment"[Title/Abstract] OR "Terminal Condition"[Title/Abstract] OR "Terminal Disease"[Title/Abstract] OR "Terminal Illness"[Title/Abstract] OR "Terminally ill"[Title/Abstract] OR "Terminally Sick"[Title/Abstract] OR "Terminal Stadi"[Title/Abstract] OR "Terminal Stag"[Title/Abstract] OR "Terminal Phase"[Title/Abstract] OR "Death"[MeSH Terms] OR "Deathbed"[Title/Abstract] OR "Dying"[Title/Abstract] OR "Last Day"[Title/Abstract] OR "Last days"[Title/Abstract] OR "Final day"[Title/Abstract] OR "Final days"[Title/Abstract] OR "Last Minute"[Title/Abstract] OR "Final minute"[Title/Abstract] OR "Last Hour"[Title/Abstract] OR "Final hour"[Title/Abstract] OR "Last Week"[Title/Abstract] OR "Final week"[Title/Abstract] OR "Final moment"[Title/Abstract] OR "End of life"[Title/Abstract] OR "End-of-life"[Title/Abstract] OR "Palliative Medicine"[MeSH Terms] OR "Hospice and palliative care nursing"[MeSH Terms] OR "Palliative care"[MeSH Terms] OR "Palliative"[Title/Abstract] OR "Hospice Care"[MeSH Terms] OR "Hospice"[Title/Abstract] OR "Life Treathening"[Title/Abstract] OR "Palliation"[Title/Abstract] OR "Supportive Care"[Title/Abstract] OR "Supportive Treatment"[Title/Abstract] OR "Supportive Management"[Title/Abstract] OR "Patient comfort"[MeSH Terms] OR "Comfort"[Title/Abstract] OR "Symptom relief"[Title/Abstract])
#3	("Pharmacology"[MeSH Terms] OR "Pharmac"[Title/Abstract] OR "Drug therapy"[MeSH Terms] OR "Drug therap"[Title/Abstract] OR "Analgesics, Opioid"[MeSH Terms] OR "Opioid"[Title/Abstract] OR "Opiate alkaloids"[MeSH Terms] OR "Opiat"[Title/Abstract] OR "Narcotic"[Title/Abstract] OR "Codeine"[MeSH Terms] OR "Codein"[Title/Abstract] OR "Tramadol"[MeSH Terms] OR "Tramadol"[Title/Abstract] OR "Morphine"[MeSH Terms] OR "Morphin"[Title/Abstract] OR "Morfin"[Title/Abstract] OR "Oxycodone"[MeSH Terms] OR "Oxycodon"[Title/Abstract] OR "Oxicodon"[Title/Abstract] OR "Hydromorphone"[MeSH Terms] OR "Hydromorphin"[Title/Abstract] OR "Hydromorphon"[Title/Abstract] OR "Buprenorphine"[MeSH Terms] OR "Buprenorphin"[Title/Abstract] OR "Buprenorfin"[Title/Abstract] OR "Fentanyl"[MeSH Terms] OR "Fentanyl"[Title/Abstract] OR "Phentanyl"[Title/Abstract] OR "Fentanil"[Title/Abstract] OR "Pirinitramide"[MeSH Terms] OR "Piritramid"[Title/Abstract] OR "Dipidolor"[Title/Abstract] OR "Dipydolor"[Title/Abstract] OR "Naltrexone"[MeSH Terms] OR "Methylnaltrexon"[Title/Abstract] OR "Naltrexon"[Title/Abstract] OR "Methadone"[MeSH Terms] OR "Methadon"[Title/Abstract] OR "Glucocorticoids"[MeSH Terms] OR "Glucocorticoid"[Title/Abstract] OR "Corticosteroid"[Title/Abstract] OR "Betamethasone"[MeSH Terms] OR "Betamethason"[Title/Abstract] OR "Dexamethasone"[MeSH Terms] OR "Dexamethason"[Title/Abstract] OR "Methylprednisolone"[MeSH Terms] OR "Methylprednisolon"[Title/Abstract] OR "Prednisolone"[MeSH Terms] OR "Prednisolon"[Title/Abstract] OR "Prednisone"[MeSH Terms] OR "Prednison"[Title/Abstract] OR "Hydrocortisone"[MeSH Terms] OR "Hydrocortison"[Title/Abstract] OR "Administration, Inhalation"[MeSH Terms] OR "Inhalation Therap"[Title/Abstract] OR "Inhalation drug"[Title/Abstract] OR "Albuterol"[MeSH Terms] OR "Albuterol"[Title/Abstract] OR "Levalbuterol"[MeSH Terms] OR "Levalbuterol"[Title/Abstract] OR "Salbutamol"[Title/Abstract] OR "Formoterol Fumarate"[MeSH Terms] OR "Formoterol"[Title/Abstract] OR "Salmeterol Xinafoate"[MeSH Terms] OR "Salmeterol"[Title/Abstract] OR "Fenoterol"[MeSH Terms] OR "Fenoterol"[Title/Abstract] OR "Ipratropium"[MeSH Terms] OR "Ipratropium"[Title/Abstract] OR "Diuretics"[MeSH Terms] OR "Diuretic"[Title/Abstract] OR "Bumetanide"[MeSH Terms] OR "Bumetanid"[Title/Abstract] OR "Furosemide"[MeSH Terms] OR "Furosemide"[Title/Abstract] OR "Torsemide"[MeSH Terms] OR "Torsemid"[Title/Abstract] OR "Oxygen Inhalation Therapy"[MeSH Terms] OR "Oxygen"[Title/Abstract] OR "Benzodiazepines"[MeSH Terms] OR "Benzodiazepin"[Title/Abstract] OR "Diazepam"[MeSH Terms] OR "Diazepam"[Title/Abstract] OR "Alprazolam"[MeSH Terms] OR "Alprazolam"[Title/Abstract] OR "Bromazepam"[MeSH Terms] OR "Bromazepam"[Title/Abstract] OR "Clobazam"[MeSH Terms] OR "Clobazam"[Title/Abstract] OR "Clonazepam"[MeSH Terms] OR "Clonazepam"[Title/Abstract] OR "Flunitrazepam"[MeSH Terms] OR "Flunitrazepam"[Title/Abstract] OR "Flurazepam"[MeSH Terms] OR "Flurazepam"[Title/Abstract] OR "Lorazepam"[MeSH Terms] OR "Lorazepam"[Title/Abstract] OR "Lormetazepam"[Supplementary Concept] OR "Lormetazepam"[Title/Abstract] OR "Midazolam"[MeSH Terms] OR "Midazolam"[Title/Abstract] OR "Nitrazepam"[MeSH Terms] OR "Nitrazepam"[Title/Abstract] OR "Nordazepam"[MeSH Terms] OR "Nordazepam"[Title/Abstract] OR "Oxazepam"[MeSH Terms] OR "Oxazepam"[Title/Abstract] OR "Prazepam"[MeSH Terms] OR "Prazepam"[Title/Abstract])
#4	("Dyspnea"[MeSH Terms] OR "Dyspn"[Title/Abstract] OR "Difficult breathing"[Title/Abstract] OR "Difficulty breathing"[Title/Abstract] OR "Breathles"[Title/Abstract] OR "Hypoxia"[MeSH Terms] OR "Hypox"[Title/Abstract] OR "Anox"[Title/Abstract] OR "Gaspig"[Title/Abstract] OR "Bronchial spasm"[MeSH Terms] OR "Bronchospas"[Title/Abstract] OR "Bronchial spas"[Title/Abstract] OR "Airway obstruction"[MeSH Terms] OR "Airway obstruc"[Title/Abstract] OR "Obstructed airway"[Title/Abstract] OR "Suffocat"[Title/Abstract] OR "Respiratory insufficiency"[MeSH Terms] OR "Respiratory insufficiency"[Title/Abstract] OR "Respiratory failure"[Title/Abstract] OR "Heart failure"[MeSH Terms] OR "Heart failure"[Title/Abstract] OR "Heart decompensation"[Title/Abstract] OR "Cardiac failure"[Title/Abstract] OR "Cardiac decompensation"[Title/Abstract] OR "Lung diseases"[MeSH Terms] OR "Lung disease"[Title/Abstract] OR "Respiratory disease"[Title/Abstract])
#5	("Veterinary" [Subheading] OR "Veterinary"[Title/Abstract] OR "Animals"[MeSH Terms:noexp] OR "Models, Animal"[MeSH Terms] OR "Rodentia"[MeSH Terms] OR "Animal Experimentation"[MeSH Terms] OR "Animals, laboratory"[MeSH Terms] OR "Animal"[Title/Abstract] OR "Mice"[Title/Abstract] OR "Mouse"[Title/Abstract] OR "Rats"[Title/Abstract] OR "Rat"[Title/Abstract] OR "In Vitro Techniques"[MeSH Terms] OR "In vitro"[Title/Abstract] OR "Cells"[MeSH Terms] OR "Cell"[Title] OR "Child"[MeSH Terms] OR "Infant"[MeSH Terms] OR "Infant"[Title/Abstract] OR "Adolescent"[MeSH Terms] OR "Adolescent"[Title/Abstract] OR "Juvenile"[Title/Abstract] OR "Pediatrics"[MeSH Terms] OR "Pediatric"[Title/Abstract] OR "Paediatric"[Title/Abstract] OR "Substance-Related Disorders"[MeSH Terms] OR "Addiction Medicine"[MeSH Terms] OR "Addiction Medicine"[Title/Abstract] OR "Surgical procedures, Operative"[MeSH Terms] OR "Surgical procedure"[Title/Abstract] OR "Operating Rooms"[MeSH Terms] OR "Operating room"[Title/Abstract] OR "Specialties, surgical"[MeSH Terms] OR "Intensive care units"[MeSH Terms] OR "Intensive care"[Title/Abstract] OR "Antineoplastic Agents"[MeSH Terms] OR

	"Chemotherap*" [Title/Abstract] OR "Psychiatric Department, Hospital" [MeSH Terms] OR "Psychiatric Department*" [Title/Abstract] OR "Psychiatric Service*" [Title/Abstract] OR "Psychiatric Ward*" [Title/Abstract] OR "Letter" [Publication Type] OR "Correspondence as topic" [MeSH Terms] OR "Editorial" [Publication Type] OR "Comment" [Publication Type] OR "Case Reports" [Publication Type] OR "Letter" [Title/Abstract] OR "Comment" [Title] OR "Case report*" [Title/Abstract] OR "Case" [Title] OR "Address" [Publication Type] OR "Clinical trial, veterinary" [Publication type]
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EMBASE: (#1 AND #2 AND #3 AND #4) NOT #5

#1	('Nursing home'/exp OR 'Nursing home\$':ti,ab OR 'Institutionalized elderly'/exp OR 'Institutionalized NEAR/3 elderly OR 'Home for the aged'/exp OR 'home\$ NEAR/4 aged OR 'Long-term care' NEAR/3 facilit* OR 'Long term care' NEAR/3 facilit* OR 'Aged care facilit*':ti,ab OR 'Very elderly'/exp OR 'elderly':ti,ab OR 'Aged hospital patient'/exp OR 'Aged hospital patient':ti,ab OR 'Geriatric care'/exp OR 'Geriatrics'/de OR 'Geriatr*':ti,ab OR 'Older patient\$':ti,ab OR 'Older person\$':ti,ab OR 'Older adult\$':ti,ab OR 'Advanced age':ti,ab OR 'Octogenarian\$':ti,ab OR 'Nonagenarian\$':ti,ab OR 'Centenarian\$':ti,ab OR 'Frailty'/exp OR 'Frail elderly'/exp OR 'Frail':ti,ab OR 'Frailty':ti,ab OR 'Vulnerable population'/exp OR 'Dementia'/de OR 'Dement*':ti,ab OR 'Cognitive aging'/de OR 'Cognitive decline':ti,ab OR 'Cognitive impairment':ti,ab OR 'Cognitive defect'/de OR 'Cognitive dysfunction*':ti,ab OR 'Cognitive disfunction*':ti,ab OR 'Cognitive disorder\$':ti,ab OR 'Memory disorder'/de OR 'Neurocognitive disorders\$':ti,ab OR 'Neurocognitive decline':ti,ab OR 'Multiple chronic conditions'/exp OR 'Multimorbidit*':ti,ab OR 'Polypatholog*':ti,ab OR 'Multipatholog*':ti,ab OR 'Aged'/de OR 'Aging'/de OR 'Aging':ti,ab)
#2	('Terminal care'/de OR 'Terminal care':ti,ab OR 'Terminal treatment\$':ti,ab OR 'Terminal condition\$':ti,ab OR 'Terminal disease'/exp OR 'Terminal disease\$':ti,ab OR 'Terminal illness':ti,ab OR 'Terminally ill patient'/exp OR 'Terminally ill':ti,ab OR 'Terminally sick':ti,ab OR 'Terminal stadi*':ti,ab OR 'Terminal stag*':ti,ab OR 'Terminal phase\$':ti,ab OR 'Dying'/exp OR 'Deathbed':ti,ab OR 'Dying':ti,ab OR 'last NEAR/1 day\$ NEAR/3 life OR final NEAR/1 day\$ NEAR/3 life OR last NEAR/1 minute\$ NEAR/3 life OR final NEAR/1 minute\$ NEAR/3 life OR last NEAR/1 hour\$ NEAR/3 life OR final NEAR/1 hour\$ NEAR/3 life OR last NEAR/1 week\$ NEAR/3 life OR final NEAR/1 week\$ NEAR/3 life OR 'Final moment\$':ti,ab OR 'End-of life'/exp OR 'End of life':ti,ab OR 'End-of-life':ti,ab OR 'Palliative therapy'/de OR 'Palliative':ti,ab OR 'Palliative nursing'/exp OR 'Hospice'/exp OR 'Hospice care'/exp OR 'Hospice':ti,ab OR 'Palliation':ti,ab OR 'Supportive care'/exp OR 'Supportive care':ti,ab OR 'Supportive treatment\$':ti,ab OR 'Supportive Management':ti,ab OR 'Patient comfort'/exp OR 'Patient comfort':ti,ab OR 'Symptom relief':ti,ab)
#3	('clinical pharmacology'/exp OR 'clinical pharmacology':ti,ab OR 'drug therapy'/exp OR 'drug therapy':ti,ab OR 'analgesic agent'/exp OR 'analges*':ti,ab OR 'opiate'/exp OR 'opiate':ti,ab OR 'opioid\$':ti,ab OR 'opiat*':ti,ab OR 'narcotic\$':ti,ab OR 'codeine'/exp OR 'codeine':ti,ab OR 'tramadol'/exp OR 'morphine'/exp OR 'morphine':ti,ab OR 'morphin*':ti,ab OR 'morphine':ti,ab OR 'oxycodone'/exp OR 'oxycodone':ti,ab OR 'oxycodon*':ti,ab OR 'oxycodone':ti,ab OR 'hydromorphone':ti,ab OR 'hydromorphone':ti,ab OR 'buprenorphine':ti,ab OR 'buprenorphin*':ti,ab OR 'buprenorfine':ti,ab OR 'buprenorfin*':ti,ab OR 'fentanyl derivative'/exp OR 'fentanyl':ti,ab OR 'phentanyl':ti,ab OR 'fentanil':ti,ab OR 'Pirinitramide':ti,ab OR 'Pirinitramid*':ti,ab OR 'dipidolor':ti,ab OR 'dipydolor':ti,ab OR 'naltrexone'/exp OR 'naltrexon\$':ti,ab OR 'naltrexon*':ti,ab OR 'methylnaltrexon\$':ti,ab OR 'methylnaltrexon*':ti,ab OR 'methadone'/exp OR 'methadone':ti,ab OR 'methadon*':ti,ab OR 'glucocorticoid'/exp OR 'glucocorticoid\$':ti,ab OR 'corticosteroid'/exp OR 'corticosteroid\$':ti,ab OR 'Betamethason\$':ti,ab OR 'Betamethason*':ti,ab OR 'Dexamethason\$':ti,ab OR 'Methylprednisolon\$':ti,ab OR 'Methylprednisolon*':ti,ab OR 'prednisolon\$':ti,ab OR 'prednisolon*':ti,ab OR 'Prednison\$':ti,ab OR 'Prednison*':ti,ab OR 'Hydrocortison\$':ti,ab OR 'Hydrocortison\$':ti,ab OR 'inhalational drug administration'/exp OR 'inhalational drug administration':ti,ab OR 'inhalational therap*':ti,ab OR 'inhalational therapy':ti,ab OR 'inhalational drug*':ti,ab OR 'inhalational drugs':ti,ab OR 'salbutamol'/exp OR 'salbutamol':ti,ab OR 'albuterol':ti,ab OR 'levalbuterol'/exp OR 'levalbuterol':ti,ab OR 'formoterol fumarate'/exp OR 'formoterol':ti,ab OR 'salmeterol xinafoate'/exp OR 'salmeterol':ti,ab OR 'fenoterol'/exp OR 'fenoterol':ti,ab OR 'ipratropium bromide'/exp OR 'ipratropium':ti,ab OR 'diuretic agent'/exp OR 'diuretic*':ti,ab OR 'bumetanid*':ti,ab OR 'bumetanide':ti,ab OR 'furosemide':ti,ab OR 'furosemid*':ti,ab OR 'torsemide':ti,ab OR 'torsemid*':ti,ab OR 'torasemide':ti,ab OR 'torasemid*':ti,ab OR 'oxygen therapy'/exp OR 'oxygen therapy':ti,ab OR 'oxygen*':ti,ab OR 'benzodiazepine derivative'/exp OR 'benzodiazepine':ti,ab OR 'benzodiazepin*':ti,ab OR 'diazepam':ti,ab OR 'alprazolam':ti,ab OR 'bromazepam':ti,ab OR 'clobazam':ti,ab OR 'clonazepam':ti,ab OR 'flunitrazepam':ti,ab OR 'flurazepam':ti,ab OR 'lorazepam':ti,ab OR 'lormetazepam':ti,ab OR 'midazolam':ti,ab OR 'nitrazepam':ti,ab OR 'nordazepam':ti,ab OR 'oxazepam':ti,ab OR 'prazepam':ti,ab)
#4	('dyspnea'/exp OR 'dyspnea':ti,ab OR 'dyspn*':ti,ab OR 'Difficult breathing':ti,ab OR 'Difficulty breathing':ti,ab OR 'Breathles*':ti,ab OR 'hypoxemia'/exp OR 'hypox*':ti,ab OR 'anox*':ti,ab OR 'gasping'/exp OR 'gasping':ti,ab OR 'bronchospasm'/exp OR 'bronchospasm':ti,ab OR 'bronchospas*':ti,ab OR 'bronchial spasm':ti,ab OR 'bronchial spas*':ti,ab OR 'airway obstruct*':ti,ab OR 'obstructed airway\$':ti,ab OR 'suffocate':ti,ab OR 'suffocat*':ti,ab OR 'respiratory failure'/exp OR 'respiratory failure':ti,ab OR 'respiratory insufficiency':ti,ab OR 'heart failure'/exp OR 'heart failure':ti,ab OR 'heart decompensation':ti,ab OR 'cardiac failure':ti,ab OR 'cardiac decompensation':ti,ab OR 'lung disease'/exp OR 'lung disease\$':ti,ab OR 'respiratory diseases\$':ti,ab)
#5	('Veterinary medicine'/exp OR 'Veterinary study'/exp OR 'Veterinary':ti,ab OR 'Animal'/de OR 'Animal model'/exp OR 'Rodent'/exp OR 'Animal experiment'/exp OR 'Experimental animal'/exp OR 'Animal tissue, cells or cell components'/exp OR 'Animal\$':ti,ab OR 'Mouse'/exp OR 'Mouse':ti,ab OR 'Mice':ti,ab OR 'Rat'/exp OR 'Rat\$':ti,ab OR 'In vitro study'/exp OR 'In vitro':ti,ab OR 'Cells'/exp OR 'Cell*':ti,ab OR 'Child'/exp OR 'Infant'/exp OR 'Infant\$':ti,ab OR 'Adolescent'/exp OR 'Adolescent\$':ti,ab OR 'Juvenile'/exp OR 'Juvenile\$':ti,ab OR 'Young adult'/exp OR 'Pediatrics'/exp OR 'Pediatrics':ti,ab OR 'Paediatric\$':ti,ab OR 'Drug dependence':exp OR 'Addiction medicine'/exp OR 'Addiction Medicine':ti,ab OR 'Surgery'/exp OR 'Surgical procedure\$':ti,ab OR 'Operating room'/exp OR 'Operating room\$':ti,ab OR 'Intensive care'/exp OR 'Intensive care unit\$':ti,ab OR 'Cancer chemotherapy'/exp OR 'Palliative NEAR/2 chemotherapy OR 'Psychiatric department'/exp OR 'Psychiatric department\$':ti,ab OR 'Psychiatric service\$':ti,ab OR 'Psychiatric ward\$':ti,ab OR 'Letter'/exp OR 'Note'/exp OR 'Editorial'/exp OR 'Case report'/exp OR 'Letter\$':ti,ab OR 'Editorial':ti,ab OR 'Case':ti OR 'Case report':ti,ab)

Nausea and vomiting: clinical research question 3

MEDLINE/PubMed: (#1 AND #2 AND #3 AND #4) NOT #5

#1	("Nursing Homes"[MeSH Terms] OR "Nursing Home"[Title/Abstract] OR "Housing for the Elderly"[MeSH Terms] OR "Housing for the Elderly"[Title/Abstract] OR "Homes for the Aged"[MeSH Terms] OR "Homes for the Aged"[Title/Abstract] OR "Long-term care"[MeSH Terms] OR "Long-term care"[Title/Abstract] OR "Long term care"[Title/Abstract] OR "Aged care faciliti*[Title/Abstract] OR "Aged, 80 and over"[MeSH Terms] OR "Elderly"[Title/Abstract] OR "Geriatrics"[MeSH Terms] OR "Geriatr*[Title/Abstract] OR "Older patient*[Title/Abstract] OR "Older person*[Title/Abstract] OR "Older adult*[Title/Abstract] OR "Advanced age"[Title/Abstract] OR "Octogenarian*[Title/Abstract] OR "Nonagenarian*[Title/Abstract] OR "Centenarian*[Title/Abstract] OR "Frailty"[MeSH Terms] OR "Frail elderly"[Mesh] OR "Frail*[Title/Abstract] OR "Vulnerable"[Title/Abstract] OR "Dementia"[MeSH Terms] OR "Dement*[Title/Abstract] OR "Cognitive disease*[Title/Abstract] OR "Cognitive decline"[Title/Abstract] OR "Cognitive impairment"[Title/Abstract] OR "Multimorbidity"[MeSH Terms] OR "Multimorbidit*[Title/Abstract] OR "Polypatholog*[Title/Abstract] OR "Multipatholog*[Title/Abstract] OR "Cognitive dysfunction"[MeSH Terms] OR "Cognitive dysfunction*[Title/Abstract] OR "Cognitive disfunction*[Title/Abstract] OR "Cognitive disorder"[Title/Abstract] OR "Neurocognitive disorders"[MeSH Terms] OR "Neurocognitive disorder*[Title/Abstract] OR "Neurocognitive decline"[Title/Abstract] OR "Neurocognitive impairment"[Title/Abstract] OR "Neurocognitive disease"[Title/Abstract] OR "Aged"[MeSH Terms] OR "Aging"[Title/Abstract])
#2	("Terminal Care"[MeSH Terms] OR "Terminal care"[Title/Abstract] OR "Terminal Treatment*[Title/Abstract] OR "Terminal Condition"[Title/Abstract] OR "Terminal Disease*[Title/Abstract] OR "Terminal illness"[Title/Abstract] OR "Terminally ill"[Title/Abstract] OR "Terminally Sick*[Title/Abstract] OR "Terminal Stadi*[Title/Abstract] OR "Terminal Stag*[Title/Abstract] OR "Terminal Phase*[Title/Abstract] OR "Death"[MeSH Terms] OR "Deathbed"[Title/Abstract] OR "Dying"[Title/Abstract] OR "Last Day"[Title/Abstract] OR "Last days"[Title/Abstract] OR "Final day"[Title/Abstract] OR "Final days"[Title/Abstract] OR "Last Minute*[Title/Abstract] OR "Final minute*[Title/Abstract] OR "Last Hour*[Title/Abstract] OR "Final hour*[Title/Abstract] OR "Last Week*[Title/Abstract] OR "Final week*[Title/Abstract] OR "Final moment*[Title/Abstract] OR "End of life"[Title/Abstract] OR "End-of-life"[Title/Abstract] OR "Palliative Medicine"[MeSH Terms] OR "Hospice and palliative care nursing"[MeSH Terms] OR "Palliative care"[MeSH Terms] OR "Palliative"[Title/Abstract] OR "Hospice Care"[MeSH Terms] OR "Hospice"[Title/Abstract] OR "Life Treathening"[Title/Abstract] OR "Palliation"[Title/Abstract] OR "Supportive Care"[Title/Abstract] OR "Supportive Treatment"[Title/Abstract] OR "Supportive Management"[Title/Abstract] OR "Patient comfort"[MeSH Terms] OR "Comfort*[Title/Abstract] OR "Symptom relief"[Title/Abstract])
#3	("Pharmacology"[MeSH Terms] OR "Pharmac*[Title/Abstract] OR "Drug therapy"[MeSH Terms] OR "Drug therap*[Title/Abstract] OR "Antiemetics"[MeSH Terms] OR "Antiemetic*[Title/Abstract] OR "Antiemetic*[Title/Abstract] OR "Dopamine antagonists"[MeSH Terms] OR "Dopamin*[Title/Abstract] OR "Domperidone"[MeSH Terms] OR "Domperidon*[Title/Abstract] OR "Metoclopramide"[MeSH Terms] OR "Metoclopramid*[Title/Abstract] OR "Alizapride"[Supplementary Concept] OR "Alizaprid*[Title/Abstract] OR "Gastropokinetic*[Title/Abstract] OR "Serotonin 5-HT3 Receptor Antagonists"[MeSH Terms] OR "5-HT3"[Title/Abstract] OR "5 HT3"[Title/Abstract] OR "Serotonin antagonist*[Title/Abstract] OR "Granisetron"[MeSH Terms] OR "Granisetron"[Title/Abstract] OR "Ondansetron"[MeSH Terms] OR "Ondansetron"[Title/Abstract] OR "Palonosetron"[MeSH Terms] OR "Palonosetron"[Title/Abstract] OR "Tropisetron"[MeSH Terms] OR "Tropisetron"[Title/Abstract] OR "Antipsychotic Agents"[MeSH Terms] OR "Antipsychotic*[Title/Abstract] OR "Neuroleptic*[Title/Abstract] OR "Phenothiazines"[MeSH Terms] OR "Phenothiazin*[Title/Abstract] OR "Methotrimeprazine"[MeSH Terms] OR "Methotrimeprazin*[Title/Abstract] OR "Levopromazin*[Title/Abstract] OR "Prothipendyl"[Title/Abstract] OR "Thioxanthenes"[MeSH Terms] OR "Thioxanthen*[Title/Abstract] OR "Flupenthixol"[MeSH Terms] OR "Flupenthixol"[Title/Abstract] OR "Fluanxol"[Title/Abstract] OR "Clopenthixol"[MeSH Terms] OR "Clopenthixol"[Title/Abstract] OR "Zuclopenthixol"[Title/Abstract] OR "Clotiapi*[Title/Abstract] OR "Butyrophenones"[MeSH Terms] OR "Butyrophenon*[Title/Abstract] OR "Bromperidol*[Title/Abstract] OR "Droperidol"[MeSH Terms] OR "Droperidol*[Title/Abstract] OR "Haloperidol"[MeSH Terms] OR "Haloperidol*[Title/Abstract] OR "Haldol"[Title/Abstract] OR "Aripiprazol*[Title/Abstract] OR "Asenapine"[Supplementary Concept] OR "Asenapin*[Title/Abstract] OR "Cariprazine" [Supplementary Concept] OR "Cariprazin*[Title/Abstract] OR "Clozapine"[MeSH Terms] OR "Clozapine"[Title/Abstract] OR "Olanzapine"[MeSH Terms] OR "Olanzapine"[Title/Abstract] OR "Paliperidone Palmitate"[MeSH Terms] OR "Paliperidone Palmitate"[Title/Abstract] OR "Paliperidon*[Title/Abstract] OR "Quetiapine Fumarate"[MeSH Terms] OR "Quetiapin*[Title/Abstract] OR "Risperidone"[MeSH Terms] OR "Risperidon*[Title/Abstract] OR "Sertindol*[Title/Abstract] OR "Pimozide"[MeSH Terms] OR "Pimozide"[Title/Abstract] OR "Pipamperon*[Title/Abstract] OR "Amisulpride"[MeSH Terms] OR "Amisulprid*[Title/Abstract] OR "Sulpiride"[MeSH Terms] OR "Sulpirid*[Title/Abstract] OR "Tiapride Hydrochloride"[MeSH Terms] OR "Tiapride Hydrochloride"[Title/Abstract] OR "Tiaprid*[Title/Abstract] OR "Neurokinin-1 Receptor Antagonists"[MeSH Terms] OR "Neurokinin-1 Receptor Antagonist*[Title/Abstract] OR "Neurokinin 1 Receptor Antagonist*[Title/Abstract] OR "NK1-Antagonist*[Title/Abstract] OR "NK1 Antagonist*[Title/Abstract] OR "Aprepitant"[MeSH Terms] OR "Aprepitant"[Title/Abstract] OR "Fosaprepitant" [Supplementary Concept] OR "Fosaprepitant" [Title/Abstract] OR "Glucocorticoids"[MeSH Terms] OR "Glucocorticoid*[Title/Abstract] OR "Corticosteroid*[Title/Abstract] OR "Betamethasone"[MeSH Terms] OR "Betamethason*[Title/Abstract] OR "Dexamethasone"[MeSH Terms] OR "Dexamethason*[Title/Abstract] OR "Methylprednisolone"[MeSH Terms] OR "Methylprednisolon*[Title/Abstract] OR "Prednisolone"[MeSH Terms] OR "Prednisolon*[Title/Abstract] OR "Prednisone"[MeSH Terms] OR "Prednison*[Title/Abstract] OR "Hydrocortisone"[MeSH Terms] OR "Hydrocortison*[Title/Abstract] OR "Somatostatin"[MeSH Terms] OR "Somatostatin*[Title/Abstract] OR "Octreotide"[MeSH Terms] OR "Octreotid*[Title/Abstract] OR "Lanreotide" [Supplementary Concept] OR "Lanreotid*[Title/Abstract] OR "Cyclizine"[MeSH Terms] OR "Cyclizin*[Title/Abstract])
#4	("Nausea"[MeSH Terms] OR "Nause*[Title/Abstract] OR "Vomiting"[MeSH Terms] OR "Vomit*[Title/Abstract] OR "Emesis"[Title/Abstract] OR "Emit*[Title/Abstract])
#5	("Veterinary" [Subheading] OR "Veterinary"[Title/Abstract] OR "Animals"[MeSH Terms:noexp] OR "Models, Animal"[MeSH Terms] OR "Rodentia"[MeSH Terms] OR "Animal Experimentation"[MeSH Terms] OR "Animals, laboratory"[MeSH Terms] OR "Animal*[Title/Abstract] OR "Mice"[Title/Abstract] OR "Mouse"[Title/Abstract] OR "Rats"[Title/Abstract] OR "Rat"[Title/Abstract] OR "In Vitro Techniques"[MeSH Terms] OR "In vitro"[Title/Abstract] OR "Cells"[MeSH Terms] OR "Cell*[Title] OR "Child"[MeSH Terms] OR "Infant"[MeSH Terms] OR "Infant*[Title/Abstract] OR "Adolescent"[MeSH Terms] OR "Adolescent*[Title/Abstract] OR "Juvenile*[Title/Abstract] OR "Pediatrics"[MeSH Terms] OR "Pediatric*[Title/Abstract] OR "Paediatric*[Title/Abstract] OR "Substance-Related Disorders"[MeSH Terms] OR "Addiction Medicine"[MeSH Terms] OR "Addiction Medicine"[Title/Abstract] OR "Surgical procedures, Operative"[MeSH Terms] OR "Surgical procedure*[Title/Abstract] OR "Operating Rooms"[MeSH Terms] OR "Operating room*[Title/Abstract] OR "Specialties, surgical"[MeSH Terms] OR "Intensive care units"[MeSH Terms] OR "Intensive care"[Title/Abstract] OR "Antineoplastic Agents"[MeSH Terms] OR "Chemotherap*[Title/Abstract] OR "Psychiatric Department, Hospital"[MeSH Terms] OR "Psychiatric Department*[Title/Abstract] OR "Psychiatric Service*[Title/Abstract] OR "Psychiatric Ward*[Title/Abstract] OR "Letter"[Publication Type] OR "Correspondence as topic"[MeSH Terms] OR "Editorial"[Publication Type] OR "Comment"[Publication Type] OR "Case Reports"[Publication Type] OR "Letter"[Title/Abstract] OR "Comment"[Title] OR "Case report*[Title/Abstract] OR "Case"[Title] OR "Address"[Publication Type] OR "Clinical trial, veterinary"[Publication type])

EMBASE: (#1 AND #2 AND #3 AND #4) NOT #5

#1	('Nursing home'/exp OR 'Nursing home\$':ti,ab OR 'Institutionalized elderly'/exp OR Institutionalized NEAR/3 elderly OR 'Home for the aged'/exp OR home\$ NEAR/4 aged OR 'Long-term care' NEAR/3 facilit* OR 'Long term care' NEAR/3 facilit* OR 'Aged care facilit*':ti,ab OR 'Very elderly'/exp OR 'elderly':ti,ab OR 'Aged hospital patient'/exp OR 'Aged hospital patient':ti,ab OR 'Geriatric care'/exp OR 'Geriatrics'/de OR 'Geriatr*':ti,ab OR 'Older patient\$':ti,ab OR 'Older person\$':ti,ab OR 'Older adult\$':ti,ab OR 'Advanced age':ti,ab OR 'Octogenarian\$':ti,ab OR 'Nonagenarian\$':ti,ab OR 'Centenarian\$':ti,ab OR 'Frailty'/exp OR 'Frail elderly'/exp OR 'Frail':ti,ab OR 'Frailty':ti,ab OR 'Vulnerable population'/exp OR 'Dementia'/de OR 'Dement*':ti,ab OR 'Cognitive aging'/de OR 'Cognitive decline':ti,ab OR 'Cognitive impairment':ti,ab OR 'Cognitive defect'/de OR 'Cognitive dysfunction*':ti,ab OR 'Cognitive disfunction*':ti,ab OR 'Cognitive disorder\$':ti,ab OR 'Memory disorder'/de OR 'Neurocognitive disorder\$':ti,ab OR 'Neurocognitive decline':ti,ab OR 'Multiple chronic conditions'/exp OR 'Multimorbidit*':ti,ab OR 'Polypatholog*':ti,ab OR 'Multipatholog*':ti,ab OR 'Aged'/de OR 'Aging'/de OR 'Aging':ti,ab)
#2	('Terminal care'/de OR 'Terminal care':ti,ab OR 'Terminal treatment\$':ti,ab OR 'Terminal condition\$':ti,ab OR 'Terminal disease'/exp OR 'Terminal disease\$':ti,ab OR 'Terminal illness':ti,ab OR 'Terminally ill patient'/exp OR 'Terminally ill':ti,ab OR 'Terminally sick':ti,ab OR 'Terminal stadi*':ti,ab OR 'Terminal stag*':ti,ab OR 'Terminal phase\$':ti,ab OR 'Dying'/exp OR 'Deathbed':ti,ab OR 'Dying':ti,ab OR last NEAR/1 day\$ NEAR/3 life OR final NEAR/1 day\$ NEAR/3 life OR last NEAR/1 minute\$ NEAR/3 life OR final NEAR/1 minute\$ NEAR/3 life OR last NEAR/1 hour\$ NEAR/3 life OR final NEAR/1 hour\$ NEAR/3 life OR last NEAR/1 week\$ NEAR/3 life OR final NEAR/1 week\$ NEAR/3 life OR 'Final moment\$':ti,ab OR 'End-of life'/exp OR 'End of life':ti,ab OR 'End-of-life':ti,ab OR 'Palliative therapy'/de OR 'Palliative':ti,ab OR 'Palliative nursing'/exp OR 'Hospice'/exp OR 'Hospice care'/exp OR 'Hospice':ti,ab OR 'Palliation':ti,ab OR 'Supportive care'/exp OR 'Supportive care':ti,ab OR 'Supportive treatment\$':ti,ab OR 'Supportive Management':ti,ab OR 'Patient comfort'/exp OR 'Patient comfort':ti,ab OR 'Symptom relief':ti,ab)
#3	('clinical pharmacology'/exp OR 'clinical pharmacology':ti,ab OR 'drug therapy'/exp OR 'drug therapy':ti,ab OR 'antiemetic agent'/exp OR 'antiemetic':ti,ab OR 'antiemetic':ti,ab OR 'dopamine receptor blocking agent'/exp OR 'dopamine receptor blocking agent':ti,ab OR 'dopamine':ti,ab OR 'dopamin*':ti,ab OR 'domperidone'/exp OR 'domperidone':ti,ab OR 'domperidon*':ti,ab OR 'metoclopramide'/exp OR 'metoclopramide':ti,ab OR 'metoclopramid*'/exp OR 'alzapride':ti,ab OR 'alizaprid*':ti,ab OR 'prokinetic agent'/exp OR 'prokinetic agent':ti,ab OR 'gastroprokinetic':ti,ab OR 'gastroprokineti*':ti,ab OR 'serotonin 3 antagonist'/exp OR '5-HT3':ti,ab OR '5 HT3':ti,ab OR 'serotonin antagonist':ti,ab OR 'granisetron':ti,ab OR 'ondansetron':ti,ab OR 'palonosetron':ti,ab OR 'tropisetron':ti,ab OR 'neuroleptic agent'/exp OR 'antipsychotic\$':ti,ab OR 'antipsychotic*':ti,ab OR 'neuroleptic\$':ti,ab OR 'neuroleptic*':ti,ab OR 'phenothiazine derivative'/exp OR 'phenothiazin*':ti,ab OR 'levomepromazine':ti,ab OR 'levomepromazin*':ti,ab OR 'Methotrimeprazine':ti,ab OR 'Methotrimeprazin*':ti,ab OR 'prothipendyl'/exp OR 'prothipendyl':ti,ab OR 'thioxanthene derivative'/exp OR 'thioxanthene':ti,ab OR 'thioxanthen*':ti,ab OR 'flupentixol':ti,ab OR 'flupenthixol':ti,ab OR 'fluanaxol':ti,ab OR 'clopenthixol':ti,ab OR 'zuclopenthixol':ti,ab OR 'clotiapine':ti,ab OR 'clotiapin*':ti,ab OR 'butyrophenone derivative'/exp OR 'butyrophenone':ti,ab OR 'butyrophenon*':ti,ab OR 'bromperidol':ti,ab OR 'droperidol':ti,ab OR 'haloperidol':ti,ab OR 'haloperidol*':ti,ab OR 'haldol':ti,ab OR 'aripiprazole'/exp OR 'aripiprazole':ti,ab OR 'aripiprazol*':ti,ab OR 'asenapine'/exp OR 'asenapine':ti,ab OR 'asenapin*':ti,ab OR 'cariprazine'/exp OR 'cariprazine':ti,ab OR 'cariprazin*':ti,ab OR 'clozapine'/exp OR 'clozapine':ti,ab OR 'olanzapine'/exp OR 'olanzapine':ti,ab OR 'paliperidone'/exp OR 'paliperidone':ti,ab OR 'paliperidon*':ti,ab OR 'quetiapine':ti,ab OR 'quetiapin*':ti,ab OR 'risperidone':ti,ab OR 'risperidon*':ti,ab OR 'sertindole':ti,ab OR 'sertindol*':ti,ab OR 'pimozide':ti,ab OR 'pipamperone':ti,ab OR 'pipamperon*':ti,ab OR 'amisulpride':ti,ab OR 'amisulprid*':ti,ab OR 'sulpiride':ti,ab OR 'sulpirid*':ti,ab OR 'tiapride':ti,ab OR 'tiaprid*':ti,ab OR 'neurokinin 1 receptor antagonist'/exp OR 'neurokinin 1 receptor antagonist':ti,ab OR 'neurokinin-1 receptor antagonist':ti,ab OR 'NK1-antagonist':ti,ab OR 'NK1 antagonist':ti,ab OR 'NK1-antagonis*':ti,ab OR 'NK1 antagonis*':ti,ab OR 'aprepitant':ti,ab OR 'fosaprepitant':ti,ab OR 'glucocorticoid'/exp OR 'glucocorticoid\$':ti,ab OR 'corticosteroid'/exp OR 'corticosteroid\$':ti,ab OR 'Betamethason\$':ti,ab OR 'Betamethason*':ti,ab OR 'Dexamethason\$':ti,ab OR 'Methylprednisolon\$':ti,ab OR 'Methylprednisolon*':ti,ab OR 'prednisolon\$':ti,ab OR 'prednisolon*':ti,ab OR 'Prednison\$':ti,ab OR 'Prednison*':ti,ab OR 'Prednison*':ti,ab OR 'Hydrocortison*':ti,ab OR 'somatostatin'/exp OR 'somatostatin':ti,ab OR 'octreotide'/exp OR 'octreotide':ti,ab OR 'angiopeptin'/exp OR 'angiopeptin':ti,ab OR 'lanreotide':ti,ab OR 'lanreotid*':ti,ab OR 'cyclizine':ti,ab OR 'cyclizin*':ti,ab)
#4	('nausea and vomiting'/exp OR 'nausea and vomiting':ti,ab OR 'nausea':ti,ab OR 'vomiting':ti,ab OR 'emesis':ti,ab OR 'emit*':ti,ab)
#5	('Veterinary medicine'/exp OR 'Veterinary study'/exp OR 'Veterinary':ti,ab OR 'Animal'/de OR 'Animal model'/exp OR 'Rodent'/exp OR 'Animal experiment'/exp OR 'Experimental animal'/exp OR 'Animal tissue, cells or cell components'/exp OR 'Animal\$':ti,ab OR 'Mouse'/exp OR 'Mouse':ti,ab OR 'Mice':ti,ab OR 'Rat'/exp OR 'Rat\$':ti,ab OR 'In vitro study'/exp OR 'In vitro':ti,ab OR 'Cells'/exp OR 'Cell*':ti,ab OR 'Child'/exp OR 'Infant'/exp OR 'Infant\$':ti,ab OR 'Adolescent'/exp OR 'Adolescent\$':ti,ab OR 'Juvenile'/exp OR 'Juvenile\$':ti,ab OR 'Young adult'/exp OR 'Pedicatrics'/exp OR 'Pediatric\$':ti,ab OR 'Paediatric\$':ti,ab OR 'Drug dependence'/exp OR 'Addiction medicine'/exp OR 'Addiction Medicine':ti,ab OR 'Surgery'/exp OR 'Surgical procedure\$':ti,ab OR 'Operating room'/exp OR 'Operating room\$':ti,ab OR 'Intensive care'/exp OR 'Intensive care unit\$':ti,ab OR 'Cancer chemotherapy'/exp OR Palliative NEAR/2 chemotherapy OR 'Psychiatric department'/exp OR 'Psychiatric department\$':ti,ab OR 'Psychiatric service\$':ti,ab OR 'Psychiatric ward\$':ti,ab OR 'Letter'/exp OR 'Note'/exp OR 'Editorial'/exp OR 'Case report'/exp OR 'Letter\$':ti,ab OR 'Editorial':ti,ab OR 'Case':ti OR 'Case report':ti,ab)

Death rattle: clinical research question 4

MEDLINE/PubMed: (#1 AND #2 AND #3 AND #4) NOT #5

#1	("Nursing Homes"[MeSH Terms] OR "Nursing Home*" [Title/Abstract] OR "Housing for the Elderly"[MeSH Terms] OR "Housing for the Elderly"[Title/Abstract] OR "Homes for the Aged"[MeSH Terms] OR "Homes for the Aged"[Title/Abstract] OR "Long-term care"[MeSH Terms] OR "Long-term care"[Title/Abstract] OR "Long term care"[Title/Abstract] OR "Aged care facilit*" [Title/Abstract] OR "Aged, 80 and over"[MeSH Terms] OR "Elderly"[Title/Abstract] OR "Geriatrics"[MeSH Terms] OR "Geriatr*" [Title/Abstract] OR "Older patient*" [Title/Abstract] OR "Older person*" [Title/Abstract] OR "Older adult*" [Title/Abstract] OR "Advanced age" [Title/Abstract] OR "Octogenarian*" [Title/Abstract] OR "Nonagenarian*" [Title/Abstract] OR "Centenarian*" [Title/Abstract] OR "Frailty"[MeSH Terms] OR "Frail elderly"[Mesh] OR "Frail*" [Title/Abstract] OR "Vulnerable"[Title/Abstract] OR "Dementia"[MeSH Terms] OR "Dement*" [Title/Abstract] OR "Cognitive disease*" [Title/Abstract] OR "Cognitive decline" [Title/Abstract] OR "Cognitive impairment" [Title/Abstract] OR "Multimorbidity"[MeSH Terms] OR "Multimorbidit*" [Title/Abstract] OR "Polypatholog*" [Title/Abstract] OR "Multipatholog*" [Title/Abstract] OR "Cognitive dysfunction"[MeSH Terms] OR "Cognitive dysfunction*" [Title/Abstract] OR "Cognitive disfunction*" [Title/Abstract] OR "Cognitive disorder*" [Title/Abstract] OR "Neurocognitive disorders"[MeSH Terms] OR "Neurocognitive disorder*" [Title/Abstract] OR "Neurocognitive decline" [Title/Abstract] OR "Neurocognitive impairment" [Title/Abstract] OR "Neurocognitive disease" [Title/Abstract] OR "Aged" [MeSH Terms] OR "Aging" [Title/Abstract])
#2	("Terminal Care"[MeSH Terms] OR "Terminal care" [Title/Abstract] OR "Terminal Treatment*" [Title/Abstract] OR "Terminal Condition" [Title/Abstract] OR "Terminal Disease*" [Title/Abstract] OR "Terminal illness" [Title/Abstract] OR "Terminally ill" [Title/Abstract] OR "Terminally Sick*" [Title/Abstract] OR "Terminal Stadi*" [Title/Abstract] OR "Terminal Stag*" [Title/Abstract] OR "Terminal Phase*" [Title/Abstract] OR "Death" [MeSH Terms] OR "Deathbed" [Title/Abstract] OR "Dying" [Title/Abstract] OR "Last Day" [Title/Abstract] OR "Last days" [Title/Abstract] OR "Final day" [Title/Abstract] OR "Final days" [Title/Abstract] OR "Last Minute*" [Title/Abstract] OR "Final minute*" [Title/Abstract] OR "Last Hour*" [Title/Abstract] OR "Final hour*" [Title/Abstract] OR "Last Week*" [Title/Abstract] OR "Final week*" [Title/Abstract] OR "Final moment*" [Title/Abstract] OR "End of life" [Title/Abstract] OR "End-of-life" [Title/Abstract] OR "Palliative Medicine"[MeSH Terms] OR "Hospice and palliative care nursing"[MeSH Terms] OR "Palliative care" [MeSH Terms] OR "Palliative" [Title/Abstract] OR "Hospice Care" [MeSH Terms] OR "Hospice" [Title/Abstract] OR "Life Treathening" [Title/Abstract] OR "Palliation" [Title/Abstract] OR "Supportive Care" [Title/Abstract] OR "Supportive Treatment" [Title/Abstract] OR "Supportive Management" [Title/Abstract] OR "Patient comfort" [MeSH Terms] OR "Comfort*" [Title/Abstract] OR "Symptom relief" [Title/Abstract])
#3	("Pharmacology"[MeSH Terms] OR "Pharmac*" [Title/Abstract] OR "Drug therapy"[MeSH Terms] OR "Drug therap*" [Title/Abstract] OR "Muscarinic antagonists"[MeSH Terms] OR "Muscarinic antagonists" [Title/Abstract] OR "Antimuscarinic" [Title/Abstract] OR "Cholinergic Antagonists"[MeSH Terms] OR "Cholinergic Antagonist*" [Title/Abstract] OR "Anticholinerg*" [Title/Abstract] OR "Scopolamine"[MeSH Terms] OR "Scopol*" [Title/Abstract] OR "Hyoscin*" [Title/Abstract] OR "Atropine derivatives"[MeSH Terms] OR "Atropine" [MeSH Terms] OR "Atropin*" [Title/Abstract] OR "Butylscopolammonium Bromide" [MeSH Terms] OR "Butylscop*" [Title/Abstract] OR "Busc*" [Title/Abstract] OR "Butylbrom*" [Title/Abstract] OR "Hyoscinbutyl*" [Title/Abstract] OR "Glycopyrrolate" [MeSH Terms] OR "Glycopyr*" [Title/Abstract])
#4	("Sputum"[MeSH Terms] OR "Sputum" [Title/Abstract] OR "Death Rattle" [Title/Abstract] OR "Lung secret*" [Title/Abstract] OR "Pulmonary secret*" [Title/Abstract] OR "Pulmonary secret*" [Title/Abstract] OR "Respiratory tract secret*" [Title/Abstract] OR "Bronchial secret*" [Title/Abstract] OR "Retained secret*" [Title/Abstract] OR "Airway secret*" [Title/Abstract] OR "Respiratory sound*" [Title/Abstract] OR "Noisy respir*" [Title/Abstract] OR "Noisy breath*" [Title/Abstract] OR "Terminal breath*" [Title/Abstract] OR "Rattling breath*" [Title/Abstract] OR "Gasping breath*" [Title/Abstract])
#5	("Veterinary" [Subheading] OR "Veterinary" [Title/Abstract] OR "Animals"[MeSH Terms:noexp] OR "Models, Animal"[MeSH Terms] OR "Rodentia"[MeSH Terms] OR "Animal Experimentation"[MeSH Terms] OR "Animals, laboratory"[MeSH Terms] OR "Animal*" [Title/Abstract] OR "Mice" [Title/Abstract] OR "Mouse" [Title/Abstract] OR "Rats" [Title/Abstract] OR "Rat" [Title/Abstract] OR "In Vitro Techniques"[MeSH Terms] OR "In vitro" [Title/Abstract] OR "Cells"[MeSH Terms] OR "Cell*" [Title] OR "Child"[MeSH Terms] OR "Infant"[MeSH Terms] OR "Infant*" [Title/Abstract] OR "Adolescent"[MeSH Terms] OR "Adolescent*" [Title/Abstract] OR "Juvenile*" [Title/Abstract] OR "Pediatrics"[MeSH Terms] OR "Pediatric*" [Title/Abstract] OR "Paediatric*" [Title/Abstract] OR "Substance-Related Disorders"[MeSH Terms] OR "Addiction Medicine"[MeSH Terms] OR "Addiction Medicine" [Title/Abstract] OR "Surgical procedures, Operative"[MeSH Terms] OR "Surgical procedure*" [Title/Abstract] OR "Operating Rooms"[MeSH Terms] OR "Operating room*" [Title/Abstract] OR "Specialties, surgical"[MeSH Terms] OR "Intensive care units"[MeSH Terms] OR "Intensive care" [Title/Abstract] OR "Antineoplastic Agents"[MeSH Terms] OR "Chemotherap*" [Title/Abstract] OR "Psychiatric Department, Hospital"[MeSH Terms] OR "Psychiatric Department*" [Title/Abstract] OR "Psychiatric Service*" [Title/Abstract] OR "Psychiatric Ward*" [Title/Abstract] OR "Letter" [Publication Type] OR "Correspondence as topic" [MeSH Terms] OR "Editorial" [Publication Type] OR "Comment" [Publication Type] OR "Case Reports" [Publication Type] OR "Letter" [Title/Abstract] OR "Comment" [Title] OR "Case report*" [Title/Abstract] OR "Case" [Title] OR "Address" [Publication Type] OR "Clinical trial, veterinary" [Publication type])

EMBASE: (#1 AND #2 AND #3 AND #4) NOT #5

#1	('Nursing home'/exp OR 'Nursing home\$':ti,ab OR 'Institutionalized elderly'/exp OR Institutionalized NEAR/3 elderly OR 'Home for the aged'/exp OR home\$ NEAR/4 aged OR 'Long-term care' NEAR/3 facilit* OR 'Long term care' NEAR/3 facilit* OR 'Aged care facilit*':ti,ab OR 'Very elderly'/exp OR 'elderly':ti,ab OR 'Aged hospital patient'/exp OR 'Aged hospital patient':ti,ab OR 'Geriatric care'/exp OR 'Geriatrics'/de OR 'Geriatr*':ti,ab OR 'Older patient\$':ti,ab OR 'Older person\$':ti,ab OR 'Older adult\$':ti,ab OR 'Advanced age':ti,ab OR 'Octogenarian\$':ti,ab OR 'Nonagenarian\$':ti,ab OR 'Centenarian\$':ti,ab OR 'Frailty'/exp OR 'Frail elderly'/exp OR 'Frail':ti,ab OR 'Frailty':ti,ab OR 'Vulnerable population'/exp OR 'Dementia'/de OR 'Dement*':ti,ab OR 'Cognitive aging'/de OR 'Cognitive decline':ti,ab OR 'Cognitive impairment':ti,ab OR 'Cognitive defect'/de OR 'Cognitive dysfunction*':ti,ab OR 'Cognitive disfunction*':ti,ab OR 'Cognitive disorder\$':ti,ab OR 'Memory disorder'/de OR 'Neurocognitive disorders\$':ti,ab OR 'Neurocognitive decline':ti,ab OR 'Multiple chronic conditions'/exp OR 'Multimorbidit*':ti,ab OR 'Polypatholog*':ti,ab OR 'Multipatholog*':ti,ab OR 'Aged'/de OR 'Aging'/de OR 'Aging':ti,ab)
#2	('Terminal care'/de OR 'Terminal care':ti,ab OR 'Terminal treatment\$':ti,ab OR 'Terminal condition\$':ti,ab OR 'Terminal disease'/exp OR 'Terminal disease\$':ti,ab OR 'Terminal illness':ti,ab OR 'Terminally ill patient'/exp OR 'Terminally ill':ti,ab OR 'Terminally sick':ti,ab OR 'Terminal stadi*':ti,ab OR 'Terminal stag*':ti,ab OR 'Terminal phase\$':ti,ab OR 'Dying'/exp OR 'Deathbed':ti,ab OR 'Dying':ti,ab OR last NEAR/1 day\$ NEAR/3 life OR final NEAR/1 day\$ NEAR/3 life OR last NEAR/1 minute\$ NEAR/3 life OR final NEAR/1 minute\$ NEAR/3 life OR last NEAR/1 hour\$ NEAR/3 life OR final NEAR/1 hour\$ NEAR/3 life OR last NEAR/1 week\$ NEAR/3 life OR final NEAR/1 week\$ NEAR/3 life OR 'Final moment\$':ti,ab OR 'End-of life'/exp OR 'End of life':ti,ab OR 'End-of-life':ti,ab OR 'Palliative therapy'/de OR 'Palliative':ti,ab OR 'Palliative nursing'/exp OR 'Hospice'/exp OR 'Hospice care'/exp OR 'Hospice':ti,ab OR 'Palliation':ti,ab OR 'Supportive care'/exp OR 'Supportive care':ti,ab OR 'Supportive treatment\$':ti,ab OR 'Supportive Management':ti,ab OR 'Patient comfort'/exp OR 'Patient comfort':ti,ab OR 'Symptom relief':ti,ab)
#3	('clinical pharmacology'/exp OR 'clinical pharmacology':ti,ab OR 'drug therapy'/exp OR 'drug therapy':ti,ab OR 'analgesic agent'/exp OR 'muscarinic receptor blocking agent'/exp OR 'muscarinic receptor blocking agent':ti,ab OR 'muscarinic antagonists':ti,ab OR 'antimuscarinic':ti,ab OR 'cholinergic receptor blocking agent'/exp OR 'cholinergic receptor blocking agent':ti,ab OR 'cholinergic antagonists':ti,ab OR 'anticholinerg*':ti,ab OR 'scopolamine'/exp OR 'scopolamin\$':ti,ab OR 'scopolamin*':ti,ab OR 'hyoscine':ti,ab OR 'hyoscin*':ti,ab OR 'atropine derivative'/exp OR 'atropine derivative':ti,ab OR 'atropine':ti,ab OR 'atropin*':ti,ab OR 'scopolamine butyl bromide':ti,ab OR 'butylscopolammonium bromide':ti,ab OR 'butylscop*':ti,ab OR 'busc*':ti,ab OR 'butylbrom*':ti,ab OR 'hyoscinbutyl*':ti,ab OR 'glycopyrronium':ti,ab OR 'glycopyrrolate':ti,ab OR 'glycopyr*':ti,ab)
#4	('sputum'/exp OR 'sputum':ti,ab OR 'death rattle'/exp OR 'death rattle':ti,ab OR 'lung secretion'/exp OR 'lung secret*':ti,ab OR 'pulmonary secret*':ti,ab OR 'respiratory secret*':ti,ab OR 'bronchial secret*':ti,ab OR 'retained secret*':ti,ab OR 'airway secret*':ti,ab OR 'respiratory sound\$':ti,ab OR 'noisy respir*':ti,ab OR 'noisy breath\$':ti,ab OR 'noisy breath*':ti,ab OR 'noisy breathing'/exp OR 'terminal breath\$':ti,ab OR 'terminal breath*':ti,ab OR 'rattling breath\$':ti,ab OR 'rattling breath*':ti,ab OR 'gasping breath\$':ti,ab OR 'gasping breath*':ti,ab)
#5	('Veterinary medicine'/exp OR 'Veterinary study'/exp OR 'Veterinary':ti,ab OR 'Animal'/de OR 'Animal model'/exp OR 'Rodent'/exp OR 'Animal'/exp OR 'Animal experiment'/exp OR 'Experimental animal'/exp OR 'Animal tissue, cells or cell components'/exp OR 'Animal\$':ti,ab OR 'Mouse'/exp OR 'Mouse':ti,ab OR 'Mice':ti,ab OR 'Rat'/exp OR 'Rat\$':ti,ab OR 'In vitro study'/exp OR 'In vitro':ti,ab OR 'Cells'/exp OR 'Cell*':ti,ab OR 'Child'/exp OR 'Infant'/exp OR 'Infant\$':ti,ab OR 'Adolescent'/exp OR 'Adolescent\$':ti,ab OR 'Juvenile'/exp OR 'Juvenile\$':ti,ab OR 'Young adult'/exp OR 'Pedicatrics'/exp OR 'Pediatric\$':ti,ab OR 'Paediatric\$':ti,ab OR 'Drug dependence':exp OR 'Addiction medicine'/exp OR 'Addiction Medicine':ti,ab OR 'Surgery'/exp OR 'Surgical procedure\$':ti,ab OR 'Operating room'/exp OR 'Operating rooms\$':ti,ab OR 'Intensive care'/exp OR 'Intensive care unit\$':ti,ab OR 'Cancer chemotherapy'/exp OR Palliative NEAR/2 chemotherapy OR 'Psychiatric department'/exp OR 'Psychiatric department\$':ti,ab OR 'Psychiatric service\$':ti,ab OR 'Psychiatric ward\$':ti,ab OR 'Letter'/exp OR 'Note'/exp OR 'Editorial'/exp OR 'Case report'/exp OR 'Letter\$':ti,ab OR 'Editorial':ti,ab OR 'Case':ti OR 'Case report':ti,ab)

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#1	("Nursing Homes"[MeSH Terms] OR "Nursing Home"[Title/Abstract] OR "Housing for the Elderly"[MeSH Terms] OR "Housing for the Elderly"[Title/Abstract] OR "Homes for the Aged"[MeSH Terms] OR "Homes for the Aged"[Title/Abstract] OR "Long-term care"[MeSH Terms] OR "Long-term care"[Title/Abstract] OR "Long term care"[Title/Abstract] OR "Aged care faciliti*[Title/Abstract] OR "Aged, 80 and over"[MeSH Terms] OR "Elderly"[Title/Abstract] OR "Geriatrics"[MeSH Terms] OR "Geriatr*[Title/Abstract] OR "Older patient*[Title/Abstract] OR "Older person*[Title/Abstract] OR "Older adult*[Title/Abstract] OR "Advanced age"[Title/Abstract] OR "Octogenarian*[Title/Abstract] OR "Nonagenarian*[Title/Abstract] OR "Centenarian*[Title/Abstract] OR "Frailty"[MeSH Terms] OR "Frail elderly"[Mesh] OR "Frail*[Title/Abstract] OR "Vulnerable"[Title/Abstract] OR "Dementia"[MeSH Terms] OR "Dement*[Title/Abstract] OR "Cognitive disease*[Title/Abstract] OR "Cognitive decline"[Title/Abstract] OR "Cognitive impairment"[Title/Abstract] OR "Multimorbidity"[MeSH Terms] OR "Multimorbidit*[Title/Abstract] OR "Polypatholog*[Title/Abstract] OR "Multipatholog*[Title/Abstract] OR "Cognitive dysfunction"[MeSH Terms] OR "Cognitive dysfunction*[Title/Abstract] OR "Cognitive disfunction*[Title/Abstract] OR "Cognitive disorder*[Title/Abstract] OR "Neurocognitive disorders"[MeSH Terms] OR "Neurocognitive disorder*[Title/Abstract] OR "Neurocognitive decline"[Title/Abstract] OR "Neurocognitive impairment"[Title/Abstract] OR "Neurocognitive disease"[Title/Abstract] OR "Aged"[MeSH Terms] OR "Aging"[Title/Abstract])
#2	("Terminal Care"[MeSH Terms] OR "Terminal care"[Title/Abstract] OR "Terminal Treatment*[Title/Abstract] OR "Terminal Condition"[Title/Abstract] OR "Terminal Disease*[Title/Abstract] OR "Terminal illness"[Title/Abstract] OR "Terminally ill"[Title/Abstract] OR "Terminally Sick*[Title/Abstract] OR "Terminal Stadi*[Title/Abstract] OR "Terminal Stag*[Title/Abstract] OR "Terminal Phase*[Title/Abstract] OR "Death"[MeSH Terms] OR "Deathbed"[Title/Abstract] OR "Dying"[Title/Abstract] OR "Last Day"[Title/Abstract] OR "Last days"[Title/Abstract] OR "Final day"[Title/Abstract] OR "Final days"[Title/Abstract] OR "Last Minute*[Title/Abstract] OR "Final minute*[Title/Abstract] OR "Last Hour*[Title/Abstract] OR "Final hour*[Title/Abstract] OR "Last Week*[Title/Abstract] OR "Final week*[Title/Abstract] OR "Final moment*[Title/Abstract] OR "End of life"[Title/Abstract] OR "End-of-life"[Title/Abstract] OR "Palliative Medicine"[MeSH Terms] OR "Hospice and palliative care nursing"[MeSH Terms] OR "Palliative care"[MeSH Terms] OR "Palliative"[Title/Abstract] OR "Hospice Care"[MeSH Terms] OR "Hospice"[Title/Abstract] OR "Life Treathening"[Title/Abstract] OR "Palliation"[Title/Abstract] OR "Supportive Care"[Title/Abstract] OR "Supportive Treatment"[Title/Abstract] OR "Supportive Management"[Title/Abstract] OR "Patient comfort"[MeSH Terms] OR "Comfort*[Title/Abstract] OR "Symptom relief"[Title/Abstract])
#3	("Pharmacology"[MeSH Terms] OR "Pharmac*[Title/Abstract] OR "Drug therapy"[MeSH Terms] OR "Drug therap*[Title/Abstract] OR "Analgesics"[MeSH Terms] OR "Analgesic*[Title/Abstract] OR "Antipyretics"[MeSH Terms] OR "Antipyretic*[Title/Abstract] OR "Anti-Inflammatory Agents, Non-Steroidal"[MeSH Terms] OR "Non-Steroidal Anti-Inflammatory"[Title/Abstract] OR "NSAID*[Title/Abstract] OR "Acetaminophen"[MeSH Terms] OR "Acetaminophen"[Title/Abstract] OR "Paracetamol"[Title/Abstract] OR "Aspirin"[MeSH Terms] OR "Aspirin*[Title/Abstract] OR "Dipyrrone"[MeSH Terms] OR "Dipyrrone"[Title/Abstract] OR "Metamizol*[Title/Abstract] OR "Methamizol*[Title/Abstract] OR "Novalg*[Title/Abstract] OR "Dipyrronium"[Title/Abstract] OR "Pyrargin"[Title/Abstract] OR "Novaminsulfon*[Title/Abstract] OR "Sulpyrin*[Title/Abstract] OR "Optalgin*[Title/Abstract] OR "Noramidopyrin*[Title/Abstract] OR "Novamid*[Title/Abstract] OR "Methampyrone*[Title/Abstract] OR "Algopyrin*[Title/Abstract] OR "Analgin*[Title/Abstract] OR "Naron*[Title/Abstract] OR "Nefopam"[MeSH Terms] OR "Nefopam"[Title/Abstract] OR "Acedofenac" [Supplementary Concept] OR "Diclofenac"[Mesh Terms] OR "Diclofenac"[Title/Abstract] OR "Ibuprofen"[MeSH Terms] OR "Ibuprofen"[Title/Abstract] OR "Naproxen"[MeSH Terms] OR "Naproxen"[Title/Abstract] OR "Meloxicam"[MeSH Terms] OR "Meloxicam"[Title/Abstract] OR "Piroxicam"[MeSH Terms] OR "Piroxicam"[Title/Abstract] OR "Tenoxicam" [Supplementary Concept] OR "Tenoxicam"[Title/Abstract] OR "Celecoxib"[MeSH Terms] OR "Celecoxib"[Title/Abstract] OR "Etoricoxib"[MeSH Terms] OR "Etoricoxib"[Title/Abstract] OR "Parecoxib" [Supplementary Concept] OR "Parecoxib"[Title/Abstract] OR "Glucocorticoids"[MeSH Terms] OR "Glucocorticoid*[Title/Abstract] OR "Corticosteroid*[Title/Abstract] OR "Betamethasone"[MeSH Terms] OR "Betamethason*[Title/Abstract] OR "Dexamethasone"[MeSH Terms] OR "Dexamethason*[Title/Abstract] OR "Methylprednisolone"[MeSH Terms] OR "Methylprednisolon*[Title/Abstract] OR "Prednisolone"[MeSH Terms] OR "Prednisolon*[Title/Abstract] OR "Prednisone"[MeSH Terms] OR "Prednison*[Title/Abstract] OR "Hydrocortisone"[MeSH Terms] OR "Hydrocortison*[Title/Abstract])
#4	("Fever"[MeSH Terms] OR "Fever"[Title/Abstract] OR "Hyperthermia"[MeSH Terms] OR "Hyperthermia"[MeSH Terms] OR "Pyrexia*[Title/Abstract] OR "Hot Temperature"[MeSH Terms] OR "Temperature"[Title/Abstract])
#5	("Veterinary" [Subheading] OR "Veterinary"[Title/Abstract] OR "Animals"[MeSH Terms:noexp] OR "Models, Animal"[MeSH Terms] OR "Rodentia"[MeSH Terms] OR "Animal Experimentation"[MeSH Terms] OR "Animals, laboratory"[MeSH Terms] OR "Animal*[Title/Abstract] OR "Mice"[Title/Abstract] OR "Mouse"[Title/Abstract] OR "Rats"[Title/Abstract] OR "Rat"[Title/Abstract] OR "In Vitro Techniques"[MeSH Terms] OR "In vitro"[Title/Abstract] OR "Cells"[MeSH Terms] OR "Cell*[Title] OR "Child"[MeSH Terms] OR "Infant"[MeSH Terms] OR "Infant*[Title/Abstract] OR "Adolescent"[MeSH Terms] OR "Adolescent*[Title/Abstract] OR "Juvenile*[Title/Abstract] OR "Pediatrics"[MeSH Terms] OR "Pediatric*[Title/Abstract] OR "Paediatric*[Title/Abstract] OR "Substance-Related Disorders"[MeSH Terms] OR "Addiction Medicine"[MeSH Terms] OR "Addiction Medicine"[Title/Abstract] OR "Surgical procedures, Operative"[MeSH Terms] OR "Surgical procedure*[Title/Abstract] OR "Operating Rooms"[MeSH Terms] OR "Operating room*[Title/Abstract] OR "Specialties, surgical"[MeSH Terms] OR "Intensive care units"[MeSH Terms] OR "Intensive care"[Title/Abstract] OR "Antineoplastic Agents"[MeSH Terms] OR "Chemotherapy*[Title/Abstract] OR "Psychiatric Department, Hospital"[MeSH Terms] OR "Psychiatric Department*[Title/Abstract] OR "Psychiatric Service*[Title/Abstract] OR "Psychiatric Ward*[Title/Abstract] OR "Letter"[Publication Type] OR "Correspondence as topic"[MeSH Terms] OR "Editorial"[Publication Type] OR "Comment"[Publication Type] OR "Case Reports"[Publication Type] OR "Letter"[Title/Abstract] OR "Comment"[Title] OR "Case report*[Title/Abstract] OR "Case"[Title] OR "Address"[Publication Type] OR "Clinical trial, veterinary"[Publication type])

EMBASE: (#1 AND #2 AND #3 AND #4) NOT #5

#1	('Nursing home'/exp OR 'Nursing home\$':ti,ab OR 'Institutionalized elderly'/exp OR Institutionalized NEAR/3 elderly OR 'Home for the aged'/exp OR home\$ NEAR/4 aged OR 'Long-term care' NEAR/3 facilit* OR 'Long term care' NEAR/3 facilit* OR 'Aged care facilit*':ti,ab OR 'Very elderly'/exp OR 'elderly':ti,ab OR 'Aged hospital patient'/exp OR 'Aged hospital patient':ti,ab OR 'Geriatric care'/exp OR 'Geriatrics'/de OR 'Geriatr*':ti,ab OR 'Older patient\$':ti,ab OR 'Older person\$':ti,ab OR 'Older adult\$':ti,ab OR 'Advanced age':ti,ab OR 'Octogenarian\$':ti,ab OR 'Nonagenarian\$':ti,ab OR 'Centenarian\$':ti,ab OR 'Frailty'/exp OR 'Frail elderly'/exp OR 'Frail':ti,ab OR 'Frailty':ti,ab OR 'Vulnerable population'/exp OR 'Dementia'/de OR 'Dement*':ti,ab OR 'Cognitive aging'/de OR 'Cognitive decline':ti,ab OR 'Cognitive impairment':ti,ab OR 'Cognitive defect'/de OR 'Cognitive dysfunction*':ti,ab OR 'Cognitive disfunction*':ti,ab OR 'Cognitive disorder\$':ti,ab OR 'Memory disorder'/de OR 'Neurocognitive disorder\$':ti,ab OR 'Neurocognitive decline':ti,ab OR 'Multiple chronic conditions'/exp OR 'Multimorbidit*':ti,ab OR 'Polypatholog*':ti,ab OR 'Multipatholog*':ti,ab OR 'Aged'/de OR 'Aging'/de OR 'Aging':ti,ab)
#2	('Terminal care'/de OR 'Terminal care':ti,ab OR 'Terminal treatment\$':ti,ab OR 'Terminal condition\$':ti,ab OR 'Terminal disease'/exp OR 'Terminal disease\$':ti,ab OR 'Terminal illness':ti,ab OR 'Terminally ill patient'/exp OR 'Terminally ill':ti,ab OR 'Terminally sick':ti,ab OR 'Terminal stadi*':ti,ab OR 'Terminal stag*':ti,ab OR 'Terminal phase\$':ti,ab OR 'Dying'/exp OR 'Deathbed':ti,ab OR 'Dying':ti,ab OR last NEAR/1 day\$ NEAR/3 life OR final NEAR/1 day\$ NEAR/3 life OR last NEAR/1 minute\$ NEAR/3 life OR final NEAR/1 minute\$ NEAR/3 life OR last NEAR/1 hour\$ NEAR/3 life OR final NEAR/1 hour\$ NEAR/3 life OR last NEAR/1 week\$ NEAR/3 life OR final NEAR/1 week\$ NEAR/3 life OR 'Final moment\$':ti,ab OR 'End-of life'/exp OR 'End of life':ti,ab OR 'End-of-life':ti,ab OR 'Palliative therapy'/de OR 'Palliative':ti,ab OR 'Palliative nursing'/exp OR 'Hospice'/exp OR 'Hospice care'/exp OR 'Hospice':ti,ab OR 'Palliation':ti,ab OR 'Supportive care'/exp OR 'Supportive care':ti,ab OR 'Supportive treatment\$':ti,ab OR 'Supportive Management':ti,ab OR 'Patient comfort'/exp OR 'Patient comfort':ti,ab OR 'Symptom relief':ti,ab)
#3	('clinical pharmacology'/exp OR 'clinical pharmacology':ti,ab OR 'drug therapy'/exp OR 'drug therapy':ti,ab OR 'analgesic agent'/exp OR 'analges*':ti,ab OR 'Non-Steroidal Anti-Inflammatory':ti,ab OR 'NSAID\$':ti,ab OR 'paracetamol'/exp OR 'paracetamol':ti,ab OR 'acetaminophen':ti,ab OR 'acetylsalicylic acid'/exp OR 'aspirin':ti,ab OR 'dipyrrone'/exp OR 'dipyrrone':ti,ab OR 'metamizol\$':ti,ab OR 'methamizol\$':ti,ab OR 'novalg*':ti,ab OR 'dipyrionium':ti,ab OR 'biopyrrin'/exp OR 'biopyrrin':ti,ab OR 'pyralgin':ti,ab OR 'novaminsulfon':ti,ab OR 'Sulpyrin\$':ti,ab OR 'optalgin'/exp OR 'optalgin':ti,ab OR 'Noramidopyrin\$':ti,ab OR 'Novamid*':ti,ab OR 'Methampyrone\$':ti,ab OR 'Algopyrin\$':ti,ab OR 'Analgin\$':ti,ab OR 'naron\$':ti,ab OR 'naron*':ti,ab OR 'nefopam':ti,ab OR 'Aceclofenac':ti,ab OR 'diclofenac':ti,ab OR 'Ketorolac':ti,ab OR 'ibuprofen':ti,ab OR 'naproxen':ti,ab OR 'meloxicam':ti,ab OR 'piroxicam':ti,ab OR 'tenoxicam':ti,ab OR 'celecoxib':ti,ab OR 'Etoricoxib':ti,ab OR 'Parecoxib':ti,ab OR 'glucocorticoid'/exp OR 'glucocorticoid\$':ti,ab OR 'corticosteroid'/exp OR 'corticosteroid\$':ti,ab OR 'Betamethason\$':ti,ab OR 'Betamethason*':ti,ab OR 'Dexamethason\$':ti,ab OR 'Methylprednisolon\$':ti,ab OR 'Methylprednisolon*':ti,ab OR 'prednisolon\$':ti,ab OR 'prednisolon*':ti,ab OR 'Prednison\$':ti,ab OR 'Prednison*':ti,ab OR 'Hydrocortison\$':ti,ab OR 'Hydrocortison\$':ti,ab)
#4	('fever'/exp OR 'fever':ti,ab OR 'hyperthermia'/exp OR 'hyperthermia':ti,ab OR 'pyrexia':ti,ab OR 'pyrexia':ti,ab OR 'high temperature'/exp OR 'temperature':ti,ab)
#5	('Veterinary medicine'/exp OR 'Veterinary study'/exp OR 'Veterinary':ti,ab OR 'Animal'/de OR 'Animal model'/exp OR 'Rodent'/exp OR 'Animal experiment'/exp OR 'Experimental animal'/exp OR 'Animal tissue, cells or cell components'/exp OR 'Animal\$':ti,ab OR 'Mouse'/exp OR 'Mouse':ti,ab OR 'Mice':ti,ab OR 'Rat'/exp OR 'Rat\$':ti,ab OR 'In vitro study'/exp OR 'In vitro':ti,ab OR 'Cells'/exp OR 'Cell*':ti,ab OR 'Child'/exp OR 'Infant'/exp OR 'Infant\$':ti,ab OR 'Adolescent'/exp OR 'Adolescent\$':ti,ab OR 'Juvenile'/exp OR 'Juvenile\$':ti,ab OR 'Young adult'/exp OR 'Pedicatrics'/exp OR 'Pediatric\$':ti,ab OR 'Paediatric\$':ti,ab OR 'Drug dependence':exp OR 'Addiction medicine'/exp OR 'Addiction Medicine':ti,ab OR 'Surgery'/exp OR 'Surgical procedure\$':ti,ab OR 'Operating room'/exp OR 'Operating room\$':ti,ab OR 'Intensive care'/exp OR 'Intensive care unit\$':ti,ab OR 'Cancer chemotherapy'/exp OR Palliative NEAR/2 chemotherapy OR 'Psychiatric department'/exp OR 'Psychiatric department\$':ti,ab OR 'Psychiatric service\$':ti,ab OR 'Psychiatric ward\$':ti,ab OR 'Letter'/exp OR 'Note'/exp OR 'Editorial'/exp OR 'Case report'/exp OR 'Letter\$':ti,ab OR 'Editorial':ti,ab OR 'Case':ti OR 'Case report':ti,ab)

Psychological symptoms (agitation+delirium+anxiety): clinical research question 6

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#2	("Terminal Care"[MeSH Terms] OR "Terminal care"[Title/Abstract] OR "Terminal Treatment*[Title/Abstract] OR "Terminal Condition"[Title/Abstract] OR "Terminal Disease*[Title/Abstract] OR "Terminal illness"[Title/Abstract] OR "Terminally ill"[Title/Abstract] OR "Terminally Sick*[Title/Abstract] OR "Terminal Stadi*[Title/Abstract] OR "Terminal Stag*[Title/Abstract] OR "Terminal Phase*[Title/Abstract] OR "Death"[MeSH Terms] OR "Deathbed"[Title/Abstract] OR "Dying"[Title/Abstract] OR "Last Day"[Title/Abstract] OR "Last days"[Title/Abstract] OR "Final day"[Title/Abstract] OR "Final days"[Title/Abstract] OR "Last Minute*[Title/Abstract] OR "Final minute*[Title/Abstract] OR "Last Hour*[Title/Abstract] OR "Final hour*[Title/Abstract] OR "Last Week*[Title/Abstract] OR "Final week*[Title/Abstract] OR "Final moment*[Title/Abstract] OR "End of life"[Title/Abstract] OR "End-of-life"[Title/Abstract] OR "Palliative Medicine"[MeSH Terms] OR "Hospice and palliative care nursing"[MeSH Terms] OR "Palliative care"[MeSH Terms] OR "Palliative"[Title/Abstract] OR "Hospice Care"[MeSH Terms] OR "Hospice"[Title/Abstract] OR "Life Treathening"[Title/Abstract] OR "Palliation"[Title/Abstract] OR "Supportive Care"[Title/Abstract] OR "Supportive Treatment"[Title/Abstract] OR "Supportive Management"[Title/Abstract] OR "Patient comfort"[MeSH Terms] OR "Comfort*[Title/Abstract] OR "Symptom relief"[Title/Abstract])
#3	("Pharmacology"[MeSH Terms] OR "Pharmacol*[Title/Abstract] OR "Drug therapy"[MeSH Terms] OR "Drug therap*[Title/Abstract] OR "Benzodiazepines"[MeSH Terms] OR "Benzodiazepin*[Title/Abstract] OR "Anti-Anxiety Agents"[MeSH Terms] OR "Anti-Anxiety"[Title/Abstract] OR "Diazepam"[MeSH Terms] OR "Diazepam"[Title/Abstract] OR "Alprazolam"[MeSH Terms] OR "Alprazolam"[Title/Abstract] OR "Bromazepam"[MeSH Terms] OR "Bromazepam"[Title/Abstract] OR "Clobazam"[MeSH Terms] OR "Clobazam"[Title/Abstract] OR "Clonazepam"[MeSH Terms] OR "Clonazepam"[Title/Abstract] OR "Flunitrazepam"[MeSH Terms] OR "Flunitrazepam"[Title/Abstract] OR "Flurazepam"[MeSH Terms] OR "Flurazepam"[Title/Abstract] OR "Lorazepam"[MeSH Terms] OR "Lorazepam"[Title/Abstract] OR "Lormetazepam"[Supplementary Concept] OR "Lormetazepam"[Title/Abstract] OR "Midazolam"[MeSH Terms] OR "Midazolam"[Title/Abstract] OR "Nitrazepam"[MeSH Terms] OR "Nitrazepam"[Title/Abstract] OR "Nordazepam"[MeSH Terms] OR "Nordazepam"[Title/Abstract] OR "Oxazepam"[MeSH Terms] OR "Oxazepam"[Title/Abstract] OR "Prazepam"[MeSH Terms] OR "Prazepam"[Title/Abstract] OR "Zolpidem"[MeSH Terms] OR "Zolpidem"[Title/Abstract] OR "Zopiclon"[Title/Abstract] OR "Ethylloflazepat*[Title/Abstract] OR "Valerian"[MeSH Terms] OR "Valerian"[Title/Abstract] OR "Passiflora"[MeSH Terms] OR "Passiflora"[Title/Abstract] OR "Antipsychotic Agents"[MeSH Terms] OR "Antipsychotic*[Title/Abstract] OR "Neuroleptic*[Title/Abstract] OR "Phenothiazines"[MeSH Terms] OR "Phenothiazin*[Title/Abstract] OR "Methotrimeprazine"[MeSH Terms] OR "Methotrimeprazin*[Title/Abstract] OR "Levopromazin*[Title/Abstract] OR "Prothipendyl"[Title/Abstract] OR "Thioxanthenes"[MeSH Terms] OR "Thioxanthen*[Title/Abstract] OR "Flupenthixol"[MeSH Terms] OR "Flupenthixol"[Title/Abstract] OR "Fluanxol"[Title/Abstract] OR "Clopenthixol"[MeSH Terms] OR "Clopenthixol"[Title/Abstract] OR "Zuclopenthixol"[Title/Abstract] OR "Clotiapi*[Title/Abstract] OR "Butyrophenones"[MeSH Terms] OR "Butyrophenon*[Title/Abstract] OR "Bromperidol*[Title/Abstract] OR "Droperidol"[MeSH Terms] OR "Droperidol*[Title/Abstract] OR "Haloperidol"[MeSH Terms] OR "Haloperidol*[Title/Abstract] OR "Haldol"[Title/Abstract] OR "Aripiprazol*[Title/Abstract] OR "Asenapine"[Supplementary concept] OR "Asenapin*[Title/Abstract] OR "Cariprazine"[Supplementary Concept] OR "Cariprazin*[Title/Abstract] OR "Clozapine"[MeSH Terms] OR "Clozapine"[Title/Abstract] OR "Olanzapine"[MeSH Terms] OR "Olanzapine"[Title/Abstract] OR "Paliperidone Palmitate"[MeSH Terms] OR "Paliperidone Palmitate"[Title/Abstract] OR "Paliperidon*[Title/Abstract] OR "Quetiapine Fumarate"[MeSH Terms] OR "Quetiapin*[Title/Abstract] OR "Risperidone"[MeSH Terms] OR "Risperidon*[Title/Abstract] OR "Sertindol*[Title/Abstract] OR "Pimozide"[MeSH Terms] OR "Pimozide"[Title/Abstract] OR "Pipamperon*[Title/Abstract] OR "Amisulpride"[MeSH Terms] OR "Amisulprid*[Title/Abstract] OR "Sulpiride"[MeSH Terms] OR "Sulpirid*[Title/Abstract] OR "Tiapride Hydrochloride"[MeSH Terms] OR "Tiapride Hydrochloride"[Title/Abstract] OR "Tiaprid*[Title/Abstract] OR "Donepezil"[MeSH Terms] OR "Donepezil"[Title/Abstract] OR "Rivastigmine"[MeSH Terms] OR "Rivastigmin*[Title/Abstract] OR "Galantamine"[MeSH Terms] OR "Galantamine"[Title/Abstract])
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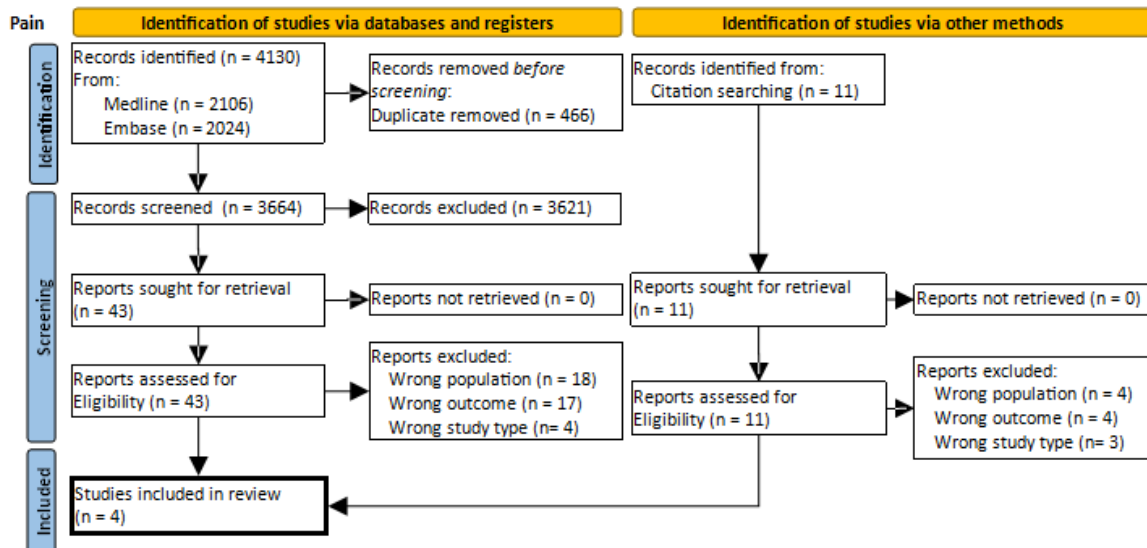
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EMBASE: (#1 AND #2 AND #3 AND #4) NOT #5

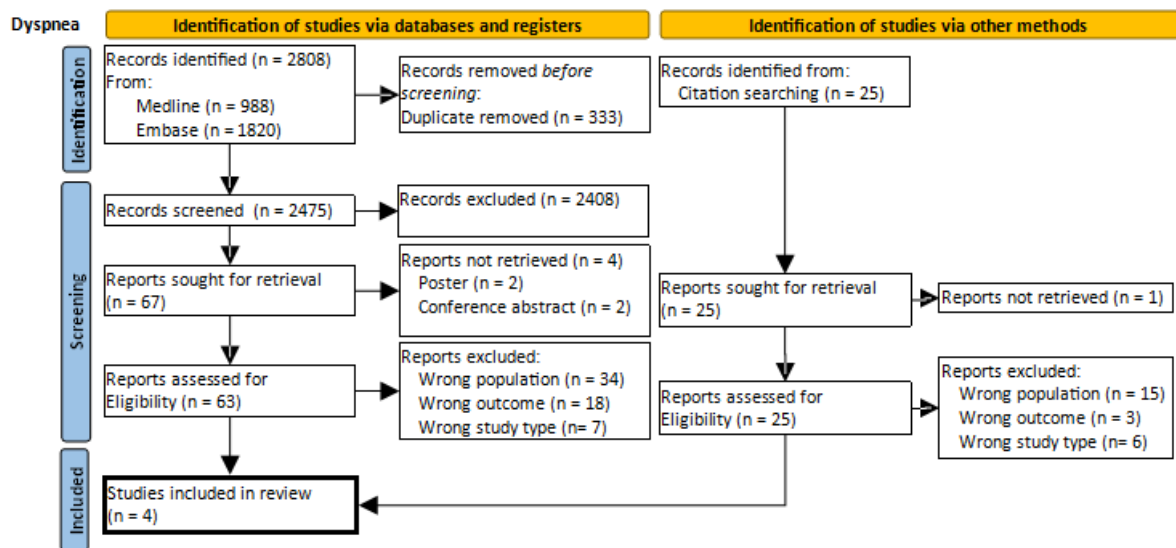
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Appendix I: Prisma flowcharts ADAPTE procedure

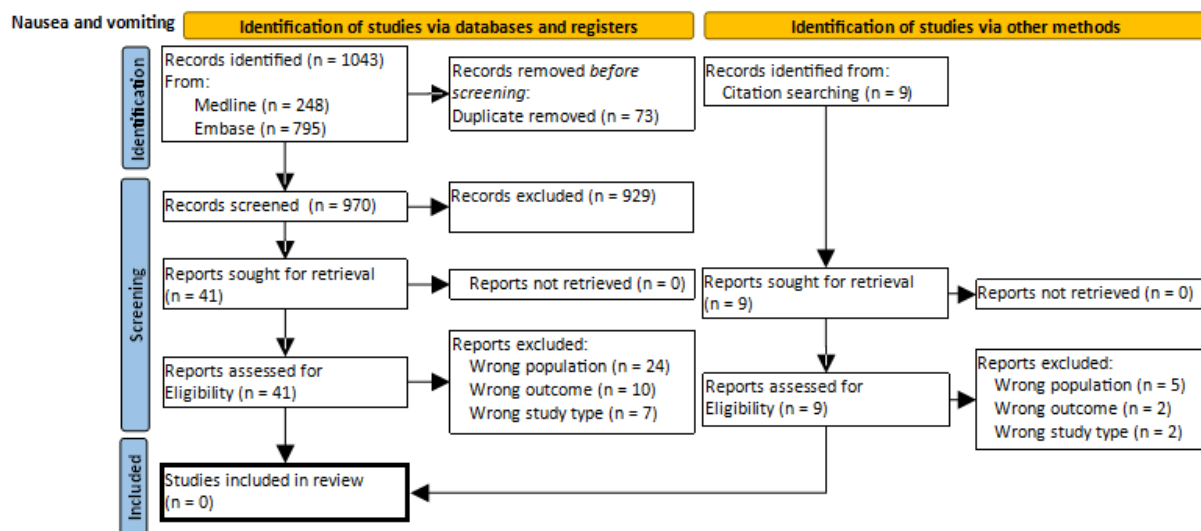
Pain: clinical research question 1



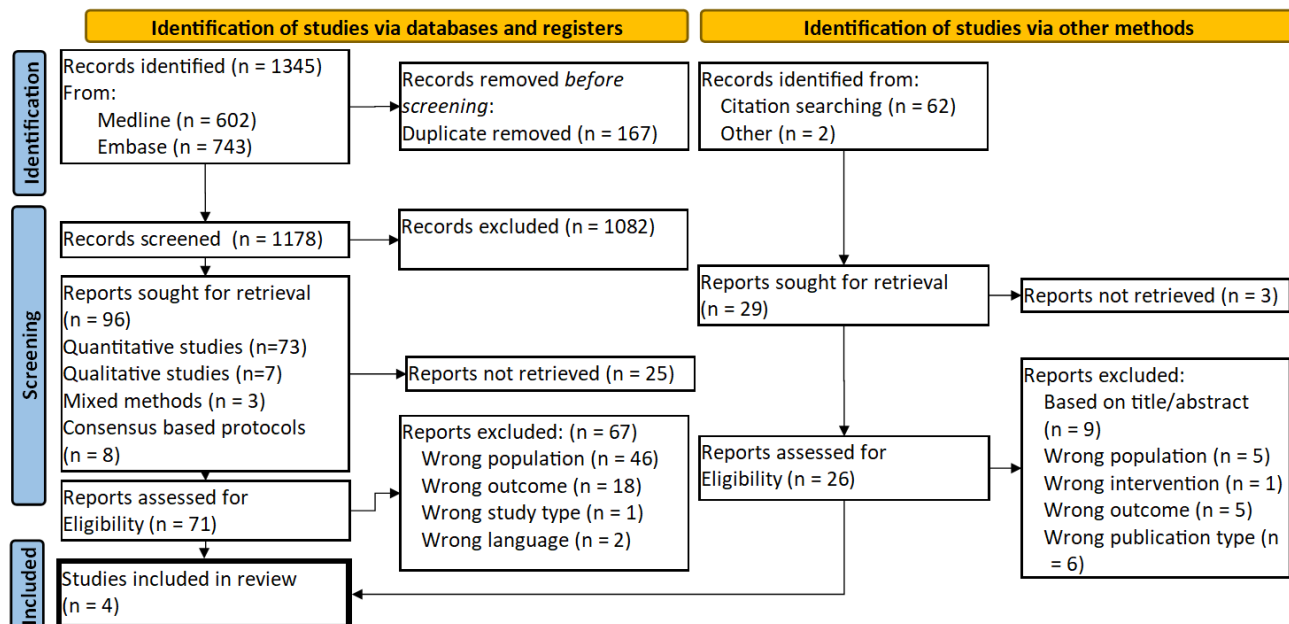
Dyspnea: clinical research question 2



Nausea and vomiting: clinical research question 3



Psychological symptoms (agitation+delirium+anxiety): clinical research question 6 (a+b+c)



Appendix J: Clinical evidence tables of the systematic literature review within ADAPTE procedure

Pain: clinical research question 1

Author	Study design	Setting and population	Results		
			Medication	Symptom control	Adverse effects/ survival
Pain					
Fürst et al. (Sweden, 2020)	Quantitative, observational cohort study	<u>Setting:</u> Specialized in-patient palliative care unit (in hospital) <u>Population:</u> n=47 Age: Mean 75.6 y (SD 12.1) Oncology diagnosis: 91% Opioid naive: 0% Survival: Median 5 days (IQR 9) Frailty characteristics: Mean ECOG performance scale 3.3 (SD 1)	<u>Drug: Opioids</u> (morphine, hydromorphone, oxycodone or methadone) used in CSCI (doses in MEDD, mg/24h) <u>Daily dose:</u> Increased from median 123 (IQR 151, range 22.5- 1020), mean 184 (SD 181) to median 150 (IQR 210, range 30- 870), mean 205 (SD 182) from day 0 to day 3 (p < 0.05).	<u>Symptom (scale): Pain</u> (IPOS) <u>Frequency:</u> Percentage of patients with severe or overwhelming pain decreased from 45% to 19% from day 0 to day 3 (p < 0.001) <u>Symptom score:</u> Mean IPOS decreased from 2.2 (SD 1.2) to 1.5 (SD 1.2) from day 0 to day 3 (p ≤ 0.001) <u>Assessments by:</u> Bedside nurses	- No significant adverse effects
Golcic et al. (Croatia, 2018)	Quantitative, retrospective study of medical records	<u>Setting:</u> Hospice <u>Population:</u> n=667 Age: mean 77.6 y Oncology diagnosis: 84.3% Opioid naive: / Average stay in hospice: 17d 83.8% died in hospice Frailty characteristics: Croatian patient categorization system 3.24 (SD 0.92); 52.1% bedridden at admission	<u>Drug: Fentanyl patch</u> (doses in OME, mg/24h) (43.8%, n=292) <u>Daily dose:</u> Mean initial opioid dose 105.21 (SD 91.80), last opioid dose 144.30 (SD 97.30) <u>Drug: Buprenorphine</u> (doses in OME, mg/24h) (13.9%, n=93) <u>Daily dose:</u> Mean initial opioid dose 102.11 (SD 89.16), last opioid dose 133.76 (SD 89.97) (using the 1/80 conversion) <u>Drug: peroral opioids</u> (morphine, oxycodone, fentanyl, tramadol, and methadone) (doses in OME, mg/24h) (13.2%, n=88) <u>Daily dose:</u> Mean initial opioid dose 15.07 (SD 17.47), last opioid dose 18.15 (SD 18.63) <u>Additional pain medication</u> <u>Paracetamol:</u> Used in +/- 23% of fentanyl and buprenorphine groups with an average dose of 1g/d. <u>NSAIDs:</u> used in +/- 36% of fentanyl and buprenorphine groups. Paracetamol or NSAID use did not influence opioid dose.	No symptom assessment, but no differences were observed In terms of dose adaptations, use of additional analgesics or performance score between fentanyl and buprenorphine groups.	- No adverse effects mentioned. - No differences in survival between the different opioid patches or per-oral use.
Golcic et al. (Croatia, 2020)	Quantitative, retrospective cohort study	<u>Setting:</u> Hospice <u>Population:</u> n=137 divided into groups >65y and <65y. > 65y: n=109, average age 75.2y (SD 6.3) <65y: n=28, average age 58.6 (SD 6.7) Oncology diagnosis: 100% Opioid naive: 31.2% and 14.3% for >65y and <65y groups respectively (p=0.07) Time at hospice: 17.54d (SD 19.7) 83.9% died while in hospice.	<u>Drug: Opioids</u> (tramadol (n=69), fentanyl transdermal patches (n=45), buprenorphine transdermal patches (n=20), oral morphine (n=14)) (doses in OME mg/24h) <u>Starting dose:</u> / <u>Daily dose:</u> (>65y vs <65y) Trend towards lower average opioid doses on admission: 95.42 vs 115.19 (p=0.36) Lower average dose of opioid during the last week in hospice OME 109.95 vs 165.61 (=0.03) <u>Frequency:</u> Trend towards fewer older patients using opioids on admission and during last week. Admission: 68.8% vs 85 7% (p=0.07), last week: 75.23 vs 89.3% (p=0.11) <u>Other medication:</u> (>65y vs <65y) Lower use of <u>NSAIDs</u> 11% vs 28.6% (p=0.02)	<u>Symptom (scale): Pain</u> (frequency of pain: EORTC QLQ-C15- PAL, pain symptom scores: ESAS) <u>Frequency:</u> (>65y vs <65y) Average scores on admission: 54.28 (SD 24.68) vs 60.71 (SD 29.82) (p=0.24) <u>Symptom score(ESAS):</u> (>65y vs <65y) Average scores on admission: 3.59 (SD 2.55) vs 5.14 (SD 2.33) (p=0.01) <u>Assessment by:</u> Not specified	- No adverse effects mentioned. - Use of NSAIDs on admission was linked with longer survival (26.7vs 15.9d) (p=0.001)

		Frailty characteristics (PPS): 32.1 (SD 11.5) and 37.1 (SD 15.6) for >65y and <65y groups respectively	Higher use of paracetamol 33% vs 10.7% (p=0.02) (average dose +/-1g/d)		
Rashidi et al. (Australia, 2011)	Quantitative, retrospective study of medical records	Setting: 2 palliative care units (one in hospital, one community based) Population: n=205 ≥80y: n=105, mean age 85.59y (SD 4.65) 50-70y: n=100, mean age 62.76y (SD 5.76) Oncology diagnosis: ≥80y 70.5% and 89% in group 50-70y Opioid naïve: / Frailty characteristics: Mean AKPS score 2 days before death: ≥80y 23.49 (SD 12.24) and 27.67 in group 50-70y (SD 15.31)	Drug: Opioids (morphine, ...), modes of administration not specified (doses in Parenteral morphine equivalent mg/24h) Daily doses: (≥80y vs 50-70y) 2d before death: Mean 29.87 vs 61.88 (p=0.024) 1d before death: Mean 32.37 vs 67.85 (p=0.008) Day of death: Mean 20.51 vs 40.77 (p=0.161) Total dose over last 3d: Mean 82.75 vs 170.49 mg/72h (p=0.027) Drug: Midazolam, modes of administration not specified (mg/24h) Daily dose: (≥80y vs 50-70y) 2d before death: Mean 3.26 vs 4.89 (p=0.129) 1d before death: Mean 4.84 vs 8.50 (p=0.011) Day of death: Mean 4.04 vs 5.74 (p=0.179) Total dose over last 3d: Mean 12.14 vs 19.13 mg/72h (p=0.022)	Symptom (scale): pain (PCOC) Symptom scores: (≥80y vs 50-70y) Mean pain score 2d before death: 2.70 (SD 2.90) vs 2.00 (SD 2.32) (p=0.082) Mean pain score day of death: 1.50 (SD 1.81) vs 1.84 (SD 2.99) (p=0.474) Symptom (scale): dyspnea (PCOC) Symptom scores: (≥80y vs 50-70y) Mean breathing problems 2d before death 2.71 (SD 3.13) vs 1.94 (SD 2.69) (p=0.087) Mean breathing problems day of death 2.71 (SD 3.21) vs 2.31 (SD 3.37) (p=0.517) Assessments by: Patient self-report or by 2 involved clinical staff members	- No significant adverse effects observed - Mean survival 13d (range 0-108) vs 19d (range 0-73), (p≤0.001) (≥80y vs 50-70y)

Dyspnea: clinical research question 2

Author	Study design	Setting and population	Results		
			Medication	Symptom control	Adverse effects/survival
Dyspnea					
Benitez-Rosario et al. (Spain, 2018)	Quantitative, retrospective cohort study	<u>Setting:</u> Tertiary palliative care unit (in hospital) <u>Population:</u> n=72 < 75y, n=36; 75-80y n=15; ≥ 80y n=21 Oncology diagnosis: 93% Opioid naïve: / 60% died within first 6 days. Frailty characteristics: median PPS 30 (IQR 20-40)	<u>Drug:</u> Fentanyl, 85% intravenous and 15% subcutaneous continuous administration (mcg/h) <u>Indication:</u> Most frequent reason was previous transdermal fentanyl treatment. <u>Starting dose:</u> Older patients started on lower doses compared to younger patients (p<0.01) <u>Daily dose:</u> Median 25 (IQR 12-37). Median fentanyl dose higher in patients who survived until day 6 compared to those who died before day 6, 37 vs 18 (p<0.05). Median dose of responders 25 mcg/h (IQR 12-62.5), median dose of non-responders 25 mcg/h (IQR 12-63.4) (p=0.5) <u>Other medication</u> Midazolam: n=27, 10mg/d Dexamethasone: n=40, 8mg/d	<u>Symptom (scale):</u> Dyspnea, (Responder, partial responder, non-responder) <u>Symptom scores:</u> 76% responders (n=55) (47 after 24hours, additional 8 after 48 hours) 12% partial responders (n=9) (7 after 24 hours, additional 2 during follow-up) 11% non-responders (n=8) (7 excluded after 24 hours, one excluded after 48 hours) Sustained responses were not statistically related to fentanyl doses or other collected variables. <u>Assessment based on:</u> Patient self-reports or based on the need for fentanyl rescue doses	- No adverse effects observed. - The duration before exclusion was not statistically related to fentanyl dosing (p=0.4)
Matsuda et al. (Japan, 2017)	Quantitative, retrospective study of medical records	<u>Setting:</u> Hospital <u>Population:</u> n=25 Age: mean 75y, (range 72-80) Median survival 47 hours (IQR 26-133) Oncology diagnosis: 0% Opioid naïve: /	<u>Drug:</u> Morphine, administered by CSCI (mg/h) <u>Indication:</u> Dyspnea <u>Starting dose:</u> Median 0.25 (IQR 0.25-0.25) <u>Daily dose:</u> After 2 hours: Median 0.25 (IQR 0.25-0.5) After 4 hours: Median 0.5 (IQR 0.25-0.75)	<u>Symptom (scale):</u> Dyspnea, (NRS) <u>Frequency:</u> / <u>Symptom scores:</u> Significant decrease from mean 7.08 (SD 2.33) at baseline to 5.32 (SD 2.58) after 4 hours. (p=0.04) <u>Assessment by:</u> not specified	- No significant change in respiratory rate

		Frailty characteristics: ECOG performance scale 4 for all patients			
Mori et al. (Japan, 2021)	Quantitative prospective observational study	<u>Setting:</u> 23 palliative care units <u>Population:</u> n=164 Age: mean 73y (SD 12) Median survival: 5d Oncology diagnosis: 100% Opioid naïve: 35% Frailty characteristics: ECOG performance scale 3-4 for all patients	<u>Drug:</u> Oxycodone, administered by CSCI (mg/24h) (n=24) <u>Indication:</u> Dyspnea <u>Starting dose:</u> Opioid-naïve: mean 11 (SD 5.7) Opioid tolerant: mean 28 (SD 26) <u>Daily dose after 24h:</u> Opioid-naïve: mean 12 (SD 6.6) Opioid tolerant: mean 41 (SD 39) <u>Drug:</u> Morphine, administered by CSCI (mg/24h) (n=138) <u>Indication:</u> Dyspnea <u>Starting dose:</u> Opioid-naïve 8.9 (SD 3.8) Opioid tolerant 30 (SD 38) <u>Daily dose after 24h:</u> CSCI Morphine: Opioid-naïve 11 (SD 5.6) Opioid tolerant 32 (SD 36) <u>Other:</u> 40% received corticosteroids	<u>Symptom (scale):</u> Dyspnea, (IPOS) <u>Frequency:</u> Dyspnea relief in 8/19 after 24h in oxycodone group (42%) and 58/130 in morphine group (45%). <u>Symptom scores:</u> Oxycodone: decreased from 2.9 (SD 0.7) to 1.6 (SD 0.7) 24h after initiation (p<0.001). Morphine: Decreased from 3.0 (SD 0.7) to 1.9 (SD 1.1) 24h after initiation (p<0.001). No significant difference in change between the two groups (p=0.815). <u>Assessment by:</u> Responsible palliative care physicians	- No significant adverse effects in the oxycodone group. In the morphine group there were 3 mild-moderate adverse effects (nausea, vomiting and somnolence) and 2 severe/serious adverse effects (hypotension and apnea)
Rashidi et al. (Australia, 2011)	Quantitative, retrospective study of medical records	<u>Setting:</u> 2 palliative care units (one in hospital, one community based) <u>Population:</u> n=205 ≥80y: n=105, mean age 85.59y (SD 4.65) 50-70y: n=100, mean age 62.76y (SD 5.76) Oncology diagnosis: ≥80y 70.5% and 89% in group 50-70y Opioid naïve: / Frailty characteristics: Mean AKPS score 2 days before death: ≥80y 23.49 (SD 12.24) and 27.67 in group 50-70y (SD 15.31)	<u>Drug:</u> Opioids (morphine, ...), modes of administration not specified (doses in Parenteral morphine equivalent mg/24h) <u>Daily doses:</u> (≥80y vs 50-70y) 2d before death: Mean 29.87 vs 61.88 (p=0.024) 1d before death: Mean 32.37 vs 67.85 (p=0.008) Day of death: Mean 20.51 vs 40.77 (p=0.161) Total dose over last 3d: Mean 82.75 vs 170.49 mg/72h (p=0.027) <u>Drug:</u> Midazolam, modes of administration not specified (mg/24h) <u>Daily dose:</u> (≥80y vs 50-70y) 2d before death: Mean 3.26 vs 4.89 (p=0.129) 1d before death: Mean 4.84 vs 8.50 (p=0.011) Day of death: Mean 4.04 vs 5.74 (p=0.179) Total dose over last 3d: Mean 12.14 vs 19.13 mg/72h (p=0.022)	<u>Symptom (scale):</u> pain (PCOC) <u>Symptom scores:</u> (≥80y vs 50-70y) Mean pain score 2d before death: 2.70 (SD 2.90) vs 2.00 (SD 2.32) (p=0.082) Mean pain score day of death: 1.50 (SD 1.81) vs 1.84 (SD 2.99) (p=0.474) <u>Symptom (scale):</u> dyspnea (PCOC) <u>Symptom scores:</u> (≥80y vs 50-70y) Mean breathing problems 2d before death 2.71 (SD 3.13) vs 1.94 (SD 2.69) (p=0.087) Mean breathing problems day of death 2.71 (SD 3.21) vs 2.31 (SD 3.37) (p=0.517) <u>Assessments by:</u> Patient self-report or by 2 involved clinical staff members	- No significant adverse effects observed - Mean survival 13d (range 0-108) vs 19d (range 0-73), (p≤0.001) (≥80y vs 50-70y)

Nausea and vomiting: clinical research question 3

No studies were found.

Psychological symptoms (agitation+delirium+anxiety): clinical research question 6 (a+b+c)

Quantitative non-randomized trials

Author (country and year of publication)	Population	Study characteristics	Outcome	Level of evidence (MMAT 2018)
Schildmann (2021) Germany	Setting: Nursing homes	Study type: Quantitative non-randomized study Multicenter retrospective cohort study	Symptom definition: Agitation, anxiety, delirium/hallucinations not further defined.	- Sampling strategy is relevant for research question but there is a high number of patients dying in hospital and not withheld in this sample. - Outcome measurements restricted to nursing notes without access to medical notes. - Appropriate statistical analysis.
	Number of participants: 512 residents	Study duration: January 2015-December 2017	Symptom control (measures): Use of sedatives - Indications: agitation (53%), anxiety (32%), no indication noted (33%) - Lorazepam median daily dose: 1mg (range 0.5-6 mg) 10% of patients receiving sedatives and 2% of all deceased residents received total daily doses judged as moderately sedating)	
	Age: Median 89 years (range 55-107) Gender: Female: 70% Male: 30%	Power analysis:	- Symptoms and consciousness level were not systematically recorded. - Proportion of residents receiving sedatives in the last week of life differed significantly between nursing homes (range 14-36%, $p<0.001$)	
	Population characteristics: - Majority had multiple diagnoses 54% (n=273) had dementia 54% (n=272) had cardiovascular disease - Only 19% (n=95) had cancer 21% died in hospital 3% (n=16) was followed by a specialist palliative care team	Intervention: Use of sedative - Lorazepam 19% (n=98) - Lormetazepam 1% (n=512) - Haloperidol 0.8% (n=4) - Diazepam, midazolam, levomepromazine (0.2%, n=1 (each)) Use of sedative with continuous effect 9% (n=110) (or 42% of all prescriptions)	Group receiving sedatives residents were younger ($p<0.001$), dementia was less present ($p=0.006$), were more frequently followed by a SPC team ($p=0.011$)	Comments: - Notes of general practitioners and PCT were not available for examination. Only nurses notes. - Stricter regulation about the use of sedatives in nursing homes can be a contributing factor to the relatively low percentage of use of sedatives in this setting) - Oral lorazepam seemed to be the drug of choice in nursing homes as compared to midazolam in palliative care units. - Missing data on symptoms and the level of sedation. - High number dying in hospital so lost to follow up
	Frailty characteristics No or little need for care 4% (n=20) Medium to high need for care 55% (n=282) Very high need for care 41% (n=207)	Comparison: No use of sedatives No use of sedatives with continuous effect	- Use of sedatives with continuous effect - Indications: agitation (54%), anxiety (50%), no indication noted (30%) - Median days of treatment was 3 (range 1-7) - Lorazepam with continuous effect median daily dose: 2mg (range 1-6) 2% (n=9) received doses judged at least moderately sedating. - Residents were younger and kidney disease was more prevalent ($p<0.001$ and $p=0.015$ respectively)	
			(Adverse) side effects: No information available. Life span: No information available.	

Author (country and year of publication)	Population	Study characteristics	Outcome	Level of evidence (MMAT 2018)
Tatokoro (2022) Japan	Setting: In patient palliative care unit	Study type: Quantitative non-randomized study Retrospective cohort study	Symptom definition: No clear definitions stated.	- Representative target population - Outcome measures not clearly defined, lack of measurements of dosage for benzodiazepines and antipsychotics
	Number of participants: 1032 patients	Study duration: April 2012-October 2019		
	Age: Median 79y (IQR 71-86)	Power analysis: Absent	Symptom control (measures): - Symptoms with the highest prevalence were appetite loss (94%), delirium (83%), fatigue (83%), oedema (81%), difficulty sleeping (79%), and pain (77%) - As age increased, significant decreasing trends were observed in the prevalence of pain, dyspnoea, fatigue, constipation, nausea, drowsiness, difficulty sleeping, anxiety, and dysuria, but no trends were observed across age groups in appetite loss, oedema, sputum production, or delirium - As age increased, a significant decreasing trend was observed in the dose of opioids (P < 0.0001): The median doses of opioids for those aged <70 years, 70–79 years, 80–89 years, and ≥90 years were 118, 72, 48, and 48 mg OME/day, respectively - As age increased, a significant decreasing trend was observed in the need for sedation (P < 0.0001) - As age increased, a significant decreasing trend was observed in the need for benzodiazepines: for those aged <70 years, 70–79 years, 80–89 years, and ≥90 years 84.8%, 81.4%, 71.9% and 69.7% respectively (p<0.0001). - As age increased, a non-significant decreasing trend was observed in the need for antipsychotics: for those aged <70 years, 70–79 years, 80–89 years, and ≥90 years 27.4%, 35.1%, 34.9% 31.1% (p=0.26)	Comments: - Somewhat atypical population with most prevalent cancer diagnosis being pancreatic cancer - Absence of information about symptom control or medication dosage for use of benzodiazepines and antipsychotics.
	Gender: Male: 51.8%			
	Population characteristics: - Terminally ill cancer patients - Most common cancer = pancreatic cancer	Intervention: - Treatment of pain and dyspnoea with opioids. - Use of benzodiazepines, antipsychotics, antiemetics, anticholinergics In specific age group		
	Frailty characteristics - Functional state using ECOG PS and palliative performance status. 90% of all patients had relatively poor performance status: 3 or higher on ECOG.	Comparison: Same intervention in other age groups: <70y 70-79y 80-89y 90y or older	(Adverse) side effects: No information available Life span: No significant trends in survival time between age groups. Mean survival time: 20 days	

Author (country and year of publication)	Population	Study characteristics	Outcome	Level of evidence (MMAT 2018)
Ellsworth (2021) USA*	Setting: Long term care hospice unit (often part of community living centers)	Study type: Quantitative non-randomized study Retrospective case-control study	Symptom definition: – Delirium defined as a disturbance in attention that develops over a relatively short period with often an additional cognitive disturbance such as disorientation, memory deficit, language deficits, visuospatial deficit of deficit in perception. – Terminal delirium is defined as delirium that occurs in the dying process and implies that reversal is less likely.	– Participants representative for target population in this study, but not sure that they are representative for our target population. – Treatment is taken as a surrogate for diagnosis making the measurements potentially inappropriate. – Outcome data lacking dosage and indication to treat. – Confounders are accounted for in the analysis.
	Number of participants: – 307 cases analyzed, 31 excluded – n=276	Study duration: October 1, 2013-September 30, 2015		
	Age: Overall mean age: 75.5y Cases: Mean 76y (SD 12.1) Controls 74.8y (SD 2.8) (p0.46)	Power analysis: /	Symptom control (measures): Primary outcome: Determination of risk factors for the development of delirium Secondary outcomes: – Non-pharmacological interventions – Indication for delirium – Current usage patterns in treatment for terminal delirium • Most used antipsychotic drug was haloperidol ▪ 94% (n=175) in the case group received haloperidol at least once in the last 2 weeks ▪ 4.4% (n=4) received haloperidol for another indication, namely nausea and vomiting • Atypical antipsychotics were infrequently used: risperidone, olanzapine, quetiapine, aripiprazole • 52.2% (n=97) veterans requiring any type of antipsychotics for the treatment of terminal delirium. • 40.3% (n=75) received only as-needed doses. Mean as-needed doses per veteran 5.8. • 7.5%(n=14) required only scheduled doses. • With as needed and scheduled doses combined, each veteran received a mean 14.9 doses. • Administration was split evenly throughout the day with about 30% during day-evening- night.	Comments: – Lack of information about dosage administered. – Lack of information about case-detection. – Lack of transparency about case definition/ identification
	Gender: Not explicitly stated			
	Population characteristics: – 2 weeks prior to death evaluation – All veterans ▣ 24% and 18.9% of patients suffered from dementia – Most common diagnosis was cancer	Intervention: Use of antipsychotic in the last 2 weeks of life Comparison: No-use of antipsychotic in the last 2 weeks of life	(Adverse) side effects: No information available.	
	Frailty characteristics: Absent			

* This is a non randomised trial but the outcome of interest to our clinical question is purely descriptive

Quantitative descriptive studies

Author (country and year of publication)	Population	Study characteristics	Outcome	Level of evidence (MMAT 2018)
Gambles (2011) UK	Setting: General hospitals	Study type: Quantitative descriptive study Retrospective epidemiological study	Symptom definition: Agitation and restlessness without further defining	– Sample strategy relevant for research questions – Representative and large study population – Low risk of nonresponse bias. Acceptable statistical analysis
	Number of participants: – 3893 patients – 30 consecutive patients from 155 hospitals	Study duration: 1 October – 31 December 2008		
	Age: Median 81 y Group 1 (receiving medication): Median 79y (IQR 70-86) Group 2 (no medication): Median 83y (IQR 76-88)	Power analysis: /	Symptom control (measures): 51% of all patients were given medication for agitation/restlessness Midazolam – PRN only (n=612): • n=570 (93%) • 2.5 mg median total dose last 24h (90%CI: 2.5-10 mg) – CSCI only (n=802): • n=696 (87%) • 10 mg median total dose last 24h in mg (90%CI: 5-20 mg) – PRN +CSCI (n=568): • n=554 (98%) • 15 median total dose last 24h (90%CI: 7.5-40 mg) Haloperidol – PRN only (n=612): • n=22 (4%) • 1.5 mg median total dose last 24h (90%CI: 0.5-2.85 mg) – CSCI only (n=802): • n=127 (16%) • 3 mg median total dose last 24h (90%CI: 1.5-5) – PRN +CSCI (n=568): • n=94 (17%) • 3 mg median total dose last 24h (90%CI: 1.5-10 mg) Levomepromazine – PRN only (n=612): • n=17 (3%) • 6.25 mg median total dose last 24h (90%CI: 4.75-80 mg) – CSCI only (n=802): • n=99 (12%) • 6.25 mg median total dose last 24h (90%CI: 5-25 mg) – PRN +CSCI (n=568): • N=82 (14%) • 12.5 mg median total dose last 24h (90%CI: 6.25-117.5)	Comments: – Large study population of predominantly older patients – Large non-cancer population – Absence of frailty characteristics – Lack of information about choice of drug related to indication – Lack of a clear definition of agitation Lack of response to medication
	Gender: Female 55% Group 1: female n=1047 (53%) Group 2: female n=1094 (57%)	Intervention: Explore the use of medication for agitation and restlessness in the final days of life supported by the Liverpool Care pathway: midazolam, haloperidol, levomepromazine		
	Population characteristics: 66% diagnosis other than cancer Frailty characteristics: absent	Comparison: Not receiving medication for agitation and restlessness.		
			(Adverse) side effects: No information available about adverse side effects	

Appendix K: Quality of the selected evidence ADAPTE procedure – MMAT

Pain: clinical research question 1

Quantitative non-randomized studies		Clear research question? (S1)	Collected data addresses research questions? (S2)	Representative population? (3.1)	Appropriate measurements regarding outcome and intervention? (3.2)	Complete outcome data? (3.3)	Confounders accounted for? (3.4)	Intervention administered as intended? (3.5)	Comments
Fürst et al. (Sweden, 2020)	Pain	Yes	Yes	Yes	Yes	Yes	Yes	Yes	. Required survival >3days after starting CSCI. . Sedated patients were excluded. Includes some patients <65y.
Golcic et al. (Croatia, 2020)	Pain	Yes	Yes	Yes	Yes	Yes	Yes	Yes	. Only 83.9% of patients died in the hospice.
Rashidi et al. (Australia, 2011)	Dyspnea and pain	Yes	Yes	Yes	Yes	Yes	Yes	Yes	. Significantly different scores appetite, bowel problems and nausea while evaluation of gastrointestinal complaints is very challenging in the dying phase. . No reason for the prescriptions was given.
Quantitative descriptive studies		Clear research question? (S1)	Collected data addresses research questions? (S2)	Relevant sampling strategy? (4.1)	Representative sampling strategy? (4.2)	Appropriate measurements? (4.3)	Low risk of nonresponsive bias? (4.4)	Appropriate statistical analysis? (4.5)	Comments
Golcic et al. (Croatia, 2018)	Pain	No	Unclear	Yes	Yes	Yes	Yes	Yes	. When assuming the research question as stated in table 1-2 all requirements are met. . Oncology population with prior exposure to opioids. . No pain score was measured. . Effect on pain not reported, no side effects e.g. constipation or delirium reported. . Only 83.8% of patients died in the hospice

Dyspnea: clinical research question 2

Quantitative non-randomized studies		Clear research question? (S1)	Collected data addresses research questions? (S2)	Representative population? (3.1)	Appropriate measurements regarding outcome and intervention? (3.2)	Complete outcome data? (3.3)	Confounders accounted for? (3.4)	Intervention administered as intended? (3.5)	Comments
Mori et al. (2021, Japan)	Dyspnea	Yes	Yes	Unclear	Yes	Yes	No	Yes	. Time until death is only mentioned as a median duration. Some patients might have survived for a longer period than the scope of our review.
Rashidi et al. (Australia, 2011)	Dyspnea and pain	Yes	Yes	Yes	Yes	Yes	Yes	Yes	. Significantly different scores appetite, bowel problems and nausea while evaluation of gastrointestinal complaints is very challenging in the dying phase. . No reason for the prescriptions was given.
Quantitative descriptive studies		Clear research question? (S1)	Collected data addresses research questions? (S2)	Relevant sampling strategy? (4.1)	Representative sampling strategy? (4.2)	Appropriate measurements? (4.3)	Low risk of nonresponsive bias? (4.4)	Appropriate statistical analysis? (4.5)	Comments
Matsuda et al. (Japan, 2017)	Dyspnea	Yes	Yes	Yes	Yes	Yes	Yes	Yes	. Last observation carried forward method for 4 out of 25 patients unable to rate dyspnea on NRS after 2 or 4 hours (due to sleep or decreased consciousness). . All patients with interstitial pneumonia.

Nausea and vomiting: clinical research question 3

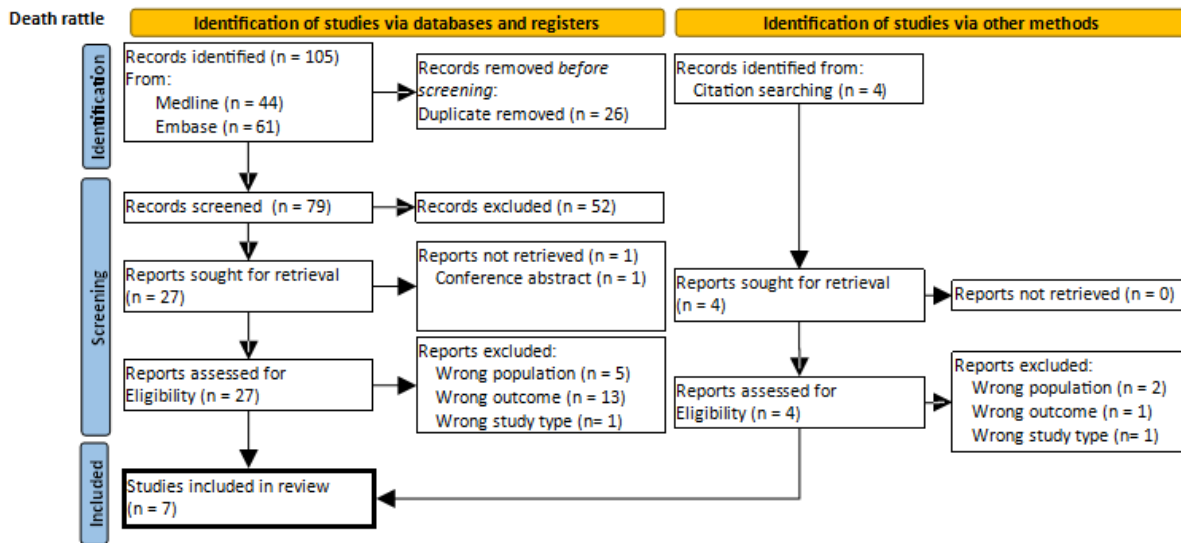
No studies were found

Psychological symptoms (agitation + delirium + anxiety): clinical research question 6 (a+b+c)

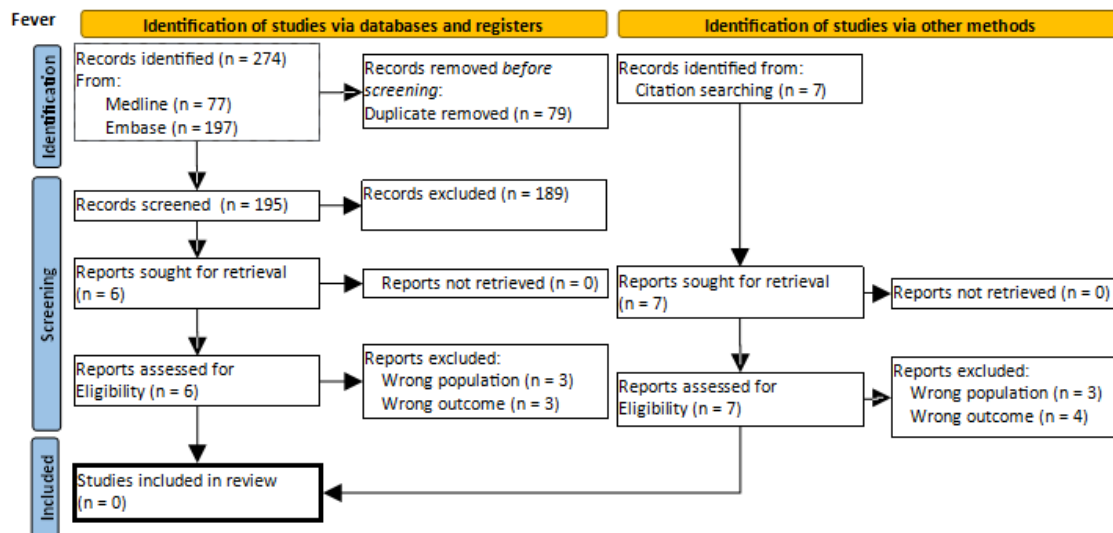
Quantitative non-randomized studies	Clear research question? (S1)	Collected data addresses research questions? (S2)	Representative population? (3.1)	Appropriate measurements regarding outcome and intervention? (3.2)	Complete outcome data? (3.3)	Confounders accounted for? (3.4)	Intervention administered as intended? (3.5)	Comments
Ellsworth et al. (USA, 2021)	Yes	yes	Yes	No	No	Yes	/	. Unsure if participants of this study are representative for our target population. . Treatment is taken as a surrogate for diagnosis making the measurements potentially inappropriate. . Outcome data lacking dosage and indication to treat. . Lack of information about case- detection. . Lack of transparency about case definition/ identification
Schildmann et al. (Germany, 2021)	Yes	Yes	Yes/No	No	No	Yes	/	. Sampling strategy is relevant for research question but there is a high number of patients dying in hospital and not withheld in this sample. . Outcome measurements restricted to nursing notes without access to medical notes. . Missing data on symptoms and the level of sedation.
Tatokoro et al. (Japan, 2022)	Yes	Yes	Yes	No	No	No	/	. Outcome measures not clearly defined, lack of measurements of dosage for benzodiazepines and antipsychotics . Somewhat atypical population with most prevalent cancer diagnosis being pancreatic cancer . Absence of information about symptom control or medication dosage for use of benzodiazepines and antipsychotics.
Quantitative descriptive studies	Clear research question? (S1)	Collected data addresses research questions? (S2)	Relevant sampling strategy? (4.1)	Representative sampling strategy? (4.2)	Appropriate measurements ? (4.3)	Low risk of nonresponsiv e bias? (4.4)	Appropriate statistical analysis? (4.5)	Comments
Gambles et al. (UK, 2011)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	. Lack of a clear definition of agitation . Lack of response to medication

Appendix L: Prisma flowcharts DE NOVO procedure

Death rattle: clinical research question 4



Fever: clinical research question 5



Appendix M: Clinical evidence tables within DE NOVO procedure

Death rattle: clinical research question 4

	Study design	Setting and population	Results		
			Medication	Symptom control	Adverse effects/ survival
Death rattle					
Clark et al. (Australia, 2008)	Quantitative, randomized double blinded cross-over trial <u>Group 1:</u> One injection of octreotide <u>Group 2:</u> One injection of hyoscine hydrobromide If death rattle persisted >4h after injection, the alternate medication could be administered.	<u>Setting:</u> Palliative care unit (in hospital) <u>Population:</u> n=10 Age: median 79y (range 63-88) Oncology diagnosis: 100% Frailty characteristics: Not specified.	<u>Group 1</u> (n=5) <u>Drug:</u> Octreotide , subcutaneous <u>Indication:</u> Death rattle <u>Starting dose:</u> 200µg <u>Group 2</u> (n=5) <u>Drug:</u> Hyoscine hydrobromide , subcutaneous <u>Indication:</u> Death rattle <u>Starting dose:</u> 400µg All patients in group 1 received additional hyoscine hydrobromide, median time 3h (range 1-6). 4 out of 5 patients from group 2 received additional octreotide, median time 3h (range 1-8).	<u>Symptom (scale):</u> Death rattle (none, mild, moderate, severe, very severe) <u>Symptom scores:</u> At start of medication: moderate (n=1), severe (n=7) and very severe (n=2). 1h after first injection: - Group 1: Unchanged (n=4), reduced (n=1) - Group 2: Unchanged (n=3), worsened (n=1), reduced (n=1) 1h after second injection - Group 1: unchanged (n=3), reduced (n=2) - Group 2: reduced (n=3), worsened (n=1) Clinically significant change in death rattle score (n=4)(intensity score improved ≥ 2) <u>Assessments by:</u> Bedside nurses	- No significant adverse effects mentioned.
Heisler et al. (USA, 2013)	Quantitative, randomized double-blinded, placebo controlled trial <u>Group 1:</u> One administration of atropine <u>Group 2:</u> One administration of saline solution	<u>Setting:</u> Hospice <u>Population:</u> n=160 Age: mean 77.2y (SD 11.5) Oncology diagnosis: 43% 20% died during study (n=32) Frailty characteristics: Not specified	<u>Group 1</u> (n=84) <u>Drug:</u> Atropine , presumably oral <u>Indication:</u> Death rattle score ≥ 1 <u>Starting dose:</u> 2 drops of atropine 1% solution <u>Group 2</u> (n=76) <u>Drug:</u> Placebo, presumably oral administration <u>Indication:</u> Death rattle score ≥ 1 <u>Starting dose:</u> 2 drops of saline solution	<u>Symptom (scale):</u> Death rattle (Grade according to Back et al) <u>Symptom scores:</u> (group 1 vs group 2) At start of medication: Group 1: score 1 in n=14, score 2 in n=43, score 3 in n=17 Group 2: score 1 in n=12, score 2 in n=36, score 3 in n=15 After 2 hours: reduction of noise score in 37.8% (n=26) vs 41.3% (n=28) (p=0.73). Unchanged in n=31 vs n=32 and increased in n=6 vs n=14 After 4 hours: reduction of noise score in 39.7% (n=27) vs 51.7% (n=31) (p=0.21) Unchanged in n=20 and n=30 and increased in n=9 and n=11 <u>Assessment by:</u> Bedside nurses	- Slight increase in heartrate +1.1/min vs +3.1/min in atropine vs placebo group (p=0.47) No reports of other adverse effects. - Time from treatment to death was not different. (19.2 vs 17.0h in atropine vs placebo group)

Hugel et al. (2006, UK)	Quantitative, retrospective cohort study	<u>Setting:</u> Palliative care unit <u>Population:</u> n=72 Glycopyrronium group: n=36, mean age 70.7y (SD 10.4) Hyoscine hydrobromide group: n=36, mean age 69.8y (SD 9.6) Oncology diagnosis: 100% Frailty characteristics: not specified	Group 1 (n=36) <u>Drug:</u> Glycopyrronium, subcutaneous <u>Indication:</u> Death rattle <u>Starting dose:</u> 200µg <u>Daily dose:</u> 0.6mg/24h with 200µg bolus injections if needed. If ≥2 boluses were needed maintenance infusion was increased to 1.2mg/24h (n=7) Group 2 (n=36) <u>Drug:</u> Hyoscine hydrobromide, subcutaneous <u>Indication:</u> Death rattle <u>Starting dose:</u> 400µg <u>Daily dose:</u> 1.2mg/24h with 400µg bolus injections if needed. If ≥2 boluses were needed maintenance infusion was increased to 2.4mg/24h (n=3).	<u>Symptom (scale):</u> Death rattle (Graded “yes” or “no” every 4h) <u>Symptom scores:</u> Group 1: 100% had some response, 72% was death rattle free at death, 28 % transient response, 0% no response. Group 2: 78% had some response, 58% RTS free at death, 20% transient response, 22% no response (none of non-responders received highest dose due to death within 24h). Patients in group 1 were significantly more likely to have a response than patients in group 2 (p < 0.01). Median time to response was 4h in both groups. Median time to permanent response was 6h in group 1 vs 4h in group 2 (p=0.245). <u>Assessment by:</u> Not specified	(Group 1 vs 2) - Median number of agitated episodes 1 vs 0. Patients without agitation episode 15 vs 24 (p=0.386 when taking the number of observations into account). - Median time on LCP until death 48 vs 30h (p<0.05). - Median time from onset of death rattle to death 24 vs 12h (p<0.01).
Mercadante et al. (Italy, 2018)	Quantitative, randomized controlled trial	<u>Setting:</u> Hospice <u>Population:</u> n=132. Age: mean 74.5 (SD 12.73) Oncology diagnosis: 100% 68.2% palliative sedation Mean survival 41.28h (SD 27.44) Frailty characteristics: Not specified <u>Group 1:</u> Hyoscine butylbromide started prophylactically (before death rattle developed) <u>Group 2:</u> Hyoscine butylbromide started after development of death rattle	<u>Drug:</u> Hyoscine butylbromide, subcutaneous <u>Indication:</u> Group 1: Reduced consciousness and initiation on the LCP without death rattle at the time of treatment initiation. (n=51) Group 2: Development of death rattle score ≥1 (n=81) <u>Starting dose:</u> 20mg <u>Daily dose:</u> 60mg/24h <u>Other:</u> Opioids: mean OME 129(SD 161.5)mg/d	<u>Symptom (scale):</u> Death rattle (Grade according to Back et al) <u>Frequency:</u> (group 1 vs 2) 5.9% vs 60.5% developed death rattle score ≥1(p=0.001). Median death rattle free time 36h vs 12h (IQR 6-30) (p=0.0001) <u>Symptom scores:</u> Group 2 had a decrease in death rattle score in 20.4% (n=10), after median time of 12h (IQR 6-24h) <u>Assessment by:</u> not specified	- No adverse effects were mentioned. - Survival: (Group 1 vs 2) 45.26h vs 41.10h (p<0.05)
Van Esch et al. (the Netherlands, 2018) AND Van Esch et al. (the Netherlands, 2021)	Quantitative, randomized double-blinded, placebo-controlled trial	<u>Setting:</u> 6 hospices <u>Population:</u> n=157 Group 1: n=79, mean age 78y (SD 69-86) Group 2: n=78, mean age 75y (SD 64-83) Oncology diagnosis: 86% Frailty characteristics: Not specified <u>Group 1:</u> Received scopolamine butylbromide prophylactically <u>Group 2:</u> Received Placebo prophylactically	Group 1(n=79) <u>Drug:</u> hyoscine butylbromide, subcutaneous <u>Indication:</u> Recognition of the dying phase, without presence of death rattle. <u>Starting dose:</u> 20mg <u>Daily dose:</u> 20mg 4 times a day Group 2 (n=78) <u>Drug:</u> Placebo, subcutaneous <u>Indication:</u> Recognition of the dying phase, without presence of death rattle. <u>Starting dose:</u> 1 injection <u>Daily dose:</u> 4 injections per day	<u>Symptom (scale):</u> Death rattle (Grade according to Back et al) <u>Frequency:</u> (group 1 vs 2) 19% vs 37% developed death rattle score ≥2(p=0.01). Median death rattle free time 36h vs 12h (IQR 6-30) (p=0.0001). <u>Symptom scores:</u> Death rattle grade ≥2 at two consecutive time points occurred in 13% vs 27% (p=0.02) <u>Assessment by:</u> Trained bedside nurses	(Group 1 vs 2) - Restlessness (“yes” or “no”): 28% vs 23% (p=0.48) According to VICs: 10% vs 9% (p=0.98) - Dry mouth: 10% vs 15% (p=0.34) - Urinary retention 23% vs 17% (p=0.60) - No differences in occurrence of pain, dyspnea, nausea or vomiting. - Median survival 42.8h (IQR 20.9-80.1) vs 29.5h (IQR 21.1-41.7) (p=0.04)
Wildiers et al. (Belgium, 2002)	Quantitative, retrospective study of medical records	<u>Setting:</u> Palliative care unit (in hospital) <u>Population:</u> n=25 Age: mean 68.1y Oncology diagnosis: 100% Frailty characteristics: not specified	<u>Drug:</u> hyoscine hydrobromide, subcutaneous or intravenous <u>Indication:</u> Death rattle <u>Starting dose:</u> 0.25mg <u>Daily dose:</u> N=20 received 0.25mg every 4h through subcutaneous bolus. N=5 received 1-2.5mg/24h through continuous IV infusion (mostly 1.5mg/24h).	<u>Symptom (scale):</u> Death rattle (Not specified) <u>Frequency:</u> Treatment effective in 72% (n=18) (Intermittent dosing n=16, continuous n=2). Persisting death rattle in 24% (n=6). With alternative explanations for death rattle in all patients (Intermittent dosing n=3, continuous n=3). <u>Symptom scores:</u> / <u>Assessment by:</u> not specified	- No confusion or other side effects. - Survival: <24h: n=12; <48h: n=19; >4d: n=3 (max 600h)

Wildiers et al. (2009, Belgium)	Quantitative, Randomized trial <u>Group 1:</u> Received atropine <u>Group 2:</u> Received Scopolamine hydrobromide <u>Group 3:</u> Received Hyoscine butylbromide	<u>Setting:</u> 6 palliative care units (in hospital) <u>Population:</u> n=333 Group 1: n=115, mean age 70.7 Group 2: n=106, mean age 72.6 Group 3: n=112, mean age 74.3 Oncology diagnosis: 94.8% Frailty characteristics: Not specified	<u>Drug:</u> Atropine , subcutaneous or intravenous <u>Indication:</u> Death rattle score ≥ 1 <u>Starting dose:</u> 0.5mg (Subcutaneous) <u>Daily dose:</u> 3mg/24h (Subcutaneous or intravenous) <u>Drug:</u> hyoscine hydrobromide , subcutaneous or IV <u>Indication:</u> Death rattle score ≥ 1 <u>Starting dose:</u> 0.25mg(Subcutaneous) <u>Daily dose:</u> 1.5mg/24h (continuous IV or SC) or 0.25mg bolus every 4 hours <u>Drug:</u> hyoscine hydrobromide , subcutaneous or IV <u>Indication:</u> Death rattle score ≥ 1 <u>Starting dose:</u> 20mg (Subcutaneous) <u>Daily dose:</u> 60mg/24h (Continuous IV or SC) or 10mg every 4 hours If rattle persisted > 12h at a score ≥ 2 , starting bolus was re-administered and maintenance dose doubled. If rattle persisted > 24h treating physician was free to decide further therapy.	<u>Symptom (scale):</u> Death rattle (Grade according to Back et al) <u>Frequency:</u> (Group 1 vs 2 vs 3) 42%, 37% and 42% showed a decrease in death rattle score after 1h(p=0.72). 76%, 68% and 60% showed decrease in death rattle score after 24h. <u>Symptom scores:</u> Treatment was more effective when started at rattle score 1 vs 2 vs 3 (p<0.000001). Less effective in patients with primary lung tumor or lung metastases (p=0.009) <u>Assessment by:</u> not specified_	- No important differences were observed. - Median survival was 23.9h. Patients with higher rattle scores had median shorter survival: score 1 38.6h, score 2 20h, score 3 24.3h.
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Fever: clinical research question 5

No studies were found.

Appendix N: Quality of selected evidence DE NOVO procedure – MMAT

Death rattle: clinical research question 4

Randomized controlled trials (RCT)		Clear research question? (S1)	Collected data addresses research questions? (S2)	Appropriate randomization? (2.1)	Comparable groups at baseline? (2.2)	Complete outcome data? (2.3)	Outcome assessors blinded? (2.4)	Did participants adhere to the intervention? (2.5)	Comments
Clark et al. (Australia, 2008)	Death rattle	Yes	Yes	Yes	Yes	Yes	Yes	Yes	. Main goal of the study was to achieve consent and conduct of a RCT at the end of life. . New questionnaire was developed. . Only descriptive data.
Heisler et al. (USA, 2013)	Death rattle	Yes	Yes	Yes	Yes	No	Yes	Yes	. 20% of patients died during study protocol. . Single dose study with short follow-up. . Trial was stopped prematurely because of futility.
Mercadante et al. (Italy, 2018)	Death rattle	Yes	Yes	Yes	Yes	Yes	No	Yes	. 40% palliative sedation in prophylactic group vs 61.4%.
Van Esch et al. (the Netherlands, 2018 and 2021)	Death rattle	Yes	Yes	Yes	No	Yes	Yes	Yes	. Scopolamine group had more patients with history of COPD, lung cancer and recent smoking.
Wildiers et al. (2009, Belgium)	Death rattle	Yes	Yes	Yes	Yes	Yes	No	Yes	. No placebo control group. . Not blinded.
Quantitative non-randomized studies		Clear research question? (S1)	Collected data addresses research questions? (S2)	Representative population? (3.1)	Appropriate measurements regarding outcome and intervention? (3.2)	Complete outcome data? (3.3)	Confounders accounted for? (3.4)	Intervention administered as intended? (3.5)	Comments
Hugel et al. (2006, UK)	Death rattle	Yes	Yes	Yes	Yes	No	No	Yes	. Time period of collected data differs between the two groups, 1999 vs 2002. . Significant difference in survival. . Possible observer bias. . 10.7% non-recorded observations in hyoscine group vs 1.4%.
Quantitative descriptive studies		Clear research question? (S1)	Collected data addresses research questions? (S2)	Relevant sampling strategy? (4.1)	Representative sampling strategy? (4.2)	Appropriate measurements? (4.3)	Low risk of nonresponsive bias? (4.4)	Appropriate statistical analysis? (4.5)	Comments
Wildiers et al. (Belgium, 2002)	Death rattle	No	/	/	/	/	/	/	. Vague research question.

Fever: clinical research question 5

No studies were found.

Appendix O: Clinical research question 7

KLINISCHE VRAAG 7: VEILIGE EN PRAKTISCHE MANIER VAN COMBINEREN VAN VERSCHILLENDE GENEESMIDDELEN

i.k.v. Kom Op Tegen Kanker fonds project:

“Behandeling van symptomen in de stervensfase bij kwetsbare ouderen”

Rapport

Versie 1.0

December 13, 2021

Schrijvers

Apr. Jolien Langui
Apr. Winne Van Mele
Apr. Glenn Van Wynsberge

Reviewers

Apr. Andreas Capiou
Prof. Dr. Apr. Annemie Somers

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1 CONTEXT EN DOELSTELLING

Uitgaande van het Kom Op Tegen Kanker fonds werd door een arts-specialist in opleiding binnen de geriatrie (VUB) en een doctoraatsstudent (VUB) een project gestart rond het ontwikkelen van een klinische richtlijn voor de behandeling van symptomen in de stervensfase bij kwetsbare ouderen. Aansluitend op de literatuurreview waarbij het projectteam een antwoord tracht te formuleren op 12 klinische vragen, werd bijkomend een 13^e klinische vraag opgesteld:

“Hoe kunnen verschillende geneesmiddelen op een praktische en veilige manier (met oog voor interacties en incompatibiliteiten), gecombineerd worden toegediend in de behandeling van de terminale patiënt in het woonzorgcentrum?”

Op vraag van het projectteam was de ziekenhuisapotheek van het Universitair ziekenhuis Gent bereid om zijn medewerking te verlenen bij het uitwerken van deze 13^e klinische vraag. Drie ziekenhuisapothekers in opleiding, Apr. Jolien Langui, Apr. Glenn Van Wynsberge en Apr. Winne Van Mele vormen het schrijversteam onder supervisie van apotheker-doctoraatsstudent Andreas Capiou (UZGent/UGent) en adjunct-hoofdapotheker farmaceutische zorg Prof. Dr. Apr. Annemie Somers (UZGent/UGent).

Doelstelling van de betrokken apothekers bij het beantwoorden van deze klinische vraag is om de meest geschikte praktische aanbevelingen te formuleren voor (gecombineerd) gebruik van geneesmiddelen in de behandeling van terminale symptomen bij de oudere patiënt in het woon-zorg-centrum. Dit rapport heeft als doel om het verloop van dit deelproject te beschrijven met inbegrip van een duidelijke voorstelling van de gebruikte methodologie om tot de resultaten te komen in luik 3 van het rapport.

2 METHODOLOGIE

2.1 INLEIDING

De geformuleerde 13^e klinische vraag vormt een aanvulling op uitkomsten (zijnde geneesmiddelen) van de literatuurreview naar de symptoombehandeling tijdens de stervensfase bij kwetsbare ouderen. Een gestructureerd antwoord op deze vraag vergt data en informatie die nauwelijks beschreven staat of onderzocht werd in studies. Een zoekstrategie waarbij medisch wetenschappelijke databases doorzocht worden lijkt daarom niet ideaal om tot de meest geschikte praktische aanbevelingen te komen. In overleg met de projectgroep werd daarom besloten niet gebruik te maken van een generiek review protocol zoals bij de 12 klinische vragen omtrent de symptoombehandeling. Een pragmatische aanpak werd voorgesteld waarvoor een alternatieve zoekstrategie werd ontwikkeld (zie 2.3).

Het onderzoek verliep in vier fasen. In de eerste fase werd de geformuleerde klinische vraag geconcretiseerd. Ten eerste werd een overzicht opgesteld van de geneesmiddelen die bij het literatuuronderzoek voor de 12 klinische vragen naar voren kwamen als zijnde behandelopties bij symptomen in de stervensfase. Ten tweede werd vastgelegd welke categorieën van informatie nodig waren om te kunnen beantwoorden aan de doelstelling van de 13^e klinische vraag. De tweede fase omvatte de ontwikkeling van een pragmatische zoekstrategie als alternatief op de generieke zoekstrategie in het initiële reviewprotocol. Ter validatie werd deze zoekstrategie getest bij de uitwerking van twee geneesmiddelen waarna het bekomen resultaat werd voorgelegd aan de reviewers en de projectgroep. In een derde fase werden de gegevens verzameld conform de vooropgestelde

zoekstrategie uit fase 2. Daarnaast werd hierbij het formaat vastgelegd waarop de resultaten in dit rapport worden bijgevoegd. In een vierde en laatste fase werd door de reviewers nazicht uitgevoerd op de bekomen resultaten, gevolgd door een gezamenlijk consensusvergadering ter finalisering van de resultaten.

2.2 FASE 1: CONCRETISEREN VAN DE KLINISCHE VRAAG

2.2.1. Welke geneesmiddelen?

Onderstaande lijst van geneesmiddelen (inclusief indicaties en toedieningsvorm) is een tussentijds resultaat van de literatuurstudie bij de 12 klinische vragen naar symptoombehandeling. Deze geneesmiddelen zullen met grote waarschijnlijkheid opgenomen worden in de klinische richtlijn waarvoor onze verdere uitwerking i.k.v. de 13^e klinische vraag gevraagd werd.

Geneesmiddel	Toedieningsvorm	Indicatie(s)
Clonazepam	Orale vloeistof Subcutaan bolus Continue subcutane infusie	Angst Stuipen (preventief) Agitatie Sedatie
Fentanyl	Subcutaan bolus Continue subcutane infusie	Pijn Dyspneu
Haloperidol	Subcutaan bolus Continue subcutane infusie	Delier Psychose Agitatie Misselijkheid Braken
Hydromorfone	Subcutaan bolus Continue subcutane infusie	Pijn Dyspneu
Hyoscine butylbromide/ hydrobromide	Subcutaan bolus Continue subcutane infusie	Doodsreutel Obstipatie
Metoclopramide	Subcutaan bolus Continue subcutane infusie	Misselijkheid Braken
Midazolam	Subcutaan of sublinguaal bolus Continue subcutane infusie	Angst Stuipen (preventief) Agitatie Sedatie
Morfine sulfaat/ hydrochloride	Subcutaan bolus Continue subcutane infusie	Pijn Dyspneu
Octreotide		Misselijkheid en braken
Atropine		
Glycopyrronium bromide		
Alizapride (Litican ®)		

2.2.2. Welke categorieën van informatie?

Bij het correct, veilig en praktisch (gecombineerd) gebruik van geneesmiddelen zijn verschillende aspecten van belang. In onderstaand overzicht worden de aspecten - relevant in de context van de 13^e klinische vraag - opgelijst aan de hand van vijf categorieën en bijhorende inhoudelijke vraagstellingen.

Algemeen

- Welke specialiteiten zijn er beschikbaar op de **Belgische markt**?
- Welke specialiteiten zijn er beschikbaar in het **buitenland** (indien niet op de Belgische markt)? Wat zijn **aandachtspunten** bij invoer van een specialiteit uit het buitenland?

Toedieningsmodaliteiten

- Welke **toedieningsroutes** zijn mogelijk? Welke is de meest geschikte?
Bv. per oraal, sublinguaal, subcutaan, transdermaal, rectaal
- Zijn er bij de toediening **interacties met voeding**?
Bv. nuchter of met voeding in te nemen?
- Zijn de vaste orale toedieningsvormen **pletbaar**? Is er een alternatief in vloeibare vorm?
Bv. in geval van slikproblematiek
- Kan het geneesmiddel veilig via een **sonde** (nasogastrisch, PEG-sonde) toegediend worden?
Zo ja, zijn er bijzondere aandachtspunten?
- Kan het geneesmiddel veilig **subcutaan** (bolus, continue infusie) toegediend worden?
Zo ja, zijn er bijzondere aandachtspunten?

Geneesmiddelinteracties

- Voor welke **farmacokinetische interacties** moet men aandachtig zijn?
(deze hebben een invloed op de absorptie, distributie, metabolisatie en eliminatie van de geneesmiddelen)
Bv. interacties tussen de geneesmiddelen in de lijst (zie 2.2.1)
- Voor welke **farmacodynamische interacties** moet men aandachtig zijn? (dit betreft een mogelijk antagonistisch effect, een versterking van het effect of een verhoogd risico op bijwerkingen)
Bv. versterkt deprimerend effect op het centraal zenuwstelsel, constipatie

Contra-indicaties

- Welke combinatie van geneesmiddelen mogen **niet gelijktijdig** gebruik worden?
Bv. als gevolg van een interactie
- Welke geneesmiddelen mogen niet gebruik worden o.w.v. **comorbiditeiten** of **medische voorgeschiedenis**?
Bv. ernstige nierinsufficiëntie

Compatibiliteit in spuit voor subcutane toediening

- Welke geneesmiddelen zijn **compatibel** met elkaar in een **spuit** voor subcutane toediening?
Bv. fysische stabiliteit (neerslagvorming), chemische stabiliteit (afbraakreactie)

2.3 FASE 2: ONTWIKKELING VAN DE ZOEKSTRATEGIE + VALIDATIE

Zoals eerder vermeld werd een pragmatische zoekstrategie uitgewerkt gezien de schaarste aan wetenschappelijke publicaties omtrent correct, veilig en praktisch gebruik van de betrokken geneesmiddelen. Echter zijn er tal van *evidence-based* klinische bronnen (online medische zoekmachines en database tools, handboeken, richtlijnen, etc.) voor handen die relevante data en informatie bevatten om tot een gestructureerd antwoord te komen op de 13^e klinische vraag.

2.3.1. Overzicht evidence-based klinische bronnen

Er werd gestart met het inventariseren van de klinische bronnen die routinematig gebruikt worden in de zorg bij het beantwoorden van soortgelijke klinische vragen. Uit een lijst van 16 werden na gezamenlijk overleg 13 klinische bronnen weerhouden als zijnde relevant in de context van dit project. Naderhand werd bij iedere klinische bron bepaald welke categorieën van informatie (zie 2.2.2) er in teruggevonden kunnen worden, zoals weergegeven in onderstaande tabel. Bij de dataverzameling werd voor ieder geneesmiddel uit de lijst (zie 2.2.1) deze klinische bronnen geraadpleegd volgens de JA/NEE aanduiding in de tabel en relevante informatie verzameld. Voor het verzamelen van informatie omtrent contra-indicaties werd de wetenschappelijke bijsluiter geraadpleegd.

Subcutane toediening, al dan niet gecombineerde toediening van geneesmiddelen, staat op voorgrond in deze patiëntpopulatie. Meerdere geneesmiddelen uit de lijst (zie 2.2.1) zijn niet geregistreerd voor subcutaan gebruik met als gevolg dat er in de *evidence-based* klinische bronnen een schaarste is aan informatie m.b.t. deze toedieningsroute. Bijgevolg werd ook gezocht naar *practice-based evidence* waardoor praktijkkennis en ervaringen van zorgprofessionals ook worden meegenomen in de zoekstrategie. Bijkomende bronvermeldingen zijn in dat geval toegevoegd aan de geneesmiddelfiches.

	<i>Evidence-based</i> klinische bron	Toedienings- modaliteiten	GM-GM interacties	Compatibiliteit
1	ASHP Injectable Drug Information;	JA	NEE	JA
2	ClinicalKey - Pharmacology	JA	JA	JA
3	CredibleMeds	NEE	JA	NEE
4	Handboek Enteralia - VUmc	JA	NEE	NEE
5	Handboek Parenteralia - VUmc	JA	NEE	JA
6	Handbook of Drug Administration via Enteral Feeding Tubes	JA	NEE	NEE
7	Martindale: The Complete Drug Reference	JA	NEE	JA
8	Pletfiches - VZA	JA	NEE	NEE
9	Product Informatie Fiches (PIF) - VZA	JA	NEE	NEE
10	Samenvattingen van de Kenmerken van het Product (SKP)	JA	JA	NEE
11	Stabilis 4.0	JA	NEE	JA
12	Stockley's Drug Interaction	NEE	JA	NEE
13	UpToDate	JA	JA	NEE

2.3.2. Validatie zoekstrategie

De zoekstrategie werd uitgetest voor twee geneesmiddelen uit de lijst (zie 2.2.1). De klinische bronnen werden geraadpleegd en relevante informatie werd verzameld. Het tussentijdse resultaat van de dataverzameling werd voorgelegd aan de reviewers en projectgroep. Opmerkingen en suggesties werden meegenomen in de verfijning van de zoekstrategie. Zo werd vastgelegd dat ook data verzameld kan worden uit betrouwbare bronnen, publicaties of studies die als referentie worden vermeld in één van de 15 weerhouden klinische bronnen, op voorwaarde dat bijkomende bronvermelding voorzien wordt.

2.4 FASE 3: DATAVERZAMELING/DISCUSSIE EN FORMAAT VAN RESULTATEN

2.4.1. Dataverzameling/discussie

Per geneesmiddel uit de lijst (zie 2.2.1) werd de zoekstrategie doorlopen. De informatie vanuit de verschillende klinische bronnen werd met elkaar afgetoetst. Per categorie van informatie en per vraagstelling (zie 2.2.2) werd een zo volledig mogelijk antwoord geformuleerd. Tegenstrijdige informatie in de klinische bronnen werd voorgelegd aan het volledige schrijversteam ter discussie.

2.4.2. Formaat van resultaten

De bedoeling was om een zo volledig mogelijk rapport af te leveren, maar klinische interpretatie door de projectgroep en artsen blijft nodig om bepaalde zaken te vertalen naar de richtlijn en de klinische praktijk o.w.v. de soms beperkte of ontbreken van evidentie. De grote lijnen van de vormgeving van de resultaten werd vastgelegd na overleg met de projectgroep. Er werd besloten om praktijkgerichte documenten op te stellen, eerder dan een gedetailleerd uitgeschreven tekstdocument. Bijgevolg werden geneesmiddelfiches opgesteld die de volgende uniforme indeling hebben:

- Indicatie(s): *cfr. geneesmiddelenlijst onder 2.2.1*
- Specialiteiten op de Belgische markt: *voor de gevraagde toedieningsmodaliteiten*
- Specialiteiten op de buitenlandse markt: *indien van toepassing*
- Toedieningsmodaliteiten
- GM-GM interacties: *zie interactietabel*
- Contra-indicaties
- Compatibiliteiten: *zie compatibiliteitentabel*
- Andere: *specifieke bijzonderheden*

De informatie weergegeven in de geneesmiddelfiches en de bijhorende tabellen hebben een verschillend aanbevelend karakter. Voor interacties en contra-indicaties is een descriptief overzicht gewenst zonder expliciete aanbevelingen met als doel om de klinische interpretatie over te laten aan de arts. Mogelijke interventies bij het optreden van interacties werden wel vermeld in het overzicht. Wat betreft toedieningsmodaliteiten en compatibiliteiten is de aard van de informatie zo dat dit eerder een aanbevelend karakter heeft. Om te uitgebreide geneesmiddelfiches te vermijden werd voor de interacties en compatibiliteiten verwezen naar aparte tabellen die het toelaten om voor de volledige geneesmiddelenlijst een overzichtswaergave te bieden.

Naar geneesmiddeleninteracties werd gescreend door gebruik te maken van drie verschillende interactiedatabanken, namelijk Up to date, Stockley's drug interaction, Clinical Pharmacology. De ernst van de interacties wordt weergegeven aan de hand van verschillende levels, niveaus of symbolen. Het is vaak zo dat voor eenzelfde interactie bij de drie gebruikte databases de ernst niet gelijk ingeschat wordt of de interactie zelfs niet overal vermeld wordt. Wanneer eenzelfde ernst van interactie toegekend wordt volgens de drie geraadpleegde databases, werd een kleurencode gebruikt. Een rode kleurencode geeft aan dat deze interactie als contra-indicatie beschouwd wordt volgens de drie gebruikte interactiedatabanken. Een oranje kleurencode geeft aan dat een dosisaanpassing of bijkomende monitoring vereist is bij het gelijktijdig gebruik volgens de drie databanken. Wanneer geen kleurencode toegekend werd aan de opgelijste interacties, werd de interactie niet vermeld in elke interactiedatabank. De resultaten van deze screening werden opgenomen in een interactietabel waarin het onderliggende mechanisme, gevolg en een voorstel tot afhandeling werden vermeld. Deze interactietabel geeft enkel interactie weer tussen die geneesmiddelen die gebruikt worden voor symptoomcontrole bij einde leven zoals weergegeven onder punt 2.2.1. De interactietabel ondersteunt de detectie van potentiële geneesmiddelinteracties, maar is geen vervanging van een algemene en degelijke geneesmiddelinteractiescreening. Interacties die niet op de lijst staan, kunnen even belangrijk zijn (gelet op andere comedicaatie van de patiënt).

De geneesmiddelfiches werden toegevoegd onder 3. Resultaten. De bijhorende tabellen worden als bijlage toegevoegd aan het rapport als ook in Excel-formaat bezorgd aan het projectteam.

2.5 FASE 4: REVIEW

Na het volledig doorlopen van fase 1 tot 3 met tussentijdse overlegmomenten met projectgroep en schrijvers-/reviewteam volgt de reviewfase. Het geheel van resultaten (rapport inclusief geneesmiddelfiches, interactietabel en compatibiliteitentabel) werd voorgelegd aan de reviewers voor nazicht. Aanpassingen, opmerkingen en suggesties werden besproken tijdens een gezamenlijke vergadering ter finalisering van de documenten.

3 RESULTATEN

3.1 GM-FICHES

CLONAZEPAM	
Indicatie	Angst Stuipen (preventief) Agitatie Sedatie
Specialiteiten op de Belgische markt (10)	Rivotril® tabletten van 0,5 mg en 4 mg Rivotril® oplossing van 2,5 mg/mL
Specialiteiten buitenlandse markt	Rivotril® 1 mg/ml concentraat voor injectievloeistof (IV/IM) Clonazepamum TFZ 1 mg/ml
Toedieningsmodaliteiten	Oraal: Rivotril® tabletten en oplossing (1) - Tabletten zijn pletbaar (8) - Direct toedienen (lichtgevoelig) - Oplossing als oraal alternatief (8) - Geen invloed van voeding Subcutaan (off label): concentraat voor injectievloeistof (IV/IM) - Bolus: 0,5 mg/ml na toevoegen van 1 mL diluent (WFI*) - CSCI: - Diluent (WFI) niet toevoegen - In spuit – verdund 0,1 mg/ml in NaCl 0,9% (verdunnen met 9 ml per ampul) → 12 uur stabiel op kamertemperatuur (11) 0,2 mg/ml in NaCl 0,9% (verdunnen met 4 ml per ampul) → 12uur stabiel op kamertemperatuur (5, 11) - Gebruik van non-PVC spuit en leidingen (1, 7) Enterale sonde - Kan via enterale tubes toegediend worden (6) - Tablet dispergeren in 10 ml water - Geen PVC sonde (4)
GM-GM interacties	Zie interactietabel
Contra-indicaties (10)	- Overgevoeligheid (werkzame stof of hulpstof) - Ernstige ademhalingsinsufficiëntie - Ernstige leverinsufficiëntie - Myasthenia gravis - Acuut gesloten-hoekglaucoom
Compatibiliteiten	Zie compatibiliteitentabel
Andere	* water voor injectie

FENTANYL	
Indicatie	Pijn Dyspneu
Specialiteiten op de Belgische markt (10)	Fentanyl 0,1 mg/2 mL (IV/IM/SC) Fentanyl 0,5 mg/10 mL (IV/IM/SC)
Toedieningsmodaliteiten	Subcutaan: 0,1 mg/2 mL (IV/IM/SC) of 0,5 mg/10 mL (IV/IM/SC) - Bolus: - Onverdund - CSCI: - In spuit: onverdund, 1/4 of 1/10 verdunnen in NaCl 0,9% - 48u stabiel op kamertemperatuur (1)
GM-GM interacties	Zie interactietabel
Contra-indicaties (10)	- Overgevoeligheid (werkzame stof of hulpstof) - Ernstige ademhalingsdepressie (transdermaal) - Opioid-naïeve patiënt (2) - Paralytische ileus of gastro-intestinale obstructie (transdermaal)
Compatibiliteiten	Zie compatibiliteitentabel

HALOPERIDOL

Indicatie	Delier Psychose Agitatie Misselijkheid Braken
Specialiteiten op de Belgische markt (10)	Haloperidol tabletten 5 mg Haloperidol druppels 2 mg/mL Haloperidol (lactaat) amp. IM 5 mg/mL Haloperidol (decanoaat) amp. IM verlengde afgifte 50 mg/mL of 100 mg/mL
Toedieningsmodaliteiten	Oraal: tabletten en druppels ○ Tabletten ▪ Pletbaar (8) ▪ Lichtgevoelig, direct te gebruiken (8) ○ Oplossing als orale alternatief (8) ○ Geen invloed van voeding (6) ○ Kan toegediend worden met voedsel om gastro-intestinale irritatie te beperken. (2) Subcutaan (off label): in palliatieve setting (13), amp. IM 5 mg/mL ○ Bolus: ▪ Dosis: 2.5 mg elke 2 uur indien nodig (7) ○ CSCI: ▪ Dosis: 2,5 tot 15 mg over 24 uur. (7) ▪ Geen stabiliteitsgegevens in spuit (Glazen container: 0,1mg/ml in glucose 5% - 38 dagen stabiel buiten invloed van licht op kamertemperatuur(11)) Enterale sonde ○ Gebruik de orale vloeistoffen, verder dilutie is niet noodzakelijk. (6)
GM-GM interacties	Zie interactietabel
Contra-indicaties (10)	○ Overgevoeligheid (werkzame stof of hulpstof) ○ Ziekte van Parkinson ○ Lewy-bodydementie ○ Progressieve supranucleaire verlamming ○ Bekende QTc-intervalverlenging of het congenitale lange QT-syndroom of gelijktijdige behandeling met geneesmiddelen die het QT-interval verlengen (cfr. interactietabel) ○ Recent acuut myocardinfarct ○ Niet-gecompenseerd hartfalen ○ Voorgeschiedenis van ventriculaire aritmie of torsade de pointes ○ Niet-gecorrigeerde hypokaliëmie
Compatibiliteiten	Zie compatibiliteitentabel

HYDROMORPHONE

Indicatie	Pijn Dyspneu
Specialiteiten op de Belgische markt (10)	Capsules met onmiddellijke vrijstelling 2.6 mg Capsules met verlengde afgifte 4, 8, 16, 24 mg Palladone amp. IV/SC 2 mg/mL of 10 mg/mL
Toedieningsmodaliteiten	Oraal: capsules ○ Capsules met onmiddellijke vrijstelling: ▪ Capsule openen ▪ Inhoud is pletbaar ▪ Lichtgevoelig, direct gebruiken (8) ○ Capsules met vertraagde vrijstelling: ▪ Inhoud is niet pletbaar (8) ○ Geen invloed van voeding (6). ○ Gelijktijdig gebruik met voedsel of melk kan gastro-intestinale last beperken. (4) Subcutaan: amp. IV/SC 2 mg/mL of 10 mg/mL ○ Bolus ▪ Onverdund ▪ Dosis: 1-2 mg om de 3-4 uur (10) ▪ Gebruik korte 30G naald om lokale reactie te vermijden (2) ○ CSCI ▪ In spuit - verdund: 0,1 of 1mg/ml in NaCl 0,9% - 60 dagen stabiel buiten invloed van licht op kamertemperatuur (11) ▪ Dosis: 0,004 mg/kg lichaamsgewicht/u (0,15-0,45 mg/u) (10) ▪ Maximale infusiesnelheid is 2 ml/uur (irritatie injectieplaats is frequenter bij snelheden 1 ml/uur) (2) ▪ Subcutane regio roteren. Frequent gebruikt: subclaviculaire regio en voorste borstwand Enterale sonde (6): ○ Capsules met onmiddellijke vrijstelling: ▪ Capsules kunnen geopend worden en op zacht voedsel gestrooid worden. ▪ Geen specifieke data beschikbaar over toediening via enterale sonde. ○ Capsules met vertraagde vrijstelling: ▪ Niet aanbevolen: mogelijks blokkade van enterale sonde.
GM-GM interacties	Zie interactietabel
Contra-indicaties (10)	○ Overgevoeligheid (werkzame stof of hulpstof) ○ Ernstige ademhalingsdepressie met hypoxie en/of hypercapnie ○ Ernstige chronische luchtwegobstructie ○ Ernstige astma bronchiale ○ Paralytische ileus ○ Acute buik ○ Cor pulmonale ○ Gelijktijdige toediening van mono-amino-oxidaseremmers of korter dan twee weken na staken van het gebruik ervan (cfr. interactietabel)
Compatibiliteiten	Zie compatibiliteitentabel
Andere	N.B.: hydromorphone preparaten beschermen van licht en <u>niet</u> gekoeld bewaren (risico op precipitatie en kristallisatie) (11)

HYOSCINE BUTYLBROMIDE/HYDROBROMIDE = SCOPOLAMINE

Indicatie	Doodsreutel Obstipatie (colic)
Specialiteiten op de Belgische markt (10)	Scopolamine® HBr Amp. IM/IV/SC 0,25 mg/mL of 0,5 mg/mL
Toedieningsmodaliteiten	Subcutaan: amp. IM/IV/SC 0,25 mg/mL of 0,5 mg/mL ○ Bolus: ▪ Onverdund ▪ Dosis: 0,25 tot 0,5 mg om de 4 à 6 uur (10) ○ CSCI: ▪ In spuit – verdund: in NaCl 0,9% - 24uur stabiel op kamertemperatuur ▪ Dosis: 0,75 tot 3 mg per 24 uur (10)
GM-GM interacties	Zie interactietabel
Contra-indicaties (10)	○ Overgevoeligheid (werkzame stof of hulpstof) ○ Gesloten-hoek-glaucoom ○ Urineretentie gekoppeld aan uretro-prostaatstoornissen
Compatibiliteiten	Zie compatibiliteitentabel

METOCLOPRAMIDE

Indicatie	Misselijkheid Braken
Specialiteiten op de Belgische markt (10)	Primperan® siroop 5 mg/5mL Primperan Amp.® IM/IV 10 mg/2 mL
Toedieningsmodaliteiten	Oraal: siroop 5 mg/5mL ○ Geen invloed van voeding (6) Subcutaan (off-label): amp. IM/IV 10 mg/2mL ○ Bolus: ▪ Onverdund (5mg/mL). (9) ▪ Dosis: 5 – 10 mg elke 6 uur (10,13) ○ CSCI: ▪ In spuit: onverdund of 0,5mg/ml in NaCl 0,9% - 21 dagen stabiel op kamertemperatuur ▪ Dosis: initieel 1,25 - 1,75 mg/u, titreer zo nodig tot 5mg/uur indien getolereerd wordt (13) Enterale sonde: ○ Gebruik de siroop. Geen verdere dilutie is noodzakelijk voor intragastrische toediening. Een verlengde onderbreking van enterale voeding is niet nodig. (6)
GM-GM interacties	Zie interactietabel
Contra-indicaties (10)	○ Overgevoeligheid (werkzame stof of hulpstof) ○ Gastro-intestinale bloeding, mechanische obstructie of gastro-intestinale perforatie waarbij stimulatie van de gastro-intestinale motiliteit een risico vormt. ○ Vastgesteld of vermoed feochromocytroom in verband met het risico op ernstige hypertensie-episodes. ○ Een voorgeschiedenis van door neuroleptica of metoclopramide geïnduceerde tardieve dyskinesie. ○ Epilepsie (verhoging van frequentie en intensiteit van crises) ○ Ziekte van Parkinson ○ Combinatie met levodopa of dopaminerge agonisten ○ Bekende voorgeschiedenis van methemoglobinemie met metoclopramide of van NADH cytochroom-b5-deficiëntie. ○ Gebruik bij kinderen jonger dan 1 jaar omwille van een verhoogd risico van extrapyramidale stoornissen.
Compatibiliteiten	Zie compatibiliteitentabel

MIDAZOLAM

Indicatie	Angst Stuipen (preventief) Agitatie Sedatie
Specialiteiten op de Belgische markt (10)	Buccolam®: Buccale oplossing (indicatie convulsieve aandoeningen): 2,5mg/0,5mL, 5mg/mL, 7,5 mg/1,5mL, 10 mg/2mL Amp. IM/IV/Rect. 5 mg/5mL, 15 mg/3mL, 50 mg/10mL
Toedieningsmodaliteiten	Oraal: Buccolam®: ○ Sublinguaal toe te dienen: / ○ De oplossing dient langzaam te worden ingebracht in de ruimte tussen het tandvlees en de wang (buccale holte). (10) ○ Pediatriche toed. Vorm: / Subcutaan: Amp. IM/IV/Rect. 5mg/5mL, 15mg/3mL, 50mg/10mL ○ Bolus: / ○ CSCI: ▪ De start van het effect is vertraagd bij subcutane toediening en kan de frequentie van toedienen beïnvloeden. ▪ De dosis bij subcutane continue infusie is equivalent aan de dosis bij intraveneuze continue infusie (13) Enterale sonde ○ De orale vloeistof of de injectievloeistof kunnen gebruikt worden via enterale toedieningsweg. ○ Wanneer de injectievloeistof enteraal gebruikt wordt, dient deze oplossing in de enterale spuit opgetrokken te worden met behulp van een filternaald. (6)
GM-GM interacties	Zie interactietabel
Contra-indicaties (10)	○ Overgevoeligheid (werkzame stof of hulpstof) ○ Het gebruik voor "conscious sedation" bij patiënten met ernstige ademhalingsinsufficiëntie of acute ademhalingsdepressie
Compatibiliteiten	Zie compatibiliteitentabel

MORPHINE SULFAAT/HYDROCHLORIDE

Indicatie	Pijn Dyspneu
Specialiteiten op de Belgische markt (10)	MS Direct® 10 mg: tabletten met normale afgifte MS Contin® 10, 30, 60, 100 mg: tabletten met verlengde afgifte Morfine Amp. IM/IV/SC/IT/Epiduraal 10 mg/mL, 20 mg/mL, 30 mg/mL, 40mg/mL
Toedieningsmodaliteiten	Oraal: ○ MS Direct® 10 mg: tabletten met normale afgifte ▪ Pletbaar ▪ Lichtgevoelig, direct gebruiken ○ MS Contin® 10, 30, 60, 100 mg: tabletten met verlengde afgifte ▪ Niet pletbaar ○ Alternatief: magistrale bereiding: siroop. (8) ○ Geen invloed dan voeding (4, 6) ○ Kan toegediend worden met voedsel of melk om gastro-intestinale irritatie te beperken.(2) Subcutaan: Morfine Amp. IM/IV/SC/IT/Epiduraal 10 mg/mL, 20 mg/mL, 30 mg/mL, 40 mg/mL (off-label) ○ Bolus: / ○ CSCI: ▪ Onverdund of verdund in glucose of fysiologisch aan een maximale toedieningssnelheid van de pomp van 2 mL/uur/site. (9) ▪ In het algemeen is de subcutane dosis equivalent aan de intraveneuze dosis (13) ▪ Patiënten moeten continue respiratoire monitoring hebben en in zich in een setting bevinden die snel handelen bij respiratoire insufficiëntie toelaat. (13) ▪ Verander toedieningsplaats elke 3 tot 7 dagen of wanneer roodheid optreedt. (13) ▪ Om comfort te behouden, de infusiesnelheid van 5 mL/uur niet overschrijden. (13) Enterale sonde ○ Voor onmiddellijke pijnverlichting, gebruik orale oplossing, verdere dilutie is niet noodzakelijk.(6)
GM-GM interacties	Zie interactietabel
Contra-indicaties (10)	○ Overgevoeligheid (werkzame stof of hulpstof) ○ Acute ademhalingsinsufficiëntie. ○ Acute abdominale syndromen van onbekende etiologie. ○ Spasme van de sfincter van Oddi, contractie van de galwegen, spasme van de pylorus, ernstige hepatocellulaire insufficiëntie. ○ Craniale trauma's, intracranieële hypertensie, convulsieve toestanden, coma. ○ Alcoholintoxicatie of acuut alcoholisme, intoxicatie met barbituraten. ○ Niet toedienen als pre-operatieve medicatie aan kinderen jonger dan één (1) jaar, noch aan prematuren.
Compatibiliteiten	Zie compatibiliteitentabel

OCTREOTIDE

Indicatie	Misselijkheid en braken
Specialiteiten op de Belgische markt (10)	Amp. IV/SC 0,1 mg/mL, 0,5 mg/mL
Toedieningsmodaliteiten	Subcutaan: ○ Bolus: ▪ Onverdund, tenzij het injectie dosis volume niet accuraat toegediend kan worden zonder verdere dilutie met 0.9% NaCl. (2,9) ▪ Geen meerdere injecties met korte tussenpozen op dezelfde plek te geven. (13) ▪ De oplossing voor injectie vóór toediening op kamertemperatuur laten komen.(10) ▪ De injecties toedienen tussen de maaltijden in of bij het slapen gaan en steeds op een wisselende plaats. (9) ○ CSCI: ▪ Off-label mogelijk: doses van 300 mcg/dag tot 2000 mcg/dag beschreven (13). Enterale sonde ○ Geen informatie over enterale toediening (6)
GM-GM interacties	Zie interactietabel
Contra-indicaties (10)	Overgevoeligheid (werkzame stof of hulpstof)
Andere	Geen gegevens ASHP

ATROPINE

Indicatie	Doodsreutel
Specialiteiten op de Belgische markt (13)	Atropine Amp. IV/SC 0,25 mg/mL, 0,5 mg/mL, 1 mg/mL
Toedieningsmodaliteiten	Subcutaan: ○ Bolus: ▪ Onverdund (9) ○ CSCI: / Enterale sonde ○ Geen informatie over enterale toediening (6)
GM-GM interacties	Zie interactietabel
Contra-indicaties (15)	○ Overgevoeligheid (werkzame stof of hulpstof) ○ Geslotenkamerhoekglaucoom ○ Risico op urineretentie door een aandoening van de prostaat of urethra ○ Achalasie van de slokdarm, paralytische ileus en toxisch megacolon Al deze contra-indicaties zijn echter niet relevant bij levensbedreigende noodgevallen (zoals bradyaritmie, vergiftiging).
Compatibiliteiten	Zie compatibiliteitentabel

GLYCOPYRRONIUM BROMIDE

Indicatie	Doodsreutel
Specialiteiten op de Belgische markt (10)	Robinul®: Amp. IM/IV 0,2 mg/mL
Toedieningsmodaliteiten	Oraal: ○ Buitenlandse tabletten zijn pletbaar (6). Subcutaan (off-label): Robinul: Amp. IM/IV 0,2 mg/mL ○ Bolus: ▪ 0.2 mg elke 4 - 8 uur zo nodig, niet meer dan 4 dosissen/dag. (13) ○ CSCI: ○ Kan via SC-infusie toegediend worden in de palliatieve zorgen ter vermindering van respiratoire secreties en ter behandeling van darmkolieken. (9) Enterale sonde ○ Buitenlandse tabletten kunnen geplet worden en via enterale weg toegediend worden: tabletten dispergeren in 10 ml water binnen de 5 minuten. NB: risico dat geneesmiddel in de spuit achterblijft. (6)
GM-GM interacties	Zie interactietabel
Contra-indicaties (15)	○ Overgevoeligheid (werkzame stof of hulpstof) ○ Glaucoom ○ Prostaathypertrofie ○ Pylorisstenose
Compatibiliteiten	Zie compatibiliteitentabel

Afkortingen:

CSCI: Continuous Subcutaneous Infusion

3.2 AANVULLING OMTRENT BUITENLANDSE SPECIALITEITEN

Indien een geneesmiddel of een bepaalde toedieningsvorm van dat geneesmiddel niet geregistreerd is in België, kan op de website van het European Medicines Agency (EMA), <https://www.ema.europa.eu/en/medicines>, nagekeken worden of er in andere Europese landen specialiteiten op de markt zijn. De officina-apotheken zijn in de mogelijkheid om, indien beschikbaar, dergelijke buitenlandse specialiteiten in te voeren via internationale apotheken (Bv. Orly Pharma, Eco Pharma Supply). Bij invoeren van geneesmiddelen vanuit het buitenland is een artsenverklaring vereist, terug te vinden op https://baf.be/sites/default/files/2016-12/Invoer-buitenlandse-geneesmiddelen_artsenverklaring.pdf. Voor buitenlandse specialiteiten is er geen terugbetaling door het RIZIV.

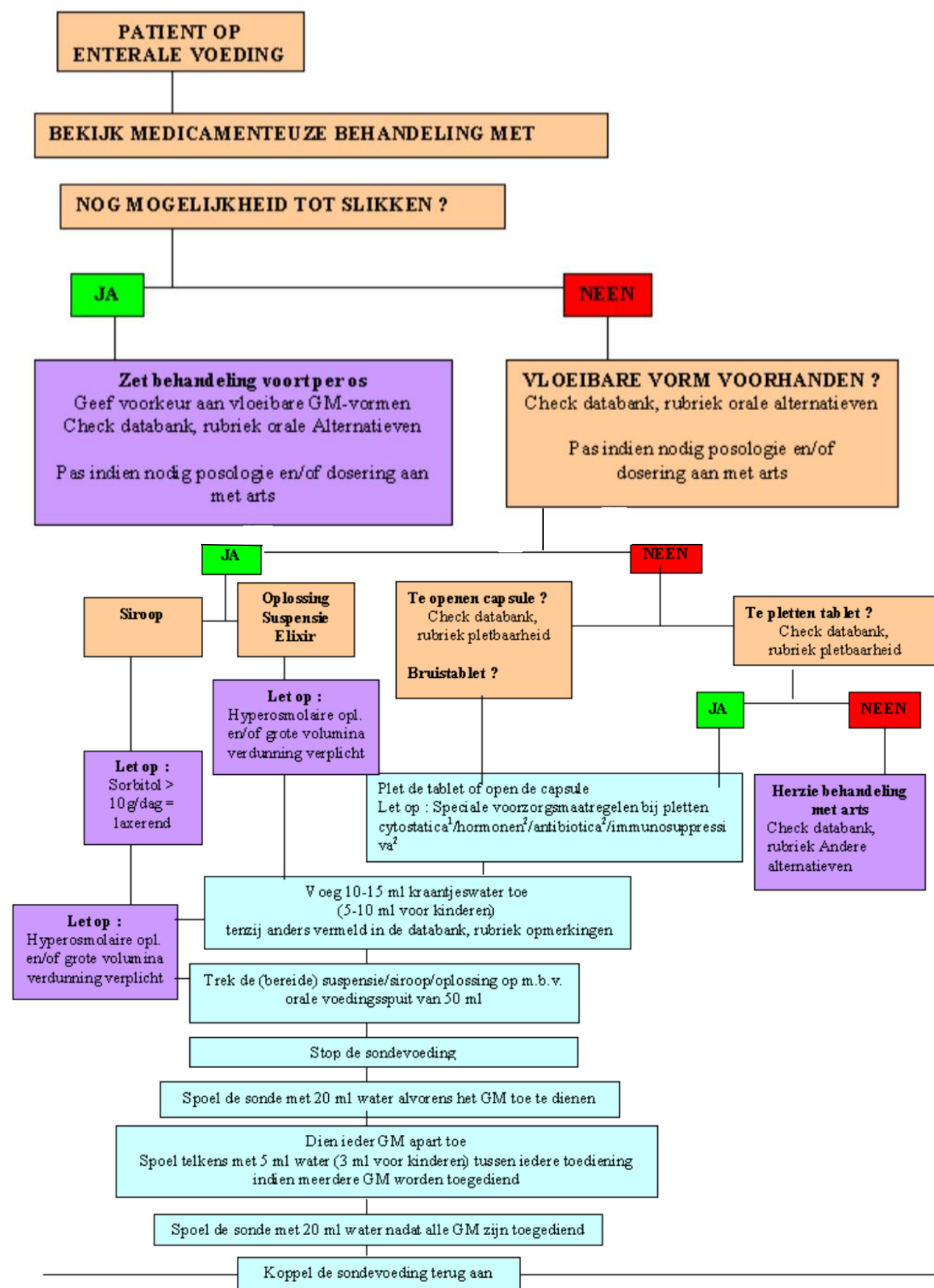
3.3 AANVULLING OMTRENT TOEDIENING VIA ENTERALE SONDE

Aandachtspunten bij het pletten van medicatie:

- Maak steeds gebruik van informatiebronnen zoals (www.pletmedicatie.be) om na te gaan of het geneesmiddel geplet kan/mag worden.
 - Tabletten, dragées: pletten/ vaak valt de tablet uiteen in water
 - Harde capsules: openen, poeder dispergeren in water
 - Bruistabletten: voor toediening dispergeren in water, goed roeren om alle gas te verwijderen (risico op reflux)
 - Oplostabletten: vallen uiteen in de mond door het speeksel, eventueel voor toediening dispergeren in water
 - Sublinguale tabletten: absorptie via slijmvliezen onder de tong, NIET via sonde (onderdosering)
 - Geneesmiddelen met een gecontroleerde vrijstelling of omhulde geneesmiddelen: NIET pletten!

Vijf gouden regels bij het pletten van medicatie

1. Vloeibare geneesmiddelenvormen hebben voorrang op het verpulveren van tabletten of het openen van capsules.
2. Verpulver elk geneesmiddel apart in een crusher of een mortier.
3. Draag handschoenen en mondmasker bij het verpulveren van antibiotica, antivirale en hormonale geneesmiddelen.
4. Verpulveren van cytostatica is een apotheekbereiding.
5. Vermeld 'via sonde' of 'pletten' op het geneesmiddelenvoorschrift.



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